

fong-spivak-seven-sketches — formula report

4029 inline formulas (1 shown: the ones that did not render), 264 display equations, 63 tables, 503 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches_110 EQ0061 (3.61) [conf 0.008]	110	■ 0.008 ● C +6	I=\left[\begin{array}{ccc} 1 & \underset{\bullet}{a} & \underset{\bullet}{b}d \\ & \downarrow & \\ & \downarrow & \bullet \end{array}\right].	$I = \begin{bmatrix} 1 & a & \bullet^2 b d \\ & \downarrow & \\ & \downarrow & \bullet \end{bmatrix}$	$I = \begin{array}{c} \begin{array}{ccccc} 1 & \xrightarrow{a} & 2 & \curvearrowright d & \\ & \searrow b & \downarrow e & & \\ & & 3 & & 4 \end{array} \end{array}$
fong-spivak-seven-sketches_111 EQ0063 [conf 0.014]	111	■ 0.014 ● C +4	G:=\stackrel{1}{\bullet} \stackrel{a}{\longrightarrow} \stackrel{2}{\bullet} \stackrel{b}{\longrightarrow} \stackrel{3}{\bullet} \quad H:=\stackrel{4}{\bullet} \stackrel{d}{\longrightarrow} \stackrel{5}{\bullet} \stackrel{e}{\curvearrowright} \text{Pe}	$G := \overset{1}{\bullet} \xrightarrow{a} \overset{2}{\bullet} \xrightarrow{b} \overset{3}{\bullet} \quad H := \overset{4}{\bullet} \xrightarrow[d]{d} \overset{5}{\bullet} \curvearrowright \text{Pe}$	$G := \begin{array}{c} \begin{array}{ccccc} 1 & \xrightarrow{a} & 2 & \xrightarrow{b} & 3 \end{array} \end{array} \quad H := \begin{array}{c} \begin{array}{ccccc} 4 & \xrightarrow[d]{d} & 5 & \curvearrowright e & \end{array} \end{array}$
fong-spivak-seven-sketches_214 EQ0133 [conf 0.025]	214	■ 0.025 ● W +1	\begin{aligned} \mu_X &:= \left(X + X \xrightarrow{[\text{id}_X, \text{id}_X]} X \xleftarrow{\text{id}_X} X \right) \\ \eta_X &:= \left(\emptyset \xrightarrow{!_X} X \xleftarrow{\text{id}_X} X \right) \\ \delta_X &:= \left(X \xrightarrow{\text{id}_X} X \xleftarrow{[\text{id}_X, \text{id}_X]} X + X \right) \\ \epsilon_X &:= \left(X \xrightarrow{\text{id}_X} X \xleftarrow{!_X} \emptyset \right) \end{aligned}	$\begin{aligned} \mu_X &:= \left(X + X \xrightarrow{[\text{id}_X, \text{id}_X]} X \longleftrightarrow \xrightarrow{id}_X X \right) \\ \eta_X &:= \left(\emptyset \xrightarrow{!_X} X \xleftarrow{id_X} X \right) \\ \delta_X &:= \left(X \xrightarrow{id_X} X \xleftarrow{[\text{id}_X, \text{id}_X]} X + X \right) \\ \epsilon_X &:= \left(X \xrightarrow{id_X} X \longleftrightarrow !_X \emptyset \right) \end{aligned}$	$\begin{aligned} \mu_X &:= (X + X \xrightarrow{[\text{id}_X, \text{id}_X]} X \xleftarrow{\text{id}_X} X) \\ \eta_X &:= (\emptyset \xrightarrow{!_X} X \xleftarrow{\text{id}_X} X) \\ \delta_X &:= (X \xrightarrow{\text{id}_X} X \xleftarrow{[\text{id}_X, \text{id}_X]} X + X) \\ \epsilon_X &:= (X \xrightarrow{\text{id}_X} X \xleftarrow{!_X} \emptyset) \end{aligned}$
fong-spivak-seven-sketches_279 EQ0188 [conf 0.038]	279	■ 0.038 ● S +2	g_{!}(\left(c_3\right))=\{ \}^{\{12\}}\{ \}_{\downarrow}\{ \}^{\{3\}} \stackrel{4}{\bullet}	$g!(c_3) = \overset{12}{\downarrow} \overset{3}{\downarrow} \overset{4}{\bullet}$	$g!(c_3) = \begin{array}{c} \begin{array}{ccc} 12 & 3 & 4 \\ \downarrow & & \downarrow \\ \bullet & & \bullet \end{array} \end{array}$
fong-spivak-seven-sketches_274 EQ0181 [conf 0.044]	274	■ 0.044 ● C +6	G=\left[\begin{array}{ccc} 1 & \underset{\bullet}{a} & \underset{\bullet}{b} \\ & \downarrow & \\ & \downarrow & \\ & \downarrow & \\ & \downarrow & \end{array}\right]	$G = \begin{vmatrix} 1 & a & \bullet^2 \\ & \downarrow & \downarrow \\ & \downarrow & \\ & \downarrow & \\ & \downarrow & \end{vmatrix}$	$G = \begin{array}{c} \begin{array}{ccccc} 1 & \xrightarrow{a} & 2 & \curvearrowright d & \\ & \searrow b & \downarrow e & & \\ & & 3 & & 4 \end{array} \end{array}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches EQ0186 [conf 0.050]	279	■ 0.050 ●	$g!\left(c_{1}\right)={ }^{12} 3{ }^{4}$	$g!\left(c_{1}\right)={ }^{12} 3{ }^4$	$g!(c_1) = \begin{array}{c} 12 \quad 3 \quad 4 \\ \swarrow \downarrow \searrow \\ \bullet \end{array}$
fong-spivak-seven-sketches EQ0189	279	■ 0.108 ● W +0	$g_{!}\left(c_{4}\right)=\operatorname{amalg}_{\bullet}^{\{12\}} I^{\{3\}} \stackrel{\downarrow}{\operatorname{stackrel{4}{\downarrow}}}$	$g!(c_4) = \Pi_{\bullet}^{12} I^3 \downarrow$	$g!(c_4) = \begin{array}{c} 12 \quad 3 \quad 4 \\ \swarrow \searrow \downarrow \\ \bullet \quad \bullet \end{array}$
fong-spivak-seven-sketches EQ0073 (3.77)	118	■ 0.114 ● U +40	$\mathcal{G}:=\left \stackrel{\text { Email }}{\underset{\text { received_by }}{\stackrel{\text { sent_by }}{\rightleftharpoons}}}\right \text { Address } \bullet \bullet$	(not rendered)	$\mathcal{G}:= \begin{array}{c} \text{Email} \xrightarrow{\text{sent_by}} \text{Address} \\ \xleftarrow{\text{received_by}} \\ \text{no equations} \end{array}$
fong-spivak-seven-sketches EQ0245	311	■ 0.129 ● C -7	$a_{\{m\}}:=\left(1+\cdots+1+\cdots\right) ; \cdots ;\left(1+1+\supset-\right) ;\left(1+\cdots\right) ; \cdots: m \rightarrow 1$	$a_m := (1 + \cdots + 1 + \cdots); \cdots; (1 + 1 + \supset -); (1 + \cdots); \cdots: m \rightarrow 1$	$a_m := (1 + \cdots + 1 + \supset -) \circ \cdots \circ (1 + 1 + \supset -) \circ (1 + \supset -) \circ \supset -: m \rightarrow 1$
fong-spivak-seven-sketches EQ0083	136	■ 0.147 ● N -1	$\text { Chassis: } \left(\text { load } \times \text { velocity }\right) \rightarrow\left(\text { torque } \times \text { speed } \times \$\right)$	Chassis: (load × velocity) → (torque × speed × \$)	Chassis: (load × velocity) → (torque × speed × \$)
fong-spivak-seven-sketches EQ0205	290	■ 0.147 ● C -7	$A, \quad A \div f, \quad A \div g, \quad A \div f \div h, \quad A \div g \text { i } i, \quad B, \quad B \div h, \quad C, \quad C: i, \quad D$	$A, \quad A \div f, \quad A \div g, \quad A \div f \div h, \quad A \div g \text { i } i, \quad B, \quad B \div h, \quad C, \quad C: i, \quad D$	$A, \quad A \circ f, \quad A \circ g, \quad A \circ f \circ h, \quad A \circ g \circ i, \quad B, \quad B \circ h, \quad C, \quad C \circ i, \quad D$
fong-spivak-seven-sketches EQ0064 (3.65)	112	■ 0.147 ● C +31	$\mathbf{G r}:=\stackrel{\text { Arrow }}{\underset{\text { target }}{\stackrel{\text { source }}{\rightarrow}}}\stackrel{\text { Vertex. }}{\rightleftarrows} \text { no equations}$	$\mathbf{Gr} := \begin{array}{c} \text{Arrow} \xrightarrow{\text{source}} \text{Vertex} \\ \xleftarrow{\text{target}} \text{no equations} \end{array}$	$\mathbf{Gr} := \begin{array}{c} \text{Arrow} \xrightarrow{\text{source}} \text{Vertex} \\ \xleftarrow{\text{target}} \\ \text{no equations} \end{array} \quad \mathbf{DDS} := \begin{array}{c} \text{State} \\ \bullet \\ \updownarrow \\ \text{next} \\ \text{no equations} \end{array}$
fong-spivak-seven-sketches EQ0214	294	■ 0.168 ● N +0	$H:=\stackrel{4}{\rightarrow} \stackrel{d}{\rightarrow} \stackrel{5}{\bullet} \triangleright e$	$H := \xrightarrow{4} \xrightarrow[d]{5} \bullet \triangleright e$	$H := \left \begin{array}{c} 4 \xrightarrow[d]{5} \bullet \triangleright e \end{array} \right $
fong-spivak-seven-sketches EQ0115	182	■ 0.203 ● C +4	$G_{\{R\}}:=\left\{\supset-, \circ-, \subset, \rightarrow\right\} \cup\left\{-a- \mid a \in R\right\},$	$G_R := \{\supset -, \circ -, \subset, \rightarrow\} \cup \{-a- \mid a \in R\},$	$G_R := \left\{\supset -, \circ -, \subset, \rightarrow\right\} \cup\left\{-\boxed{a}- \mid a \in R\right\},$
fong-spivak-seven-sketches EQ0187	279	■ 0.215 ● N +0	$g_{!}\left(c_{2}\right)=\Psi_{\bullet}^{\{12\}} J^{\{3\}} 4^{\{4\}}$	$g!(c_2) = \Psi_{\bullet}^{12} J^3 4^4$	$g!(c_2) = \begin{array}{c} 12 \quad 3 \quad 4 \\ \downarrow \swarrow \searrow \\ \bullet \quad \bullet \end{array}$
fong-spivak-seven-sketches EQ0094	148	■ 0.259 ● S +0	$\begin{array}{l} \& S \times(T \times U):=\{(s,(t, u)) \\ \mid s \in S, t \in T, u \in U\} \\ \& (S \times T) \times U:=\{((s, t), u) \mid s \in S, t \in T, u \in U\}. \end{array}$	$S \times(T \times U):=\{(s,(t, u)) \mid s \in S, t \in T, u \in U\}$ $=? \quad (S \times T) \times U:=\{((s, t), u) \mid s \in S, t \in T, u \in U\}.$	$S \times(T \times U):=\{(s,(t, u)) \mid s \in S, t \in T, u \in U\}$ $=? \quad (S \times T) \times U:=\{((s, t), u) \mid s \in S, t \in T, u \in U\}.$
fong-spivak-seven-sketches EQ0072	118	■ 0.263 ● K +0	$\Sigma_{!}: \mathcal{C}-\mathbf{I n s t} \rightarrow \mathbf{S e t} \quad \text { and } \quad \Pi_{!}: \mathcal{C}-\mathbf{I n s t} \rightarrow \mathbf{S e t}.$	$\Sigma_{!}: \mathcal{C}-\mathbf{I n s t} \rightarrow \mathbf{S e t} \quad \text { and } \quad \Pi_{!}: \mathcal{C}-\mathbf{I n s t} \rightarrow \mathbf{S e t}.$	$\Sigma_{!}: \mathcal{C}\text{-}\mathbf{Inst} \rightarrow \mathbf{Set} \quad \text { and } \quad \Pi_{!}: \mathcal{C}\text{-}\mathbf{Inst} \rightarrow \mathbf{Set}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches_0 E00020 (2.10)	055	■ 0.266 ● W +2	$\begin{array}{ll} h_{\{1\}}:=\{2 \diamond, 3 \diamond, 10 \bullet, J \bullet, Q \bullet\} & i_{\{1\}}:=\{4 \diamond, 4 \diamond, 6 \diamond, 6 \diamond, 10 \diamond\} \\ \bullet, J \bullet, Q \bullet & \\ h_{\{2\}}:=\{2 \diamond, 3 \diamond, 4 \diamond, K \star, A \diamond\} & i_{\{2\}}:=\{5 \diamond, 5 \diamond, 7 \diamond, J \diamond, Q \diamond\}. \end{array}$	$\begin{array}{ll} h_1 := \{2\Diamond, 3\Diamond, 10\bullet, J\bullet, Q\bullet\} & i_1 := \{4\Diamond, 4\Diamond, 6\Diamond, 6\Diamond, 10\Diamond\} \\ h_2 := \{2\Diamond, 3\Diamond, 4\Diamond, K\star, A\Diamond\} & i_2 := \{5\Diamond, 5\Diamond, 7\Diamond, J\Diamond, Q\Diamond\}. \end{array}$	$\begin{array}{ll} h_1 := \{2\heartsuit, 3\heartsuit, 10\spadesuit, J\spadesuit, Q\spadesuit\} & i_1 := \{4\clubsuit, 4\clubsuit, 6\heartsuit, 6\Diamond, 10\Diamond\} \\ h_2 := \{2\Diamond, 3\Diamond, 4\Diamond, K\spadesuit, A\spadesuit\} & i_2 := \{5\clubsuit, 5\heartsuit, 7\heartsuit, J\Diamond, Q\Diamond\}. \end{array}$
fong-spivak-seven-sketches_0 E00033	073	■ 0.268 ●	$X:=\operatorname{Ob}(\mathfrak{X}) \quad d(x, y):=\mathfrak{X}(x, y) .$	$X := \operatorname{Ob}(\mathfrak{X}) \quad d(x, y) := \mathfrak{X}(x, y).$	$X := \operatorname{Ob}(\mathfrak{X}) \quad d(x, y) := \mathfrak{X}(x, y).$
fong-spivak-seven-sketches_0 E00175 (7.82)	268	■ 0.278 ● S +2	$\begin{array}{l} \forall (t: \mathbb{R}) \cdot @_t(\text{bad_pos}(D)) \Rightarrow \\ \exists (r: \mathbb{R}) \cdot (0 < r < 1) \wedge \forall (r': \mathbb{R}) \cdot 0 \leq r' \leq 5 \Rightarrow @_{t+r+r'}(\text{engaged}(T)). \end{array}$	$\forall (t: \mathbb{R}) \cdot @_t(\text{bad_pos}(D)) \Rightarrow \\ \exists (r: \mathbb{R}) \cdot (0 < r < 1) \wedge \forall (r': \mathbb{R}) \cdot 0 \leq r' \leq 5 \Rightarrow @_{t+r+r'}(\text{engaged}(T)).$	$\forall (t: \mathbb{R}). @_t(\text{bad_pos}(D)) \Rightarrow \\ \exists (r: \mathbb{R}). (0 < r < 1) \wedge \forall (r': \mathbb{R}). 0 \leq r' \leq 5 \Rightarrow @_{t+r+r'}(\text{engaged}(T)).$
fong-spivak-seven-sketches_0 E00057	097	■ 0.293 ● C -3	$\mathcal{D}:=\left[\begin{array}{c} s \\ \hat{L} \\ b \\ z \end{array}\right] \quad \text{is } s = s \text{ is}$	$\mathcal{D}:=\left[\begin{array}{c} s \\ \hat{L} \\ b \\ z \end{array}\right] \quad \text{is } s = s \text{ is}$	$\mathcal{D} := \begin{array}{c} s \\ \Downarrow \\ \bullet \\ z \\ s \circ s \circ s \circ s \circ s = s \circ s \end{array}$
fong-spivak-seven-sketches_0 E00184	274	■ 0.304 ● C +18	$\begin{array}{l} (\bullet)(\circ)(*) \leq (\bullet)(\circ)(*) \quad (\bullet)(\circ)(*) \leq (\bullet\circ)(*) \\ (\bullet)(\circ)(*) \leq (\bullet)(\circ*) \quad (\bullet)(\circ)(*) \leq (\bullet\circ*) \\ (\bullet\circ)(*) \leq (\bullet\circ*) \quad (\bullet*)(\circ) \leq (\bullet*)(\circ) \\ (\bullet)(\circ*) \leq (\bullet)(\circ*) \quad (\bullet)(\circ*) \leq (\bullet\circ*) \\ (\bullet\circ)(*) \leq (\bullet\circ*) \quad (\bullet*)(\circ) \leq (\bullet\circ*) \\ (\bullet\circ*) \leq (\bullet\circ*) \end{array}$	$\begin{array}{lll} (\bullet)(\circ)(*) \leq (\bullet)(\circ)(*) & (\bullet)(\circ)(*) \leq (\bullet\circ)(*) & (\bullet)(\circ)(*) \leq (\bullet*)(\circ) \\ (\bullet)(\circ)(*) \leq (\bullet)(\circ*) & (\bullet)(\circ)(*) \leq (\bullet\circ*) & (\bullet\circ)(*) \leq (\bullet\circ)(*) \\ (\bullet\circ)(*) \leq (\bullet\circ*) & (\bullet*)(\circ) \leq (\bullet*)(\circ) & (\bullet*)(\circ) \leq (\bullet\circ*) \\ (\bullet)(\circ*) \leq (\bullet)(\circ*) & (\bullet)(\circ*) \leq (\bullet\circ*) & (\bullet\circ*) \leq (\bullet\circ*) \end{array}$	$\begin{array}{lll} (\bullet)(\circ)(*) \leq (\bullet)(\circ)(*) & (\bullet)(\circ)(*) \leq (\bullet\circ)(*) & (\bullet)(\circ)(*) \leq (\bullet*)(\circ) \\ (\bullet)(\circ)(*) \leq (\bullet)(\circ*) & (\bullet)(\circ)(*) \leq (\bullet\circ*) & (\bullet\circ)(*) \leq (\bullet\circ)(*) \\ (\bullet\circ)(*) \leq (\bullet\circ*) & (\bullet*)(\circ) \leq (\bullet*)(\circ) & (\bullet*)(\circ) \leq (\bullet\circ*) \\ (\bullet)(\circ*) \leq (\bullet)(\circ*) & (\bullet)(\circ*) \leq (\bullet\circ*) & (\bullet\circ*) \leq (\bullet\circ*) \end{array}$
fong-spivak-seven-sketches_0 E00056	097	■ 0.330 ● N +1	$e:=\begin{array}{ c } \hline s \\ \mathcal{Q} \\ z \\ s; s = z \\ \hline \end{array}$	$e := \begin{array}{ c } \hline s \\ \mathcal{Q} \\ z \\ s; s = z \\ \hline \end{array}$	$\mathcal{E} := \begin{array}{ c } \hline s \\ \Downarrow \\ \bullet \\ z \\ s \circ s = z \\ \hline \end{array}$
fong-spivak-seven-sketches_0 E00109 (5.47)	175	■ 0.331 ● N -1	$G_{\{R\}}:=\{\supset-, \circ-, \subset, \longrightarrow\} \cup \{-a- \mid a \in R\},$	$G_R := \{\supset -, \circ -, \subset, \longrightarrow\} \cup \{-a- \mid a \in R\},$	$G_R := \left\{ \supset -, \circ -, \subset, \longrightarrow \right\} \cup \left\{ -a- \mid a \in R \right\},$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches_106 EQ0059 (3.47)	106	0.333 N +1	$C:=\begin{array}{ c }\hline s\\ \mathcal{Q}\\ b\\ \hline s;s=s\end{array}$	$C := \begin{array}{ c }\hline s\\ \mathcal{Q}\\ b\\ \hline s;s=s\end{array}$	
fong-spivak-seven-sketches_178 EQ0113	178	0.335 N -1	$\text{-}\mapsto(1)\quad\text{and}\quad X\mapsto\left(\begin{array}{cc}0&1\\1&0\end{array}\right)\quad\text{and}\quad\text{-}a\text{-}\mapsto(a)$	$\text{-}\mapsto(1)\quad\text{and}\quad X\mapsto\left(\begin{array}{cc}0&1\\1&0\end{array}\right)\quad\text{and}\quad\text{-}a\text{-}\mapsto(a)$	$\text{---} \mapsto \begin{pmatrix} 1 \\ \end{pmatrix} \quad \text{and} \quad \text{X} \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \text{---} \boxed{a} \text{---} \mapsto \begin{pmatrix} a \\ \end{pmatrix}$
fong-spivak-seven-sketches_303 EQ0226	303	0.335	$\mathcal{X}(x,z)=\mathcal{X}(x,z)\otimes I\leq \mathcal{X}(x,z)\otimes \mathcal{X}(z,z)\leq \bigvee_{y\in\mathcal{X}}\mathcal{X}(x,y)\otimes \mathcal{X}(y,z),$	$\mathcal{X}(x,z)=\mathcal{X}(x,z)\otimes I\leq \mathcal{X}(x,z)\otimes \mathcal{X}(z,z)\leq \bigvee_{y\in\mathcal{X}}\mathcal{X}(x,y)\otimes \mathcal{X}(y,z),$	$\mathcal{X}(x,z)=\mathcal{X}(x,z)\otimes I\leq \mathcal{X}(x,z)\otimes \mathcal{X}(z,z)\leq \bigvee_{y\in\mathcal{X}}\mathcal{X}(x,y)\otimes \mathcal{X}(y,z),$
fong-spivak-seven-sketches_115 EQ0068	115	0.352 S +2	$\mathcal{C}\underset{\mathcal{R}}{\overset{\mathcal{L}}{\rightleftarrows}}\mathcal{D}$	$\mathcal{C}\underset{\mathcal{R}}{\overset{\mathcal{L}}{\rightleftarrows}}\mathcal{D}$	
fong-spivak-seven-sketches_290 EQ0206	290	0.362 S +0	$A,\quad A;f,\quad A;g,\quad A;j,\quad B,\quad B;h,\quad C,\quad C;i,\quad D$	$A,\quad A;f,\quad A;g,\quad A;j,\quad B,\quad B;h,\quad C,\quad C;i,\quad D$	$A,\quad A\circ f,\quad A\circ g,\quad A\circ j,\quad B,\quad B\circ h,\quad C,\quad C\circ i,\quad D$
fong-spivak-seven-sketches_156 EQ0098	156	0.366 N +0	$(x\times y)\left((x,y),\left(x^{\prime},y^{\prime}\right)\right):=x\left(x,x^{\prime}\right)\otimes y\left(y,y^{\prime}\right).$	$(x\times y)\left((x,y),\left(x^{\prime},y^{\prime}\right)\right):=x\left(x,x^{\prime}\right)\otimes y\left(y,y^{\prime}\right).$	$(\mathcal{X}\times\mathcal{Y})((x,y),(x',y')):=\mathcal{X}(x,x')\otimes\mathcal{Y}(y,y').$
fong-spivak-seven-sketches_077 EQ0036	077	0.376 W -2	$\begin{array}{l} \mathcal{C}_f(c,d)\otimes_W\mathcal{C}_f(d,e)=f(\mathcal{C}(c,d))\otimes_Wf(\mathcal{C}(d,e))\\ \leq f(\mathcal{C}(c,d)\otimes_V\mathcal{C}(d,e))\\ \leq f(\mathcal{C}(c,e))\\ =\mathcal{C}_f(c,e) \end{array}$	$\begin{array}{l} \mathcal{C}_f(c,d)\otimes_W\mathcal{C}_f(d,e)=f(\mathcal{C}(c,d))\otimes_Wf(\mathcal{C}(d,e))\\ \leq f(\mathcal{C}(c,d)\otimes_V\mathcal{C}(d,e))\\ \leq f(\mathcal{C}(c,e))\\ =\mathcal{C}_f(c,e) \end{array}$	$\begin{array}{l} \mathcal{C}_f(c,d)\otimes_W\mathcal{C}_f(d,e)=f(\mathcal{C}(c,d))\otimes_Wf(\mathcal{C}(d,e))\\ \leq f(\mathcal{C}(c,d)\otimes_V\mathcal{C}(d,e))\\ \leq f(\mathcal{C}(c,e))\\ =\mathcal{C}_f(c,e) \end{array}$
fong-spivak-seven-sketches_227 EQ0142	227	0.424 N +2	$(g\circ f)(x_1,\dots,x_{i-1},w_1,\dots,w_m,x_{i+1},\dots,x_n)=g(x_1,\dots,x_{i-1},f(w_1,\dots,w_m),x_{i+1},\dots,x_n).$	$(g\circ f)(x_1,\dots,x_{i-1},w_1,\dots,w_m,x_{i+1},\dots,x_n)=g(x_1,\dots,x_{i-1},f(w_1,\dots,w_m),x_{i+1},\dots,x_n).$	$(g\circ_i f)(x_1,\dots,x_{i-1},w_1,\dots,w_m,x_{i+1},\dots,x_n)=g(x_1,\dots,x_{i-1},f(w_1,\dots,w_m),x_{i+1},\dots,x_n).$
fong-spivak-seven-sketches_303 EQ0228	303	0.461 N +0	$\left(\alpha^{-1};\alpha\right)(x,z)=\bigvee_{(y,1)\in\mathcal{X}\times\mathbf{1}}\alpha(x,(y,1))\otimes\alpha^{-1}((y,1),z)=\bigvee_{y\in\mathcal{X}}x(x,y)\otimes x(y,z)=x(x,z).$	$\left(\alpha^{-1};\alpha\right)(x,z)=\bigvee_{(y,1)\in\mathcal{X}\times\mathbf{1}}\alpha(x,(y,1))\otimes\alpha^{-1}((y,1),z)=\bigvee_{y\in\mathcal{X}}x(x,y)\otimes x(y,z)=x(x,z).$	$(\alpha^{-1}\circ\alpha)(x,z)=\bigvee_{(y,1)\in\mathcal{X}\times\mathbf{1}}\alpha(x,(y,1))\otimes\alpha^{-1}((y,1),z)=\bigvee_{y\in\mathcal{X}}\mathcal{X}(x,y)\otimes\mathcal{X}(y,z)=\mathcal{X}(x,z).$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches EQ0227	303	0.463 N -0	$\bigvee_{y \in \mathcal{X}} x(x, y) \otimes \mathcal{X}(y, z) \leq \bigvee_{y \in \mathcal{X}} x(x, z) = \mathcal{X}(x, z).$	$\bigvee_{y \in \mathcal{X}} x(x, y) \otimes \mathcal{X}(y, z) \leq \bigvee_{y \in \mathcal{X}} x(x, z) = \mathcal{X}(x, z).$	$\bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, z) = \mathcal{X}(x, z).$
fong-spivak-seven-sketches EQ0071 (3.75)	118	0.470 K +0	$! : \mathcal{C} \rightarrow \mathbf{1}.$	$! : \mathcal{C} \rightarrow \mathbf{1}.$	$! : \mathcal{C} \rightarrow \mathbf{1}.$
fong-spivak-seven-sketches EQ0039 (2.77)	079	0.472 N +0	$(\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_2, y_2)) \otimes (\mathcal{X} \times \mathcal{Y})((x_2, y_2), (x_3, y_3)) \leq (\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_3, y_3)).$	$(\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_2, y_2)) \otimes (\mathcal{X} \times \mathcal{Y})((x_2, y_2), (x_3, y_3)) \leq (\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_3, y_3)).$	$(\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_2, y_2)) \otimes (\mathcal{X} \times \mathcal{Y})((x_2, y_2), (x_3, y_3)) \leq (\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_3, y_3)).$
fong-spivak-seven-sketches EQ0163	257	0.487	$\Omega_{\text{Grph}} = (0, 0; 0) \subset 0 \underset{\{(V, 0; 0)\} \stackrel{\{(0, V; 0)\}}{\longrightarrow} \{(V, V; A)\}^{\{(V, V; 0)\}}}{\underset{\{(V, 0; 0)\}}{\overset{\{(0, V; 0)\}}{\longrightarrow}}} 0$	$\Omega_{\text{Grph}} = (0, 0; 0) \subset 0 \overset{(0, V; 0)}{\underset{(V, 0; 0)}{\longleftarrow}} V \overset{(V, V; 0)}{\underset{(V, V; A)}{\longrightarrow}}$	$\Omega_{\text{Grph}} = (0, 0; 0) \overset{(0, V; 0)}{\underset{(V, 0; 0)}{\longleftarrow}} V \overset{(V, V; 0)}{\underset{(V, V; A)}{\longrightarrow}}$
fong-spivak-seven-sketches EQ0258	317	0.494 N -1	$\left[\text{id}_A, !_A \right] \stackrel{\circ}{\circ} \iota_A = \left[\text{id}_A \stackrel{\circ}{\circ} \iota_A, !_A \stackrel{\circ}{\circ} \iota_A \right] = [\iota_A, !_{A+\emptyset}] = [\iota_A, \iota_{\emptyset}] = \text{id}_{A+\emptyset}.$	$[\text{id}_A, !_A] \stackrel{\circ}{\circ} \iota_A = [\text{id}_A \stackrel{\circ}{\circ} \iota_A, !_A \stackrel{\circ}{\circ} \iota_A] = [\iota_A, !_{A+\emptyset}] = [\iota_A, \iota_{\emptyset}] = \text{id}_{A+\emptyset}.$	$[\text{id}_A, !_A] \stackrel{\circ}{\circ} \iota_A = [\text{id}_A \stackrel{\circ}{\circ} \iota_A, !_A \stackrel{\circ}{\circ} \iota_A] = [\iota_A, !_{A+\emptyset}] = [\iota_A, \iota_{\emptyset}] = \text{id}_{A+\emptyset}.$
fong-spivak-seven-sketches EQ0074	119	0.500 W -1	$\begin{array}{ c } \hline 1 \\ \hline \text{Bob-Grace-Pat-Emmy} \\ \hline \text{Sue-Doug} \end{array}$	$\begin{array}{ c } \hline 1 \\ \hline \text{Bob-Grace-Pat-Emmy} \\ \hline \text{Sue-Doug} \end{array}$	$\begin{array}{ c } \hline 1 \\ \hline \text{Bob-Grace-Pat-Emmy} \\ \hline \text{Sue-Doug} \end{array}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches_1 EQ0223	299	■ 0.512 ● W +0	$\begin{aligned} \Phi = & M_X^3 * M_\Phi * M_Y^2 = \\ & \left(\begin{array}{cccc} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{array} \right) \left(\begin{array}{ccc} \infty & \infty & \infty \\ 11 & \infty & \infty \\ \infty & \infty & \infty \\ \infty & 9 & \infty \end{array} \right) \left(\begin{array}{ccc} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{array} \right) \\ & = \left(\begin{array}{ccc} 17 & 20 & \infty \\ 11 & 14 & \infty \\ 14 & 17 & \infty \\ 20 & 9 & \infty \end{array} \right) \left(\begin{array}{ccc} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{array} \right) \\ & = \left(\begin{array}{ccc} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{array} \right) \end{aligned}$	$\Phi = M_X^3 * M_\Phi * M_Y^2 = \left(\begin{array}{cccc} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{array} \right) \left(\begin{array}{ccc} \infty & \infty & \infty \\ 11 & \infty & \infty \\ \infty & \infty & \infty \\ \infty & 9 & \infty \end{array} \right) \left(\begin{array}{ccc} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{array} \right)$ $= \left(\begin{array}{ccc} 17 & 20 & \infty \\ 11 & 14 & \infty \\ 14 & 17 & \infty \\ 20 & 9 & \infty \end{array} \right) \left(\begin{array}{ccc} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{array} \right)$ $= \left(\begin{array}{ccc} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{array} \right)$	
fong-spivak-seven-sketches_1 EQ0021	058	■ 0.515 ● C +3	$\left. \left. \left. \frac{\bar{x}}{y} \right\} = \frac{x \otimes y}{z} \right\} = \left\{ \frac{x}{y \otimes z} = \left\{ \frac{y}{z} \right. \right. \right.$	$\left. \left. \left. \frac{\bar{x}}{y} \right\} = \frac{x \otimes y}{z} \right\} = \left\{ \frac{x}{y \otimes z} = \left\{ \frac{y}{z} \right. \right. \right.$	$\left. \left. \left. \frac{x}{y} \right\} = \frac{x \otimes y}{z} \right\} = \left\{ \frac{x}{y \otimes z} = \left\{ \frac{y}{z} \right. \right. \right.$
fong-spivak-seven-sketches_1 EQ0069	115	■ 0.520 ● W +2	$\text{Set} \stackrel{-\times B}{\Rightarrow} \text{Set} \quad \text{Set}(A \times B, C) \cong \text{Set}(A, C^B)$	$\text{Set} \stackrel{-\times B}{\Rightarrow} \text{Set} \quad \text{Set}(A \times B, C) \cong \text{Set}(A, C^B)$	$\text{Set} \stackrel{-\times B}{\Rightarrow} \text{Set} \quad \text{Set}(A \times B, C) \cong \text{Set}(A, C^B)$
fong-spivak-seven-sketches_1 EQ0116	186	■ 0.520 ●	$\mathrm{B}(-\mathrm{C}) = \{(x, (x, x)) \mid x \in R\}.$	$\mathrm{B}(-\mathrm{C}) = \{(x, (x, x)) \mid x \in R\}.$	$\mathrm{B}(-\mathrm{C}) = \{(x, (x, x)) \mid x \in R\}.$
fong-spivak-seven-sketches_1 EQ0257	317	■ 0.522 ● N +0	$\mathrm{id}_A + \mathrm{id}_B = [\mathrm{id}_A \circ \iota_A, \mathrm{id}_B \circ \iota_B] = [\iota_A, \iota_B] = \mathrm{id}_{A+B}.$	$\mathrm{id}_A + \mathrm{id}_B = [\mathrm{id}_A \circ \iota_A, \mathrm{id}_B \circ \iota_B] = [\iota_A, \iota_B] = \mathrm{id}_{A+B}.$	$\mathrm{id}_A + \mathrm{id}_B = [\mathrm{id}_A \circ \iota_A, \mathrm{id}_B \circ \iota_B] = [\iota_A, \iota_B] = \mathrm{id}_{A+B}.$
fong-spivak-seven-sketches_1 EQ0054 (3.11)	095	■ 0.526 ● C +43	$\mathbf{3} := \text{Free} \left(\bullet \xrightarrow{f_1} \bullet \xrightarrow{f_2} \bullet \right)$	$\mathbf{3} := \text{Free} \left(\bullet \xrightarrow{f_1} \bullet \xrightarrow{f_2} \bullet \right)$	$\mathbf{3} := \text{Free} \left(\begin{array}{ccccc} v_1 & & f_1 & & v_2 & & f_2 & & v_3 \end{array} \right)$
fong-spivak-seven-sketches_1 EQ0005	033	■ 0.539 ● C +4	$p = g(f(p)) \quad \text{and} \quad q = f(g(q)).$	$p = g(f(p)) \quad \text{and} \quad q = f(g(q)).$	$p = g(f(p)) \quad \text{and} \quad q = f(g(q)).$
fong-spivak-seven-sketches_1 EQ0078 (4.3)	132	■ 0.549 ● N +2	$\Phi: x^{\mathrm{op}} \times y \rightarrow \text{Bool}.$	$\Phi: x^{\mathrm{op}} \times y \rightarrow \text{Bool}.$	$\Phi: \mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \rightarrow \mathbf{Bool}.$

Conf. MathPix confidence:

■ ≥0.9

■ 0.5–0.9

■ <0.5

— no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches-EQ0091	142	0.550 C +3	$\widehat{F}(p, q) := Q(F(p), q)$ \text { and } \check{F}(q, p) := Q(q, F(p))	$\widehat{F}(p, q) := Q(F(p), q)$ and $\check{F}(q, p) := Q(q, F(p))$	$\widehat{F}(p, q) := Q(F(p), q)$ and $\check{F}(q, p) := Q(q, F(p))$
fong-spivak-seven-sketches-EQ0104 (5.17)	165	0.566 N +0	$G + G^{\prime} := \left(\left(V \sqcup V^{\prime} \right), [\text { in }, \text { in' }], [\text { out }, \text { out' }], (\iota \sqcup \iota^{\prime}) \right)$.	$G + G' := ((V \sqcup V'), [in, in'], [out, out'], (\iota \sqcup \iota'))$.	$G + G' := ((V \sqcup V'), [in, in'], [out, out'], (\iota \sqcup \iota'))$.
fong-spivak-seven-sketches-EQ0221	298	0.568 N +0	$x \left(x^{\prime}, x \right) \otimes y \left(y, y^{\prime} \right) \leq \Phi(x, y) \multimap \Phi \left(x^{\prime}, y^{\prime} \right)$	$x(x', x) \otimes y(y, y') \leq \Phi(x, y) \multimap \Phi(x', y')$	$\mathcal{X}(x', x) \otimes \mathcal{Y}(y, y') \leq \Phi(x, y) \multimap \Phi(x', y')$
fong-spivak-seven-sketches-EQ0253	313	0.570 N +0	$\left\{ (y, z) \mid \text { there exists } x \in R^m \text { such that } y = S(g)(x) \text { and } S(h)(x) = z \right\}$.	$\{(y, z) \mid \text { there exists } x \in R^m \text { such that } y = S(g)(x) \text { and } S(h)(x) = z\}$.	$\{(y, z) \mid \text { there exists } x \in R^m \text { such that } y = S(g)(x) \text { and } S(h)(x) = z\}$.
fong-spivak-seven-sketches-EQ0043	082	0.593 N -1	$2 \mathrm{H}_{2} \rightarrow (2 \mathrm{Na} \rightarrow (2 \mathrm{NaOH} + \mathrm{H}_{2}))$	$2\mathrm{H}_2\mathrm{O} \rightarrow (2\mathrm{Na} \rightarrow (2\mathrm{NaOH} + \mathrm{H}_2))$	$2\mathrm{H}_2\mathrm{O} \rightarrow (2\mathrm{Na} \multimap (2\mathrm{NaOH} + \mathrm{H}_2))$
fong-spivak-seven-sketches-EQ0179	272	0.598 N +0	$h, \quad 1, \quad 2, \quad 3 \quad .$	$h, \quad 1, \quad 2, \quad 3.$	$h, \quad 1, \quad 2, \quad 3.$
fong-spivak-seven-sketches-EQ0002 (1.15)	021	0.658 N +2	$A = \bigcup_{p \in P} A_p$ \quad and \quad \quad if $p \neq q$ then $A_p \cap A_q = \emptyset$.	$A = \bigcup_{p \in P} A_p$ and if $p \neq q$ then $A_p \cap A_q = \emptyset$.	$A = \bigcup_{p \in P} A_p$ and if $p \neq q$ then $A_p \cap A_q = \emptyset$.
fong-spivak-seven-sketches-EQ0158 (7.37)	250	0.658 N +0	$\operatorname{Sec}_f(U) := \{s : U \rightarrow X \mid (s; f)(u) = u \text { for all } u \in U\}$.	$\operatorname{Sec}_f(U) := \{s : U \rightarrow X \mid (s; f)(u) = u \text { for all } u \in U\}$.	$\operatorname{Sec}_f(U) := \{s : U \rightarrow X \mid (s \circ f)(u) = u \text { for all } u \in U\}$.
fong-spivak-seven-sketches-EQ0147 (7.2)	235	0.661 N +1	$((A = \mathrm{go}) \rightarrow \ddot{S} > 1) \wedge ((A = \mathrm{stop}) \Rightarrow \ddot{S} = 0)$.	$((A = \mathrm{go}) \Rightarrow \ddot{S} > 1) \wedge ((A = \mathrm{stop}) \Rightarrow \ddot{S} = 0)$.	$((A = \mathrm{go}) \Rightarrow \ddot{S} > 1) \wedge ((A = \mathrm{stop}) \Rightarrow \ddot{S} = 0)$.
fong-spivak-seven-sketches-EQ0191	282	0.677 C -8	$\begin{array}{l} t + u \leq (v + w) + u \quad (\text { monotonicity, } t \leq v + w, u \leq u) \\ = v + (w + u) \quad (\text { associativity }) \\ \leq v + (x + z) \quad (\text { monotonicity, } v \leq v, w + u \leq x + v) \\ = (v + x) + z \quad (\text { associativity }) \\ \leq y + z. \quad (\text { monotonicity, } v + x \leq y, z \leq z) \end{array}$	$\begin{array}{l} t + u \leq (v + w) + u \quad (\text { monotonicity, } t \leq v + w, u \leq u) \\ = v + (w + u) \quad (\text { associativity }) \\ \leq v + (x + z) \quad (\text { monotonicity, } v \leq v, w + u \leq x + v) \\ = (v + x) + z \quad (\text { associativity }) \\ \leq y + z. \quad (\text { monotonicity, } v + x \leq y, z \leq z) \end{array}$	$\begin{array}{l} t + u \leq (v + w) + u \quad (\text { monotonicity, } t \leq v + w, u \leq u) \\ = v + (w + u) \quad (\text { associativity }) \\ \leq v + (x + z) \quad (\text { monotonicity, } v \leq v, w + u \leq x + v) \\ = (v + x) + z \quad (\text { associativity }) \\ \leq y + z. \quad (\text { monotonicity, } v + x \leq y, z \leq z) \end{array}$

Conf. MathPix confidence:
■ ≥0.9
■ 0.5–0.9
■ <0.5
— no value

LaTeX source MathPix’s reading, unchanged;
Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):
● C component
● W weak
● S stable
● N noise
● K clean
● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches EQ0224	300	■ 0.683 ●	$\begin{aligned} \backslash\mathrm{begin}\{\mathrm{aligned}\} \backslash\mathrm{Phi}: \backslash\mathrm{Psi} \&=\backslash\mathrm{left}(\mathrm{M}_{\mathrm{X}}^{\mathrm{3}} * \mathrm{M}_{\backslash\mathrm{Phi}} * \mathrm{M}_{\backslash\mathrm{Y}}^{\mathrm{2}}\backslash\mathrm{right}) * \backslash\mathrm{left}(\mathrm{M}_{\backslash\mathrm{Y}}^{\mathrm{2}} * \mathrm{M}_{\backslash\mathrm{Psi}} * \mathrm{M}_{\backslash\mathrm{Z}}^{\mathrm{3}}\backslash\mathrm{right}) \backslash\backslash \&=\mathrm{M}_{\mathrm{X}}^{\mathrm{3}} * \mathrm{M}_{\backslash\mathrm{Phi}} * \mathrm{M}_{\backslash\mathrm{Y}}^{\mathrm{4}} * \mathrm{M}_{\backslash\mathrm{Psi}} * \mathrm{M}_{\backslash\mathrm{Z}}^{\mathrm{3}} \backslash\backslash \&=\backslash\mathrm{left}(\mathrm{M}_{\mathrm{X}}^{\mathrm{3}} * \mathrm{M}_{\backslash\mathrm{Phi}} * \mathrm{M}_{\backslash\mathrm{Y}}^{\mathrm{2}}\backslash\mathrm{right}) * \mathrm{M}_{\backslash\mathrm{Psi}} * \mathrm{M}_{\backslash\mathrm{Z}}^{\mathrm{3}} \backslash\backslash \&=\mathrm{Phi} * \mathrm{M}_{\backslash\mathrm{Psi}} * \mathrm{M}_{\backslash\mathrm{Z}}^{\mathrm{3}} \backslash\backslash \&=\backslash\mathrm{left}(\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{ccc}\} 17 \&20 \&20 \backslash\backslash 11 \&14 \&14 \backslash\backslash 14 \&17 \&17 \backslash\backslash 12 \&9 \&15 \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right})\backslash\mathrm{left}(\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{cccc}\} \backslash\mathrm{infty} \&\backslash\mathrm{infty} \&\backslash\mathrm{infty} \&\backslash\mathrm{infty} \backslash\backslash \backslash\mathrm{infty} \&\backslash\mathrm{infty} \&0 \&\backslash\mathrm{infty} \backslash\backslash 4 \&\backslash\mathrm{infty} \&\backslash\mathrm{infty} \&4 \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}) \backslash\mathrm{left}(\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{cccc}\} 0 \&2 \&4 \&5 \backslash\backslash 4 \&0 \&2 \&3 \backslash\backslash 2 \&4 \&0 \&1 \backslash\backslash 1 \&3 \&5 \&0 \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}) \backslash\backslash \&=\backslash\mathrm{left}(\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{ccc}\} 17 \&20 \&20 \backslash\backslash 11 \&14 \&14 \backslash\backslash 14 \&17 \&17 \backslash\backslash 12 \&9 \&15 \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right})\backslash\mathrm{left}(\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{cccc}\} \backslash\mathrm{infty} \&\backslash\mathrm{infty} \&\backslash\mathrm{infty} \&\backslash\mathrm{infty} \backslash\backslash 2 \&4 \&0 \&1 \backslash\backslash 4 \&6 \&8 \&4 \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}) \backslash\backslash \&=\backslash\mathrm{left}(\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{cccc}\} 22 \&24 \&20 \&21 \backslash\backslash 16 \&18 \&14 \&15 \backslash\backslash 19 \&21 \&17 \&18 \backslash\backslash 11 \&13 \&9 \&10 \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}) \backslash\mathrm{end}\{\mathrm{aligned}\}$	$\begin{aligned} \Phi: \Psi &= \left(M_X^3 * M_\Phi * M_Y^2\right) * \left(M_Y^2 * M_\Psi * M_Z^3\right) \\ &= M_X^3 * M_\Phi * M_Y^4 * M_\Psi * M_Z^3 \\ &= \left(M_X^3 * M_\Phi * M_Y^2\right) * M_\Psi * M_Z^3 \\ &= \Phi * M_\Psi * M_Z^3 \\ &= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty & \infty \\ \infty & \infty & 0 & \infty \\ 4 & \infty & \infty & 4 \end{pmatrix} \begin{pmatrix} 0 & 2 & 4 & 5 \\ 4 & 0 & 2 & 3 \\ 2 & 4 & 0 & 1 \\ 1 & 3 & 5 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty & \infty \\ 2 & 4 & 0 & 1 \\ 4 & 6 & 8 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 22 & 24 & 20 & 21 \\ 16 & 18 & 14 & 15 \\ 19 & 21 & 17 & 18 \\ 11 & 13 & 9 & 10 \end{pmatrix} \end{aligned}$	$\begin{aligned} \Phi \circ \Psi &= \left(M_X^3 * M_\Phi * M_Y^2\right) * \left(M_Y^2 * M_\Psi * M_Z^3\right) \\ &= M_X^3 * M_\Phi * M_Y^4 * M_\Psi * M_Z^3 \\ &= \left(M_X^3 * M_\Phi * M_Y^2\right) * M_\Psi * M_Z^3 \\ &= \Phi * M_\Psi * M_Z^3 \\ &= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty & \infty \\ \infty & \infty & 0 & \infty \\ 4 & \infty & \infty & 4 \end{pmatrix} \begin{pmatrix} 0 & 2 & 4 & 5 \\ 4 & 0 & 2 & 3 \\ 2 & 4 & 0 & 1 \\ 1 & 3 & 5 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty & \infty \\ 2 & 4 & 0 & 1 \\ 4 & 6 & 8 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 22 & 24 & 20 & 21 \\ 16 & 18 & 14 & 15 \\ 19 & 21 & 17 & 18 \\ 11 & 13 & 9 & 10 \end{pmatrix} \end{aligned}$
fong-spivak-seven-sketches EQ0120 (5.81)	187	■ 0.690 ● N +0	$\mathbf{SFG}_R^+ := \text{Free} \left(G_R \sqcup G_R^{\mathrm{op}} \right).$	$\mathbf{SFG}_R^+ := \text{Free} \left(G_R \sqcup G_R^{\mathrm{op}} \right).$	$\mathbf{SFG}_R^+ := \text{Free} \left(G_R \sqcup G_R^{\mathrm{op}} \right).$
fong-spivak-seven-sketches EQ0081 (◁)	133	■ 0.690 ● N +0	$x \left(x^{\backslash\mathrm{prime}}, x \right) \otimes \Phi(x, y) \otimes y(y, y') \leq \Phi(x', y').$	$x \left(x', x \right) \otimes \Phi(x, y) \otimes y(y, y') \leq \Phi(x', y').$	$\mathcal{X}(x', x) \otimes \Phi(x, y) \otimes \mathcal{Y}(y, y') \leq \Phi(x', y').$
fong-spivak-seven-sketches EQ0140 (6.81)	220	■ 0.725 ● W +0	$\begin{aligned} \backslash\mathrm{begin}\{\mathrm{aligned}\} \backslash\mathrm{psi}_{\mathrm{V}, \mathrm{V}^{\backslash\mathrm{prime}}}\!: \backslash\mathrm{operatorname}\{\mathrm{Circ}\}(\mathrm{V}) \&\mathrm{times} \backslash\mathrm{operatorname}\{\mathrm{Circ}\} \backslash\mathrm{left}(\mathrm{V}^{\backslash\mathrm{prime}}\backslash\mathrm{right}) \&\backslash\mathrm{longrightarrow} \backslash\mathrm{operatorname}\{\mathrm{Circ}\}\backslash\mathrm{left}(\mathrm{V}+\mathrm{V}^{\backslash\mathrm{prime}}\backslash\mathrm{right}) \&\backslash\backslash \backslash\mathrm{left}((\mathrm{V}, \mathrm{A}, \mathrm{s}, \mathrm{t}, \backslash\mathrm{ell}), \backslash\mathrm{left}(\mathrm{V}^{\backslash\mathrm{prime}}, \mathrm{A}^{\backslash\mathrm{prime}}, \mathrm{s}^{\backslash\mathrm{prime}}, \mathrm{t}^{\backslash\mathrm{prime}}, \backslash\mathrm{ell}^{\backslash\mathrm{prime}}\backslash\mathrm{right})\backslash\mathrm{right}) \&\backslash\mathrm{longmapsto} \backslash\mathrm{left}(\mathrm{V}+\mathrm{V}^{\backslash\mathrm{prime}}, \mathrm{A}+\mathrm{A}^{\backslash\mathrm{prime}}, \mathrm{s}+\mathrm{s}^{\backslash\mathrm{prime}}, \mathrm{t}+\mathrm{t}^{\backslash\mathrm{prime}}, \backslash\mathrm{ell}^{\backslash\mathrm{prime}}\backslash\mathrm{right}) \backslash\mathrm{right}) \&\backslash\mathrm{end}\{\mathrm{aligned}\}$	$\begin{aligned} \psi_{V, V'}: \mathrm{Circ}(V) \times \mathrm{Circ}(V') &\longrightarrow \mathrm{Circ}(V + V'); \\ ((V, A, s, t, \ell), (V', A', s', t', \ell')) &\longmapsto (V + V', A + A', s + s', t + t', [\ell, \ell']). \end{aligned}$	$\begin{aligned} \psi_{V, V'}: \mathrm{Circ}(V) \times \mathrm{Circ}(V') &\longrightarrow \mathrm{Circ}(V + V'); \\ ((V, A, s, t, \ell), (V', A', s', t', \ell')) &\longmapsto (V + V', A + A', s + s', t + t', [\ell, \ell']). \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches-EQ0222	298	0.731 N +0	$\backslash\mathrm{cal}\{X\}\backslash\left(x^{\backslash\mathrm{prime}},x\backslash\mathrm{right})\backslash\mathrm{otimes}\backslash\Phi(x,y)\backslash\mathrm{otimes}y\backslash\left(y,y^{\backslash\mathrm{prime}}\backslash\mathrm{right})\backslash\mathrm{leq}\backslash\Phi\backslash\left(x^{\backslash\mathrm{prime}},y^{\backslash\mathrm{prime}}\backslash\mathrm{right})\backslash\right).$	$\mathcal{X}(x',x)\otimes\Phi(x,y)\otimes y(y,y')\leq\Phi(x',y').$	$\mathcal{X}(x',x)\otimes\Phi(x,y)\otimes\mathcal{Y}(y,y')\leq\Phi(x',y').$
fong-spivak-seven-sketches-EQ0132	212	0.742	$\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{rrrr}\}\backslash\mathrm{operatorname}\{\mathrm{in}\}(\backslash\mu):=2,\&\backslash\mathrm{operatorname}\{\mathrm{in}\}(\backslash\eta):=0,\&\backslash\mathrm{operatorname}\{\mathrm{in}\}(\backslash\delta):=1,\&\backslash\mathrm{operatorname}\{\mathrm{in}\}(\backslash\epsilon):=1,\backslash\backslash\backslash\mathrm{operatorname}\{\mathrm{out}\}(\backslash\mu):=1,\&\backslash\mathrm{operatorname}\{\mathrm{out}\}(\backslash\eta):=1,\&\backslash\mathrm{operatorname}\{\mathrm{out}\}(\backslash\delta):=2,\&\backslash\mathrm{operatorname}\{\mathrm{out}\}(\backslash\epsilon):=0.\backslash\mathrm{end}\{\mathrm{array}\}$	$\begin{array}{cccc} \mathrm{in}(\mu):=2, & \mathrm{in}(\eta):=0, & \mathrm{in}(\delta):=1, & \mathrm{in}(\epsilon):=1, \\ \mathrm{out}(\mu):=1, & \mathrm{out}(\eta):=1, & \mathrm{out}(\delta):=2, & \mathrm{out}(\epsilon):=0. \end{array}$	$\mathrm{in}(\mu):=2,$ $\mathrm{in}(\eta):=0,$ $\mathrm{in}(\delta):=1,$ $\mathrm{in}(\epsilon):=1,$ $\mathrm{out}(\mu):=1,$ $\mathrm{out}(\eta):=1,$ $\mathrm{out}(\delta):=2,$ $\mathrm{out}(\epsilon):=0.$
fong-spivak-seven-sketches-EQ0097	155	0.755 N +0	$\backslash[\backslash\alpha:\backslash\beta:=\backslash\iota_{A\sqcup C}^*\backslash\left(\backslash\left(\backslash\iota_{\backslash\iota_{A\sqcup C}}\backslash\left(\backslash\iota_{A\sqcup B}\backslash\right)!\backslash\alpha\right)\backslash\mathrm{vee}\backslash\left(\backslash\iota_{\backslash\iota_{B\sqcup C}}\backslash\left(\backslash\beta\right)!\backslash\right)\backslash\right)].$	$\alpha:\beta:=\iota_{A\sqcup C}^*((\iota_{A\sqcup B})!(\alpha)\vee(\iota_{B\sqcup C})!(\beta)).$	$\alpha\circ\beta:=\iota_{A\sqcup C}^*((\iota_{A\sqcup B})!(\alpha)\vee(\iota_{B\sqcup C})!(\beta)).$
fong-spivak-seven-sketches-EQ0230	304	0.759 W +0	$\backslash\mathrm{begin}\{\mathrm{aligned}\}\&\backslash\mathrm{quad}\backslash\mathrm{bigvee}_{a,b,c,d,e\in X}\backslash\alpha^{\backslash-1}(x,(a,1))\backslash\mathrm{otimes}\backslash\left(\backslash\mathrm{mathrm}\{U\}_{\backslash X}\backslash\mathrm{times}\backslash\eta_{\backslash X}\backslash\right)((a,1),(b,c,d))\backslash\backslash\&=\backslash\mathrm{bigvee}_{a,b,c,d,e\in X}\backslash\alpha^{\backslash-1}(x,(a,1))\backslash\mathrm{otimes}\backslash\mathrm{mathrm}\{U\}_{\backslash X}(a,b)\backslash\mathrm{otimes}\backslash\eta_{\backslash X}(1,c,d)\backslash\mathrm{quad}\backslash\mathrm{otimes}\backslash\left(\backslash\epsilon_{\backslash X}\backslash\mathrm{times}\backslash\mathrm{mathrm}\{U\}_{\backslash X}\backslash\right)((b,c,d),(1,e))\backslash\mathrm{otimes}\backslash\alpha((1,e),y)\backslash\backslash\&=\backslash\mathrm{bigvee}_{a,b,c,d,e\in X}x(x,a)\backslash\mathrm{otimes}X(a,b)\backslash\mathrm{otimes}X(c,d)\backslash\mathrm{otimes}X(b,c)\backslash\mathrm{otimes}X(d,e)\backslash\mathrm{otimes}X(e,y)\backslash\backslash\&=X(x,y)\backslash\mathrm{end}\{\mathrm{aligned}\}\backslash\mathrm{quad}\backslash\mathrm{otimes}\backslash\epsilon_{\backslash X}(b,c,1)\backslash\mathrm{otimes}\backslash\mathrm{mathrm}\{U\}_{\backslash X}(d,e)\backslash\mathrm{otimes}\backslash\alpha((1,e),y)$	$\begin{array}{l} \bigvee_{a,b,c,d,e\in X} \alpha^{-1}(x,(a,1))\otimes (U_X\times \eta_X)((a,1),(b,c,d)) \\ = \bigvee_{a,b,c,d,e\in X} \alpha^{-1}(x,(a,1))\otimes U_X(a,b)\otimes \eta_X(1,c,d) \quad \otimes (\epsilon_X\times U_X)((b,c,d),(1,e))\otimes \alpha((1,e),y) \\ = \bigvee_{a,b,c,d,e\in X} x(x,a)\otimes X(a,b)\otimes X(c,d)\otimes X(b,c)\otimes X(d,e)\otimes X(e,y) \\ = X(x,y) \end{array}$	$\begin{array}{l} \bigvee_{a,b,c,d,e\in X} \alpha^{-1}(x,(a,1))\otimes (U_X\times \eta_X)((a,1),(b,c,d)) \\ \quad \otimes (\epsilon_X\times U_X)((b,c,d),(1,e))\otimes \alpha((1,e),y) \\ = \bigvee_{a,b,c,d,e\in X} \alpha^{-1}(x,(a,1))\otimes U_X(a,b)\otimes \eta_X(1,c,d) \\ \quad \otimes \epsilon_X(b,c,1)\otimes U_X(d,e)\otimes \alpha((1,e),y) \\ = \bigvee_{a,b,c,d,e\in X} X(x,a)\otimes X(a,b)\otimes X(c,d)\otimes X(b,c)\otimes X(d,e)\otimes X(e,y) \\ = X(x,y) \end{array}$
fong-spivak-seven-sketches-EQ0242	310	0.775 C -4	$c_{\backslash n}:=\backslash c:(1+\backslash c): (1+1+\backslash c):\backslash\mathrm{cdots}:(1+\backslash\mathrm{cdots}+1+\backslash c):1\backslash\mathrm{rightarrow}n$	$c_n:=-c:(1+{-c}):(1+1+{-c}):\cdots:(1+\cdots+1+{-c}):1\rightarrow n$	$c_n:=-c\circ(1+{-c})\circ(1+1+{-c})\circ\cdots\circ(1+\cdots+1+{-c}):1\rightarrow n$
fong-spivak-seven-sketches-EQ0156	248	0.780 N +0	$\backslash\mathrm{mathbf}\{0\backslash p\}_{\backslash 1}:=\backslash\{\backslash\mathrm{varnothing},\backslash\{1\},X\}\backslash\mathrm{quad}\backslash\mathrm{text}\{\mathrm{and}\}\backslash\mathrm{quad}\backslash\mathrm{mathbf}\{0\backslash p\}_{\backslash 2}:=\backslash\{\backslash\mathrm{varnothing},\backslash\{2\},X\}$	$\mathbf{Op}_1:=\{\emptyset,\{1\},X\}\quad\text{and}\quad\mathbf{Op}_2:=\{\emptyset,\{2\},X\}$	$\mathbf{Op}_1:=\{\emptyset,\{1\},X\}\quad\text{and}\quad\mathbf{Op}_2:=\{\emptyset,\{2\},X\}$
fong-spivak-seven-sketches-EQ0219	298	0.795 C -7	$\backslash\mathrm{begin}\{\mathrm{aligned}\}\backslash\Lambda(\backslash\mathrm{text}\{\mathrm{monoid}\},\backslash\mathrm{text}\{\mathrm{nothing}\})\&=\backslash\Lambda(\backslash\mathrm{text}\{\mathrm{monoid}\},\backslash\mathrm{text}\{\mathrm{this\ book}\})\backslash\backslash\&=\backslash\Lambda(\backslash\mathrm{text}\{\mathrm{preorder},\mathrm{this\ book}\})\backslash\backslash\backslash\Lambda(\backslash\mathrm{text}\{\mathrm{preorder},\mathrm{nothing}\})\&=\backslash\Lambda(\backslash\mathrm{text}\{\mathrm{category},\mathrm{nothing}\})=\backslash\mathrm{text}\{\mathrm{false.}\}\backslash\mathrm{end}\{\mathrm{aligned}\}\backslash\mathrm{quad}=\backslash\Lambda(\backslash\mathrm{text}\{\mathrm{category},\mathrm{this\ book}\})=\backslash\mathrm{text}\{\mathrm{true}\}$	$\begin{array}{l} \Lambda(\text{monoid},\text{nothing})=\Lambda(\text{monoid},\text{this book}) \\ \hspace{10em}=\Lambda(\text{preorder},\text{this book})\hspace{10em}=\Lambda(\text{category},\text{this book})=\text{true} \\ \Lambda(\text{preorder},\text{nothing})=\Lambda(\text{category},\text{nothing})=\text{false.} \end{array}$	$\begin{array}{l} \Lambda(\text{monoid},\text{nothing})=\Lambda(\text{monoid},\text{this book}) \\ \hspace{10em}=\Lambda(\text{preorder},\text{this book})\hspace{10em}=\Lambda(\text{category},\text{this book})=\text{true} \\ \Lambda(\text{preorder},\text{nothing})=\Lambda(\text{category},\text{nothing})=\text{false.} \end{array}$
fong-spivak-seven-sketches-EQ0077	124	0.797 C +3	$\backslash\mathrm{begin}\{\mathrm{aligned}\}\&\backslash\lim_{\backslash J}\backslash\mathrm{mathcal}\{J\}D:=\backslash\left(\backslash\left(d_{\backslash 1},\backslash\mathrm{ldots},d_{\backslash n}\backslash\right)\backslash\mathrm{mid}d_{\backslash i}\backslash\mathrm{in}D\backslash\left(v_{\backslash i}\backslash\right)\backslash\mathrm{text}\{\mathrm{for\ all}\}\backslash 1\backslash\mathrm{leq}i\backslash\mathrm{leq}n\backslash\mathrm{right}.\backslash\mathrm{text}\{\mathrm{and}\}\backslash\backslash\&\backslash\mathrm{left}.\backslash\mathrm{qqquad}\backslash\mathrm{text}\{\mathrm{for\ all}\}\backslash a:v_{\backslash i}\backslash\mathrm{rightarrow}v_{\backslash j}\backslash\mathrm{in}A,\backslash\mathrm{text}\{\mathrm{we\ have}\}\backslash D(a)\backslash\left(d_{\backslash i}\backslash\right)=d_{\backslash j}\backslash\right)\backslash\backslash.\backslash\mathrm{end}\{\mathrm{aligned}\}$	$\lim_{\mathcal{J}}D:=\{(d_1,\dots,d_n)\mid d_i\in D(v_i)\text{ for all }1\leq i\leq n\text{ and for all }a:v_i\rightarrow v_j\in A,\text{ we have }D(a)(d_i)=d_j\}.$	$\lim_{\mathcal{J}}D:=\{(d_1,\dots,d_n)\mid d_i\in D(v_i)\text{ for all }1\leq i\leq n\text{ and for all }a:v_i\rightarrow v_j\in A,\text{ we have }D(a)(d_i)=d_j\}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches-EQ0131	211	0.808 S 1	$m \text{ legs } \{\lesssim \in\} n \text{ legs }$	$m \text{ legs } \{\lesssim \in\} n \text{ legs }$	$m \text{ legs } \left\{ \begin{array}{c} \text{diagram} \end{array} \right\} n \text{ legs }$
fong-spivak-seven-sketches-EQ0193	286	0.825	$\begin{aligned} (x \times y) \left(\left(x_1, y_1 \right), \left(x_2, y_2 \right) \right) \otimes (x \times y) \left(\left(x_2, y_2 \right), \left(x_3, y_3 \right) \right) \\ = x(x_1, x_2) \otimes y(y_1, y_2) \otimes x(x_2, x_3) \otimes y(y_2, y_3) \\ = x(x_1, x_2) \otimes x(x_2, x_3) \otimes y(y_1, y_2) \otimes y(y_2, y_3) \\ \leq x(x_1, x_3) \otimes y(y_1, y_3) \\ = (x \times y) \left(\left(x_1, y_1 \right), \left(x_3, y_3 \right) \right). \end{aligned}$	$(x \times y) \left(\left(x_1, y_1 \right), \left(x_2, y_2 \right) \right) \otimes (x \times y) \left(\left(x_2, y_2 \right), \left(x_3, y_3 \right) \right) \\ = x(x_1, x_2) \otimes y(y_1, y_2) \otimes x(x_2, x_3) \otimes y(y_2, y_3) \\ = x(x_1, x_2) \otimes x(x_2, x_3) \otimes y(y_1, y_2) \otimes y(y_2, y_3) \\ \leq x(x_1, x_3) \otimes y(y_1, y_3) \\ = (x \times y) \left(\left(x_1, y_1 \right), \left(x_3, y_3 \right) \right).$	$\begin{aligned} (\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_2, y_2)) \otimes (\mathcal{X} \times \mathcal{Y})((x_2, y_2), (x_3, y_3)) \\ = \mathcal{X}(x_1, x_2) \otimes \mathcal{Y}(y_1, y_2) \otimes \mathcal{X}(x_2, x_3) \otimes \mathcal{Y}(y_2, y_3) \\ = \mathcal{X}(x_1, x_2) \otimes \mathcal{X}(x_2, x_3) \otimes \mathcal{Y}(y_1, y_2) \otimes \mathcal{Y}(y_2, y_3) \\ \leq \mathcal{X}(x_1, x_3) \otimes \mathcal{Y}(y_1, y_3) \\ = (\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_3, y_3)). \end{aligned}$
fong-spivak-seven-sketches-EQ0172	264	0.827 N 0	$\mathbb{I} := \{[d, u] \subseteq \mathbb{R} \mid d \leq u\}.$	$\mathbb{I} := \{[d, u] \subseteq \mathbb{R} \mid d \leq u\}.$	$\mathbb{I} := \{[d, u] \subseteq \mathbb{R} \mid d \leq u\}.$
fong-spivak-seven-sketches-EQ0093	143	0.828 C 9	$\operatorname{Col}(\Phi)(a, b) := \begin{cases} x(a, b) & \text{if } a, b \in X \\ \Phi(a, b) & \text{if } a \in X, b \in Y \\ \emptyset & \text{if } a \in Y, b \in X \\ y(a, b) & \text{if } a, b \in Y \end{cases}$	$\operatorname{Col}(\Phi)(a, b) := \begin{cases} x(a, b) & \text{if } a, b \in X \\ \Phi(a, b) & \text{if } a \in X, b \in Y \\ \emptyset & \text{if } a \in Y, b \in X \\ y(a, b) & \text{if } a, b \in Y \end{cases}$	$\operatorname{Col}(\Phi)(a, b) := \begin{cases} \mathcal{X}(a, b) & \text{if } a, b \in \mathcal{X} \\ \Phi(a, b) & \text{if } a \in \mathcal{X}, b \in \mathcal{Y} \\ \emptyset & \text{if } a \in \mathcal{Y}, b \in \mathcal{X} \\ \mathcal{Y}(a, b) & \text{if } a, b \in \mathcal{Y} \end{cases}$
fong-spivak-seven-sketches-EQ0022	058	0.830 W 0	$\frac{\frac{\frac{x}{y}}{z}}{\frac{y}{z}} = \frac{\frac{x}{y}}{\frac{z}{z}}$	$\frac{\frac{x}{y}}{z} = \frac{\frac{x}{y}}{z}$	$\frac{\frac{x}{y}}{z} = \frac{\frac{x}{y}}{z}$
fong-spivak-seven-sketches-EQ0146	229	0.839 K 0	$\operatorname{Circ}(\varphi) : \operatorname{Circ}(2) \times \operatorname{Circ}(2) \times \operatorname{Circ}(2) \rightarrow \operatorname{Circ}(0).$	$\operatorname{Circ}(\varphi) : \operatorname{Circ}(2) \times \operatorname{Circ}(2) \times \operatorname{Circ}(2) \rightarrow \operatorname{Circ}(0).$	$\operatorname{Circ}(\varphi) : \operatorname{Circ}(2) \times \operatorname{Circ}(2) \times \operatorname{Circ}(2) \rightarrow \operatorname{Circ}(0).$
fong-spivak-seven-sketches-EQ0086 (4.25)	140	0.842 K 0	$\operatorname{U}_x(x, y) := \mathcal{X}(x, y).$	$\operatorname{U}_x(x, y) := \mathcal{X}(x, y).$	$\operatorname{U}_x(x, y) := \mathcal{X}(x, y).$
fong-spivak-seven-sketches-EQ0136	219	0.843 N 1	$C := \{ \text{light}, \text{switch}, \text{battery} \} \sqcup \{ x\Omega \mid x \in \mathbb{R}^+ \}.$	$C := \{ \text{light}, \text{switch}, \text{battery} \} \sqcup \{ x\Omega \mid x \in \mathbb{R}^+ \}.$	$C := \{ \text{light}, \text{switch}, \text{battery} \} \sqcup \{ x\Omega \mid x \in \mathbb{R}^+ \}.$
fong-spivak-seven-sketches-EQ0125	197	0.843 N 1	$\mathcal{C} := \stackrel{A}{\bullet} \stackrel{f}{\rightrightarrows} \stackrel{B}{\bullet}$	$\mathcal{C} := \stackrel{A}{\bullet} \stackrel{f}{\rightrightarrows} \stackrel{B}{\bullet}$	$\mathcal{C} := \stackrel{A}{\bullet} \stackrel{f}{\rightrightarrows} \stackrel{B}{\bullet}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketchesEQ0003	224	0.849 C +7	S:=\begin{array}{ ccc }\hline 11 & 12 & 13 \\ \bullet & & \\ 21 & 22 & 23 \\ \bullet & \bullet & \bullet \end{array}	$S := \left[\begin{array}{ccc} 11 & 12 & 13 \\ \bullet & & \\ 21 & 22 & 23 \\ \bullet & \bullet & \bullet \end{array} \right]$	S := 11 12 13 21 22 23
fong-spivak-seven-sketchesEQ0121	188	0.868 N +1	\mathrm{B}\backslash\left(g\circ\mathrm{B}\backslash\left(h^{\mathrm{op}}\right)\right)=\{(x,y)\mid S(g)(x)=S(h)(y)\}.	$B\left(g\circ\left(h^{\mathrm{op}}\right)\right)=\{(x,y)\mid S(g)(x)=S(h)(y)\}.$	$B(g\circ(h^{\mathrm{op}}))=\{(x,y)\mid S(g)(x)=S(h)(y)\}.$
fong-spivak-seven-sketchesEQ0107	174	0.873	\begin{array}{r}\dot{x}+3\ddot{y}-2z=0\\ \ddot{y}+5\dot{z}=0\end{array}	$\begin{array}{l} \dot{x}+3\ddot{y}-2z=0 \\ \ddot{y}+5\dot{z}=0 \end{array}$	$\begin{array}{l} \dot{x}+3\ddot{y}-2z=0 \\ \ddot{y}+5\dot{z}=0 \end{array}$
fong-spivak-seven-sketchesEQ0010 (1.108)	042	0.877 C +4	$p\leq g(f(p))\quad\text{and}\quad f(g(q))\leq q.$	$p\leq g(f(p))\quad\text{and}\quad f(g(q))\leq q.$	$p\leq g(f(p))\quad\text{and}\quad f(g(q))\leq q.$
fong-spivak-seven-sketchesEQ0079 (4.5)	132	0.881 K +0	$b\wedge c\leq d\quad\text{iff}\quad b\leq(c\Rightarrow d).$	$b\wedge c\leq d\quad\text{iff}\quad b\leq(c\Rightarrow d).$	$b\wedge c\leq d\quad\text{iff}\quad b\leq(c\Rightarrow d).$
fong-spivak-seven-sketchesEQ0217	296	0.890 C -4	\begin{array}{l}\lim_{J\rightarrow\infty}D:=\{(d_1,\dots,d_n)\mid d_i\in D(v_i)\text{ for all }1\leq i\leq n\text{ and} \\ \text{for all }a:v_i\rightarrow v_j\in A,\text{ we have }D(a)(d_i)=d_j\}.\end{array}	$\lim_{J\rightarrow\infty}D:=\{(d_1,\dots,d_n)\mid d_i\in D(v_i)\text{ for all }1\leq i\leq n\text{ and} \\ \text{for all }a:v_i\rightarrow v_j\in A,\text{ we have }D(a)(d_i)=d_j\}.$	$\lim_{J\rightarrow\infty}D:=\{(d_1,\dots,d_n)\mid d_i\in D(v_i)\text{ for all }1\leq i\leq n\text{ and} \\ \text{for all }a:v_i\rightarrow v_j\in A,\text{ we have }D(a)(d_i)=d_j\}.$
fong-spivak-seven-sketchesEQ0153 (7.18)	242	0.894 C +3	$X=\{(\text{true},\text{true}),(\text{true},\text{false}),(\text{false},\text{true}),(\text{false},\text{false}),(\text{false},\text{true})\}\subseteq\mathbb{B}\times\mathbb{B}.$	$X=\{(\text{true},\text{true}),(\text{true},\text{false}),(\text{false},\text{true})\}\subseteq\mathbb{B}\times\mathbb{B}.$	$X=\{(\text{true},\text{true}),(\text{true},\text{false}),(\text{false},\text{true})\}\subseteq\mathbb{B}\times\mathbb{B}.$
fong-spivak-seven-sketchesEQ0105	167	0.897 N +0	$a\subset\bullet>b$	$a\subset\bullet>b$	$a\subset\bullet>b$
fong-spivak-seven-sketchesEQ0143 (6.95)	227	0.902 K +0	$(g\circ_i f):X_1\times\cdots\times X_{i-1}\times W_1\times\cdots\times W_m\times X_{i+1}\times\cdots\times X_n\longrightarrow Y$	$(g\circ_i f):X_1\times\cdots\times X_{i-1}\times W_1\times\cdots\times W_m\times X_{i+1}\times\cdots\times X_n\longrightarrow Y$	$(g\circ_i f):X_1\times\cdots\times X_{i-1}\times W_1\times\cdots\times W_m\times X_{i+1}\times\cdots\times X_n\longrightarrow Y$
fong-spivak-seven-sketchesEQ0129	205	0.913 N +0	$\{(v,d)\mid d\in D(v),\text{ where }v=v_1\text{ or }v=v_2\}.$	$\{(v,d)\mid d\in D(v),\text{ where }v=v_1\text{ or }v=v_2\}.$	$\{(v,d)\mid d\in D(v),\text{ where }v=v_1\text{ or }v=v_2\}.$
fong-spivak-seven-sketchesEQ0164	257	0.917 C +22	$\stackrel{A}{\longrightarrow}\bullet\stackrel{B}{\longrightarrow}\bullet\stackrel{C}{\longrightarrow}\bullet$	$\begin{array}{c} \xrightarrow{A} \bullet \quad \xrightarrow{B} \bullet \quad \xrightarrow{C} \bullet \end{array}$	A B C $\subseteq$ A B C $\xrightarrow{f}$ A B C $\xleftarrow{g}$ A B C $\xrightarrow{h}$ A B C $\xrightarrow{i}$ A B C $\xrightarrow{D}$

Conf. MathPix confidence:

≥0.9

0.5–0.9

<0.5

— no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ
Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches EQ0256	315	0.919 C -10	$\begin{aligned} f(m) *_R f(n) &= (\underbrace{1_R + \cdots + 1_R}_m \text{ summands} + \underbrace{1_R + \cdots + 1_R}_n \text{ summands}) *_R (\underbrace{1_R + \cdots + 1_R}_n \text{ summands} + \underbrace{1_R + \cdots + 1_R}_m \text{ summands}) \\ &= \underbrace{1_R *_R (\underbrace{1_R + \cdots + 1_R}_n \text{ summands}) + \cdots + 1_R *_R (\underbrace{1_R + \cdots + 1_R}_n \text{ summands})}_m \text{ summands} + \underbrace{1_R *_R (\underbrace{1_R + \cdots + 1_R}_m \text{ summands}) + \cdots + 1_R *_R (\underbrace{1_R + \cdots + 1_R}_m \text{ summands})}_n \text{ summands} \\ &= \underbrace{1_R + \cdots + 1_R}_{mn \text{ summands}} = f(m * n). \end{aligned}$	$f(m) *_R f(n) = (\underbrace{1_R + \cdots + 1_R}_m \text{ summands} + \underbrace{1_R + \cdots + 1_R}_n \text{ summands}) *_R (\underbrace{1_R + \cdots + 1_R}_n \text{ summands} + \underbrace{1_R + \cdots + 1_R}_m \text{ summands})$ $= \underbrace{1_R *_R (\underbrace{1_R + \cdots + 1_R}_n \text{ summands}) + \cdots + 1_R *_R (\underbrace{1_R + \cdots + 1_R}_n \text{ summands})}_m \text{ summands} + \underbrace{1_R *_R (\underbrace{1_R + \cdots + 1_R}_m \text{ summands}) + \cdots + 1_R *_R (\underbrace{1_R + \cdots + 1_R}_m \text{ summands})}_n \text{ summands}$ $= \underbrace{1_R + \cdots + 1_R}_{mn \text{ summands}} = f(m * n).$	$f(m) *_R f(n) = (\underbrace{1_R + \cdots + 1_R}_m \text{ summands} + \underbrace{1_R + \cdots + 1_R}_n \text{ summands}) *_R (\underbrace{1_R + \cdots + 1_R}_n \text{ summands} + \underbrace{1_R + \cdots + 1_R}_m \text{ summands})$ $= \underbrace{1_R *_R (\underbrace{1_R + \cdots + 1_R}_n \text{ summands}) + \cdots + 1_R *_R (\underbrace{1_R + \cdots + 1_R}_n \text{ summands})}_m \text{ summands} + \underbrace{1_R *_R (\underbrace{1_R + \cdots + 1_R}_m \text{ summands}) + \cdots + 1_R *_R (\underbrace{1_R + \cdots + 1_R}_m \text{ summands})}_n \text{ summands}$ $= \underbrace{1_R + \cdots + 1_R}_{mn \text{ summands}} = f(m * n).$
fong-spivak-seven-sketches EQ0096	149	0.919 S +0	$(f \times g) ; \left(f^{\prime} \times g^{\prime} \right) \times \left(f^{\prime} \times g^{\prime} \right) = (f ; g) \times \left(f^{\prime} \times g^{\prime} \right) .$	$(f \times g) ; (f' \times g') = (f ; g) \times (f' ; g') .$	$(f \times g) \circ (f' \times g') = (f \circ g) \times (f' \circ g') .$
fong-spivak-seven-sketches EQ0259	321	0.923	$\begin{aligned} \varphi_{S^{\prime}, T^{\prime}} \left(\left(\operatorname{im}_f \times \operatorname{im}_g \right) (A \times B) \right) &= \varphi_{S^{\prime}, T^{\prime}} \left(\{ f(a) \mid a \in A \} \times \{ g(b) \mid b \in B \} \right) \\ &= \{ (f(a), g(b)) \mid a \in A, b \in B \} \\ &= \operatorname{im}_{f \times g} (A \times B) \\ &= \operatorname{im}_{f \times g} (\varphi_{S, T} (A, B)). \end{aligned}$	$\varphi_{S', T'} ((\operatorname{im}_f \times \operatorname{im}_g)(A \times B)) = \varphi_{S', T'}(\{f(a) \mid a \in A\} \times \{g(b) \mid b \in B\})$ $= \{(f(a), g(b)) \mid a \in A, b \in B\}$ $= \operatorname{im}_{f \times g}(A \times B)$ $= \operatorname{im}_{f \times g}(\varphi_{S, T}(A, B)) .$	$\varphi_{S', T'} \left((\operatorname{im}_f \times \operatorname{im}_g)(A \times B) \right) = \varphi_{S', T'}(\{f(a) \mid a \in A\} \times \{g(b) \mid b \in B\})$ $= \{(f(a), g(b)) \mid a \in A, b \in B\}$ $= \operatorname{im}_{f \times g}(A \times B)$ $= \operatorname{im}_{f \times g}(\varphi_{S, T}(A, B)).$
fong-spivak-seven-sketches EQ0087	140	0.928 N +0	$\mathrm{U}_{\mathcal{P}}; \Phi = \Phi = \Phi; \mathrm{U}_{\mathcal{Q}}$	$\mathrm{U}_{\mathcal{P}}; \Phi = \Phi = \Phi; \mathrm{U}_{\mathcal{Q}}$	$\mathrm{U}_{\mathcal{P}} \circ \Phi = \Phi = \Phi \circ \mathrm{U}_{\mathcal{Q}}$
fong-spivak-seven-sketches EQ0199	288	0.928 W +0	$\begin{aligned} ((M * N) * P)(w, z) &= \bigvee_{y \in Y} \left(\bigvee_{x \in X} M(w, x) \otimes N(x, y) \right) \otimes P(y, z) \\ &\cong \bigvee_{y \in Y, x \in X} M(w, x) \otimes N(x, y) \otimes P(y, z) \\ &\cong \bigvee_{x \in X} M(w, x) \otimes \left(\bigvee_{y \in Y} N(x, y) \otimes P(y, z) \right) \\ &= (M * (N * P))(w, z). \end{aligned}$	$((M * N) * P)(w, z) = \bigvee_{y \in Y} \left(\bigvee_{x \in X} M(w, x) \otimes N(x, y) \right) \otimes P(y, z)$ $\cong \bigvee_{y \in Y, x \in X} M(w, x) \otimes N(x, y) \otimes P(y, z)$ $\cong \bigvee_{x \in X} M(w, x) \otimes \left(\bigvee_{y \in Y} N(x, y) \otimes P(y, z) \right)$ $= (M * (N * P))(w, z).$	$((M * N) * P)(w, z) = \bigvee_{y \in Y} \left(\bigvee_{x \in X} M(w, x) \otimes N(x, y) \right) \otimes P(y, z)$ $\cong \bigvee_{y \in Y, x \in X} M(w, x) \otimes N(x, y) \otimes P(y, z)$ $\cong \bigvee_{x \in X} M(w, x) \otimes \left(\bigvee_{y \in Y} N(x, y) \otimes P(y, z) \right)$ $= (M * (N * P))(w, z).$
fong-spivak-seven-sketches EQ0251	313	0.930 N +0	$\mathrm{B} \left(g ; \left(h^{\mathrm{op}} \right) \right) = \{ (x, y) \mid S(g)(x) = S(h)(y) \} .$	$\mathrm{B} (g; (h^{\mathrm{op}})) = \{ (x, y) \mid S(g)(x) = S(h)(y) \} .$	$\mathrm{B}(g \circ (h^{\mathrm{op}})) = \{ (x, y) \mid S(g)(x) = S(h)(y) \} .$
Conf. MathPix confidence: 0.919 0.5-0.9 0					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image				
fong-spivak-seven-sketchesEQ0034	073	0.932 N -1	$d_{\{L\}}(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v),$	$d_L(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v),$	$d_L(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v),$				
fong-spivak-seven-sketchesEQ0128	205	0.939 C +3	$\operatorname{colim}_{\mathcal{J}} D := \{(v, d) \mid v \in V \text{ and } d \in D(v)\} / \sim$	$\operatorname{colim}_{\mathcal{J}} D := \{(v, d) \mid v \in V \text{ and } d \in D(v)\} / \sim$	$\operatorname{colim}_{\mathcal{J}} D := \{(v, d) \mid v \in V \text{ and } d \in D(v)\} / \sim$				
fong-spivak-seven-sketchesEQ0053 (3.8)	094	0.941 S +1	$\mathbf{2} := \operatorname{Free} \left(\begin{array}{ c } \hline v_1 & f_1 & v_2 \\ \hline \bullet & & \bullet \\ \hline \end{array} \right)$	$\mathbf{2} := \operatorname{Free} \left(\begin{array}{ c } \hline v_1 & f_1 & v_2 \\ \hline \bullet & & \bullet \\ \hline \end{array} \right)$	$\mathbf{2} := \operatorname{Free} \left(\begin{array}{ c } \hline v_1 & f_1 & v_2 \\ \hline \bullet & & \bullet \\ \hline \end{array} \right)$				
fong-spivak-seven-sketchesEQ0075	119	0.943 C -7	$\frac{1}{\text{Em}_6}$	<i>(not rendered)</i>	<table><tr><td>1</td><td></td></tr><tr><td>Em_6</td><td></td></tr></table>	1		Em_6	
1									
Em_6									
fong-spivak-seven-sketchesEQ0026	060	0.948 N -1	$\mathrm{H}_{\{2\}} \mathrm{H}_0, \quad \mathrm{NaCl}, \quad 2 \mathrm{NaOH}, \quad \mathrm{CH}_4 + 3 \mathrm{O}_2$	$\mathrm{H}_2\mathrm{O}, \quad \mathrm{NaCl}, \quad 2\mathrm{NaOH}, \quad \mathrm{CH}_4 + 3\mathrm{O}_2$	$\mathrm{H}_2\mathrm{O}, \quad \mathrm{NaCl}, \quad 2\mathrm{NaOH}, \quad \mathrm{CH}_4 + 3\mathrm{O}_2$				
fong-spivak-seven-sketchesEQ0111	176	0.950 N +1	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array} \right)$	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array} \right)$	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array} \right)$				
fong-spivak-seven-sketchesEQ0159 (7.43)	252	0.951 N +0	$\operatorname{Sec}_{\mathcal{f}}(U) := \{g\colon U \rightarrowtail X \mid g \text{ is continuous and } (g \circ f)(u)=u \text{ for all } u \in U\},$	$\operatorname{Sec}_f(U) := \{g\colon U \rightarrow X \mid g \text{ is continuous and } (g \circ f)(u) = u \text{ for all } u \in U\},$	$\operatorname{Sec}_f(U) := \{g\colon U \rightarrowtail X \mid g \text{ is continuous and } (g \circ f)(u) = u \text{ for all } u \in U\},$				
fong-spivak-seven-sketchesEQ0007	038	0.952 N +1	$f(a) \vee f(b) \not\equiv f(a \vee b) \text{ .}$	$f(a) \vee f(b) \not\equiv f(a \vee b).$	$f(a) \vee f(b) \not\equiv f(a \vee b).$				
fong-spivak-seven-sketchesEQ0030 (2.25)	062	0.954	$\mathrm{H}_{\{2\}} \mathrm{H}_0 + \mathrm{NaCl} \rightarrow^? \mathrm{H}_2\mathrm{O}$	$\mathrm{H}_2\mathrm{O} + \mathrm{NaCl} \rightarrow^? \mathrm{H}_2\mathrm{O}$	$\mathrm{H}_2\mathrm{O} + \mathrm{NaCl} \rightarrow^? \mathrm{H}_2\mathrm{O}$				
fong-spivak-seven-sketchesEQ0194	287	0.955 K +0	$A = (A \cap \bar{B}) \cup (A \cap B) \subseteq \bar{B} \cup C \text{ .}$	$A = (A \cap \bar{B}) \cup (A \cap B) \subseteq \bar{B} \cup C.$	$A = (A \cap \bar{B}) \cup (A \cap B) \subseteq \bar{B} \cup C.$				
fong-spivak-seven-sketchesEQ0171 (7.73)	263	0.958 K +0	$\forall (t\colon T) \text{ . } \exists (s\colon S) \text{ . } f(s)=t$	$\forall (t\colon T). \exists (s\colon S). f(s) = t$	$\forall (t\colon T). \exists (s\colon S). f(s) = t$				
fong-spivak-seven-sketchesEQ0040 (2.80)	081	0.960 K +0	$(a \otimes v) \leq w \quad \text{iff} \quad a \leq (v \multimap w) \text{ .}$	$(a \otimes v) \leq w \quad \text{iff} \quad a \leq (v \multimap w).$	$(a \otimes v) \leq w \quad \text{iff} \quad a \leq (v \multimap w).$				
fong-spivak-seven-sketchesEQ0262	328	0.963 K +0	$\bigcup_{i \in I} A_i = \bigcup_{i \in I} \left(B_i \cap \bigcap_{Y \in \mathbf{Op}_{\{0\}} Y} Y \right) = \left(\bigcup_{i \in I} B_i \right) \cap \bigcap_{Y \in \mathbf{Op}_{\{0\}} Y} Y \text{ .}$	$\bigcup_{i \in I} A_i = \bigcup_{i \in I} (B_i \cap Y) = \left(\bigcup_{i \in I} B_i \right) \cap Y \in \mathbf{Op}_{? \cap Y}.$	$\bigcup_{i \in I} A_i = \bigcup_{i \in I} (B_i \cap Y) = \left(\bigcup_{i \in I} B_i \right) \cap Y \in \mathbf{Op}_{? \cap Y}.$				
fong-spivak-seven-sketchesEQ0092 (4.40)	143	0.967 U +0	$\mathcal{P}(p, G(q)) \cong \mathcal{Q}(F(p), q)$	$\mathcal{P}(p, G(q)) \cong \mathcal{Q}(F(p), q)$	$\mathcal{P}(p, G(q)) \cong \mathcal{Q}(F(p), q)$				
fong-spivak-seven-sketchesEQ0218	297	0.967 C -3	$\lim_{\mathcal{J}} D := \{(d_1, d_2) \mid d_1 \in A \text{ and } d_2 \in B\} \text{ .}$	$\lim_{\mathcal{J}} D := \{(d_1, d_2) \mid d_1 \in A \text{ and } d_2 \in B\} \text{ .}$	$\lim_{\mathcal{J}} D := \{(d_1, d_2) \mid d_1 \in A \text{ and } d_2 \in B\}.$				
fong-spivak-seven-sketchesEQ0168	260	0.967 K +0	$s\colon S \mid p(s) \dashv q(s) \quad \text{or} \quad p(s) \vdash_{s\colon S} q(s) \quad \vdash_{s\colon S} q(s) \text{ .}$	$s\colon S \mid p(s) \vdash q(s) \quad \text{or} \quad p(s) \vdash_{s\colon S} q(s).$	$s\colon S \mid p(s) \vdash q(s) \quad \text{or} \quad p(s) \vdash_{s\colon S} q(s).$				
fong-spivak-seven-sketchesEQ0066 (3.69)	114	0.968 N +0	$f(p) \leq q \text{ if and only if } p \leq g(q) \text{ .}$	$f(p) \leq q \text{ if and only if } p \leq g(q).$	$f(p) \leq q \text{ if and only if } p \leq g(q).$				
fong-spivak-seven-sketchesEQ0119 (5.78)	187	0.969 N +0	$B_{\{1\}} ; B_{\{2\}} := \{(x, z) \mid \text{there exists } y \in R^{\{n\}} \text{ such that } (x, y) \in B_{\{1\}} \text{ and } (y, z) \in B_{\{2\}}\} \text{ .}$	$B_1; B_2 := \{(x, z) \mid \text{there exists } y \in R^n \text{ such that } (x, y) \in B_1 \text{ and } (y, z) \in B_2\} \text{ .}$	$B_1 \circ B_2 := \{(x, z) \mid \text{there exists } y \in R^n \text{ such that } (x, y) \in B_1 \text{ and } (y, z) \in B_2\}.$				

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches-1 EQ0157	249	■ 0.969 ● N +2	$\left.s_{\mathrm{i}}\right _{U_{\mathrm{i}}\cap U_{\mathrm{j}}}=\left.s_{\mathrm{j}}\right _{U_{\mathrm{i}}\cap U_{\mathrm{j}}}$.	$s_i _{U_i\cap U_j} = s_j _{U_i\cap U_j}$.	$s_i _{U_i\cap U_j} = s_j _{U_i\cap U_j}$.
fong-spivak-seven-sketches-1 EQ0100	157	■ 0.973 ● N +0	$\eta_x\colon \mathbf{1}\times \mathcal{X}^{\mathrm{op}}\times \mathcal{X}\rightarrow \mathcal{V};$ $\times \mathcal{H}\rightarrow \mathcal{V}$;	$\eta_x:\mathbf{1}\times \mathcal{X}^{\mathrm{op}}\times \mathcal{X}\rightarrow \mathcal{V};$	$\eta_{\mathcal{X}}\colon \mathbf{1}\times \mathcal{X}^{\mathrm{op}}\times \mathcal{X}\rightarrow \mathcal{V};$
fong-spivak-seven-sketches-1 EQ0045	085	■ 0.974 ● W -1	$d(U,V)\coloneqq \sup_{u\in U}\inf_{v\in V}d(u,v)\quad X(U,V)\coloneqq \bigwedge_{u\in U}\bigvee_{v\in V}x(u,v)$.	$d(U,V)\coloneqq \sup_{u\in U}\inf_{v\in V}d(u,v)\quad X(U,V)\coloneqq \bigwedge_{u\in U}\bigvee_{v\in V}x(u,v).$	$d(U,V)\coloneqq \sup_{u\in U}\inf_{v\in V}d(u,v)\quad \mathcal{X}(U,V)\coloneqq \bigwedge_{u\in U}\bigvee_{v\in V}\mathcal{X}(u,v).$
fong-spivak-seven-sketches-1 EQ0046	085	■ 0.974 ● N +1	$v\rightarrowtail w\coloneqq \bigvee_{\{a\in V\mid a\otimes v\leq w\}}a$.	$v\rightarrowtail w\coloneqq \bigvee_{\{a\in V\mid a\otimes v\leq w\}}a.$	$v\multimap w\coloneqq \bigvee_{\{a\in V\mid a\otimes v\leq w\}}a.$
fong-spivak-seven-sketches-1 EQ0085	139	■ 0.976 ● C +6	$(\Phi;\Psi)(p,r)\coloneqq \bigvee_{q\in Q}(\Phi(p,q)\otimes \Psi(q,r))$.	$(\Phi;\Psi)(p,r)=\bigvee_{q\in Q}(\Phi(p,q)\otimes \Psi(q,r)).$	$(\Phi\circledcirc \Psi)(p,r)=\bigvee_{q\in Q}(\Phi(p,q)\otimes \Psi(q,r)).$
fong-spivak-seven-sketches-1 EQ0195	287	■ 0.979 ● K +0	$A\cap B\subseteq (\overline{B}\cup C)\cap B=(\overline{B}\cap B)\cup (C\cap B)=C\cap B\subseteq C$.	$A\cap B\subseteq (\overline{B}\cup C)\cap B=(\overline{B}\cap B)\cup (C\cap B)=C\cap B\subseteq C.$	$A\cap B\subseteq (\overline{B}\cup C)\cap B=(\overline{B}\cap B)\cup (C\cap B)=C\cap B\subseteq C.$
fong-spivak-seven-sketches-1 EQ0173	265	■ 0.979 ● W +0	$\mathbb{R}\cong \{[d,u]\in \mathbb{I}\mid d=u\}$.	$\mathbb{R}\cong \{[d,u]\in \mathbb{I}\mid d=u\}.$	$\mathbb{R}\cong \{[d,u]\in \mathbb{I}\mid d=u\}.$
fong-spivak-seven-sketches-1 EQ0137	220	■ 0.982 ● K +0	$\text{Circ}\colon (\mathbf{FinSet},+)\rightarrowtail (\mathbf{Set},\times)$	Circ: $(\mathbf{FinSet},+)\longrightarrow (\mathbf{Set},\times)$	Circ: $(\mathbf{FinSet},+)\longrightarrow (\mathbf{Set},\times)$
fong-spivak-seven-sketches-1 EQ0169	260	■ 0.983 ● N +1	$\forall (t:T).p(s,t)$.	$\forall (t:T).p(s,t).$	$\forall (t:T).p(s,t).$
fong-spivak-seven-sketches-1 EQ0018	046	■ 0.985 ● N +1	$\operatorname{fix}_j\coloneqq \{p\in P\mid j(p)\cong p\}$.	$\operatorname{fix}_j\coloneqq \{p\in P\mid j(p)\cong p\}.$	$\operatorname{fix}_j\coloneqq \{p\in P\mid j(p)\cong p\}.$
fong-spivak-seven-sketches-1 EQ0103	165	■ 0.985 ●	$\iota^{\mathrm{\prime}}(x)=\begin{cases}\iota(x)&\text{if } \iota(x)\in I\\ \iota'(\iota(x))&\text{if } \iota(x)\in \underline{n}\\ \iota'(x)&\text{if } x\in O.\end{cases}$	$\iota''(x)=\begin{cases}\iota(x)&\text{if } \iota(x)\in I\\ \iota'(\iota(x))&\text{if } \iota(x)\in \underline{n}\\ \iota'(x)&\text{if } x\in O.\end{cases}$	$\iota''(x)=\begin{cases}\iota(x)&\text{if } \iota(x)\in I\\ \iota'(\iota(x))&\text{if } \iota(x)\in \underline{n}\\ \iota'(x)&\text{if } x\in O.\end{cases}$
fong-spivak-seven-sketches-1 EQ0210	292	■ 0.986 ● N +1	$F(a)\coloneqq a^{\mathrm{\prime}},\quad G(a)\coloneqq a^{\mathrm{\prime}},\quad F(b)\coloneqq b^{\mathrm{\prime}},\quad G(b)\coloneqq b^{\mathrm{\prime}},\quad F(f)\coloneqq f_1,\text{ and }G(f)\coloneqq f_2.$	$F(a)\coloneqq a',\quad G(a)\coloneqq a',\quad F(b)\coloneqq b',\quad G(b)\coloneqq b',\quad F(f)\coloneqq f_1,\text{ and }G(f)\coloneqq f_2.$	$F(a)\coloneqq a',\quad G(a)\coloneqq a',\quad F(b)\coloneqq b',\quad G(b)\coloneqq b',\quad F(f)\coloneqq f_1,\text{ and }G(f)\coloneqq f_2.$
fong-spivak-seven-sketches-1 EQ0117 (5.75)	186	■ 0.988 ● N +0	$\mathbf{B}(g)=\{(x,S(g)(x))\mid x\in R^m\}\subseteq R^m\times R^n$.	$\mathbf{B}(g)=\{(x,S(g)(x))\mid x\in R^m\}\subseteq R^m\times R^n.$	$\mathbf{B}(g)=\{(x,S(g)(x))\mid x\in R^m\}\subseteq R^m\times R^n.$
fong-spivak-seven-sketches-1 EQ0031	067	■ 0.989 ● K +0	$\text{Cost}\coloneqq ([0,\infty],\geq,0,+),$	Cost $\coloneqq ([0,\infty],\geq,0,+),$	Cost $\coloneqq ([0,\infty],\geq,0,+),$
fong-spivak-seven-sketches-1 EQ0204	289	■ 0.989 ● N +0	$\stackrel{\text{v}_1}{\bullet}\xrightarrow{f_1}\stackrel{\text{v}_2}{\bullet}\xrightarrow{f_2}\dots\stackrel{f_{n-1}}{\longrightarrow}\stackrel{\text{v}_n}{\bullet}$	$\stackrel{v_1}{\bullet}\xrightarrow{f_1}\stackrel{v_2}{\bullet}\xrightarrow{f_2}\dots\stackrel{f_{n-1}}{\longrightarrow}\stackrel{v_n}{\bullet}$	$\stackrel{v_1}{\bullet}\xrightarrow{f_1}\stackrel{v_2}{\bullet}\xrightarrow{f_2}\dots\stackrel{f_{n-1}}{\longrightarrow}\stackrel{v_n}{\bullet}$
fong-spivak-seven-sketches-1 EQ0213	294	■ 0.989 ● N +1	$G\coloneqq \stackrel{1}{\bullet}\xrightarrow{a}\stackrel{2}{\bullet}\xrightarrow{b}\stackrel{3}{\bullet}$	$G\coloneqq \stackrel{1}{\bullet}\xrightarrow{a}\stackrel{2}{\bullet}\xrightarrow{b}\stackrel{3}{\bullet}$	$G\coloneqq \stackrel{1}{\bullet}\xrightarrow{a}\stackrel{2}{\bullet}\xrightarrow{b}\stackrel{3}{\bullet}$
fong-spivak-seven-sketches-1 EQ0250	313	■ 0.990 ● N -1	$\{(x,y)\mid \text{there exists } z\in R^n\text{ such that } S(g)(x)=z\text{ and } z=S(h)(y)\}.$	$\{(x,y)\mid \text{there exists } z\in R^n\text{ such that } S(g)(x)=z\text{ and } z=S(h)(y)\}.$	$\{(x,y)\mid \text{there exists } z\in R^n\text{ such that } S(g)(x)=z\text{ and } z=S(h)(y)\}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured
[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches- EQ0038	978	■ 0.990 ● N +1	$\mathrm{cal{X}}\left(x_{1}, x_{2}\right) \leq y\left(F\left(x_{1}\right), F\left(x_{2}\right)\right),$	$\mathcal{X}\left(x_{1}, x_{2}\right) \leq y\left(F\left(x_{1}\right), F\left(x_{2}\right)\right),$	$\mathcal{X}\left(x_{1}, x_{2}\right) \leq \mathcal{Y}\left(F\left(x_{1}\right), F\left(x_{2}\right)\right),$
fong-spivak-seven-sketches- EQ0102	162	■ 0.991 ● N +0	$R ; S:=\{(i, k) \in \underline{m} \times \underline{p} \mid \exists(j \in \underline{n}).(i, j) \in R \text { and } (j, k) \in S\}.$	$R ; S:=\{(i, k) \in \underline{m} \times \underline{p} \mid \exists(j \in \underline{n}).(i, j) \in R \text { and } (j, k) \in S\}.$	$R \circ S:=\{(i, k) \in \underline{m} \times \underline{p} \mid \exists(j \in \underline{n}).(i, j) \in R \text { and } (j, k) \in S\}.$
fong-spivak-seven-sketches- EQ0080 (4.6)	133	■ 0.991 ● N +1	$\Phi: x^{\mathrm{op}} \times y \rightarrow \mathrm{cal{V}}.$	$\Phi: x^{\mathrm{op}} \times y \rightarrow \mathcal{V}.$	$\Phi: \mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \rightarrow \mathcal{V}.$
fong-spivak-seven-sketches- EQ0134	216	■ 0.992 ● N +0	$\varphi_{c_1, c_2}: F\left(c_1\right) \otimes_{\mathcal{D}} F\left(c_2\right) \rightarrow F\left(c_1 \otimes_{\mathcal{C}} c_2\right),$	$\varphi_{c_1, c_2}: F\left(c_1\right) \otimes_{\mathcal{D}} F\left(c_2\right) \rightarrow F\left(c_1 \otimes_{\mathcal{C}} c_2\right),$	$\varphi_{c_1, c_2}: F\left(c_1\right) \otimes_{\mathcal{D}} F\left(c_2\right) \rightarrow F\left(c_1 \otimes_{\mathcal{C}} c_2\right),$
fong-spivak-seven-sketches- EQ0148	236	■ 0.992 ● C +8	$\{p \in P \mid \text { likes_cats }(p)\}.$	<i>(not rendered)</i>	$\{p \in P \mid \text { likes_cats }(p)\}.$
fong-spivak-seven-sketches- EQ0144	228	■ 0.992 ● N +1	$(f \circ i g):=f \circ(\mathrm{id}, \ldots, \mathrm{id}, g, \mathrm{id}, \ldots, \mathrm{id})$	$(f \circ i g):=f \circ(\mathrm{id}, \ldots, \mathrm{id}, g, \mathrm{id}, \ldots, \mathrm{id})$	$(f \circ_i g):=f \circ(\mathrm{id}, \ldots, \mathrm{id}, g, \mathrm{id}, \ldots, \mathrm{id})$
fong-spivak-seven-sketches- EQ0123 (5.86)	189	■ 0.993 ● N -1	$\{(0,(x, x)) \mid x \in R\} \subseteq R^0 \times R^2 \quad \text { and } \quad\left\{((x, x), 0) \mid x \in R\right\} \subseteq R^2 \times R^0.$	$\{(0,(x, x)) \mid x \in R\} \subseteq R^0 \times R^2 \quad \text { and } \quad\left\{((x, x), 0) \mid x \in R\right\} \subseteq R^2 \times R^0.$	$\{(0,(x, x)) \mid x \in R\} \subseteq R^0 \times R^2 \quad \text { and } \quad\left\{((x, x), 0) \mid x \in R\right\} \subseteq R^2 \times R^0.$
fong-spivak-seven-sketches- EQ0229	304	■ 0.993 ● W +0	$x \rightarrow^{\alpha^{-1}} x \times \mathbf{1} \xrightarrow{U_x \times \eta_x} x \times x^{\mathrm{op}} \times x \xrightarrow{\epsilon_x \times U_x} \mathbf{1} \times x \xrightarrow{\alpha} x$	$x \xrightarrow{\alpha^{-1}} x \times \mathbf{1} \xrightarrow{U_x \times \eta_x} x \times x^{\mathrm{op}} \times x \xrightarrow{\epsilon_x \times U_x} \mathbf{1} \times x \xrightarrow{\alpha} x$	$\mathcal{X} \xrightarrow{\alpha^{-1}} \mathcal{X} \times \mathbf{1} \xrightarrow{U_{\mathcal{X}} \times \eta_{\mathcal{X}}} \mathcal{X} \times \mathcal{X}^{\mathrm{op}} \times \mathcal{X} \xrightarrow{\epsilon_{\mathcal{X}} \times U_{\mathcal{X}}} \mathbf{1} \times \mathcal{X} \xrightarrow{\alpha} \mathcal{X}$
fong-spivak-seven-sketches- EQ0225	301	■ 0.993 ●	$\begin{aligned} & ((\Phi ; \Psi) ; \Upsilon)(p, s) \& \\ & =\bigvee_{r \in \mathrm{cal{R}}}\left(\Phi(p, q) \otimes \Psi(q, r)\right) \otimes \Upsilon(r, s) \\ & =\bigvee_{r \in \mathrm{cal{R}}, q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \otimes \Upsilon(r, s) \\ & =\bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \bigvee_{r \in \mathcal{R}}(\Psi(q, r) \otimes \Upsilon(r, s)) \\ & =(\Phi ;(\Psi ; \Upsilon))(p, s) \end{aligned}$	$\begin{aligned} ((\Phi ; \Psi) ; \Upsilon)(p, s) &= \bigvee_{r \in \mathcal{R}}\left(\bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r)\right) \otimes \Upsilon(r, s) \\ &= \bigvee_{r \in \mathcal{R}, q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \otimes \Upsilon(r, s) \\ &= \bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \bigvee_{r \in \mathcal{R}}(\Psi(q, r) \otimes \Upsilon(r, s)) \\ &= (\Phi ;(\Psi ; \Upsilon))(p, s) \end{aligned}$	$\begin{aligned} ((\Phi \circ \Psi) \circ \Upsilon)(p, s) &= \bigvee_{r \in \mathcal{R}}\left(\bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r)\right) \otimes \Upsilon(r, s) \\ &= \bigvee_{r \in \mathcal{R}, q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \otimes \Upsilon(r, s) \\ &= \bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \bigvee_{r \in \mathcal{R}}(\Psi(q, r) \otimes \Upsilon(r, s)) \\ &= (\Phi \circ (\Psi \circ \Upsilon))(p, s) \end{aligned}$
fong-spivak-seven-sketches- EQ0089 (4.29)	140	■ 0.994 ● N +0	$\mathrm{cal{P}}\left(p, p_{1}\right) \otimes \Phi\left(p_1, q\right) \leq \mathcal{P}\left(p, p_1\right) \otimes \Phi\left(p_1, q\right) \otimes \mathcal{Q}(q, q) \leq \Phi(p, q) .$	$\mathcal{P}\left(p, p_1\right) \otimes \Phi\left(p_1, q\right)=\mathcal{P}\left(p, p_1\right) \otimes \Phi\left(p_1, q\right) \otimes I \leq \mathcal{P}\left(p, p_1\right) \otimes \Phi\left(p_1, q\right) \otimes \mathcal{Q}(q, q) \leq \Phi(p, q) .$	$\mathcal{P}\left(p, p_1\right) \otimes \Phi\left(p_1, q\right)=\mathcal{P}\left(p, p_1\right) \otimes \Phi\left(p_1, q\right) \otimes I \leq \mathcal{P}\left(p, p_1\right) \otimes \Phi\left(p_1, q\right) \otimes \mathcal{Q}(q, q) \leq \Phi(p, q) .$
fong-spivak-seven-sketches- EQ0165 (7.56)	257	■ 0.995 ● K +0	$(U \wedge V):=U \cap V \quad \text { and } \quad(U \vee V):=U \cup V.$	$(U \wedge V):=U \cap V \quad \text { and } \quad(U \vee V):=U \cup V.$	$(U \wedge V):=U \cap V \quad \text { and } \quad(U \vee V):=U \cup V.$
fong-spivak-seven-sketches- EQ0150 (7.14)	241	■ 0.995 ● N +0	$\Omega_{\text {Set }}:=\mathbb{B}=\{\text { true }, \text { false }\}.$	$\Omega_{\text {Set }}:=\mathbb{B}=\{\text { true }, \text { false }\}.$	$\Omega_{\text {Set }}:=\mathbb{B}=\{\text { true }, \text { false }\}.$
fong-spivak-seven-sketches- EQ0160 (7.50)	255	■ 0.996 ● N +0	$\Omega(U):=\left\{U^{\prime} \in \mathbf{Op} \mid U^{\prime} \subseteq U\right\}.$	$\Omega(U):=\left\{U^{\prime} \in \mathbf{Op} \mid U^{\prime} \subseteq U\right\}.$	$\Omega(U):=\left\{U^{\prime} \in \mathbf{Op} \mid U^{\prime} \subseteq U\right\}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches- E00260	324	■ 0.996 ● S +0	$p ; h_3=r ; g ; h_3=r ; h_2 ; g^{\backslash\prime}=q ; f^{\backslash\prime} ; g^{\backslash\prime}$	$p;h_3=r;g;h_3=r;h_2;g'=q;f';g'$	$p\circ h_3=r\circ g\circ h_3=r\circ h_2\circ g'=q\circ f'\circ g'$
fong-spivak-seven-sketches- E00088 (4.28)	140	■ 0.996 ● W +1	$\Phi(p,q)=I\otimes\Phi(p,q)\leq\mathcal{P}(p,p)\otimes\Phi(p,q)\leq\bigvee_{p_1\in P}(\mathcal{P}(p,p_1)\otimes\Phi(p_1,q))=(U_P\Phi)(p,q).$	$\Phi(p,q)=I\otimes\Phi(p,q)\leq\mathcal{P}(p,p)\otimes\Phi(p,q)\leq\bigvee_{p_1\in P}(\mathcal{P}(p,p_1)\otimes\Phi(p_1,q))=(U_P\Phi)(p,q).$	$\Phi(p,q)=I\otimes\Phi(p,q)\leq\mathcal{P}(p,p)\otimes\Phi(p,q)\leq\bigvee_{p_1\in P}(\mathcal{P}(p,p_1)\otimes\Phi(p_1,q))=(U_P\circ\Phi)(p,q).$
fong-spivak-seven-sketches- E00185	276	■ 0.996 ● N -1	$f^{\ast}(P)=\{\{x\in\mathbb{Z}\mid x\leq 0\},\{x\in\mathbb{Z}\mid x\geq 1\}\}\quad\text{and}\quad f^{\ast}(Q)=\{\{0\},\{x\in\mathbb{Z}\mid x\neq 0\}\}.$	$f^{\ast}(P)=\{\{x\in\mathbb{Z}\mid x\leq 0\},\{x\in\mathbb{Z}\mid x\geq 1\}\}\quad\text{and}\quad f^{\ast}(Q)=\{\{0\},\{x\in\mathbb{Z}\mid x\neq 0\}\}.$	$f^{\ast}(P)=\{\{x\in\mathbb{Z}\mid x\leq 0\},\{x\in\mathbb{Z}\mid x\geq 1\}\}\quad\text{and}\quad f^{\ast}(Q)=\{\{0\},\{x\in\mathbb{Z}\mid x\neq 0\}\}.$
fong-spivak-seven-sketches- E00166 (7.57)	257	■ 0.996 ● U +0	$(U\Rightarrow V):=\bigcup_{\{R\in\mathbf{Op}\mid R\cap U\subseteq V\}}R.$	$(U\Rightarrow V):=\bigcup_{\{R\in\mathbf{Op}\mid R\cap U\subseteq V\}}R.$	$(U\Rightarrow V):=\bigcup_{\{R\in\mathbf{Op}\mid R\cap U\subseteq V\}}R.$
fong-spivak-seven-sketches- E00013 (1.116)	044	■ 0.997 ● N +0	$f(p):=\bigwedge\{q\in Q\mid p\leq g(q)\};$	$f(p):=\bigwedge\{q\in Q\mid p\leq g(q)\};$	$f(p):=\bigwedge\{q\in Q\mid p\leq g(q)\};$
fong-spivak-seven-sketches- E00154	243	■ 0.997 ● W +2	$\begin{array}{cc c} P&Q&P\wedge Q&P=(P\wedge Q)\\\text{true}&\text{true}&?&?\\\text{true}&\text{false}&?&?\\\text{false}&\text{true}&?&?\\\text{false}&\text{false}&?&?\\\end{array}$	$\begin{array}{cc c} P&Q&P\wedge Q&P=(P\wedge Q)\\\text{true}&\text{true}&?&?\\\text{true}&\text{false}&?&?\\\text{false}&\text{true}&?&?\\\text{false}&\text{false}&?&?\\\end{array}$	$\begin{array}{cc c} P&Q&P\wedge Q&P=(P\wedge Q)\\\text{true}&\text{true}&?&?\\\text{true}&\text{false}&?&?\\\text{false}&\text{true}&?&?\\\text{false}&\text{false}&?&?\\\end{array}$
fong-spivak-seven-sketches- E00041	081	■ 0.998 ●	$a+x\geq y\quad\text{iff}\quad a\geq y-x\quad\text{iff}\quad\max(0,a)\geq\max(0,y-x)\quad\text{iff}\quad a\geq(x\multimap y)$	$a+x\geq y\quad\text{iff}\quad a\geq y-x\quad\text{iff}\quad\max(0,a)\geq\max(0,y-x)\quad\text{iff}\quad a\geq(x\multimap y)$	$a+x\geq y\quad\text{iff}\quad a\geq y-x\quad\text{iff}\quad\max(0,a)\geq\max(0,y-x)\quad\text{iff}\quad a\geq(x\multimap y)$
fong-spivak-seven-sketches- E00082 (4.14)	135	■ 0.998 ● N +0	$\Phi(B,x)=11,\quad\Phi(A,z)=20,\quad\Phi(C,y)=17.$	$\Phi(B,x)=11,\quad\Phi(A,z)=20,\quad\Phi(C,y)=17.$	$\Phi(B,x)=11,\quad\Phi(A,z)=20,\quad\Phi(C,y)=17.$
fong-spivak-seven-sketches- E00124	190	■ 0.998 ● S +0	$v=\int\frac{1}{m}F(t)\,dt+u(t)+p\int au(t)+bv(t)\,dt.$	$v=\int\frac{1}{m}F(t)dt+u(t)+p\int au(t)+bv(t)dt.$	$v=\int\frac{1}{m}F(t)dt+u(t)+p\int au(t)+bv(t)dt.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					
LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					
[refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches_1 EQ0198	288	■0.998 ● N +0	$\begin{aligned} I_X * M(x, y) &= \bigvee_{x' \in X} I_X(x, x') \otimes M(x', y) \\ &= (I_X(x, x) \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} I_X(x, x') \otimes M(x', y) \right) \\ &= (I \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} 0 \otimes M(x', y) \right) \\ &= M(x, y) \vee 0 = M(x, y). \end{aligned}$	$I_X * M(x, y) = \bigvee_{x' \in X} I_X(x, x') \otimes M(x', y)$ $= (I_X(x, x) \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} I_X(x, x') \otimes M(x', y) \right)$ $= (I \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} 0 \otimes M(x', y) \right)$ $= M(x, y) \vee 0 = M(x, y).$	$I_X * M(x, y) = \bigvee_{x' \in X} I_X(x, x') \otimes M(x', y)$ $= (I_X(x, x) \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} I_X(x, x') \otimes M(x', y) \right)$ $= (I \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} 0 \otimes M(x', y) \right)$ $= M(x, y) \vee 0 = M(x, y).$
fong-spivak-seven-sketches_1 EQ0070	117	■0.999 ● N -2	$\Sigma_F(I)(\text{Airline Seat}) = I(\text{Economy}) \sqcup I(\text{First Class}).$	$\Sigma_F(I)(\text{Airline Seat}) = I(\text{Economy}) \sqcup I(\text{First Class}).$	$\Sigma_F(I)(\text{Airline Seat}) = I(\text{Economy}) \sqcup I(\text{First Class}).$
fong-spivak-seven-sketches_1 EQ0084 (4.20)	138	■0.999 ● N +0	$(\Phi; \Psi)(p, r) := \bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r).$	$(\Phi; \Psi)(p, r) := \bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r).$	$(\Phi \circ \Psi)(p, r) := \bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r).$
fong-spivak-seven-sketches_1 EQ0008 (1.96)	039	■0.999 ● N +0	$f(p) \leq q \quad \text{if and only if} \quad p \leq g(q).$	$f(p) \leq q \quad \text{if and only if} \quad p \leq g(q).$	$f(p) \leq q \quad \text{if and only if} \quad p \leq g(q).$
fong-spivak-seven-sketches_1 EQ0035	076	■0.999 ● W +1	$\begin{array}{l} I_W \leq f(I_V) \quad (f \text{ is monoidal monotone}) \\ \leq f(\mathcal{C}(c, c)) \quad (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ = \mathcal{C}_f(c, c) \quad (\text{definition of } \mathcal{C}_f) \end{array}$	$\begin{array}{l} I_W \leq f(I_V) \quad (f \text{ is monoidal monotone}) \\ \leq f(\mathcal{C}(c, c)) \quad (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ = \mathcal{C}_f(c, c) \quad (\text{definition of } \mathcal{C}_f) \end{array}$	$\begin{array}{l} I_W \leq f(I_V) \quad (f \text{ is monoidal monotone}) \\ \leq f(\mathcal{C}(c, c)) \quad (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ = \mathcal{C}_f(c, c) \quad (\text{definition of } \mathcal{C}_f) \end{array}$
fong-spivak-seven-sketches_1 EQ0042	082	■0.999 ● N -1	$a + c \rightarrow d \quad \text{if and only if} \quad a \rightarrow (c \rightarrow d).$	$a + c \rightarrow d \quad \text{if and only if} \quad a \rightarrow (c \rightarrow d).$	$a + c \rightarrow d \quad \text{if and only if} \quad a \rightarrow (c \rightarrow d).$
fong-spivak-seven-sketches_1 EQ0065 (3.66)	113	■0.999 ● N +0	$\text{Gr} \xrightarrow{F} \text{DDS} \xrightarrow{I} \text{Set}.$	$\text{Gr} \xrightarrow{F} \text{DDS} \xrightarrow{I} \text{Set}.$	$\text{Gr} \xrightarrow{F} \text{DDS} \xrightarrow{I} \text{Set}.$
fong-spivak-seven-sketches_1 EQ0126 (6.15)	199	■0.999 ● S +0	$[f, g](x) = \begin{cases} f(x) & \text{if } x = \iota_A(a) \text{ for some } a \in A; \\ g(x) & \text{if } x = \iota_B(b) \text{ for some } b \in B. \end{cases}$	$[f, g](x) = \begin{cases} f(x) & \text{if } x = \iota_A(a) \text{ for some } a \in A; \\ g(x) & \text{if } x = \iota_B(b) \text{ for some } b \in B. \end{cases}$	$[f, g](x) = \begin{cases} f(x) & \text{if } x = \iota_A(a) \text{ for some } a \in A; \\ g(x) & \text{if } x = \iota_B(b) \text{ for some } b \in B. \end{cases}$
fong-spivak-seven-sketches_1 EQ0139	220	■0.999 ●	$\begin{aligned} \text{Circ}(f) : \text{Circ}(V) &\longrightarrow \text{Circ}(V'); \\ (V, A, s, t, \ell) &\longmapsto (V', A, (s; f), (t; f), \ell). \end{aligned}$	$\begin{aligned} \text{Circ}(f) : \text{Circ}(V) &\longrightarrow \text{Circ}(V'); \\ (V, A, s, t, \ell) &\longmapsto (V', A, (s; f), (t; f), \ell). \end{aligned}$	$\begin{aligned} \text{Circ}(f) : \text{Circ}(V) &\longrightarrow \text{Circ}(V'); \\ (V, A, s, t, \ell) &\longmapsto (V', A, (s \circ f), (t \circ f), \ell). \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches EQ0032	068	■ 1.000 ● S -1	$d(x) := \left\{ \begin{array}{ll} \text{false} & \text{if } x > 0 \\ \text{true} & \text{if } x = 0 \end{array} \right. \quad u(x) := \left\{ \begin{array}{ll} \text{false} & \text{if } x = \infty \\ \text{true} & \text{if } x < \infty \end{array} \right.$	$d(x) := \left\{ \begin{array}{ll} \text{false} & \text{if } x > 0 \\ \text{true} & \text{if } x = 0 \end{array} \right. \quad u(x) := \left\{ \begin{array}{ll} \text{false} & \text{if } x = \infty \\ \text{true} & \text{if } x < \infty \end{array} \right.$	$d(x) := \left\{ \begin{array}{ll} \text{false} & \text{if } x > 0 \\ \text{true} & \text{if } x = 0 \end{array} \right. \quad u(x) := \left\{ \begin{array}{ll} \text{false} & \text{if } x = \infty \\ \text{true} & \text{if } x < \infty \end{array} \right.$
fong-spivak-seven-sketches EQ0176	272	■ 1.000 ● K +0	$\varnothing, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}.$	$\varnothing, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}.$	$\varnothing, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}.$
fong-spivak-seven-sketches EQ0252	313	■ 1.000 ● W +0	$\begin{aligned} & \mathrm{B}(g^{\mathrm{op}}) = \{(y, x) \in R^n \times R^m \mid y = S(g)(x)\} \\ & \mathrm{B}(h) = \{(x, z) \in R^m \times R^p \mid S(h)(x) = z\} \end{aligned}$	$\mathrm{B}(g^{\mathrm{op}}) = \{(y, x) \in R^n \times R^m \mid y = S(g)(x)\}$ $\mathrm{B}(h) = \{(x, z) \in R^m \times R^p \mid S(h)(x) = z\}$	$\mathrm{B}(g^{\mathrm{op}}) = \{(y, x) \in R^n \times R^m \mid y = S(g)(x)\}$ $\mathrm{B}(h) = \{(x, z) \in R^m \times R^p \mid S(h)(x) = z\}$
fong-spivak-seven-sketches EQ0001 (1.8)	018	■ 1.000 ● K +0	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B).$	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B).$	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B).$
fong-spivak-seven-sketches EQ0004	024	■ 1.000 ● N +0	$f(11)=a, \quad f(12)=a, \quad f(13)=b, \quad f(21)=c, \quad f(22)=d, \quad f(23)=d$	$f(11) = a, \quad f(12) = a, \quad f(13) = b, \quad f(21) = c, \quad f(22) = d, \quad f(23) = d$	$f(11) = a, \quad f(12) = a, \quad f(13) = b, \quad f(21) = c, \quad f(22) = d, \quad f(23) = d$
fong-spivak-seven-sketches EQ0006	035	■ 1.000 ● N +0	$\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\} \subseteq \mathbb{R}$	$\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\} \subseteq \mathbb{R}$	$\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\} \subseteq \mathbb{R}$
fong-spivak-seven-sketches EQ0009	039	■ 1.000 ● K +0	$\lceil x/3 \rceil \leq y \text{ if and only if } x \leq 3y.$	$\lceil x/3 \rceil \leq y \text{ if and only if } x \leq 3y.$	$\lceil x/3 \rceil \leq y \text{ if and only if } x \leq 3y.$
fong-spivak-seven-sketches EQ0011	043	■ 1.000 ● C +4	$g(\bigwedge A) \cong \bigwedge g(A).$	$g(\bigwedge A) \cong \bigwedge g(A).$	$g(\bigwedge A) \cong \bigwedge g(A).$
fong-spivak-seven-sketches EQ0012	043	■ 1.000 ● C +3	$f(\bigvee A) \cong \bigvee f(A)$	$f(\bigvee A) \cong \bigvee f(A)$	$f(\bigvee A) \cong \bigvee f(A).$
fong-spivak-seven-sketches EQ0014	044	■ 1.000 ● N +0	$p_0 \leq \bigwedge \{g(q) \in P \mid p_0 \leq g(q)\} \cong g\left(\bigwedge \{q \in Q \mid p_0 \leq g(q)\}\right) = g(f(p_0)),$	$p_0 \leq \bigwedge \{g(q) \in P \mid p_0 \leq g(q)\} \cong g\left(\bigwedge \{q \in Q \mid p_0 \leq g(q)\}\right) = g(f(p_0)),$	$p_0 \leq \bigwedge \{g(q) \in P \mid p_0 \leq g(q)\} \cong g\left(\bigwedge \{q \in Q \mid p_0 \leq g(q)\}\right) = g(f(p_0)),$
fong-spivak-seven-sketches EQ0015	044	■ 1.000 ● N +0	$f(g(q_0)) = \bigwedge \{q \in Q \mid g(q_0) \leq g(q)\} \leq \bigwedge \{q_0\} = q_0,$	$f(g(q_0)) = \bigwedge \{q \in Q \mid g(q_0) \leq g(q)\} \leq \bigwedge \{q_0\} = q_0,$	$f(g(q_0)) = \bigwedge \{q \in Q \mid g(q_0) \leq g(q)\} \leq \bigwedge \{q_0\} = q_0,$
fong-spivak-seven-sketches EQ0016	045	■ 1.000 ● N -1	$f_!(A') := \{b \in B \mid \text{there exists } a \in A' \text{ such that } f(a) = b\}$	$f_!(A') := \{b \in B \mid \text{there exists } a \in A' \text{ such that } f(a) = b\}$	$f_!(A') := \{b \in B \mid \text{there exists } a \in A' \text{ such that } f(a) = b\}$
fong-spivak-seven-sketches EQ0017	045	■ 1.000 ● N -1	$f_*(A') := \{b \in B \mid \text{for all } a \text{ such that } f(a) = b, \text{ we have } a \in A'\}$	$f_*(A') := \{b \in B \mid \text{for all } a \text{ such that } f(a) = b, \text{ we have } a \in A'\}$	$f_*(A') := \{b \in B \mid \text{for all } a \text{ such that } f(a) = b, \text{ we have } a \in A'\}$
fong-spivak-seven-sketches EQ0019 (2.7)	054	■ 1.000 ● N +0	$m * e = m, \quad e * m = m, \quad (m * n) * p = m * (n * p)$	$m * e = m, \quad e * m = m, \quad (m * n) * p = m * (n * p)$	$m * e = m, \quad e * m = m, \quad (m * n) * p = m * (n * p)$
fong-spivak-seven-sketches EQ0023 (2.16)	059	■ 1.000 ●	$t \leq v + w \quad w + u \leq x + z \quad v + x \leq y$	$t \leq v + w \quad w + u \leq x + z \quad v + x \leq y$	$t \leq v + w \quad w + u \leq x + z \quad v + x \leq y$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches EQ0024 (2.17)	059	■ 1.000 ● K +0	$t+u \leq y+z$.	$t + u \leq y + z.$	$t + u \leq y + z.$
fong-spivak-seven-sketches EQ0025 (2.18)	059	■ 1.000 ● N +0	$t+u \leq v+w+u \leq v+x+z \leq y+z$.	$t + u \leq v + w + u \leq v + x + z \leq y + z.$	$t + u \leq v + w + u \leq v + x + z \leq y + z.$
fong-spivak-seven-sketches EQ0027 (2.24)	061	■ 1.000 ● N +0	$2 \mathrm{H}_{_2} \mathrm{H}_{_2} + 2 \mathrm{Na} \rightarrow 2 \mathrm{NaOH} + \mathrm{H}_2$ $\mathrm{NaOH} + \mathrm{H}_{_2}$.	$2\mathrm{H}_2\mathrm{O} + 2\mathrm{Na} \rightarrow 2\mathrm{NaOH} + \mathrm{H}_2.$	$2\mathrm{H}_2\mathrm{O} + 2\mathrm{Na} \rightarrow 2\mathrm{NaOH} + \mathrm{H}_2.$
fong-spivak-seven-sketches EQ0028 (2.22)	061	■ 1.000 ● N +0	$y+k \rightarrow y^{\prime}+k^{\prime} \quad x+y^{\prime} \rightarrow z^{\prime} \quad z^{\prime}+k^{\prime} \rightarrow z+k$ $\rightarrow z^{\prime} \quad z^{\prime}+k^{\prime}$ $\rightarrow z+k$	$y + k \rightarrow y' + k' \quad x + y' \rightarrow z' \quad z' + k' \rightarrow z + k$	$y + k \rightarrow y' + k' \quad x + y' \rightarrow z' \quad z' + k' \rightarrow z + k$
fong-spivak-seven-sketches EQ0029 (2.23)	061	■ 1.000 ● N +0	$x+y+k \rightarrow x+y^{\prime}+k^{\prime} \rightarrow z^{\prime}+k^{\prime} \rightarrow z+k$ $\rightarrow z^{\prime}+k^{\prime} \rightarrow z+k$.	$x + y + k \rightarrow x + y' + k' \rightarrow z' + k' \rightarrow z + k.$	$x + y + k \rightarrow x + y' + k' \rightarrow z' + k' \rightarrow z + k.$
fong-spivak-seven-sketches EQ0037 (2.66)	077	■ 1.000 ● S +1	$f(x) := \begin{cases} \text{true} & \text{if } x = 0 \\ \text{false} & \text{if } x > 0 \end{cases}$	$f(x) := \begin{cases} \text{true} & \text{if } x = 0 \\ \text{false} & \text{if } x > 0 \end{cases}$	$f(x) := \begin{cases} \text{true} & \text{if } x = 0 \\ \text{false} & \text{if } x > 0 \end{cases}$
fong-spivak-seven-sketches EQ0044 (2.88)	082	■ 1.000 ● C +8	$\left(v \otimes \bigvee_{a \in A} a\right) \cong \bigvee_{a \in A} (v \otimes a)$.	$\left(v \otimes \bigvee_{a \in A} a\right) \cong \bigvee_{a \in A} (v \otimes a).$	$\left(v \otimes \bigvee_{a \in A} a\right) \cong \bigvee_{a \in A} (v \otimes a).$
fong-spivak-seven-sketches EQ0047 (2.99)	085	■ 1.000 ● N +0	$(M * N)(i, k) = \sum_j M(i, j) * N(j, k)$.	$(M * N)(i, k) = \sum_j M(i, j) * N(j, k).$	$(M * N)(i, k) = \sum_j M(i, j) * N(j, k).$
fong-spivak-seven-sketches EQ0048 (2.101)	086	■ 1.000 ● N +1	$(M * N)(x, z) := \bigvee_{y \in Y} M(x, y) \otimes N(y, z)$.	$(M * N)(x, z) := \bigvee_{y \in Y} M(x, y) \otimes N(y, z).$	$(M * N)(x, z) := \bigvee_{y \in Y} M(x, y) \otimes N(y, z).$
fong-spivak-seven-sketches EQ0049	086	■ 1.000 ● C +6	$\left(\begin{array}{cc} \text{false} & \text{false} \\ \text{false} & \text{true} \\ \text{true} & \text{true} \end{array}\right) * \left(\begin{array}{ccc} \text{true} & \text{true} & \text{false} \\ \text{true} & \text{false} & \text{true} \end{array}\right) = \left(\begin{array}{ccc} \text{false} & \text{false} & \text{false} \\ \text{true} & \text{false} & \text{true} \\ \text{true} & \text{true} & \text{true} \end{array}\right)$	$\left(\begin{array}{ccc} \text{false} & \text{false} & \text{false} \\ \text{true} & \text{false} & \text{true} \\ \text{true} & \text{true} & \text{true} \end{array}\right)$	$\left(\begin{array}{ccc} \text{false} & \text{false} & \text{false} \\ \text{true} & \text{false} & \text{true} \\ \text{true} & \text{true} & \text{true} \end{array}\right)$
fong-spivak-seven-sketches EQ0050	086	■ 1.000 ● N +0	$I_X(x, y) := \begin{cases} I & \text{if } x = y \\ 0 & \text{if } x \neq y \end{cases}$	$I_X(x, y) := \begin{cases} I & \text{if } x = y \\ 0 & \text{if } x \neq y. \end{cases}$	$I_X(x, y) := \begin{cases} I & \text{if } x = y \\ 0 & \text{if } x \neq y. \end{cases}$
fong-spivak-seven-sketches EQ0051	087	■ 1.000 ● N +0	$M_Y^2 = \begin{array}{c ccc} & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}$	$M_Y^2 = \begin{array}{c ccc} & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}$	$M_Y^2 = \begin{array}{c ccc} & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches_087 EQ0052	087	1.000	$M_{Y}^{\wedge 3}=\begin{array}{c ccc}\nearrow&x&y&z\\\hline x&0&4&3\\y&3&0&6\\z&7&4&0\end{array}$	$M_Y^3 = \begin{array}{c ccc}\nearrow&x&y&z\\\hline x&0&4&3\\y&3&0&6\\z&7&4&0\end{array}$	$M_Y^3 = \begin{array}{c ccc}\nearrow&x&y&z\\\hline x&0&4&3\\y&3&0&6\\z&7&4&0\end{array}$
fong-spivak-seven-sketches_096 EQ0055	096	1.000 N +0	$\{A, B, C, D, f, g, h, i, f ; h\}$	$\{A, B, C, D, f, g, h, i, f; h\}$	$\{A, B, C, D, f, g, h, i, f \circ h\}$
fong-spivak-seven-sketches_104 EQ0058	104	1.000 S +0	$f^{\wedge \{ \prime \}} ; h^{\wedge \{ \prime \}} = F(f) ; F(h) = F(f ; h) = F(g ; i) = F(g) ; F(i) = g^{\wedge \{ \prime \}} ; i^{\wedge \{ \prime \}} \\ i) = F(g) ; F(i) = g^{\wedge \{ \prime \}} ; i^{\wedge \{ \prime \}}$	$f'; h' = F(f); F(h) = F(f; h) = F(g; i) = F(g); F(i) = g'; i'$	$f' \circ h' = F(f) \circ F(h) = F(f \circ h) = F(g \circ i) = F(g) \circ F(i) = g' \circ i'$
fong-spivak-seven-sketches_107 EQ0060 (3.50)	107	1.000 N +1	$F(f) ; \alpha_d = \alpha_c ; G(f) .$	$F(f); \alpha_d = \alpha_c; G(f).$	$F(f) \circ \alpha_d = \alpha_c \circ G(f).$
fong-spivak-seven-sketches_110 EQ0062	110	1.000 C +4	$\alpha_{\text{Vertex}} : G(\text{Vertex}) \rightarrow H(\text{Vertex}) \quad \text{and} \quad \alpha_{\text{Arrow}} : G(\text{Arrow}) \rightarrow H(\text{Arrow}),$	$\alpha_{\text{Vertex}} : G(\text{Vertex}) \rightarrow H(\text{Vertex}) \quad \text{and} \quad \alpha_{\text{Arrow}} : G(\text{Arrow}) \rightarrow H(\text{Arrow}),$	$\alpha_{\text{Vertex}} : G(\text{Vertex}) \rightarrow H(\text{Vertex}) \quad \text{and} \quad \alpha_{\text{Arrow}} : G(\text{Arrow}) \rightarrow H(\text{Arrow}),$
fong-spivak-seven-sketches_114 EQ0067	114	1.000 C +4	$\alpha_{c, d}: \mathcal{C}(c, R(d)) \xrightarrow{\cong} \mathcal{D}(L(c), d)$	$\alpha_{c, d}: \mathcal{C}(c, R(d)) \xrightarrow{\cong} \mathcal{D}(L(c), d)$	$\alpha_{c, d}: \mathcal{C}(c, R(d)) \xrightarrow{\cong} \mathcal{D}(L(c), d)$
fong-spivak-seven-sketches_121 EQ0076	121	1.000 K +0	$X \times Y := \{(x, y) \mid x \in X, y \in Y\},$	$X \times Y := \{(x, y) \mid x \in X, y \in Y\},$	$X \times Y := \{(x, y) \mid x \in X, y \in Y\},$
fong-spivak-seven-sketches_141 EQ0090	141	1.000 C +3	$(\Phi ; \Psi) ; \Upsilon = \Phi ; (\Psi ; \Upsilon) .$	$(\Phi; \Psi); \Upsilon = \Phi; (\Psi; \Upsilon).$	$(\Phi \circ \Psi) \circ \Upsilon = \Phi \circ (\Psi \circ \Upsilon).$
fong-spivak-seven-sketches_148 EQ0095	148	1.000 N +0	$\alpha_{s, t, u}: \{(s, (t, u)) \mid s \in S, t \in T, u \in U\} \xrightarrow{\cong} \{((s, t), u) \mid s \in S, t \in T, u \in U\} .$	$\alpha_{s, t, u}: \{(s, (t, u)) \mid s \in S, t \in T, u \in U\} \xrightarrow{\cong} \{((s, t), u) \mid s \in S, t \in T, u \in U\} .$	$\alpha_{s, t, u}: \{(s, (t, u)) \mid s \in S, t \in T, u \in U\} \xrightarrow{\cong} \{((s, t), u) \mid s \in S, t \in T, u \in U\} .$
fong-spivak-seven-sketches_156 EQ0099	156	1.000 N +1	$(\Phi \times \Psi) (\left(\left(x_1, y_1 \right), \left(x_2, y_2 \right) \right) := \Phi \left(x_1, x_2 \right) \otimes \Psi \left(y_1, y_2 \right) .$	$(\Phi \times \Psi) ((x_1, y_1), (x_2, y_2)) := \Phi(x_1, x_2) \otimes \Psi(y_1, y_2).$	$(\Phi \times \Psi)((x_1, y_1), (x_2, y_2)) := \Phi(x_1, x_2) \otimes \Psi(y_1, y_2).$
fong-spivak-seven-sketches_162 EQ0101 (5.4)	162	1.000 W -1	$i \longmapsto \begin{cases} f(i) & \text{if } 1 \leq i \leq m; \\ m' + g(i) & \text{if } m + 1 \leq i \leq m + n. \end{cases}$	$i \mapsto \begin{cases} f(i) & \text{if } 1 \leq i \leq m; \\ m' + g(i) & \text{if } m + 1 \leq i \leq m + n. \end{cases}$	$i \mapsto \begin{cases} f(i) & \text{if } 1 \leq i \leq m; \\ m' + g(i) & \text{if } m + 1 \leq i \leq m + n. \end{cases}$
fong-spivak-seven-sketches_168 EQ0106 (5.26)	168	1.000 S +0	$G:=\left\{\rho_{m,n}\mid m,n\in\mathbb{N}\right\},\quad\quad\quad s\left(\rho_{m,n}\right):=m,\quad\quad\quad t\left(\rho_{m,n}\right):=n,$	$G := \{\rho_{m,n} \mid m, n \in \mathbb{N}\}, \quad s(\rho_{m,n}) := m, \quad t(\rho_{m,n}) := n,$	$G := \{\rho_{m,n} \mid m, n \in \mathbb{N}\}, \quad s(\rho_{m,n}) := m, \quad t(\rho_{m,n}) := n,$
fong-spivak-seven-sketches_174 EQ0108	174	1.000 W +0	$\begin{array}{l} D \ x + 3 D^2 y - 2 z = 0 \\ D^2 y + 5 D z = 0 \end{array}$	$\begin{array}{l} D x + 3 D^2 y - 2 z = 0 \\ D^2 y + 5 D z = 0 \end{array}$	$\begin{array}{l} D x + 3 D^2 y - 2 z = 0 \\ D^2 y + 5 D z = 0 \end{array}$
fong-spivak-seven-sketches_176 EQ0110 (5.48)	176	1.000 N +1	$M ; N(a, c) := \sum_{b \in \underline{n}} M(a, b) \times N(b, c),$	$M; N(a, c) := \sum_{b \in \underline{n}} M(a, b) \times N(b, c),$	$M \circ N(a, c) := \sum_{b \in \underline{n}} M(a, b) \times N(b, c),$
fong-spivak-seven-sketches_176 EQ0112	176	1.000 K +0	$A=\left(\begin{array}{lll} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array}\right) \quad B=\left(\begin{array}{llll} 2 & 5 & 6 & 1 \end{array}\right) .$	$A = \left(\begin{array}{ccc} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array} \right) \quad B = \left(\begin{array}{cccc} 2 & 5 & 6 & 1 \end{array} \right) .$	$A = \left(\begin{array}{ccc} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array} \right) \quad B = \left(\begin{array}{cccc} 2 & 5 & 6 & 1 \end{array} \right) .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches EQ0114	179	1.000 N +0	$\sum_{j \in \underline{n}} \alpha(i, j) * \beta(j, k) \text{ .}$	$\sum_{j \in \underline{n}} \alpha(i, j) * \beta(j, k).$	$\sum_{j \in \underline{n}} \alpha(i, j) * \beta(j, k).$
fong-spivak-seven-sketches EQ0118 (5.76)	186	1.000	$\mathrm{B}\!\left(g^{\mathrm{op}}\right)\!:=\!\left\{\!\left(S(g)(x), x\right) \mid x \in R^m\right\} \subseteq R^n \times R^m \text{ .}$	$B(g^{\mathrm{op}}) := \{(S(g)(x), x) \mid x \in R^m\} \subseteq R^n \times R^m.$	$B(g^{\mathrm{op}}) := \{(S(g)(x), x) \mid x \in R^m\} \subseteq R^n \times R^m.$
fong-spivak-seven-sketches EQ0122	188	1.000 N +0	$\mathrm{B}\!\left(\!\left(g^{\mathrm{op}}\right) ; h\right) = \left\{\!\left(S(g)(x), S(h)(x)\right) \mid x \in R^m\right\} \text{ .}$	$B((g^{\mathrm{op}}); h) = \{(S(g)(x), S(h)(x)) \mid x \in R^m\}.$	$B((g^{\mathrm{op}}) \mathbin{\mathbb{S}} h) = \{(S(g)(x), S(h)(x)) \mid x \in R^m\}.$
fong-spivak-seven-sketches EQ0127	203	1.000 N +1	$y(\theta)\!=\!y(g(\theta))\!=\!x(f(\theta))\!=\!x(\theta) \text{ .}$	$y(0) = y(g(0)) = x(f(0)) = x(0).$	$y(0) = y(g(0)) = x(f(0)) = x(0).$
fong-spivak-seven-sketches EQ0130	206	1.000 K +0	$\{(v, d) \mid d \in D(v)\}$	$\{(v, d) \mid d \in D(v)\}$	$\{(v, d) \mid d \in D(v)\}$
fong-spivak-seven-sketches EQ0135 (6.76)	219	1.000 N +0	$F\!\left(\!\left[\iota_N, \iota_P\right]\!\right)\!\left(\varphi_{N,P}(s,t)\right)$	$F([\iota_N, \iota_P])(\varphi_{N,P}(s, t))$	$F([\iota_N, \iota_P])(\varphi_{N,P}(s, t))$
fong-spivak-seven-sketches EQ0138	220	1.000 N +0	$\operatorname{Circ}(V)\!:=\!\{(V, A, s, t, \ell) \mid \text{where } s, t\colon A \rightarrow V, \ell\colon E \rightarrow C\} \text{ .}$	$\operatorname{Circ}(V) := \{(V, A, s, t, \ell) \mid \text{ where } s, t \colon A \rightarrow V, \ell \colon E \rightarrow C\}.$	$\operatorname{Circ}(V) := \{(V, A, s, t, \ell) \mid \text{ where } s, t \colon A \rightarrow V, \ell \colon E \rightarrow C\}.$
fong-spivak-seven-sketches EQ0141	225	1.000 C +7	$\operatorname{circ}_{\mathbf{i}}\!:\! \mathcal{O}\!\left(s_1, \ldots, s_m; t_{\mathbf{i}}\right) \times \mathcal{O}\!\left(t_1, \ldots, t_n\right) \rightarrow \mathcal{O}\!\left(t_1, \ldots, t_{i-1}, s_1, \ldots, s_m, t_{i+1}, \ldots, t_n; t\right);$	$\circ_i : \mathcal{O}(s_1, \dots, s_m; t_i) \times \mathcal{O}(t_1, \dots, t_n; t) \rightarrow \mathcal{O}(t_1, \dots, t_{i-1}, s_1, \dots, s_m, t_{i+1}, \dots, t_n; t);$	$\circ_i : \mathcal{O}(s_1, \dots, s_m; t_i) \times \mathcal{O}(t_1, \dots, t_n; t) \rightarrow \mathcal{O}(t_1, \dots, t_{i-1}, s_1, \dots, s_m, t_{i+1}, \dots, t_n; t);$
fong-spivak-seven-sketches EQ0145	229	1.000 N +2	$F\!:\! \mathcal{O}\!\left(t_1, \ldots, t_n\right) \rightarrow \mathcal{P}\!\left(f\!\left(t_1\right), \ldots, f\!\left(t_n\right); f(t)\right)$	$F : \mathcal{O}(t_1, \dots, t_n; t) \rightarrow \mathcal{P}(f(t_1), \dots, f(t_n); f(t))$	$F : \mathcal{O}(t_1, \dots, t_n; t) \rightarrow \mathcal{P}(f(t_1), \dots, f(t_n); f(t))$
fong-spivak-seven-sketches EQ0149 (7.10)	239	1.000 N +0	$\mathcal{C}(A \times C, D) \cong \mathcal{C}(A, D^{\mathcal{C}})$	$\mathcal{C}(A \times C, D) \cong \mathcal{C}(A, D^C)$	$\mathcal{C}(A \times C, D) \cong \mathcal{C}(A, D^C)$
fong-spivak-seven-sketches EQ0151	241	1.000 W +1	$\ulcorner m \urcorner(y) := \begin{cases} \text{true} & \text{if } m(x) = y \text{ for some } x \in X \\ \text{false} & \text{otherwise} \end{cases}$	$\ulcorner m \urcorner(y) := \begin{cases} \text{true} & \text{if } m(x) = y \text{ for some } x \in X \\ \text{false} & \text{otherwise} \end{cases}$	$\ulcorner m \urcorner(y) := \begin{cases} \text{true} & \text{if } m(x) = y \text{ for some } x \in X \\ \text{false} & \text{otherwise} \end{cases}$
fong-spivak-seven-sketches EQ0152 (7.15)	241	1.000 N +0	$X=\{y \in Y \mid p(y)=\text{true}\} \text{ .}$	$X = \{y \in Y \mid p(y) = \text{true}\}.$	$X = \{y \in Y \mid p(y) = \text{true}\}.$
fong-spivak-seven-sketches EQ0155	246	1.000 N +0	$B(p \ ; \ \epsilon)\!:=\!\left\{p^{\prime} \in \mathbb{R}^2 \mid d(p, p^{\prime}) < \epsilon\right\}^7$	$B(p; \epsilon) := \{p' \in \mathbb{R}^2 \mid d(p, p') < \epsilon\}^7$	$B(p; \epsilon) := \{p' \in \mathbb{R}^2 \mid d(p, p') < \epsilon\}^7$
fong-spivak-seven-sketches EQ0161 (7.51)	255	1.000 N +0	$\begin{aligned} &\Omega(U) \ \& \ \rightarrow \ \Omega(V) \ \backslash \ U^{\prime} \\ &\hspace{1.5cm} U' \mapsto U' \cap V. \end{aligned}$	$\Omega(U) \rightarrow \Omega(V)$ $U' \mapsto U' \cap V.$	$\Omega(U) \rightarrow \Omega(V)$ $U' \mapsto U' \cap V.$
fong-spivak-seven-sketches EQ0162	256	1.000 W +0	$V \cap U_{\mathbf{j}}\!=\!\left(\bigcup_{i \in I} V_i\right) \cap U_{\mathbf{j}}\!=\!\bigcup_{i \in I} (V_i \cap U_{\mathbf{j}})\!=\!\bigcup_{i \in I} (V_j \cap U_i)\!=\!\left(\bigcup_{i \in I} U_i\right) \cap V_j\!=\!V_j$	$V \cap U_j = \left(\bigcup_{i \in I} V_i\right) \cap U_j = \bigcup_{i \in I} (V_i \cap U_j) = \bigcup_{i \in I} (V_j \cap U_i) = \left(\bigcup_{i \in I} U_i\right) \cap V_j = V_j$	$V \cap U_j = \left(\bigcup_{i \in I} V_i\right) \cap U_j = \bigcup_{i \in I} (V_i \cap U_j) = \bigcup_{i \in I} (V_j \cap U_i) = \left(\bigcup_{i \in I} U_i\right) \cap V_j = V_j$
fong-spivak-seven-sketches EQ0167 (7.63)	259	1.000 K +0	$\left(\left \Omega^{\mathcal{S}}\right , \leq^{\mathcal{S}}\right)$	$(\Omega^S , \leq^S)$	$(\Omega^S , \leq^S)$

Conf. MathPix confidence: ≥ 0.9 0.5–0.9 < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches EQ0170	261	1.000 W +2	$p(n, z) = \begin{cases} \text{true} & \text{if } n \leq z \\ \text{false} & \text{if } n > z . \end{cases}$	$p(n, z) = \begin{cases} \text{true} & \text{if } n \leq z \\ \text{false} & \text{if } n > z . \end{cases}$	$p(n, z) = \begin{cases} \text{true} & \text{if } n \leq z \\ \text{false} & \text{if } n > z . \end{cases}$
fong-spivak-seven-sketches EQ0174	265	1.000 N -2	$S(\text{left}(o_{[a, b]}) \text{right}) \cong \lim_{\epsilon > 0} S(\text{left}(o_{[a - \epsilon, b + \epsilon]}) \text{right}) .$	$S(o_{[a, b]}) \cong \lim_{\epsilon > 0} S(o_{[a - \epsilon, b + \epsilon]}) .$	$S(o_{[a, b]}) \cong \lim_{\epsilon > 0} S(o_{[a - \epsilon, b + \epsilon]}) .$
fong-spivak-seven-sketches EQ0177	272	1.000	$(h, 1), \quad \text{quad}(h, 2), \quad \text{quad}(h, 3), \quad \text{quad}(1, 1), \quad \text{quad}(1, 2), \quad \text{quad}(1, 3) .$	$(h, 1), \quad (h, 2), \quad (h, 3), \quad (1, 1), \quad (1, 2), \quad (1, 3).$	$(h, 1), \quad (h, 2), \quad (h, 3), \quad (1, 1), \quad (1, 2), \quad (1, 3).$
fong-spivak-seven-sketches EQ0178	272	1.000 N +0	$(h, 1), \quad \text{quad}(1, 1), \quad \text{quad}(1, 2), \quad \text{quad}(2, 2), \quad \text{quad}(3, 2) .$	$(h, 1), \quad (1, 1), \quad (1, 2), \quad (2, 2), \quad (3, 2).$	$(h, 1), \quad (1, 1), \quad (1, 2), \quad (2, 2), \quad (3, 2).$
fong-spivak-seven-sketches EQ0180	273	1.000 W +0	$\begin{array}{lllll} (11, 11) & \& (11, 12) & \& (12, 11) & \& \\ (12, 12) & \& (13, 13) & \& (21, 21) & \& (22, 22) & \& (12, 23) & \& \\ (23, 22) & \& (23, 23) & \end{array}$	$\begin{array}{lllll} (11, 11) & (11, 12) & (12, 11) & (12, 12) & (13, 13) \\ (21, 21) & (22, 22) & (12, 23) & (23, 22) & (23, 23) \end{array}$	$\begin{array}{lllll} (11, 11) & (11, 12) & (12, 11) & (12, 12) & (13, 13) \\ (21, 21) & (22, 22) & (12, 23) & (23, 22) & (23, 23) \end{array}$
fong-spivak-seven-sketches EQ0182	274	1.000 K +0	$1 \leq 1, \quad \text{quad} \quad 1 \leq 2, \quad \text{quad} \quad 1 \leq 3, \quad \text{quad} \quad 2 \leq 2, \quad \text{quad} \quad 2 \leq 3, \quad \text{quad} \quad 3 \leq 3, \quad \text{quad} \quad 4 \leq 4$	$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 2 \leq 2, \quad 2 \leq 3, \quad 3 \leq 3, \quad 4 \leq 4$	$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 2 \leq 2, \quad 2 \leq 3, \quad 3 \leq 3, \quad 4 \leq 4$
fong-spivak-seven-sketches EQ0183	274	1.000 S +0	$(\bullet)(\circ)(*), \quad \text{quad}(\bullet\circ)(*), \quad \text{quad}(\bullet\circ)(\circ), \quad \text{quad}(\bullet)(\circ*), \quad \text{quad}(\bullet\circ*)$	$(\bullet)(\circ)(*), \quad (\bullet\circ)(*), \quad (\bullet\circ)(\circ), \quad (\bullet)(\circ*), \quad (\bullet\circ*)$	$(\bullet)(\circ)(*), \quad (\bullet\circ)(*), \quad (\bullet\circ)(\circ), \quad (\bullet)(\circ*), \quad (\bullet\circ*)$
fong-spivak-seven-sketches EQ0190	280	1.000 N +0	$g(q) \leq g^{\prime}(f(g(q))) \leq g^{\prime}(q) .$	$g(q) \leq g^{\prime}(f(g(q))) \leq g^{\prime}(q).$	$g(q) \leq g^{\prime}(f(g(q))) \leq g^{\prime}(q).$
fong-spivak-seven-sketches EQ0192	283	1.000 C +4	$\begin{gathered} g(\text{false}) + g(\text{false}) = \infty + \infty \geq \infty = g(\text{false} \wedge \text{false}) \\ g(\text{false}) + g(\text{true}) = \infty + 0 \geq \infty = g(\text{false} \wedge \text{true}) \\ g(\text{true}) + g(\text{false}) = 0 + \infty \geq \infty = g(\text{true} \wedge \text{false}) \\ g(\text{true}) + g(\text{true}) = 0 + 0 \geq 0 = g(\text{true} \wedge \text{true}). \end{gathered}$	$\begin{array}{l} g(\text{false}) + g(\text{false}) = \infty + \infty \geq \infty = g(\text{false} \wedge \text{false}) \\ g(\text{false}) + g(\text{true}) = \infty + 0 \geq \infty = g(\text{false} \wedge \text{true}) \\ g(\text{true}) + g(\text{false}) = 0 + \infty \geq \infty = g(\text{true} \wedge \text{false}) \\ g(\text{true}) + g(\text{true}) = 0 + 0 \geq 0 = g(\text{true} \wedge \text{true}) \end{array}$	$\begin{array}{l} g(\text{false}) + g(\text{false}) = \infty + \infty \geq \infty = g(\text{false} \wedge \text{false}) \\ g(\text{false}) + g(\text{true}) = \infty + 0 \geq \infty = g(\text{false} \wedge \text{true}) \\ g(\text{true}) + g(\text{false}) = 0 + \infty \geq \infty = g(\text{true} \wedge \text{false}) \\ g(\text{true}) + g(\text{true}) = 0 + 0 \geq 0 = g(\text{true} \wedge \text{true}). \end{array}$
fong-spivak-seven-sketches EQ0196	287	1.000 W +1	$\text{left}(\begin{array}{ll} 1 & \& 0 \\ 0 & \& 1 \end{array} \text{right}), \quad \text{quad} \quad \text{left}(\begin{array}{cc} \text{true} & \\ \text{false} & \end{array} \text{right}), \quad \text{quad} \quad \text{left}(\begin{array}{cc} 0 & \infty \\ \infty & 0 \end{array} \text{right}) .$	$\left(\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right), \quad \left(\begin{array}{cc} \text{true} & \text{false} \\ \text{false} & \text{true} \end{array} \right), \quad \text{and} \quad \left(\begin{array}{cc} 0 & \infty \\ \infty & 0 \end{array} \right).$	$\left(\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right), \quad \left(\begin{array}{cc} \text{true} & \text{false} \\ \text{false} & \text{true} \end{array} \right), \quad \text{and} \quad \left(\begin{array}{cc} 0 & \infty \\ \infty & 0 \end{array} \right).$
fong-spivak-seven-sketches EQ0197	287	1.000 N +0	$0 \otimes v \cong v \otimes 0 \cong \left(v \otimes \bigvee_{a \in \emptyset} a \right) = \bigvee_{a \in \emptyset} (v \otimes a) = 0.$	$0 \otimes v \cong v \otimes 0 \cong \left(v \otimes \bigvee_{a \in \emptyset} a \right) = \bigvee_{a \in \emptyset} (v \otimes a) = 0.$	$0 \otimes v \cong v \otimes 0 \cong \left(v \otimes \bigvee_{a \in \emptyset} a \right) = \bigvee_{a \in \emptyset} (v \otimes a) = 0.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches- EQ0200	288	■ 1.000 ● N +0	$\begin{aligned} M_{\{X\}} &= \left(\begin{array}{cccc} 0 & \infty & 3 & \infty \\ 2 & 0 & \infty & 5 \\ \infty & 3 & 0 & \infty \\ \infty & \infty & 6 & 0 \end{array} \right) & M_X^2 &= \left(\begin{array}{cccc} 0 & 6 & 3 & \infty \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ \infty & 9 & 6 & 0 \end{array} \right) & M_X^3 &= M_X^4 = \left(\begin{array}{cccc} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{array} \right) \end{aligned}$		
fong-spivak-seven-sketches- EQ0201	289	■ 1.000 ● W -1	$t(p) := \left(\begin{array}{l} v \text{ if } n = 0 \\ \left(a_n \right) \text{ if } n \geq 1 \end{array} \right)$	$t(p) := \begin{cases} v \text{ if } n = 0 \\ t(a_n) \text{ if } n \geq 1 \end{cases}$	$t(p) := \begin{cases} v \text{ if } n = 0 \\ t(a_n) \text{ if } n \geq 1 \end{cases}$
fong-spivak-seven-sketches- EQ0202	289	■ 1.000 ● N +0	$(p ; q) := \left(v, a_1, \ldots, a_m, b_1, \ldots, b_n, b_n \right) .$	$(p; q) := (v, a_1, \ldots, a_m, b_1, \ldots, b_n) .$	$(p \circ q) := (v, a_1, \ldots, a_m, b_1, \ldots, b_n) .$
fong-spivak-seven-sketches- EQ0203	289	■ 1.000 ●	$\left(v, a_1, \ldots, a_m, b_1, \ldots, b_n, c_1, \ldots, c_o \right) .$	$(v, a_1, \ldots, a_m, b_1, \ldots, b_n, c_1, \ldots, c_o) .$	$(v, a_1, \ldots, a_m, b_1, \ldots, b_n, c_1, \ldots, c_o) .$
fong-spivak-seven-sketches- EQ0207	291	■ 1.000 ● K +0	$\begin{array}{l} (1,1) \text{ \& } (1,2) \text{ \& } (1,3) \\ (2,1) \text{ \& } (2,2) \text{ \& } (2,3) \\ (3,1) \text{ \& } (3,2) \text{ \& } (3,3) \end{array}$	$\begin{array}{l} (1,1) \text{ \& } (1,2) \text{ \& } (1,3) \\ (2,1) \text{ \& } (2,2) \text{ \& } (2,3) \\ (3,1) \text{ \& } (3,2) \text{ \& } (3,3) \end{array}$	$\begin{array}{l} (1,1) \text{ \& } (1,2) \text{ \& } (1,3) \\ (2,1) \text{ \& } (2,2) \text{ \& } (2,3) \\ (3,1) \text{ \& } (3,2) \text{ \& } (3,3) \end{array}$
fong-spivak-seven-sketches- EQ0208	291	■ 1.000 ● N +0	$f^{-1}(1)=b, \quad f^{-1}(2)=a, \quad f^{-1}(3)=c .$	$f^{-1}(1)=b, \quad f^{-1}(2)=a, \quad f^{-1}(3)=c .$	$f^{-1}(1)=b, \quad f^{-1}(2)=a, \quad f^{-1}(3)=c .$
fong-spivak-seven-sketches- EQ0209	291	■ 1.000 ● W +0	$\begin{gathered} A^{\prime} \mapsto A, \quad f^{\prime} \mapsto f, \quad g^{\prime} \mapsto g, \quad f^{\prime} ; h^{\prime} \mapsto f ; h, \quad g^{\prime} ; i^{\prime} \mapsto f ; h, \\ B^{\prime} \mapsto B, \quad h^{\prime} \mapsto h, \quad C^{\prime} \mapsto C, \quad i^{\prime} \mapsto i, \quad D^{\prime} \mapsto D . \end{gathered}$	$\begin{aligned} A' &\mapsto A, \quad f' \mapsto f, \quad g' \mapsto g, \quad f'; h' \mapsto f; h, \quad g'; i' \mapsto f; h, \\ B' &\mapsto B, \quad h' \mapsto h, \quad C' \mapsto C, \quad i' \mapsto i, \quad D' \mapsto D. \end{aligned}$	$\begin{aligned} A' &\mapsto A, \quad f' \mapsto f, \quad g' \mapsto g, \quad f' \circ h' \mapsto f \circ h, \quad g' \circ i' \mapsto f \circ h, \\ B' &\mapsto B, \quad h' \mapsto h, \quad C' \mapsto C, \quad i' \mapsto i, \quad D' \mapsto D. \end{aligned}$
fong-spivak-seven-sketches- EQ0211	292	■ 1.000 ● S +0	$(F ; G)(f ; g)=G(F(f) ; F(g))=G(F(f)) ; G(F(g))$	$(F;G)(f;g)=G(F(f);F(g))=G(F(f));G(F(g))$	$(F \circ G)(f \circ g)=G(F(f) \circ F(g))=G(F(f)) \circ G(F(g))$
fong-spivak-seven-sketches- EQ0212	292	■ 1.000 ● S +0	$\left((F ; G) ; H \right) (c)=H\left((F ; G)(c) \right)=H(G(F(c)))=(G ; H)(F(c))=(F ; (G ; H))(c) .$	$((F;G);H)(c)=H((F;G)(c))=H(G(F(c)))=(G;H)(F(c))=(F;(G;H))(c) .$	$((F \circ G) \circ H)(c)=H((F \circ G)(c))=H(G(F(c)))=(G \circ H)(F(c))=(F \circ (G \circ H))(c) .$
fong-spivak-seven-sketches- EQ0215	295	■ 1.000 ● W +0	$\left(f_1 ; f_2 \right)^B(g)=g ; \left(f_1 ; f_2 \right)=\left(g ; f_1 \right) ; f_2=\left(f_1^B ; f_2^B \right)(g)$	$\left(f_1 ; f_2 \right)^B(g)=g ; \left(f_1 ; f_2 \right)=\left(g ; f_1 \right) ; f_2=\left(f_1^B ; f_2^B \right)(g)$	$\left(f_1 \circ f_2 \right)^B(g)=g \circ \left(f_1 \circ f_2 \right)=\left(g \circ f_1 \right) \circ f_2=\left(f_1^B \circ f_2^B \right)(g)$
fong-spivak-seven-sketches- EQ0216	296	■ 1.000 ● W +0	$\left(c_1, d_1 \right) \xrightarrow{\left(f_1, g_1 \right)} \left(c_2, d_2 \right) \xrightarrow{\left(f_2, g_2 \right)} \left(c_3, d_3 \right) \xrightarrow{\left(f_3, g_3 \right)} \left(c_4, d_4 \right) .$	$\left(c_1, d_1 \right) \xrightarrow{\left(f_1, g_1 \right)} \left(c_2, d_2 \right) \xrightarrow{\left(f_2, g_2 \right)} \left(c_3, d_3 \right) \xrightarrow{\left(f_3, g_3 \right)} \left(c_4, d_4 \right) .$	$\left(c_1, d_1 \right) \xrightarrow{\left(f_1, g_1 \right)} \left(c_2, d_2 \right) \xrightarrow{\left(f_2, g_2 \right)} \left(c_3, d_3 \right) \xrightarrow{\left(f_3, g_3 \right)} \left(c_4, d_4 \right) .$
fong-spivak-seven-sketches- EQ0220	298	■ 1.000 ● N +1	$\left(\mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \right) \left((x, y), \left(x', y' \right) \right) \leq \mathcal{V} \left(\Phi(x, y), \Phi \left(x', y' \right) \right) .$	$\left(\mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \right) \left((x, y), \left(x', y' \right) \right) \leq \mathcal{V} \left(\Phi(x, y), \Phi \left(x', y' \right) \right) .$	$\left(\mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \right) \left((x, y), \left(x', y' \right) \right) \leq \mathcal{V} \left(\Phi(x, y), \Phi \left(x', y' \right) \right) .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketchesEQ0231	306	■ 1.000 ● W -2	$\sigma_{m,n}(i) := \begin{cases} i+n & \text{if } i \leq m \\ i-m & \text{if } m+1 \leq i \end{cases}$	$\sigma_{m,n}(i) := \begin{cases} i+n & \text{if } i \leq m \\ i-m & \text{if } m+1 \leq i \end{cases}$	$\sigma_{m,n}(i) := \begin{cases} i+n & \text{if } i \leq m \\ i-m & \text{if } m+1 \leq i \end{cases}$
fong-spivak-seven-sketchesEQ0232	306	■ 1.000 ● N +0	$(f+g)(i) := \begin{cases} f(i) & \text{if } i \leq m \\ g(i-m) & \text{if } m+1 \leq i \end{cases}$	$(f+g)(i) := \begin{cases} f(i) & \text{if } i \leq m \\ g(i-m) & \text{if } m+1 \leq i \end{cases}$	$(f+g)(i) := \begin{cases} f(i) & \text{if } i \leq m \\ g(i-m) & \text{if } m+1 \leq i \end{cases}$
fong-spivak-seven-sketchesEQ0233	308	■ 1.000 ● N +0	$a^{\{0\}}, a^{\{1\}}, a^{\{2\}}, a^{\{3\}}, \ldots, a^{\{2019\}}, \ldots$	$a^0, a^1, a^2, a^3, \dots, a^{2019}, \dots$	$a^0, a^1, a^2, a^3, \dots, a^{2019}, \dots$
fong-spivak-seven-sketchesEQ0234	308	■ 1.000 ● N +0	$[], \quad \text{[a]}, \quad \text{[b]}, \quad \text{[a,a]}, \quad \text{[a,b]}, \quad \ldots, \quad \text{[b,a,b,b,a,b,a,a,a,a]}, \quad \ldots$	$[], \quad [a], \quad [b], \quad [a,a], \quad [a,b], \quad \ldots, \quad [b,a,b,b,a,b,a,a,a,a], \quad \ldots$	$[], \quad [a], \quad [b], \quad [a,a], \quad [a,b], \quad \ldots, \quad [b,a,b,b,a,b,a,a,a,a], \quad \ldots$
fong-spivak-seven-sketchesEQ0235	308	■ 1.000 ● W +0	$G := \left\{ \rho_{m,n} : m \rightarrow n \mid m, n \in \mathbb{N} \right\}, \quad s(\rho_{m,n}) := m, \quad t(\rho_{m,n}) := n;$	$G := \{ \rho_{m,n} : m \rightarrow n \mid m, n \in \mathbb{N} \}, \quad s(\rho_{m,n}) := m, \quad t(\rho_{m,n}) := n;$	$G := \{ \rho_{m,n} : m \rightarrow n \mid m, n \in \mathbb{N} \}, \quad s(\rho_{m,n}) := m, \quad t(\rho_{m,n}) := n;$
fong-spivak-seven-sketchesEQ0236	309	■ 1.000 ● K +0	$\left(\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right).$	$\left(\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right).$	$\left(\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right).$
fong-spivak-seven-sketchesEQ0237	309	■ 1.000 ●	$A * B = \left(\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array} \right) * \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right) \neq \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right) * \left(\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array} \right) = B * A$	$A * B = \left(\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array} \right) * \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right) \neq \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right) * \left(\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array} \right) = B * A$	$A * B = \left(\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array} \right) * \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right) \neq \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right) * \left(\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array} \right) = B * A$
fong-spivak-seven-sketchesEQ0238	309	■ 1.000 ● K +0	$A+B = \left(\begin{array}{cccccc} 3 & 3 & 1 & 0 & 0 & 0 \\ 2 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 5 & 6 \end{array} \right)$	$A+B = \left(\begin{array}{cccccc} 3 & 3 & 1 & 0 & 0 & 0 \\ 2 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 5 & 6 \end{array} \right)$	$A+B = \left(\begin{array}{cccccc} 3 & 3 & 1 & 0 & 0 & 0 \\ 2 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 5 & 6 \end{array} \right)$
fong-spivak-seven-sketchesEQ0239	310	■ 1.000 ● K +0	$\left(\begin{array}{ccc} 1 & 1 & 1 \end{array} \right)$	$\left(\begin{array}{ccc} 1 & 1 & 1 \end{array} \right)$	$\left(\begin{array}{ccc} 1 & 1 & 1 \end{array} \right)$
fong-spivak-seven-sketchesEQ0240	310	■ 1.000 ● K +0	$\left(\begin{array}{ccc} 1 & 1 & 1 \end{array} \right)$	$\left(\begin{array}{ccc} 1 & 1 & 1 \end{array} \right)$	$\left(\begin{array}{ccc} 1 & 1 & 1 \end{array} \right)$
fong-spivak-seven-sketchesEQ0241	310	■ 1.000 ● N +0	$g_{\{1\}} := c_{\{n\}} + \cdots + c_{\{n\}} : m \rightarrow (m \times n),$	$g_1 := c_n + \cdots + c_n : m \rightarrow (m \times n),$	$g_1 := c_n + \cdots + c_n : m \rightarrow (m \times n),$
fong-spivak-seven-sketchesEQ0243	310	■ 1.000 ● N +0	$\begin{aligned} g_{\{2\}} := & s_{\{M(1,1)\}} + \cdots + s_{\{M(1,n)\}} \\ & + s_{\{M(2,1)\}} + \cdots + s_{\{M(2,n)\}} \\ & + \cdots \\ & + s_{\{M(m,1)\}} + \cdots + s_{\{M(m,n)\}} : (m \times n) \rightarrow (m \times n), \end{aligned}$	$\begin{aligned} g_2 := & s_{M(1,1)} + \cdots + s_{M(1,n)} \\ & + s_{M(2,1)} + \cdots + s_{M(2,n)} \\ & + \cdots \\ & + s_{M(m,1)} + \cdots + s_{M(m,n)} : (m \times n) \rightarrow (m \times n), \end{aligned}$	$\begin{aligned} g_2 := & s_{M(1,1)} + \cdots + s_{M(1,n)} \\ & + s_{M(2,1)} + \cdots + s_{M(2,n)} \\ & + \cdots \\ & + s_{M(m,1)} + \cdots + s_{M(m,n)} : (m \times n) \rightarrow (m \times n), \end{aligned}$
fong-spivak-seven-sketchesEQ0244	310	■ 1.000 ● N +1	$g_{\{4\}} := a_{\{m\}} + \cdots + a_{\{m\}} : (m \times n) \rightarrow n,$	$g_4 := a_m + \cdots + a_m : (m \times n) \rightarrow n,$	$g_4 := a_m + \cdots + a_m : (m \times n) \rightarrow n,$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches EQ0246	312	1.000 K +0	$\mu:=\left(\begin{array}{ll} 1&0\\ 0&1\\ 1&0\\ 0&1\end{array}\right)$	$\mu:=\left(\begin{array}{cc} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{array}\right)$	$\mu:=\left(\begin{array}{cc} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{array}\right)$
fong-spivak-seven-sketches EQ0247	313	1.000 N +0	$U(\mu)\left(\left(a_1,\ldots,a_m\right),\left(b_1,\ldots,b_m\right)\right):=\left(a_1+b_1,\ldots,a_m+b_m\right).$	$U(\mu)\left((a_1,\ldots,a_m),(b_1,\ldots,b_m)\right):=(a_1+b_1,\ldots,a_m+b_m).$	$U(\mu)((a_1,\ldots,a_m),(b_1,\ldots,b_m)):= (a_1+b_1,\ldots,a_m+b_m).$
fong-spivak-seven-sketches EQ0248	313	1.000 N -1	$B+C:=\left\{(w,y,x,z)\in R^{m+p}\times R^{n+q}\mid (w,x)\in B\text{ and } (y,z)\in C\right\}.$	$B+C:=\{(w,y,x,z)\in R^{m+p}\times R^{n+q}\mid (w,x)\in B\text{ and } (y,z)\in C\}.$	$B+C:=\{(w,y,x,z)\in R^{m+p}\times R^{n+q}\mid (w,x)\in B\text{ and } (y,z)\in C\}.$
fong-spivak-seven-sketches EQ0249	313	1.000 W +0	$\begin{array}{l} \mathrm{B}(g) \&= \left\{ (x,z) \in R^m \times R^n \mid S(g)(x) = z \right\} \\ \mathrm{B}(h^{\mathrm{op}}) = \left\{ (z,y) \in R^n \times R^\ell \mid z = S(h)(y) \right\} \end{array}$	$B(g)=\{(x,z)\in R^m\times R^n\mid S(g)(x)=z\}$ $B(h^{\mathrm{op}})=\{(z,y)\in R^n\times R^\ell\mid z=S(h)(y)\}$	$B(g)=\{(x,z)\in R^m\times R^n\mid S(g)(x)=z\}$ $B(h^{\mathrm{op}})=\{(z,y)\in R^n\times R^\ell\mid z=S(h)(y)\}$
fong-spivak-seven-sketches EQ0254	313	1.000 N +0	$\mathrm{B}\left(\left(g^{\mathrm{op}}\right);h\right)=\left\{\left(S(g)(x),S(h)(x)\right)\mid x\in R^m\right\}.$	$B((g^{\mathrm{op}});h)=\{(S(g)(x),S(h)(x))\mid x\in R^m\}.$	$B((g^{\mathrm{op}})\S h)=\{(S(g)(x),S(h)(x))\mid x\in R^m\}.$
fong-spivak-seven-sketches EQ0255	315	1.000 C -14	$f(m)=f(\underbrace{1+1+\cdots+1}_m\text{ summands})=\underbrace{f(1)+f(1)+\cdots+f(1)}_m\text{ summands}=\underbrace{1_R+1_R+\cdots+1_R}_m\text{ summands}.$	$f(m)=f(\underbrace{1+1+\cdots+1}_m\text{ summands})=\underbrace{f(1)+f(1)+\cdots+f(1)}_m\text{ summands}=\underbrace{1_R+1_R+\cdots+1_R}_m\text{ summands}.$	$f(m)=f(\underbrace{1+1+\cdots+1}_m\text{ summands})=\underbrace{f(1)+f(1)+\cdots+f(1)}_m\text{ summands}=\underbrace{1_R+1_R+\cdots+1_R}_m\text{ summands}.$
fong-spivak-seven-sketches EQ0261 (A.3)	327	1.000	$\begin{array}{l} P \& Q \& P \wedge Q \& P=(P \wedge Q) \\ \text{true} \& \text{true} \& \text{true} \& \text{true} \\ \text{true} \& \text{false} \& \text{false} \& \text{false} \\ \text{false} \& \text{true} \& \text{false} \& \text{true} \\ \text{false} \& \text{false} \& \text{false} \& \text{true} \end{array}$	$\begin{array}{l} P \quad Q \quad \left\ \quad P \wedge Q \quad \right \quad P = (P \wedge Q) \\ \text{true} \quad \text{true} \quad \left\ \quad \text{true} \quad \right \quad \text{true} \\ \text{true} \quad \text{false} \quad \left\ \quad \text{false} \quad \right \quad \text{false} \\ \text{false} \quad \text{true} \quad \left\ \quad \text{false} \quad \right \quad \text{true} \\ \text{false} \quad \text{false} \quad \left\ \quad \text{false} \quad \right \quad \text{true} \end{array}$	$\begin{array}{l} P \quad Q \quad \left\ \quad P \wedge Q \quad \right \quad P = (P \wedge Q) \\ \text{true} \quad \text{true} \quad \left\ \quad \text{true} \quad \right \quad \text{true} \\ \text{true} \quad \text{false} \quad \left\ \quad \text{false} \quad \right \quad \text{false} \\ \text{false} \quad \text{true} \quad \left\ \quad \text{false} \quad \right \quad \text{true} \\ \text{false} \quad \text{false} \quad \left\ \quad \text{false} \quad \right \quad \text{true} \end{array}$
fong-spivak-seven-sketches EQ0263	333	1.000 N +0	$U=\bigcup_{i\in I}\left\{x\in\mathbb{R}\mid a_i<x<b_i\right\}$	$U=\bigcup_{i\in I}\{x\in\mathbb{R}\mid a_i<x<b_i\}$	$U=\bigcup_{i\in I}\{x\in\mathbb{R}\mid a_i<x<b_i\}$
fong-spivak-seven-sketches EQ0264	333	1.000 W +1	$U=\bigcup_{j\in J}\left\{x\in\mathbb{R}\mid a_j<x<b_j\right\}=\bigcup_{j\in J}\left(o_{\left[a_j,b_j\right]}\cap\mathbb{R}\right)=\left(\bigcup_{j\in J}o_{\left[a_j,b_j\right]}\right)\cap\mathbb{R}$	$U=\bigcup_{j\in J}\{x\in\mathbb{R}\mid a_j<x<b_j\}=\bigcup_{j\in J}(o_{[a_j,b_j]}\cap\mathbb{R})=\left(\bigcup_{j\in J}o_{[a_j,b_j]}\right)\cap\mathbb{R}$	$U=\bigcup_{j\in J}\{x\in\mathbb{R}\mid a_j<x<b_j\}=\bigcup_{j\in J}(o_{[a_j,b_j]}\cap\mathbb{R})=\left(\bigcup_{j\in J}o_{[a_j,b_j]}\right)\cap\mathbb{R}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Inline formulas that did not render (1 of 4029)

Identifier	Page	Conf.	LaTeX source	Rendered
fong-spivak-seven-sketches F03716	317	—	$\left[f ; h ; \iota_{E}, g \circ k ; \iota_{F}\right]=(f+g) \boxplus - [\mathbf{U+27C28}](h+k)$	<i>(not rendered)</i>
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.				

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches-200001	026	0.630	$\begin{tabular}{t l l l l } \hline \text{arrow } a & \text{source } s(a) \in V & \text{target } t(a) \in V \\ \hline a & 1 & ? \\ b & 1 & 3 \\ c & ? & ? \\ d & ? & ? \\ e & ? & ? \\ \hline \end{tabular}$	$\begin{array}{c c c} \text{arrow } a & \text{source } s(a) \in V & \text{target } t(a) \in V \\ \hline a & 1 & ? \\ b & 1 & 3 \\ c & ? & ? \\ d & ? & ? \\ e & ? & ? \\ \hline \end{array}$	$\begin{array}{c c c} \text{arrow } a & \text{source } s(a) \in V & \text{target } t(a) \in V \\ \hline a & 1 & ? \\ b & 1 & 3 \\ c & ? & ? \\ d & ? & ? \\ e & ? & ? \\ \hline \end{array}$
fong-spivak-seven-sketches-200002	064	1.000	$\begin{tabular}{t l l l l l l } \hline \wedge & \text{false} & \text{true} & * & 0 & 1 \\ \hline \text{false} & \text{false} & \text{false} & 0 & 0 & 0 \\ \text{true} & \text{false} & \text{true} & 1 & 0 & 1 \\ \hline \end{tabular}$	$\begin{array}{c c c c c c} \wedge & \text{false} & \text{true} & * & 0 & 1 \\ \hline \text{false} & \text{false} & \text{false} & 0 & 0 & 0 \\ \text{true} & \text{false} & \text{true} & 1 & 0 & 1 \\ \hline \end{array}$	$\begin{array}{c c c} \wedge & \text{false} & \text{true} \\ \hline \text{false} & \text{false} & \text{false} \\ \text{true} & \text{false} & \text{true} \\ \hline \end{array}$
fong-spivak-seven-sketches-200003	065	0.996	$\begin{tabular}{t l l l l } \hline V & \text{false} & \text{true} \\ \hline \text{false} & \text{false} & \text{true} \\ \text{true} & \text{true} & \text{true} \\ \hline \end{tabular}$	$\begin{array}{c c c} V & \text{false} & \text{true} \\ \hline \text{false} & \text{false} & \text{true} \\ \text{true} & \text{true} & \text{true} \\ \hline \end{array}$	$\begin{array}{c c c} \vee & \text{false} & \text{true} \\ \hline \text{false} & \text{false} & \text{true} \\ \text{true} & \text{true} & \text{true} \\ \hline \end{array}$
fong-spivak-seven-sketches-200004	065	1.000	$\begin{tabular}{t l l l l } \hline \max & 0 & 1 \\ \hline 0 & 0 & 1 \\ 1 & 1 & 1 \\ \hline \end{tabular}$	$\begin{array}{c c c} \max & 0 & 1 \\ \hline 0 & 0 & 1 \\ 1 & 1 & 1 \\ \hline \end{array}$	$\begin{array}{c c c} \max & 0 & 1 \\ \hline 0 & 0 & 1 \\ 1 & 1 & 1 \\ \hline \end{array}$
fong-spivak-seven-sketches-200005	066	1.000	$\begin{tabular}{t l l l l l } \hline \min & \text{no} & \text{maybe} & \text{yes} \\ \hline \text{no} & ? & ? & ? \\ \text{maybe} & ? & ? & ? \\ \text{yes} & ? & ? & ? \\ \hline \end{tabular}$	$\begin{array}{c c c c} \min & \text{no} & \text{maybe} & \text{yes} \\ \hline \text{no} & ? & ? & ? \\ \text{maybe} & ? & ? & ? \\ \text{yes} & ? & ? & ? \\ \hline \end{array}$	$\begin{array}{c c c c} \min & \text{no} & \text{maybe} & \text{yes} \\ \hline \text{no} & ? & ? & ? \\ \text{maybe} & ? & ? & ? \\ \text{yes} & ? & ? & ? \\ \hline \end{array}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																																								
fong-spivak-seven-sketches-7A0006	070	0.598	$\begin{tabular}{t}{ l l l l l l }\hline \textbf{[U+0BB4]} & \backslash(p\backslash) & 9 & \backslash(r\backslash) & \backslash(s\backslash) & \backslash(t\backslash) \\\hline \backslash(p\backslash) & \text{true} & \text{true} & \text{true} & \text{true} & \text{true} \\\hline \backslash(q\backslash) & \text{false} & \text{true} & \text{false} & \text{true} & \text{true} \\\hline \backslash(r\backslash) & \text{false} & \text{false} & \text{true} & \text{true} & \text{true} \\\hline \backslash(s\backslash) & \text{false} & \text{false} & \text{false} & \text{true} & \text{true} \\\hline \backslash(t\backslash) & \text{false} & \text{false} & \text{false} & \text{false} & \text{true} \\\hline\end{tabular}$	<table><tr><th>[U+0BB4]</th><th>p</th><th>9</th><th>r</th><th>s</th><th>t</th></tr><tr><td>p</td><td>true</td><td>true</td><td>true</td><td>true</td><td>true</td></tr><tr><td>q</td><td>false</td><td>true</td><td>false</td><td>true</td><td>true</td></tr><tr><td>r</td><td>false</td><td>false</td><td>true</td><td>true</td><td>true</td></tr><tr><td>s</td><td>false</td><td>false</td><td>false</td><td>true</td><td>true</td></tr><tr><td>t</td><td>false</td><td>false</td><td>false</td><td>false</td><td>true</td></tr></table>	[U+0BB4]	p	9	r	s	t	p	true	true	true	true	true	q	false	true	false	true	true	r	false	false	true	true	true	s	false	false	false	true	true	t	false	false	false	false	true	<table><tr><th>$\nearrow$</th><th>p</th><th>q</th><th>r</th><th>s</th><th>t</th></tr><tr><td>p</td><td>true</td><td>true</td><td>true</td><td>true</td><td>true</td></tr><tr><td>q</td><td>false</td><td>true</td><td>false</td><td>true</td><td>true</td></tr><tr><td>r</td><td>false</td><td>false</td><td>true</td><td>true</td><td>true</td></tr><tr><td>s</td><td>false</td><td>false</td><td>false</td><td>true</td><td>true</td></tr><tr><td>t</td><td>false</td><td>false</td><td>false</td><td>false</td><td>true</td></tr></table>	$\nearrow$	p	q	r	s	t	p	true	true	true	true	true	q	false	true	false	true	true	r	false	false	true	true	true	s	false	false	false	true	true	t	false	false	false	false	true
[U+0BB4]	p	9	r	s	t																																																																								
p	true	true	true	true	true																																																																								
q	false	true	false	true	true																																																																								
r	false	false	true	true	true																																																																								
s	false	false	false	true	true																																																																								
t	false	false	false	false	true																																																																								
$\nearrow$	p	q	r	s	t																																																																								
p	true	true	true	true	true																																																																								
q	false	true	false	true	true																																																																								
r	false	false	true	true	true																																																																								
s	false	false	false	true	true																																																																								
t	false	false	false	false	true																																																																								
fong-spivak-seven-sketches-7A0007	074	0.669	$\begin{tabular}{t}{ l l l l l }\hline \backslash(d\backslashleft(\backslashnearrow^{\backslashprime}\backslashright)\backslash) & \backslash(x\backslash) & \backslash(y\backslash) & \backslash(z\backslash) \\\hline \backslash(x\backslash) & 0 & 4 & 3 \\\hline y & 3 & 0 & 6 \\\hline \backslash(z\backslash) & 7 & 4 & 0 \\\hline\end{tabular}$	<table><tr><th>$d(\nearrow)$</th><th>x</th><th>y</th><th>z</th></tr><tr><td>x</td><td>0</td><td>4</td><td>3</td></tr><tr><td>y</td><td>3</td><td>0</td><td>6</td></tr><tr><td>z</td><td>7</td><td>4</td><td>0</td></tr></table>	$d(\nearrow)$	x	y	z	x	0	4	3	y	3	0	6	z	7	4	0	<table><tr><th>$d(\nearrow)$</th><th>x</th><th>y</th><th>z</th></tr><tr><td>x</td><td>0</td><td>4</td><td>3</td></tr><tr><td>y</td><td>3</td><td>0</td><td>6</td></tr><tr><td>z</td><td>7</td><td>4</td><td>0</td></tr></table>	$d(\nearrow)$	x	y	z	x	0	4	3	y	3	0	6	z	7	4	0																																								
$d(\nearrow)$	x	y	z																																																																										
x	0	4	3																																																																										
y	3	0	6																																																																										
z	7	4	0																																																																										
$d(\nearrow)$	x	y	z																																																																										
x	0	4	3																																																																										
y	3	0	6																																																																										
z	7	4	0																																																																										
fong-spivak-seven-sketches-7A0008	074	0.827	$\begin{tabular}{t}{ l l l l l l }\hline \backslash(d\backslashleft(\backslashnearrow^{\backslashprime}\backslashright)\backslash) & \backslash(A\backslash) & \backslash(B\backslash) & C & D \\\hline \backslash(A\backslash) & 0 & ? & ? & ? \\\hline B & 2 & ? & 5 & ? \\\hline \backslash(C\backslash) & ? & ? & ? & ? \\\hline D & ? & ? & ? & ? \\\hline\end{tabular}$	<table><tr><th>$d(\nearrow)$</th><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>A</td><td>0</td><td>?</td><td>?</td><td>?</td></tr><tr><td>B</td><td>2</td><td>?</td><td>5</td><td>?</td></tr><tr><td>C</td><td>?</td><td>?</td><td>?</td><td>?</td></tr><tr><td>D</td><td>?</td><td>?</td><td>?</td><td>?</td></tr></table>	$d(\nearrow)$	A	B	C	D	A	0	?	?	?	B	2	?	5	?	C	?	?	?	?	D	?	?	?	?	<table><tr><th>$d(\nearrow)$</th><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>A</td><td>0</td><td>?</td><td>?</td><td>?</td></tr><tr><td>B</td><td>2</td><td>?</td><td>5</td><td>?</td></tr><tr><td>C</td><td>?</td><td>?</td><td>?</td><td>?</td></tr><tr><td>D</td><td>?</td><td>?</td><td>?</td><td>?</td></tr></table>	$d(\nearrow)$	A	B	C	D	A	0	?	?	?	B	2	?	5	?	C	?	?	?	?	D	?	?	?	?																						
$d(\nearrow)$	A	B	C	D																																																																									
A	0	?	?	?																																																																									
B	2	?	5	?																																																																									
C	?	?	?	?																																																																									
D	?	?	?	?																																																																									
$d(\nearrow)$	A	B	C	D																																																																									
A	0	?	?	?																																																																									
B	2	?	5	?																																																																									
C	?	?	?	?																																																																									
D	?	?	?	?																																																																									
fong-spivak-seven-sketches-7A0009	075	0.577	$\begin{tabular}{t}{ l l l l l l }\hline \backslashmultirow{4}{*}{\backslash(M_{Y}:=\backslash)} & \backslash\textbackslash & \backslash(x\backslash) & \backslash(y\backslash) & \backslash(z\backslash) \\\hline \backslash(x\backslash) & 0 & 4 & 3 \\\hline \backslash(y\backslash) & 3 & 0 & 0 \\\hline \backslash(z\backslash) & 4 & 0 & 0 \\\hline\end{tabular}$	(not rendered)	$M_Y := \begin{array}{c ccc} \nearrow & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & \infty \\ z & \infty & 4 & 0 \end{array}$																																																																								
fong-spivak-seven-sketches-7A0010	075	0.608	$\begin{tabular}{t}{ l l l l l l }\hline \backslashmultirow{2}{*}{\backslash} & \backslash(A\backslash) & \backslash(B\backslash) & C & \backslash(D\backslash) \\\hline \backslash(A\backslash) & 0 & ? & ? & ? \\\hline \backslash(M_X=\backslash) & B & 2 & 0 & \backslash(\infty\backslash) \\\hline C & ? & ? & ? & ? \\\hline \backslash(D\backslash) & ? & ? & ? & ? \\\hline\end{tabular}$	(not rendered)	$M_X = \begin{array}{c ccccc} \nearrow & A & B & C & D \\ \hline A & 0 & ? & ? & ? \\ B & 2 & 0 & \infty & ? \\ C & ? & ? & ? & ? \\ D & ? & ? & ? & ? \end{array}$																																																																								
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.																																																																													

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches-080011	080	0.813	$\begin{tabular}{t}{ l l l l l l l l }\hline X & \backslash(A) & \backslash(B) & \backslash(C) & \multimap{2}{*}{y} & \multimap{2}{*}{\backslash(p)} & \backslash(q) \\ \hline \backslash(A) & 0 & 2 & 5 & & & \backslash \backslash \hline \backslash(B) & \backslash(\infty) & 0 & 3 & & \backslash(p) & 0 & 5 \\ \hline \backslash(C) & \backslash(\infty) & \backslash(\infty) & 0 & & & \backslash \hline \end{tabular}$	(not rendered)	$\begin{array}{c c} \mathcal{X} & A \quad B \quad C \\ \hline A & 0 \quad 2 \quad 5 \\ B & \infty \quad 0 \quad 3 \\ C & \infty \quad \infty \quad 0 \end{array} \qquad \begin{array}{c c} y & p \quad q \\ \hline p & 0 \quad 5 \\ q & 8 \quad 0 \end{array}$
fong-spivak-seven-sketches-080012	080	0.764	$\begin{tabular}{t}{ l l l l l l l l }\hline \backslash(x \times y) & \backslash(A, p) & \backslash(B, p) & \backslash(C, p) & \backslash(A, q) & \backslash(B, q) & \backslash(C, q) \\ \hline \backslash(A, p) & 0 & 2 & 5 & 5 & 7 & 10 \\ \backslash(B, p) & \infty & 0 & 3 & \infty & 5 & 8 \\ \backslash(C, p) & \infty & \infty & 0 & \infty & \infty & 5 \\ \hline \backslash(A, q) & 8 & 10 & 13 & 0 & 2 & 5 \\ \backslash(B, q) & \infty & 8 & 11 & \infty & 0 & 3 \\ \backslash(C, q) & \infty & \infty & 8 & \infty & \infty & 0 \end{tabular}$	$\begin{array}{c c} x \times y & \backslash(A, p) \quad \backslash(B, p) \quad \backslash(C, p) \quad \backslash(A, q) \quad \backslash(B, q) \quad \backslash(C, q) \\ \hline \backslash(A, p) & 0 \quad 2 \quad 5 \quad 5 \quad 7 \quad 10 \\ \backslash(B, p) & \infty \quad 0 \quad 3 \quad \infty \quad 5 \quad 8 \\ \backslash(C, p) & \infty \quad \infty \quad 0 \quad \infty \quad \infty \quad 5 \\ \hline \backslash(A, q) & 8 \quad 10 \quad 13 \quad 0 \quad 2 \quad 5 \\ \backslash(B, q) & \infty \quad 8 \quad 11 \quad \infty \quad 0 \quad 3 \\ \backslash(C, q) & \infty \quad \infty \quad 8 \quad \infty \quad \infty \quad 0 \end{array}$	$\begin{array}{c c} x \times y & \backslash(A, p) \quad \backslash(B, p) \quad \backslash(C, p) \quad \backslash(A, q) \quad \backslash(B, q) \quad \backslash(C, q) \\ \hline \backslash(A, p) & 0 \quad 2 \quad 5 \quad 5 \quad 7 \quad 10 \\ \backslash(B, p) & \infty \quad 0 \quad 3 \quad \infty \quad 5 \quad 8 \\ \backslash(C, p) & \infty \quad \infty \quad 0 \quad \infty \quad \infty \quad 5 \\ \hline \backslash(A, q) & 8 \quad 10 \quad 13 \quad 0 \quad 2 \quad 5 \\ \backslash(B, q) & \infty \quad 8 \quad 11 \quad \infty \quad 0 \quad 3 \\ \backslash(C, q) & \infty \quad \infty \quad 8 \quad \infty \quad \infty \quad 0 \end{array}$
fong-spivak-seven-sketches-086013	086	0.884	$\begin{tabular}{t}{ l l l l l }\hline \backslash(M_Y) & \backslash(x) & \backslash(y) & \backslash(z) \\ \hline \backslash(x) & 0 & 4 & 3 \\ \backslash(y) & 3 & 0 & \infty \\ \backslash(z) & \infty & 4 & 0 \end{tabular}$	$\begin{array}{c c} M_Y & x \quad y \quad z \\ \hline x & 0 \quad 4 \quad 3 \\ y & 3 \quad 0 \quad \infty \\ z & \infty \quad 4 \quad 0 \end{array}$	$\begin{array}{c c} M_Y & x \quad y \quad z \\ \hline x & 0 \quad 4 \quad 3 \\ y & 3 \quad 0 \quad \infty \\ z & \infty \quad 4 \quad 0 \end{array}$
fong-spivak-seven-sketches-086014	086	0.743	$\begin{tabular}{t}{ l l l l l }\hline \backslash(d_Y) & \backslash(x) & \backslash(y) & \backslash(z) \\ \hline \backslash(x) & 0 & 4 & 3 \\ \backslash(y) & 3 & 0 & 6 \\ \backslash(z) & 7 & 4 & 0 \end{tabular}$	$\begin{array}{c c} d_Y & x \quad y \quad z \\ \hline x & 0 \quad 4 \quad 3 \\ y & 3 \quad 0 \quad 6 \\ z & 7 \quad 4 \quad 0 \end{array}$	$\begin{array}{c c} d_Y & x \quad y \quad z \\ \hline x & 0 \quad 4 \quad 3 \\ y & 3 \quad 0 \quad 6 \\ z & 7 \quad 4 \quad 0 \end{array}$

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																						
fong-spivak-seven-sketches-0015	089	■1.000	\begin{tabular}[t]{ l l l l }\hline Employee & FName & WorksIn & Mngr \\ \hline 1 & Alan & 101 & 2 \\ \hline 2 & Ruth & 101 & 2 \\ \hline 3 & Kris & 102 & 3 \\ \hline \end{tabular}	<table><tr><th>Employee</th><th>FName</th><th>WorksIn</th><th>Mngr</th></tr><tr><td>1</td><td>Alan</td><td>101</td><td>2</td></tr><tr><td>2</td><td>Ruth</td><td>101</td><td>2</td></tr><tr><td>3</td><td>Kris</td><td>102</td><td>3</td></tr></table>	Employee	FName	WorksIn	Mngr	1	Alan	101	2	2	Ruth	101	2	3	Kris	102	3	<table><tr><th>Employee</th><th>FName</th><th>WorksIn</th><th>Mngr</th></tr><tr><td>1</td><td>Alan</td><td>101</td><td>2</td></tr><tr><td>2</td><td>Ruth</td><td>101</td><td>2</td></tr><tr><td>3</td><td>Kris</td><td>102</td><td>3</td></tr></table>	Employee	FName	WorksIn	Mngr	1	Alan	101	2	2	Ruth	101	2	3	Kris	102	3																						
					Employee	FName	WorksIn	Mngr																																																			
1	Alan	101	2																																																								
2	Ruth	101	2																																																								
3	Kris	102	3																																																								
Employee	FName	WorksIn	Mngr																																																								
1	Alan	101	2																																																								
2	Ruth	101	2																																																								
3	Kris	102	3																																																								
fong-spivak-seven-sketches-0016	089	■1.000	\begin{tabular}[t]{ l l l l }\hline Department & DName & Secr \\ \hline 101 & Sales & 1 \\ \hline 102 & IT & 3 \\ \hline \end{tabular}	<table><tr><th>Department</th><th>DName</th><th>Secr</th></tr><tr><td>101</td><td>Sales</td><td>1</td></tr><tr><td>102</td><td>IT</td><td>3</td></tr></table>	Department	DName	Secr	101	Sales	1	102	IT	3	<table><tr><th>Department</th><th>DName</th><th>Secr</th></tr><tr><td>101</td><td>Sales</td><td>1</td></tr><tr><td>102</td><td>IT</td><td>3</td></tr></table>	Department	DName	Secr	101	Sales	1	102	IT	3																																				
					Department	DName	Secr																																																				
101	Sales	1																																																									
102	IT	3																																																									
Department	DName	Secr																																																									
101	Sales	1																																																									
102	IT	3																																																									
fong-spivak-seven-sketches-0017	101	■1.000	\begin{tabular}[t]{ l l l l }\hline Planet of Sol & Prime number & Flying pig \\ \hline Mercury & 2 & \\ \hline Venus & 3 & \\ \hline Earth & 5 & \\ \hline Mars & 7 & \\ \hline Jupiter & 11 & \\ \hline Saturn & 13 & \\ \hline Uranus & 17 & \\ \hline Neptune & [U+22EE] & \\ \hline \end{tabular}	<table><tr><th>Planet of Sol</th><th>Prime number</th><th>Flying pig</th></tr><tr><td>Mercury</td><td>2</td><td></td></tr><tr><td>Venus</td><td>3</td><td></td></tr><tr><td>Earth</td><td>5</td><td></td></tr><tr><td>Mars</td><td>7</td><td></td></tr><tr><td>Jupiter</td><td>11</td><td></td></tr><tr><td>Saturn</td><td>13</td><td></td></tr><tr><td>Uranus</td><td>17</td><td></td></tr><tr><td>Neptune</td><td>[U+22EE]</td><td></td></tr></table>	Planet of Sol	Prime number	Flying pig	Mercury	2		Venus	3		Earth	5		Mars	7		Jupiter	11		Saturn	13		Uranus	17		Neptune	[U+22EE]		<table><tr><th>Planet of Sol</th><th>Prime number</th><th>Flying pig</th></tr><tr><td>Mercury</td><td>2</td><td></td></tr><tr><td>Venus</td><td>3</td><td></td></tr><tr><td>Earth</td><td>5</td><td></td></tr><tr><td>Mars</td><td>7</td><td></td></tr><tr><td>Jupiter</td><td>11</td><td></td></tr><tr><td>Saturn</td><td>13</td><td></td></tr><tr><td>Uranus</td><td>17</td><td></td></tr><tr><td>Neptune</td><td>:</td><td></td></tr></table>	Planet of Sol	Prime number	Flying pig	Mercury	2		Venus	3		Earth	5		Mars	7		Jupiter	11		Saturn	13		Uranus	17		Neptune	:	
					Planet of Sol	Prime number	Flying pig																																																				
Mercury	2																																																										
Venus	3																																																										
Earth	5																																																										
Mars	7																																																										
Jupiter	11																																																										
Saturn	13																																																										
Uranus	17																																																										
Neptune	[U+22EE]																																																										
Planet of Sol	Prime number	Flying pig																																																									
Mercury	2																																																										
Venus	3																																																										
Earth	5																																																										
Mars	7																																																										
Jupiter	11																																																										
Saturn	13																																																										
Uranus	17																																																										
Neptune	:																																																										
fong-spivak-seven-sketches-0018	102	■1.000	\begin{tabular}[t]{ l l l }\hline Beatle & Played \\ \hline George & Lead guitar \\ \hline John & Rhythm guitar \\ \hline Paul & Bass guitar \\ \hline Ringo & Drums \\ \hline \end{tabular}	<table><tr><th>Beatle</th><th>Played</th></tr><tr><td>George</td><td>Lead guitar</td></tr><tr><td>John</td><td>Rhythm guitar</td></tr><tr><td>Paul</td><td>Bass guitar</td></tr><tr><td>Ringo</td><td>Drums</td></tr></table>	Beatle	Played	George	Lead guitar	John	Rhythm guitar	Paul	Bass guitar	Ringo	Drums	<table><tr><th>Beatle</th><th>Played</th></tr><tr><td>George</td><td>Lead guitar</td></tr><tr><td>John</td><td>Rhythm guitar</td></tr><tr><td>Paul</td><td>Bass guitar</td></tr><tr><td>Ringo</td><td>Drums</td></tr></table>	Beatle	Played	George	Lead guitar	John	Rhythm guitar	Paul	Bass guitar	Ringo	Drums																																		
					Beatle	Played																																																					
George	Lead guitar																																																										
John	Rhythm guitar																																																										
Paul	Bass guitar																																																										
Ringo	Drums																																																										
Beatle	Played																																																										
George	Lead guitar																																																										
John	Rhythm guitar																																																										
Paul	Bass guitar																																																										
Ringo	Drums																																																										
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value																																																											
LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ																																																											
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured																																																											
[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.																																																											

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																				
fong-spivak-seven-sketches_002_019	102	1.000	\begin{tabular}[t]{ l l } \hline Beatle & Played \\ \hline George & Lead guitar \\ \hline John & Rhythm guitar \\ \hline Paul & Bass guitar \\ \hline Ringo & Drums \\ \hline \end{tabular}	<table><tr><td>Beatle</td><td>Played</td></tr><tr><td>George</td><td>Lead guitar</td></tr><tr><td>John</td><td>Rhythm guitar</td></tr><tr><td>Paul</td><td>Bass guitar</td></tr><tr><td>Ringo</td><td>Drums</td></tr></table>	Beatle	Played	George	Lead guitar	John	Rhythm guitar	Paul	Bass guitar	Ringo	Drums	<table><tr><th>Beatle</th><th>Played</th></tr><tr><td>George</td><td>Lead guitar</td></tr><tr><td>John</td><td>Rhythm guitar</td></tr><tr><td>Paul</td><td>Bass guitar</td></tr><tr><td>Ringo</td><td>Drums</td></tr></table>	Beatle	Played	George	Lead guitar	John	Rhythm guitar	Paul	Bass guitar	Ringo	Drums																
					Beatle	Played																																			
					George	Lead guitar																																			
					John	Rhythm guitar																																			
					Paul	Bass guitar																																			
Ringo	Drums																																								
Beatle	Played																																								
George	Lead guitar																																								
John	Rhythm guitar																																								
Paul	Bass guitar																																								
Ringo	Drums																																								
fong-spivak-seven-sketches_002_020	102	1.000	\begin{tabular}[t]{ l } \hline Rock-and-roll instrument \\ \hline Bass guitar \\ \hline Drums \\ \hline Keyboard \\ \hline Lead guitar \\ \hline Rhythm guitar \\ \hline \end{tabular}	<table><tr><td>Rock-and-roll instrument</td></tr><tr><td>Bass guitar</td></tr><tr><td>Drums</td></tr><tr><td>Keyboard</td></tr><tr><td>Lead guitar</td></tr><tr><td>Rhythm guitar</td></tr></table>	Rock-and-roll instrument	Bass guitar	Drums	Keyboard	Lead guitar	Rhythm guitar	<table><tr><th>Rock-and-roll instrument</th></tr><tr><td>Bass guitar</td></tr><tr><td>Drums</td></tr><tr><td>Keyboard</td></tr><tr><td>Lead guitar</td></tr><tr><td>Rhythm guitar</td></tr></table>	Rock-and-roll instrument	Bass guitar	Drums	Keyboard	Lead guitar	Rhythm guitar																								
					Rock-and-roll instrument																																				
					Bass guitar																																				
					Drums																																				
					Keyboard																																				
Lead guitar																																									
Rhythm guitar																																									
Rock-and-roll instrument																																									
Bass guitar																																									
Drums																																									
Keyboard																																									
Lead guitar																																									
Rhythm guitar																																									
fong-spivak-seven-sketches_002_021	106	1.000	\begin{tabular}[t]{ l l } \hline \mathbb{N}_{\geq 2} & \text{smallest prime factor} \\ \hline 2 & 2 \\ \hline 3 & 3 \\ \hline 4 & 2 \\ \hline \text{[U+22EE]} & \text{[U+22EE]} \\ \hline 49 & 7 \\ \hline 50 & 2 \\ \hline 51 & 3 \\ \hline \text{[U+22EE]} & \text{[U+22EE]} \\ \hline \end{tabular}	<table><tr><td>$\mathbb{N}_{\geq 2}$</td><td>smallest prime factor</td></tr><tr><td>2</td><td>2</td></tr><tr><td>3</td><td>3</td></tr><tr><td>4</td><td>2</td></tr><tr><td>[U+22EE]</td><td>[U+22EE]</td></tr><tr><td>49</td><td>7</td></tr><tr><td>50</td><td>2</td></tr><tr><td>51</td><td>3</td></tr><tr><td>[U+22EE]</td><td>[U+22EE]</td></tr></table>	$\mathbb{N}_{\geq 2}$	smallest prime factor	2	2	3	3	4	2	[U+22EE]	[U+22EE]	49	7	50	2	51	3	[U+22EE]	[U+22EE]	<table><tr><th>$\mathbb{N}_{\geq 2}$</th><th>smallest prime factor</th></tr><tr><td>2</td><td>2</td></tr><tr><td>3</td><td>3</td></tr><tr><td>4</td><td>2</td></tr><tr><td>$\vdots$</td><td>$\vdots$</td></tr><tr><td>49</td><td>7</td></tr><tr><td>50</td><td>2</td></tr><tr><td>51</td><td>3</td></tr><tr><td>$\vdots$</td><td>$\vdots$</td></tr></table>	$\mathbb{N}_{\geq 2}$	smallest prime factor	2	2	3	3	4	2	$\vdots$	$\vdots$	49	7	50	2	51	3	$\vdots$	$\vdots$
					$\mathbb{N}_{\geq 2}$	smallest prime factor																																			
					2	2																																			
					3	3																																			
					4	2																																			
					[U+22EE]	[U+22EE]																																			
					49	7																																			
					50	2																																			
					51	3																																			
					[U+22EE]	[U+22EE]																																			
$\mathbb{N}_{\geq 2}$	smallest prime factor																																								
2	2																																								
3	3																																								
4	2																																								
$\vdots$	$\vdots$																																								
49	7																																								
50	2																																								
51	3																																								
$\vdots$	$\vdots$																																								
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.																																									

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																
fong-spivak-seven-sketches_140_022	110	0.641	$\begin{tabular}{t}{ l l l l } \hline \text{Arrow} & \text{source} & \text{target} & \text{Vertex} \\ \hline (a) & 1 & 2 & 1 \\ \hline (b) & 1 & 3 & 2 \\ \hline (c) & 1 & 3 & 3 \\ \hline (d) & 2 & 2 & 4 \\ \hline (e) & 2 & 3 & \\ \hline \end{tabular}$	<table><tr><th>Arrow</th><th>source</th><th>target</th><th>Vertex</th></tr><tr><td>a</td><td>1</td><td>2</td><td>1</td></tr><tr><td>b</td><td>1</td><td>3</td><td>2</td></tr><tr><td>c</td><td>1</td><td>3</td><td>3</td></tr><tr><td>d</td><td>2</td><td>2</td><td>4</td></tr><tr><td>e</td><td>2</td><td>3</td><td></td></tr></table>	Arrow	source	target	Vertex	a	1	2	1	b	1	3	2	c	1	3	3	d	2	2	4	e	2	3		<table><tr><th>Arrow</th><th>source</th><th>target</th><th>Vertex</th></tr><tr><td>a</td><td>1</td><td>2</td><td>1</td></tr><tr><td>b</td><td>1</td><td>3</td><td>2</td></tr><tr><td>c</td><td>1</td><td>3</td><td>3</td></tr><tr><td>d</td><td>2</td><td>2</td><td>4</td></tr><tr><td>e</td><td>2</td><td>3</td><td></td></tr></table>	Arrow	source	target	Vertex	a	1	2	1	b	1	3	2	c	1	3	3	d	2	2	4	e	2	3	
Arrow	source	target	Vertex																																																		
a	1	2	1																																																		
b	1	3	2																																																		
c	1	3	3																																																		
d	2	2	4																																																		
e	2	3																																																			
Arrow	source	target	Vertex																																																		
a	1	2	1																																																		
b	1	3	2																																																		
c	1	3	3																																																		
d	2	2	4																																																		
e	2	3																																																			
fong-spivak-seven-sketches_140_023	111	0.980	$\begin{tabular}{t}{ l l l l l l l l l } \hline \multicolumn{4}{ c }{\multirow{3}{*}{}} & \text{Arrow} & \text{source} & \text{target} & \text{Vertex} \\ \hline & & & & & & & \\ \hline \multicolumn{4}{ c }{\multirow{2}{*}{(G:=\)}} & (a) & 1 & 2 & 1 \\ \hline & & & & (b) & 2 & 3 & 2 \\ \hline & & & & & & & 3 \\ \hline \text{Arrow} & \text{source} & \text{target} & & & & & \text{Vertex} \\ \hline \multicolumn{4}{ c }{\multirow{3}{*}{(H:=\)}} & (c) & 4 & 5 & 4 \\ \hline & & & & (d) & 4 & 5 & 5 \\ \hline & & & & (e) & 5 & 5 & \\ \hline \end{tabular}$	(not rendered)	$G :=$ <table><tr><th>Arrow</th><th>source</th><th>target</th><th>Vertex</th></tr><tr><td>a</td><td>1</td><td>2</td><td>1</td></tr><tr><td>b</td><td>2</td><td>3</td><td>2</td></tr><tr><td></td><td></td><td></td><td>3</td></tr></table> $H :=$<table><tr><th>Arrow</th><th>source</th><th>target</th><th>Vertex</th></tr><tr><td>c</td><td>4</td><td>5</td><td>4</td></tr><tr><td>d</td><td>4</td><td>5</td><td>5</td></tr><tr><td>e</td><td>5</td><td>5</td><td></td></tr></table>	Arrow	source	target	Vertex	a	1	2	1	b	2	3	2				3	Arrow	source	target	Vertex	c	4	5	4	d	4	5	5	e	5	5																	
Arrow	source	target	Vertex																																																		
a	1	2	1																																																		
b	2	3	2																																																		
			3																																																		
Arrow	source	target	Vertex																																																		
c	4	5	4																																																		
d	4	5	5																																																		
e	5	5																																																			
fong-spivak-seven-sketches_142_024	112	1.000	$\begin{tabular}{t}{ l l } \hline \text{State} & \text{next} \\ \hline 1 & 4 \\ \hline 2 & 4 \\ \hline 3 & 5 \\ \hline 4 & 5 \\ \hline 5 & 5 \\ \hline 6 & 7 \\ \hline 7 & 6 \\ \hline \end{tabular}$	<table><tr><th>State</th><th>next</th></tr><tr><td>1</td><td>4</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>5</td></tr><tr><td>4</td><td>5</td></tr><tr><td>5</td><td>5</td></tr><tr><td>6</td><td>7</td></tr><tr><td>7</td><td>6</td></tr></table>	State	next	1	4	2	4	3	5	4	5	5	5	6	7	7	6	<table><tr><th>State</th><th>next</th></tr><tr><td>1</td><td>4</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>5</td></tr><tr><td>4</td><td>5</td></tr><tr><td>5</td><td>5</td></tr><tr><td>6</td><td>7</td></tr><tr><td>7</td><td>6</td></tr></table>	State	next	1	4	2	4	3	5	4	5	5	5	6	7	7	6																
State	next																																																				
1	4																																																				
2	4																																																				
3	5																																																				
4	5																																																				
5	5																																																				
6	7																																																				
7	6																																																				
State	next																																																				
1	4																																																				
2	4																																																				
3	5																																																				
4	5																																																				
5	5																																																				
6	7																																																				
7	6																																																				

Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																																
fong-spivak-seven-sketches_148_025	113	1.000	\begin{tabular}[t]{ l l l l }\hline Arrow & source & target & Vertex \\ \hline 1 & 1 & 4 & 1 \\ \hline 2 & 2 & 4 & 2 \\ \hline 3 & 3 & 5 & 3 \\ \hline 4 & 4 & 5 & 4 \\ \hline 5 & 5 & 5 & 5 \\ \hline 6 & 6 & 7 & 6 \\ \hline 7 & 6 & 7 & 7 \\ \hline \end{tabular}	<table><tr><th>Arrow</th><th>source</th><th>target</th><th>Vertex</th></tr><tr><td>1</td><td>1</td><td>4</td><td>1</td></tr><tr><td>2</td><td>2</td><td>4</td><td>2</td></tr><tr><td>3</td><td>3</td><td>5</td><td>3</td></tr><tr><td>4</td><td>4</td><td>5</td><td>4</td></tr><tr><td>5</td><td>5</td><td>5</td><td>5</td></tr><tr><td>6</td><td>6</td><td>7</td><td>6</td></tr><tr><td>7</td><td>7</td><td>6</td><td>7</td></tr></table>	Arrow	source	target	Vertex	1	1	4	1	2	2	4	2	3	3	5	3	4	4	5	4	5	5	5	5	6	6	7	6	7	7	6	7	<table><tr><th>Arrow</th><th>source</th><th>target</th><th>Vertex</th></tr><tr><td>1</td><td>1</td><td>4</td><td>1</td></tr><tr><td>2</td><td>2</td><td>4</td><td>2</td></tr><tr><td>3</td><td>3</td><td>5</td><td>3</td></tr><tr><td>4</td><td>4</td><td>5</td><td>4</td></tr><tr><td>5</td><td>5</td><td>5</td><td>5</td></tr><tr><td>6</td><td>6</td><td>7</td><td>6</td></tr><tr><td>7</td><td>7</td><td>6</td><td>7</td></tr></table>	Arrow	source	target	Vertex	1	1	4	1	2	2	4	2	3	3	5	3	4	4	5	4	5	5	5	5	6	6	7	6	7	7	6	7
				Arrow	source	target	Vertex																																																														
1	1	4	1																																																																		
2	2	4	2																																																																		
3	3	5	3																																																																		
4	4	5	4																																																																		
5	5	5	5																																																																		
6	6	7	6																																																																		
7	7	6	7																																																																		
Arrow	source	target	Vertex																																																																		
1	1	4	1																																																																		
2	2	4	2																																																																		
3	3	5	3																																																																		
4	4	5	4																																																																		
5	5	5	5																																																																		
6	6	7	6																																																																		
7	7	6	7																																																																		
fong-spivak-seven-sketches_148_026	117	0.506	\begin{tabular}[t]{ l l l l }\hline Migration Functor & Pronounced & Reminiscent of & Database idea \\ \hline \Delta & Delta & Duplicate or destroy & Duplicate or destroy tables or columns \\ \hline \Sigma & Sigma & Sum & Union (sum up) data \\ \hline \Pi & Pi & Product & Pair ⁹ and query data \\ \hline \end{tabular}	<table><tr><th>Migration Functor</th><th>Pronounced</th><th>Reminiscent of</th><th>Database idea</th></tr><tr><td>Δ</td><td>Delta</td><td>Duplicate or destroy</td><td>Duplicate or destroy tables or columns</td></tr><tr><td>Σ</td><td>Sigma</td><td>Sum</td><td>Union (sum up) data</td></tr><tr><td>Π</td><td>Pi</td><td>Product</td><td>Pair⁹ and query data</td></tr></table>	Migration Functor	Pronounced	Reminiscent of	Database idea	Δ	Delta	Duplicate or destroy	Duplicate or destroy tables or columns	Σ	Sigma	Sum	Union (sum up) data	Π	Pi	Product	Pair ⁹ and query data	<table><tr><th>Migration Functor</th><th>Pronounced</th><th>Reminiscent of</th><th>Database idea</th></tr><tr><td>Δ</td><td>Delta</td><td>Duplicate or destroy</td><td>Duplicate or destroy tables or columns</td></tr><tr><td>Σ</td><td>Sigma</td><td>Sum</td><td>Union (sum up) data</td></tr><tr><td>Π</td><td>Pi</td><td>Product</td><td>Pair⁹ and query data</td></tr></table>	Migration Functor	Pronounced	Reminiscent of	Database idea	Δ	Delta	Duplicate or destroy	Duplicate or destroy tables or columns	Σ	Sigma	Sum	Union (sum up) data	Π	Pi	Product	Pair ⁹ and query data																																
Migration Functor	Pronounced	Reminiscent of	Database idea																																																																		
Δ	Delta	Duplicate or destroy	Duplicate or destroy tables or columns																																																																		
Σ	Sigma	Sum	Union (sum up) data																																																																		
Π	Pi	Product	Pair ⁹ and query data																																																																		
Migration Functor	Pronounced	Reminiscent of	Database idea																																																																		
Δ	Delta	Duplicate or destroy	Duplicate or destroy tables or columns																																																																		
Σ	Sigma	Sum	Union (sum up) data																																																																		
Π	Pi	Product	Pair ⁹ and query data																																																																		
fong-spivak-seven-sketches_148_027	118	1.000	\begin{tabular}[t]{ l l l l }\hline Email & sent_by & received_by & Address \\ \hline Em_1 & Bob & Grace & Bob \\ \hline Em_2 & Grace & Pat & Doug \\ \hline Em_3 & Bob & Emmy & Emmy \\ \hline Em_4 & Sue & Doug & Grace \\ \hline Em_5 & Doug & Sue & Pat \\ \hline Em_6 & Bob & Bob & Sue \\ \hline \end{tabular}	(not rendered)	<table><tr><th>Email</th><th>sent_by</th><th>received_by</th><th>Address</th></tr><tr><td>Em_1</td><td>Bob</td><td>Grace</td><td>Bob</td></tr><tr><td>Em_2</td><td>Grace</td><td>Pat</td><td>Doug</td></tr><tr><td>Em_3</td><td>Bob</td><td>Emmy</td><td>Emmy</td></tr><tr><td>Em_4</td><td>Sue</td><td>Doug</td><td>Grace</td></tr><tr><td>Em_5</td><td>Doug</td><td>Sue</td><td>Pat</td></tr><tr><td>Em_6</td><td>Bob</td><td>Bob</td><td>Sue</td></tr></table>	Email	sent_by	received_by	Address	Em_1	Bob	Grace	Bob	Em_2	Grace	Pat	Doug	Em_3	Bob	Emmy	Emmy	Em_4	Sue	Doug	Grace	Em_5	Doug	Sue	Pat	Em_6	Bob	Bob	Sue																																				
Email	sent_by	received_by	Address																																																																		
Em_1	Bob	Grace	Bob																																																																		
Em_2	Grace	Pat	Doug																																																																		
Em_3	Bob	Emmy	Emmy																																																																		
Em_4	Sue	Doug	Grace																																																																		
Em_5	Doug	Sue	Pat																																																																		
Em_6	Bob	Bob	Sue																																																																		
fong-spivak-seven-sketches_148_028	132	1.000	\begin{tabular}[t]{ l l l }\hline Bool-category & Bool-functor & Bool-profunctor \\ \hline Bool-category & Bool-functor & Bool-profunctor \\ \hline \end{tabular}	<table><tr><th>Bool-category</th><th>preorder</th></tr><tr><th>Bool-functor</th><th>monotone map</th></tr><tr><th>Bool-profunctor</th><th>feasibility relation</th></tr></table>	Bool-category	preorder	Bool-functor	monotone map	Bool-profunctor	feasibility relation	<table><tr><th>Bool-category</th><td>preorder</td></tr><tr><th>Bool-functor</th><td>monotone map</td></tr><tr><th>Bool-profunctor</th><td>feasibility relation</td></tr></table>	Bool-category	preorder	Bool-functor	monotone map	Bool-profunctor	feasibility relation																																																				
Bool-category	preorder																																																																				
Bool-functor	monotone map																																																																				
Bool-profunctor	feasibility relation																																																																				
Bool-category	preorder																																																																				
Bool-functor	monotone map																																																																				
Bool-profunctor	feasibility relation																																																																				

Conf. MathPix confidence:

≥0.9

0.5–0.9

<0.5

— no value

LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

C componentW weakS stableN noiseK cleannot measured

[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																												
fong-spivak-seven-sketches-000029	133	■0.986	$\begin{tabular}{t}{ l l l }\hline \backslash(c\backslash) \& \backslash(d\backslash) \& \backslash(c\backslash\Rightarrow d\backslash) \\\hline true \& true \& true \\\hline true \& false \& false \\\hline false \& true \& true \\\hline false \& false \& true \\\hline\end{tabular}$	<table><tr><th>c</th><th>d</th><th>$c \Rightarrow d$</th></tr><tr><td>true</td><td>true</td><td>true</td></tr><tr><td>true</td><td>false</td><td>false</td></tr><tr><td>false</td><td>true</td><td>true</td></tr><tr><td>false</td><td>false</td><td>true</td></tr></table>	c	d	$c \Rightarrow d$	true	true	true	true	false	false	false	true	true	false	false	true	<table><tr><th>c</th><th>d</th><th>$c \Rightarrow d$</th></tr><tr><td>true</td><td>true</td><td>true</td></tr><tr><td>true</td><td>false</td><td>false</td></tr><tr><td>false</td><td>true</td><td>true</td></tr><tr><td>false</td><td>false</td><td>true</td></tr></table>	c	d	$c \Rightarrow d$	true	true	true	true	false	false	false	true	true	false	false	true																														
c	d	$c \Rightarrow d$																																																															
true	true	true																																																															
true	false	false																																																															
false	true	true																																																															
false	false	true																																																															
c	d	$c \Rightarrow d$																																																															
true	true	true																																																															
true	false	false																																																															
false	true	true																																																															
false	false	true																																																															
fong-spivak-seven-sketches-000030	134	■0.607	$\begin{tabular}{t}{ l l l l l l }\hline \backslash(\Phi\backslash) \& \backslash(a\backslash) \& \backslash(b\backslash) \& \backslash(c\backslash) \& \backslash(d\backslash) \& \backslash(e\backslash) \\\hline \backslash(N\backslash) \& ? \& ? \& ? \& ? \& ? \& true \\\hline \backslash(E\backslash) \& true \& ? \& ? \& ? \& ? \& ? \\\hline W \& ? \& ? \& ? \& ? \& false \& ? \\\hline \backslash(S\backslash) \& ? \& ? \& ? \& ? \& ? \& ? \\\hline\end{tabular}$	<table><tr><th>Φ</th><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>N</td><td>?</td><td>?</td><td>?</td><td>?</td><td>true</td></tr><tr><td>E</td><td>true</td><td>?</td><td>?</td><td>?</td><td>?</td></tr><tr><td>W</td><td>?</td><td>?</td><td>?</td><td>false</td><td>?</td></tr><tr><td>S</td><td>?</td><td>?</td><td>?</td><td>?</td><td>?</td></tr></table>	Φ	a	b	c	d	e	N	?	?	?	?	true	E	true	?	?	?	?	W	?	?	?	false	?	S	?	?	?	?	?	<table><tr><th>Φ</th><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>N</td><td>?</td><td>?</td><td>?</td><td>?</td><td>true</td></tr><tr><td>E</td><td>true</td><td>?</td><td>?</td><td>?</td><td>?</td></tr><tr><td>W</td><td>?</td><td>?</td><td>?</td><td>false</td><td>?</td></tr><tr><td>S</td><td>?</td><td>?</td><td>?</td><td>?</td><td>?</td></tr></table>	Φ	a	b	c	d	e	N	?	?	?	?	true	E	true	?	?	?	?	W	?	?	?	false	?	S	?	?	?	?	?
Φ	a	b	c	d	e																																																												
N	?	?	?	?	true																																																												
E	true	?	?	?	?																																																												
W	?	?	?	false	?																																																												
S	?	?	?	?	?																																																												
Φ	a	b	c	d	e																																																												
N	?	?	?	?	true																																																												
E	true	?	?	?	?																																																												
W	?	?	?	false	?																																																												
S	?	?	?	?	?																																																												
fong-spivak-seven-sketches-000031	135	■0.644	$\begin{tabular}{t}{ l l l l l }\hline \backslash(\Phi\backslash) \& \backslash(x\backslash) \& \backslash(y\backslash) \& \backslash(z\backslash) \\\hline \backslash(A\backslash) \& ? \& ? \& ? \& 20 \\\hline \backslash(B\backslash) \& 11 \& ? \& ? \& ? \\\hline C \& ? \& 17 \& ? \\\hline D \& ? \& ? \& ? \\\hline\end{tabular}$	<table><tr><th>Φ</th><th>x</th><th>y</th><th>z</th></tr><tr><td>A</td><td>?</td><td>?</td><td>20</td></tr><tr><td>B</td><td>11</td><td>?</td><td>?</td></tr><tr><td>C</td><td>?</td><td>17</td><td>?</td></tr><tr><td>D</td><td>?</td><td>?</td><td>?</td></tr></table>	Φ	x	y	z	A	?	?	20	B	11	?	?	C	?	17	?	D	?	?	?	<table><tr><th>Φ</th><th>x</th><th>y</th><th>z</th></tr><tr><td>A</td><td>?</td><td>?</td><td>20</td></tr><tr><td>B</td><td>11</td><td>?</td><td>?</td></tr><tr><td>C</td><td>?</td><td>17</td><td>?</td></tr><tr><td>D</td><td>?</td><td>?</td><td>?</td></tr></table>	Φ	x	y	z	A	?	?	20	B	11	?	?	C	?	17	?	D	?	?	?																				
Φ	x	y	z																																																														
A	?	?	20																																																														
B	11	?	?																																																														
C	?	17	?																																																														
D	?	?	?																																																														
Φ	x	y	z																																																														
A	?	?	20																																																														
B	11	?	?																																																														
C	?	17	?																																																														
D	?	?	?																																																														
fong-spivak-seven-sketches-000032	135	■0.546	$\begin{tabular}{t}{ l l l l l l }\hline \backslash(M_X\backslash) \& \backslash(A\backslash) \& \backslash(B\backslash) \& \backslash(C\backslash) \& \backslash(D\backslash) \\\hline \backslash(A\backslash) \& 0 \& \backslash(\infty\backslash) \& 3 \& \backslash(\infty\backslash) \\\hline \backslash(B\backslash) \& 2 \& 0 \& \backslash(\infty\backslash) \& 5 \\\hline \backslash(C\backslash) \& \backslash(\infty\backslash) \& 3 \& 0 \& \backslash(\infty\backslash) \\\hline D \& \backslash(\infty\backslash) \& \backslash(\infty\backslash) \& 4 \& 0 \\\hline\end{tabular}$	<table><tr><th>M_X</th><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>A</td><td>0</td><td>∞</td><td>3</td><td>∞</td></tr><tr><td>B</td><td>2</td><td>0</td><td>∞</td><td>5</td></tr><tr><td>C</td><td>∞</td><td>3</td><td>0</td><td>∞</td></tr><tr><td>D</td><td>∞</td><td>∞</td><td>4</td><td>0</td></tr></table>	M_X	A	B	C	D	A	0	∞	3	∞	B	2	0	∞	5	C	∞	3	0	∞	D	∞	∞	4	0	<table><tr><th>M_X</th><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>A</td><td>0</td><td>∞</td><td>3</td><td>∞</td></tr><tr><td>B</td><td>2</td><td>0</td><td>∞</td><td>5</td></tr><tr><td>C</td><td>∞</td><td>3</td><td>0</td><td>∞</td></tr><tr><td>D</td><td>∞</td><td>∞</td><td>4</td><td>0</td></tr></table>	M_X	A	B	C	D	A	0	∞	3	∞	B	2	0	∞	5	C	∞	3	0	∞	D	∞	∞	4	0										
M_X	A	B	C	D																																																													
A	0	∞	3	∞																																																													
B	2	0	∞	5																																																													
C	∞	3	0	∞																																																													
D	∞	∞	4	0																																																													
M_X	A	B	C	D																																																													
A	0	∞	3	∞																																																													
B	2	0	∞	5																																																													
C	∞	3	0	∞																																																													
D	∞	∞	4	0																																																													

Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches_000033	135	0.637	$\begin{tabular}{t l l l l } \hline M_{\Phi} & (x) & (y) & (z) \\ \hline A & \infty & \infty & \infty \\ B & 11 & \infty & \infty \\ C & \infty & \infty & \infty \\ D & \infty & 9 & \infty \\ \hline \end{tabular}$	$\begin{array}{c ccc} M_{\Phi} & x & y & z \\ \hline A & \infty & \infty & \infty \\ B & 11 & \infty & \infty \\ C & \infty & \infty & \infty \\ D & \infty & 9 & \infty \end{array}$	$\begin{array}{c ccc} M_{\Phi} & x & y & z \\ \hline A & \infty & \infty & \infty \\ B & 11 & \infty & \infty \\ C & \infty & \infty & \infty \\ D & \infty & 9 & \infty \end{array}$
fong-spivak-seven-sketches_000034	135	0.963	$\begin{tabular}{t l l l l } \hline M_Y & (x) & (y) & (z) \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & \infty \\ z & \infty & 4 & 0 \\ \hline \end{tabular}$	$\begin{array}{c ccc} M_Y & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & \infty \\ z & \infty & 4 & 0 \end{array}$	$\begin{array}{c ccc} M_Y & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & \infty \\ z & \infty & 4 & 0 \end{array}$
fong-spivak-seven-sketches_000035	138	0.520	$\begin{tabular}{t l l l l l } \hline \Phi & (a) & (b) & (c) & (d) & (e) \\ \hline N & \text{true} & \text{false} & \text{true} & \text{false} & \text{false} \\ E & \text{true} & \text{false} & \text{true} & \text{false} & \text{true} \\ W & \text{true} & \text{true} & \text{true} & \text{true} & \text{false} \\ S & \text{true} & \text{true} & \text{true} & \text{true} & \text{true} \\ \hline \end{tabular}$	$\begin{array}{c ccccc} \Phi & a & b & c & d & e \\ \hline N & \text{true} & \text{false} & \text{true} & \text{false} & \text{false} \\ E & \text{true} & \text{false} & \text{true} & \text{false} & \text{true} \\ W & \text{true} & \text{true} & \text{true} & \text{true} & \text{false} \\ S & \text{true} & \text{true} & \text{true} & \text{true} & \text{true} \end{array}$	$\begin{array}{c ccccc} \Phi & a & b & c & d & e \\ \hline N & \text{true} & \text{false} & \text{true} & \text{false} & \text{false} \\ E & \text{true} & \text{false} & \text{true} & \text{false} & \text{true} \\ W & \text{true} & \text{true} & \text{true} & \text{true} & \text{false} \\ S & \text{true} & \text{true} & \text{true} & \text{true} & \text{true} \end{array}$
fong-spivak-seven-sketches_000036	138	0.813	$\begin{tabular}{t l l l } \hline \Psi & (a) & (b) & (c) \\ \hline a & \text{false} & \text{true} & \text{true} \\ b & \text{true} & \text{true} & \text{true} \\ c & \text{false} & \text{true} & \text{true} \\ d & \text{true} & \text{true} & \text{true} \\ e & \text{false} & \text{false} & \text{true} \\ \hline \end{tabular}$	$\begin{array}{c cc} \Psi & x & y \\ \hline a & \text{false} & \text{true} \\ b & \text{true} & \text{true} \\ c & \text{false} & \text{true} \\ d & \text{true} & \text{true} \\ e & \text{false} & \text{false} \end{array}$	$\begin{array}{c cc} \Psi & x & y \\ \hline a & \text{false} & \text{true} \\ b & \text{true} & \text{true} \\ c & \text{false} & \text{true} \\ d & \text{true} & \text{true} \\ e & \text{false} & \text{false} \end{array}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																						
fong-spivak-seven-sketches_000037	138	0.795	$\begin{tabular}{t}{ l l l }\hline\Phi\circ\Psi&\circ&\Psi\\\hline N&false&true\\\hline E&false&true\\\hline W&true&true\\\hline S&true&true\\\hline\end{tabular}$	<table><tr><td>$\Phi\circ\Psi$</td><td>x</td><td>y</td></tr><tr><td>N</td><td>false</td><td>true</td></tr><tr><td>E</td><td>false</td><td>true</td></tr><tr><td>W</td><td>true</td><td>true</td></tr><tr><td>S</td><td>true</td><td>true</td></tr></table>	$\Phi\circ\Psi$	x	y	N	false	true	E	false	true	W	true	true	S	true	true	<table><tr><td>$\Phi\circ\Psi$</td><td>x</td><td>y</td></tr><tr><td>N</td><td>false</td><td>true</td></tr><tr><td>E</td><td>false</td><td>true</td></tr><tr><td>W</td><td>true</td><td>true</td></tr><tr><td>S</td><td>true</td><td>true</td></tr></table>	$\Phi\circ\Psi$	x	y	N	false	true	E	false	true	W	true	true	S	true	true																								
$\Phi\circ\Psi$	x	y																																																									
N	false	true																																																									
E	false	true																																																									
W	true	true																																																									
S	true	true																																																									
$\Phi\circ\Psi$	x	y																																																									
N	false	true																																																									
E	false	true																																																									
W	true	true																																																									
S	true	true																																																									
fong-spivak-seven-sketches_000038	139	0.600	$\begin{tabular}{t}{ l l l l l l }\hline\Phi\circ\Psi&\circ&\Psi&\circ&\Psi&\circ\\\hline A&?&24&?&?&?\\\hline B&?&?&?&?&?\\\hline C&?&?&?&?&?\\\hline D&?&?&9&?&?\\\hline\end{tabular}$	<table><tr><td>$\Phi\circ\Psi$</td><td>p</td><td>9</td><td>r</td><td>s</td></tr><tr><td>A</td><td>?</td><td>24</td><td>?</td><td>?</td></tr><tr><td>B</td><td>?</td><td>?</td><td>?</td><td>?</td></tr><tr><td>C</td><td>?</td><td>?</td><td>?</td><td>?</td></tr><tr><td>D</td><td>?</td><td>?</td><td>9</td><td>?</td></tr></table>	$\Phi\circ\Psi$	p	9	r	s	A	?	24	?	?	B	?	?	?	?	C	?	?	?	?	D	?	?	9	?	<table><tr><td>$\Phi\circ\Psi$</td><td>p</td><td>q</td><td>r</td><td>s</td></tr><tr><td>A</td><td>?</td><td>24</td><td>?</td><td>?</td></tr><tr><td>B</td><td>?</td><td>?</td><td>?</td><td>?</td></tr><tr><td>C</td><td>?</td><td>?</td><td>?</td><td>?</td></tr><tr><td>D</td><td>?</td><td>?</td><td>9</td><td>?</td></tr></table>	$\Phi\circ\Psi$	p	q	r	s	A	?	24	?	?	B	?	?	?	?	C	?	?	?	?	D	?	?	9	?				
$\Phi\circ\Psi$	p	9	r	s																																																							
A	?	24	?	?																																																							
B	?	?	?	?																																																							
C	?	?	?	?																																																							
D	?	?	9	?																																																							
$\Phi\circ\Psi$	p	q	r	s																																																							
A	?	24	?	?																																																							
B	?	?	?	?																																																							
C	?	?	?	?																																																							
D	?	?	9	?																																																							
fong-spivak-seven-sketches_000039	144	0.574	$\begin{tabular}{t}{ l l l l l l l l l }\hline X&A&B&\Phi&x&y&y&x&y\\\hline A&0&2&A&5&8&x&0&3\\\hline B&\infty&0&B&\infty&\infty&y&4&0\\\hline\end{tabular}$	<table><tr><td>X</td><td>A</td><td>B</td><td>Φ</td><td>x</td><td>y</td><td>y</td><td>x</td><td>y</td></tr><tr><td>A</td><td>0</td><td>2</td><td>A</td><td>5</td><td>8</td><td>x</td><td>0</td><td>3</td></tr><tr><td>B</td><td>∞</td><td>0</td><td>B</td><td>∞</td><td>∞</td><td>y</td><td>4</td><td>0</td></tr></table>	X	A	B	Φ	x	y	y	x	y	A	0	2	A	5	8	x	0	3	B	∞	0	B	∞	∞	y	4	0	<table><tr><td>X</td><td>A</td><td>B</td><td>Φ</td><td>x</td><td>y</td><td>y</td><td>x</td><td>y</td></tr><tr><td>A</td><td>0</td><td>2</td><td>A</td><td>5</td><td>8</td><td>x</td><td>0</td><td>3</td></tr><tr><td>B</td><td>∞</td><td>0</td><td>B</td><td>∞</td><td>∞</td><td>y</td><td>4</td><td>0</td></tr></table>	X	A	B	Φ	x	y	y	x	y	A	0	2	A	5	8	x	0	3	B	∞	0	B	∞	∞	y	4	0
X	A	B	Φ	x	y	y	x	y																																																			
A	0	2	A	5	8	x	0	3																																																			
B	∞	0	B	∞	∞	y	4	0																																																			
X	A	B	Φ	x	y	y	x	y																																																			
A	0	2	A	5	8	x	0	3																																																			
B	∞	0	B	∞	∞	y	4	0																																																			
fong-spivak-seven-sketches_000040	177	0.492	$\begin{tabular}{t}{ l l l l }\hline generator&icon&matrix&arity\\\hline amplify by a\in R&- a -&(a)&1\rightarrow 1\\\hline add&\cup-&\begin{pmatrix} 1 \\ 1 \end{pmatrix}&2\rightarrow 1\\\hline zero&\circ-&()&0\rightarrow 1\\\hline copy&-&(1 1)&1\rightarrow 2\\\hline discard&-&()&1\rightarrow 0\\\hline\end{tabular}$	<table><tr><td>generator</td><td>icon</td><td>matrix</td><td>arity</td></tr><tr><td>amplify by $a\in R$</td><td>- a -</td><td>(a)</td><td>$1\rightarrow 1$</td></tr><tr><td>add</td><td>$\cup$-</td><td>$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$</td><td>$2\rightarrow 1$</td></tr><tr><td>zero</td><td>$\circ$-</td><td>$()$</td><td>$0\rightarrow 1$</td></tr><tr><td>copy</td><td>-</td><td>(1 1)</td><td>$1\rightarrow 2$</td></tr><tr><td>discard</td><td>-</td><td>$()$</td><td>$1\rightarrow 0$</td></tr></table>	generator	icon	matrix	arity	amplify by $a\in R$	- a -	(a)	$1\rightarrow 1$	add	$\cup$ -	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$2\rightarrow 1$	zero	$\circ$ -	$()$	$0\rightarrow 1$	copy	-	(1 1)	$1\rightarrow 2$	discard	-	$()$	$1\rightarrow 0$	<table><tr><td>generator</td><td>icon</td><td>matrix</td><td>arity</td></tr><tr><td>amplify by $a\in R$</td><td></td><td>(a)</td><td>$1\rightarrow 1$</td></tr><tr><td>add</td><td></td><td>$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$</td><td>$2\rightarrow 1$</td></tr><tr><td>zero</td><td></td><td>$()$</td><td>$0\rightarrow 1$</td></tr><tr><td>copy</td><td></td><td>$(1\ 1)$</td><td>$1\rightarrow 2$</td></tr><tr><td>discard</td><td></td><td>$()$</td><td>$1\rightarrow 0$</td></tr></table>	generator	icon	matrix	arity	amplify by $a\in R$		(a)	$1\rightarrow 1$	add		$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$2\rightarrow 1$	zero		$()$	$0\rightarrow 1$	copy		$(1\ 1)$	$1\rightarrow 2$	discard		$()$	$1\rightarrow 0$						
generator	icon	matrix	arity																																																								
amplify by $a\in R$	- a -	(a)	$1\rightarrow 1$																																																								
add	$\cup$ -	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$2\rightarrow 1$																																																								
zero	$\circ$ -	$()$	$0\rightarrow 1$																																																								
copy	-	(1 1)	$1\rightarrow 2$																																																								
discard	-	$()$	$1\rightarrow 0$																																																								
generator	icon	matrix	arity																																																								
amplify by $a\in R$		(a)	$1\rightarrow 1$																																																								
add		$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$2\rightarrow 1$																																																								
zero		$()$	$0\rightarrow 1$																																																								
copy		$(1\ 1)$	$1\rightarrow 2$																																																								
discard		$()$	$1\rightarrow 0$																																																								
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.																																																											

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																				
fong-spivak-seven-sketches-742_041	242	0.752	$\begin{array}{ l l l } \hline \backslash\ulcorner\backslash) \text{ true, true } \backslash()\urcorner(P, Q)\backslash) \\ \hline \text{true} \& \text{true} \& \text{true} \\ \hline \text{true} \& \text{true} \& \text{false} \\ \hline \text{false} \& \text{true} \& \text{false} \\ \hline \text{false} \& \text{false} \& \text{false} \\ \hline \end{array}$	<table><tr><th>P</th><th>Q</th><th>$\lceil(\text{true}, \text{true})^\top(P, Q)$</th></tr><tr><td>true</td><td>true</td><td>true</td></tr><tr><td>true</td><td>false</td><td>false</td></tr><tr><td>false</td><td>true</td><td>false</td></tr><tr><td>false</td><td>false</td><td>false</td></tr></table>	P	Q	$\lceil(\text{true}, \text{true})^\top(P, Q)$	true	true	true	true	false	false	false	true	false	false	false	false	<table><tr><th>P</th><th>Q</th><th>$\lceil(\text{true}, \text{true})^\top(P, Q)$</th></tr><tr><td>true</td><td>true</td><td>true</td></tr><tr><td>true</td><td>false</td><td>false</td></tr><tr><td>false</td><td>true</td><td>false</td></tr><tr><td>false</td><td>false</td><td>false</td></tr></table>	P	Q	$\lceil(\text{true}, \text{true})^\top(P, Q)$	true	true	true	true	false	false	false	true	false	false	false	false						
P	Q	$\lceil(\text{true}, \text{true})^\top(P, Q)$																																							
true	true	true																																							
true	false	false																																							
false	true	false																																							
false	false	false																																							
P	Q	$\lceil(\text{true}, \text{true})^\top(P, Q)$																																							
true	true	true																																							
true	false	false																																							
false	true	false																																							
false	false	false																																							
fong-spivak-seven-sketches-742_042	242	0.785	$\begin{array}{ l l l } \hline \backslash(P\backslash) \& Q \& \backslash(P \vee Q\backslash) \\ \hline \text{true} \& \text{true} \& \text{true} \\ \hline \text{true} \& \text{false} \& \text{true} \\ \hline \text{false} \& \text{true} \& \text{true} \\ \hline \text{false} \& \text{false} \& \text{false} \\ \hline \end{array}$	<table><tr><th>P</th><th>Q</th><th>$P \vee Q$</th></tr><tr><td>true</td><td>true</td><td>true</td></tr><tr><td>true</td><td>false</td><td>true</td></tr><tr><td>false</td><td>true</td><td>true</td></tr><tr><td>false</td><td>false</td><td>false</td></tr></table>	P	Q	$P \vee Q$	true	true	true	true	false	true	false	true	true	false	false	false	<table><tr><th>P</th><th>Q</th><th>$P \vee Q$</th></tr><tr><td>true</td><td>true</td><td>true</td></tr><tr><td>true</td><td>false</td><td>true</td></tr><tr><td>false</td><td>true</td><td>true</td></tr><tr><td>false</td><td>false</td><td>false</td></tr></table>	P	Q	$P \vee Q$	true	true	true	true	false	true	false	true	true	false	false	false						
P	Q	$P \vee Q$																																							
true	true	true																																							
true	false	true																																							
false	true	true																																							
false	false	false																																							
P	Q	$P \vee Q$																																							
true	true	true																																							
true	false	true																																							
false	true	true																																							
false	false	false																																							
fong-spivak-seven-sketches-742_043	274	0.813	$\begin{array}{ l l l } \hline \text{arrow } \backslash(a\backslash) \& \text{source } \backslash(s(a) \backslashin V\backslash) \& \text{target } \backslash(t(a) \backslashin V\backslash) \\ \hline 1 \& 2 \& 1 \\ \hline 1 \& 3 \& 1 \\ \hline d \& 2 \& 2 \\ \hline e \& 2 \& 3 \\ \hline \end{array}$	<table><tr><th>arrow a</th><th>source $s(a) \in V$</th><th>target $t(a) \in V$</th></tr><tr><td>a</td><td>1</td><td>2</td></tr><tr><td>b</td><td>1</td><td>3</td></tr><tr><td>c</td><td>1</td><td>3</td></tr><tr><td>d</td><td>2</td><td>2</td></tr><tr><td>e</td><td>2</td><td>3</td></tr></table>	arrow a	source $s(a) \in V$	target $t(a) \in V$	a	1	2	b	1	3	c	1	3	d	2	2	e	2	3	<table><tr><th>arrow a</th><th>source $s(a) \in V$</th><th>target $t(a) \in V$</th></tr><tr><td>a</td><td>1</td><td>2</td></tr><tr><td>b</td><td>1</td><td>3</td></tr><tr><td>c</td><td>1</td><td>3</td></tr><tr><td>d</td><td>2</td><td>2</td></tr><tr><td>e</td><td>2</td><td>3</td></tr></table>	arrow a	source $s(a) \in V$	target $t(a) \in V$	a	1	2	b	1	3	c	1	3	d	2	2	e	2	3
arrow a	source $s(a) \in V$	target $t(a) \in V$																																							
a	1	2																																							
b	1	3																																							
c	1	3																																							
d	2	2																																							
e	2	3																																							
arrow a	source $s(a) \in V$	target $t(a) \in V$																																							
a	1	2																																							
b	1	3																																							
c	1	3																																							
d	2	2																																							
e	2	3																																							
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.																																									

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																																																																																																																																																						
fong-spivak-seven-sketches_044	280	0.941	$\begin{tabular}{t}{ l l l l l l }\hline \backslash(p\backslash) \& \backslash(q\backslash) \& \backslash(f(p)\backslash) \& \backslash(g(q)\backslash) \& \backslash(f(p) \leq q\backslash) \& \backslash(p \leq ? g(q)\backslash) \& \text{same?} \\ \hline 1 \& 1 \& 1 \& 1 \& \text{yes} \& \text{yes} \& \text{yes!} \\ \hline 1 \& 2 \& 1 \& 2 \& \text{no} \& \text{no} \& \text{yes!} \\ \hline 1 \& 4 \& 1 \& 4 \& \text{yes} \& \text{yes} \& \text{yes!} \\ \hline 2 \& 1 \& 2 \& 1 \& \text{no} \& \text{no} \& \text{yes!} \\ \hline 2 \& 2 \& 2 \& 2 \& \text{yes} \& \text{yes} \& \text{yes!} \\ \hline 2 \& 4 \& 2 \& 4 \& \text{yes} \& \text{yes} \& \text{yes!} \\ \hline 3.9 \& 1 \& 4 \& 1 \& \text{no} \& \text{no} \& \text{yes!} \\ \hline 3.9 \& 2 \& 4 \& 2 \& \text{no} \& \text{no} \& \text{yes!} \\ \hline 3.9 \& 4 \& 4 \& 4 \& \text{yes} \& \text{yes} \& \text{yes!} \\ \hline 4 \& 1 \& 4 \& 1 \& \text{no} \& \text{no} \& \text{yes!} \\ \hline 4 \& 2 \& 4 \& 2 \& \text{no} \& \text{no} \& \text{yes!} \\ \hline 4 \& 4 \& 4 \& 4 \& \text{yes} \& \text{yes} \& \text{yes!} \\ \hline \end{tabular}$	<table><tr><th>p</th><th>q</th><th>$f(p)$</th><th>$g(q)$</th><th>$f(p) \leq q$</th><th>$p \leq ? g(q)$</th><th>same?</th></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>yes</td><td>yes</td><td>yes!</td></tr><tr><td>1</td><td>2</td><td>1</td><td>2</td><td>no</td><td>no</td><td>yes!</td></tr><tr><td>1</td><td>4</td><td>1</td><td>4</td><td>yes</td><td>yes</td><td>yes!</td></tr><tr><td>2</td><td>1</td><td>2</td><td>1</td><td>no</td><td>no</td><td>yes!</td></tr><tr><td>2</td><td>2</td><td>2</td><td>2</td><td>yes</td><td>yes</td><td>yes!</td></tr><tr><td>2</td><td>4</td><td>2</td><td>4</td><td>yes</td><td>yes</td><td>yes!</td></tr><tr><td>3.9</td><td>1</td><td>4</td><td>1</td><td>no</td><td>no</td><td>yes!</td></tr><tr><td>3.9</td><td>2</td><td>4</td><td>2</td><td>no</td><td>no</td><td>yes!</td></tr><tr><td>3.9</td><td>4</td><td>4</td><td>4</td><td>yes</td><td>yes</td><td>yes!</td></tr><tr><td>4</td><td>1</td><td>4</td><td>1</td><td>no</td><td>no</td><td>yes!</td></tr><tr><td>4</td><td>2</td><td>4</td><td>2</td><td>no</td><td>no</td><td>yes!</td></tr><tr><td>4</td><td>4</td><td>4</td><td>4</td><td>yes</td><td>yes</td><td>yes!</td></tr></table>	p	q	$f(p)$	$g(q)$	$f(p) \leq q$	$p \leq ? g(q)$	same?	1	1	1	1	yes	yes	yes!	1	2	1	2	no	no	yes!	1	4	1	4	yes	yes	yes!	2	1	2	1	no	no	yes!	2	2	2	2	yes	yes	yes!	2	4	2	4	yes	yes	yes!	3.9	1	4	1	no	no	yes!	3.9	2	4	2	no	no	yes!	3.9	4	4	4	yes	yes	yes!	4	1	4	1	no	no	yes!	4	2	4	2	no	no	yes!	4	4	4	4	yes	yes	yes!	<table><tr><th>p</th><th>q</th><th>$f(p)$</th><th>$g(q)$</th><th>$f(p) \leq ? q$</th><th>$p \leq ? g(q)$</th><th>same?</th></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>yes</td><td>yes</td><td>yes!</td></tr><tr><td>1</td><td>2</td><td>1</td><td>2</td><td>no</td><td>no</td><td>yes!</td></tr><tr><td>1</td><td>4</td><td>1</td><td>4</td><td>yes</td><td>yes</td><td>yes!</td></tr><tr><td>2</td><td>1</td><td>2</td><td>1</td><td>no</td><td>no</td><td>yes!</td></tr><tr><td>2</td><td>2</td><td>2</td><td>2</td><td>yes</td><td>yes</td><td>yes!</td></tr><tr><td>2</td><td>4</td><td>2</td><td>4</td><td>yes</td><td>yes</td><td>yes!</td></tr><tr><td>3.9</td><td>1</td><td>4</td><td>1</td><td>no</td><td>no</td><td>yes!</td></tr><tr><td>3.9</td><td>2</td><td>4</td><td>2</td><td>no</td><td>no</td><td>yes!</td></tr><tr><td>3.9</td><td>4</td><td>4</td><td>4</td><td>yes</td><td>yes</td><td>yes!</td></tr><tr><td>4</td><td>1</td><td>4</td><td>1</td><td>no</td><td>no</td><td>yes!</td></tr><tr><td>4</td><td>2</td><td>4</td><td>2</td><td>no</td><td>no</td><td>yes!</td></tr><tr><td>4</td><td>4</td><td>4</td><td>4</td><td>yes</td><td>yes</td><td>yes!</td></tr></table>	p	q	$f(p)$	$g(q)$	$f(p) \leq ? q$	$p \leq ? g(q)$	same?	1	1	1	1	yes	yes	yes!	1	2	1	2	no	no	yes!	1	4	1	4	yes	yes	yes!	2	1	2	1	no	no	yes!	2	2	2	2	yes	yes	yes!	2	4	2	4	yes	yes	yes!	3.9	1	4	1	no	no	yes!	3.9	2	4	2	no	no	yes!	3.9	4	4	4	yes	yes	yes!	4	1	4	1	no	no	yes!	4	2	4	2	no	no	yes!	4	4	4	4	yes	yes	yes!
p	q	$f(p)$	$g(q)$	$f(p) \leq q$	$p \leq ? g(q)$	same?																																																																																																																																																																																					
1	1	1	1	yes	yes	yes!																																																																																																																																																																																					
1	2	1	2	no	no	yes!																																																																																																																																																																																					
1	4	1	4	yes	yes	yes!																																																																																																																																																																																					
2	1	2	1	no	no	yes!																																																																																																																																																																																					
2	2	2	2	yes	yes	yes!																																																																																																																																																																																					
2	4	2	4	yes	yes	yes!																																																																																																																																																																																					
3.9	1	4	1	no	no	yes!																																																																																																																																																																																					
3.9	2	4	2	no	no	yes!																																																																																																																																																																																					
3.9	4	4	4	yes	yes	yes!																																																																																																																																																																																					
4	1	4	1	no	no	yes!																																																																																																																																																																																					
4	2	4	2	no	no	yes!																																																																																																																																																																																					
4	4	4	4	yes	yes	yes!																																																																																																																																																																																					
p	q	$f(p)$	$g(q)$	$f(p) \leq ? q$	$p \leq ? g(q)$	same?																																																																																																																																																																																					
1	1	1	1	yes	yes	yes!																																																																																																																																																																																					
1	2	1	2	no	no	yes!																																																																																																																																																																																					
1	4	1	4	yes	yes	yes!																																																																																																																																																																																					
2	1	2	1	no	no	yes!																																																																																																																																																																																					
2	2	2	2	yes	yes	yes!																																																																																																																																																																																					
2	4	2	4	yes	yes	yes!																																																																																																																																																																																					
3.9	1	4	1	no	no	yes!																																																																																																																																																																																					
3.9	2	4	2	no	no	yes!																																																																																																																																																																																					
3.9	4	4	4	yes	yes	yes!																																																																																																																																																																																					
4	1	4	1	no	no	yes!																																																																																																																																																																																					
4	2	4	2	no	no	yes!																																																																																																																																																																																					
4	4	4	4	yes	yes	yes!																																																																																																																																																																																					
fong-spivak-seven-sketches_045	282	1.000	$\begin{tabular}{t}{ l l l l }\hline \text{min} \& \text{no} \& \text{maybe} \& \text{yes} \\ \hline \text{no} \& \text{no} \& \text{no} \& \text{no} \\ \hline \text{maybe} \& \text{no} \& \text{maybe} \& \text{maybe} \\ \hline \text{yes} \& \text{no} \& \text{maybe} \& \text{yes} \\ \hline \end{tabular}$	<table><tr><th>min</th><th>no</th><th>maybe</th><th>yes</th></tr><tr><td>no</td><td>no</td><td>no</td><td>no</td></tr><tr><td>maybe</td><td>no</td><td>maybe</td><td>maybe</td></tr><tr><td>yes</td><td>no</td><td>maybe</td><td>yes</td></tr></table>	min	no	maybe	yes	no	no	no	no	maybe	no	maybe	maybe	yes	no	maybe	yes	<table><tr><th>min</th><th>no</th><th>maybe</th><th>yes</th></tr><tr><td>no</td><td>no</td><td>no</td><td>no</td></tr><tr><td>maybe</td><td>no</td><td>maybe</td><td>maybe</td></tr><tr><td>yes</td><td>no</td><td>maybe</td><td>yes</td></tr></table>	min	no	maybe	yes	no	no	no	no	maybe	no	maybe	maybe	yes	no	maybe	yes																																																																																																																																																						
min	no	maybe	yes																																																																																																																																																																																								
no	no	no	no																																																																																																																																																																																								
maybe	no	maybe	maybe																																																																																																																																																																																								
yes	no	maybe	yes																																																																																																																																																																																								
min	no	maybe	yes																																																																																																																																																																																								
no	no	no	no																																																																																																																																																																																								
maybe	no	maybe	maybe																																																																																																																																																																																								
yes	no	maybe	yes																																																																																																																																																																																								
fong-spivak-seven-sketches_046	284	0.718	$\begin{tabular}{t}{ l l l l l }\hline \backslash(d(/)\backslash) \& \backslash(A\backslash) \& \backslash(B\backslash) \& \backslash(C\backslash) \& \backslash(D\backslash) \\ \hline \backslash(A\backslash) \& 0 \& 6 \& 3 \& 11 \\ \hline B \& 2 \& 0 \& 5 \& 5 \\ \hline \backslash(C\backslash) \& 5 \& 3 \& 0 \& 8 \\ \hline D \& 11 \& 9 \& 6 \& 0 \\ \hline \end{tabular}$	<table><tr><th>$d(/)$</th><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>A</td><td>0</td><td>6</td><td>3</td><td>11</td></tr><tr><td>B</td><td>2</td><td>0</td><td>5</td><td>5</td></tr><tr><td>C</td><td>5</td><td>3</td><td>0</td><td>8</td></tr><tr><td>D</td><td>11</td><td>9</td><td>6</td><td>0</td></tr></table>	$d(/)$	A	B	C	D	A	0	6	3	11	B	2	0	5	5	C	5	3	0	8	D	11	9	6	0	<table><tr><th>$d(\nearrow)$</th><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>A</td><td>0</td><td>6</td><td>3</td><td>11</td></tr><tr><td>B</td><td>2</td><td>0</td><td>5</td><td>5</td></tr><tr><td>C</td><td>5</td><td>3</td><td>0</td><td>8</td></tr><tr><td>D</td><td>11</td><td>9</td><td>6</td><td>0</td></tr></table>	$d(\nearrow)$	A	B	C	D	A	0	6	3	11	B	2	0	5	5	C	5	3	0	8	D	11	9	6	0																																																																																																																																				
$d(/)$	A	B	C	D																																																																																																																																																																																							
A	0	6	3	11																																																																																																																																																																																							
B	2	0	5	5																																																																																																																																																																																							
C	5	3	0	8																																																																																																																																																																																							
D	11	9	6	0																																																																																																																																																																																							
$d(\nearrow)$	A	B	C	D																																																																																																																																																																																							
A	0	6	3	11																																																																																																																																																																																							
B	2	0	5	5																																																																																																																																																																																							
C	5	3	0	8																																																																																																																																																																																							
D	11	9	6	0																																																																																																																																																																																							

Conf. MathPix confidence: 0.9 0.5-0.9 <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured

[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																																																																		
fong-spivak-seven-sketches-047	284	0.808	$\begin{tabular}{t}{ l l l l l l }\hline\multirow{5}{*}{\begin{array}{l} \backslash(M_X=\backslash) \& \succ \& \backslash(A\backslash) \& \backslash(B\backslash) \& \backslash(C\backslash) \& \backslash(D\backslash) \\\hline \& A \& \emptyset \& \backslash(\infty\backslash) \& 3 \& \backslash(\infty\backslash) \\\hline \& B \& 2 \& \emptyset \& \backslash(\infty\backslash) \& 5 \\\hline \& \backslash(C\backslash) \& \backslash(\infty\backslash) \& 3 \& \emptyset \& \backslash(\infty\backslash) \\\hline \& D \& \backslash(\infty\backslash) \& \backslash(\infty\backslash) \& 6 \& \emptyset \\\hline \end{array}}\hline\end{tabular}$	(not rendered)	$M_X = \begin{array}{c ccccc} \nearrow & A & B & C & D \\ \hline A & 0 & \infty & 3 & \infty \\ B & 2 & 0 & \infty & 5 \\ C & \infty & 3 & 0 & \infty \\ D & \infty & \infty & 6 & 0 \end{array}$																																																																																																		
fong-spivak-seven-sketches-048	285	0.392	$\begin{tabular}{t}{ l l l l l }\hline X(/ \, ') \& \backslash(A\backslash) \& \backslash(B\backslash) \& \backslash(C\backslash) \& \backslash(D\backslash) \\\hline \backslash(A\backslash) \& \backslash(M\backslash) \& \backslash\{boat\backslash} \& \backslash(\varnothing\backslash) \& \backslash\{boat\backslash} \\\hline B \& \backslash(\varnothing\backslash) \& \backslash(M\backslash) \& \backslash(\varnothing\backslash) \& \backslash\{boat\backslash} \\\hline C \& \backslash\{foot,boat\backslash} \& \backslash\{boat\backslash} \& \backslash(M\backslash) \& \backslash(M\backslash) \\\hline D \& \backslash\{foot\backslash} \& \backslash(\varnothing\backslash) \& \backslash\{foot, \, car\backslash} \& \backslash(M\backslash) \\\hline \end{tabular}$	<table><tr><td>X(/')</td><td>A</td><td>B</td><td>C</td><td>D</td></tr><tr><td>A</td><td>M</td><td>{boat}</td><td>∅</td><td>{boat}</td></tr><tr><td>B</td><td>∅</td><td>M</td><td>∅</td><td>{boat}</td></tr><tr><td>C</td><td>{foot,boat}</td><td>{boat}</td><td>M</td><td>M</td></tr><tr><td>D</td><td>{foot}</td><td>∅</td><td>{foot, car}</td><td>M</td></tr></table>	X(/')	A	B	C	D	A	M	{boat}	∅	{boat}	B	∅	M	∅	{boat}	C	{foot,boat}	{boat}	M	M	D	{foot}	∅	{foot, car}	M	<table><tr><td>X(↗)</td><td>A</td><td>B</td><td>C</td><td>D</td></tr><tr><td>A</td><td>M</td><td>{boat}</td><td>∅</td><td>{boat}</td></tr><tr><td>B</td><td>∅</td><td>M</td><td>∅</td><td>{boat}</td></tr><tr><td>C</td><td>{foot,boat}</td><td>{boat}</td><td>M</td><td>M</td></tr><tr><td>D</td><td>{foot}</td><td>∅</td><td>{foot,car}</td><td>M</td></tr></table>	X(↗)	A	B	C	D	A	M	{boat}	∅	{boat}	B	∅	M	∅	{boat}	C	{foot,boat}	{boat}	M	M	D	{foot}	∅	{foot,car}	M																																																
X(/')	A	B	C	D																																																																																																			
A	M	{boat}	∅	{boat}																																																																																																			
B	∅	M	∅	{boat}																																																																																																			
C	{foot,boat}	{boat}	M	M																																																																																																			
D	{foot}	∅	{foot, car}	M																																																																																																			
X(↗)	A	B	C	D																																																																																																			
A	M	{boat}	∅	{boat}																																																																																																			
B	∅	M	∅	{boat}																																																																																																			
C	{foot,boat}	{boat}	M	M																																																																																																			
D	{foot}	∅	{foot,car}	M																																																																																																			
fong-spivak-seven-sketches-049	285	0.571	$\begin{tabular}{t}{ l l l l }\hline \backslash(M(/)\backslash) \& \backslash(A\backslash) \& \backslash(B\backslash) \& \backslash(C\backslash) \\\hline \backslash(A\backslash) \& \backslash(\infty\backslash) \& 6 \& 10 \\\hline B \& 10 \& \backslash(\infty\backslash) \& 10 \\\hline \backslash(C\backslash) \& 10 \& 6 \& \backslash(\infty\backslash) \\\hline \end{tabular}$	<table><tr><td>M(/)</td><td>A</td><td>B</td><td>C</td></tr><tr><td>A</td><td>∞</td><td>6</td><td>10</td></tr><tr><td>B</td><td>10</td><td>∞</td><td>10</td></tr><tr><td>C</td><td>10</td><td>6</td><td>∞</td></tr></table>	M(/)	A	B	C	A	∞	6	10	B	10	∞	10	C	10	6	∞	<table><tr><td>M(↗)</td><td>A</td><td>B</td><td>C</td></tr><tr><td>A</td><td>∞</td><td>6</td><td>10</td></tr><tr><td>B</td><td>10</td><td>∞</td><td>10</td></tr><tr><td>C</td><td>10</td><td>6</td><td>∞</td></tr></table>	M(↗)	A	B	C	A	∞	6	10	B	10	∞	10	C	10	6	∞																																																																		
M(/)	A	B	C																																																																																																				
A	∞	6	10																																																																																																				
B	10	∞	10																																																																																																				
C	10	6	∞																																																																																																				
M(↗)	A	B	C																																																																																																				
A	∞	6	10																																																																																																				
B	10	∞	10																																																																																																				
C	10	6	∞																																																																																																				
fong-spivak-seven-sketches-050	287	0.871	$\begin{tabular}{t}{ l l l }\hline \Rightarrow \& \text{false} \& \text{true} \\\hline \text{false} \& \text{true} \& \text{true} \\\hline \text{true} \& \text{false} \& \text{true} \\\hline \end{tabular}$	<table><tr><td>⇒</td><td>false</td><td>true</td></tr><tr><td>false</td><td>true</td><td>true</td></tr><tr><td>true</td><td>false</td><td>true</td></tr></table>	⇒	false	true	false	true	true	true	false	true	<table><tr><td>⇒</td><td>false</td><td>true</td></tr><tr><td>false</td><td>true</td><td>true</td></tr><tr><td>true</td><td>false</td><td>true</td></tr></table>	⇒	false	true	false	true	true	true	false	true																																																																																
⇒	false	true																																																																																																					
false	true	true																																																																																																					
true	false	true																																																																																																					
⇒	false	true																																																																																																					
false	true	true																																																																																																					
true	false	true																																																																																																					
fong-spivak-seven-sketches-051	289	0.306	$\begin{tabular}{t}{ l l l l l l l }\hline \succ \& \backslash(v_1\backslash) \& \backslash(f_1\backslash) \& \backslash(f_1 \ ; \ f_2\backslash) \& \backslash(v_2\backslash) \& \backslash(f_2\backslash) \& \backslash(v_3\backslash) \\\hline \backslash(v_1\backslash) \& \backslash(v_1\backslash) \& \backslash(v_1\backslash) \& \backslash(f_1\backslash) \& \backslash(f_1\backslash) \& \text{g} \backslash(f_2\backslash) \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \\\hline \backslash(f_1\backslash) \& \text{g} \backslash(f_2\backslash) \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \backslash(f_1\backslash) \& \backslash(f_1\backslash) \& \text{i} \backslash(f_2\backslash) \& \boxtimes \\\hline \boxtimes \& \boxtimes \& \boxtimes \& \backslash(f_1\backslash) \& \backslash(f_1\backslash) \& \text{i} \backslash(f_2\backslash) \& \boxtimes \& \boxtimes \\\hline \backslash(f_1 \ \stackrel{\circ}{\circlearrowleft} \ f_2\backslash) \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \backslash(f_1 \ \stackrel{\circ}{\circlearrowleft} \ f_2\backslash) \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \\\hline \backslash(v_2\backslash) \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \backslash(v_2\backslash) \& \backslash(f_2\backslash) \& \boxtimes \\\hline \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \backslash(v_2\backslash) \& \backslash(f_2\backslash) \& \boxtimes \\\hline \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \backslash(f_2\backslash) \& \backslash(v_3\backslash) \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \boxtimes \& \backslash(v_3\backslash) \\\hline \end{tabular}$	<table><tr><td>↗</td><td>v₁</td><td>f₁</td><td>f₁;f₂</td><td>v₂</td><td>f₂</td><td>v₃</td></tr><tr><td>v₁</td><td>v₁</td><td>f₁</td><td>f₁ g f₂</td><td>⊠</td><td>⊠</td><td>⊠</td></tr><tr><td>f₁</td><td>⊠</td><td>⊠</td><td>⊠</td><td>f₁</td><td>f₁ i f₂</td><td>⊠</td></tr><tr><td>f₁ ∘ f₂</td><td>⊠</td><td>⊠</td><td>⊠</td><td>⊠</td><td>⊠</td><td>f₁ ∘ f₂</td></tr><tr><td>v₂</td><td>⊠</td><td>⊠</td><td>⊠</td><td>v₂</td><td>f₂</td><td>⊠</td></tr><tr><td>f₂</td><td>⊠</td><td>⊠</td><td>⊠</td><td>⊠</td><td>⊠</td><td>f₂</td></tr><tr><td>v₃</td><td>⊠</td><td>⊠</td><td>⊠</td><td>⊠</td><td>⊠</td><td>v₃</td></tr></table>	↗	v ₁	f ₁	f ₁ ;f ₂	v ₂	f ₂	v ₃	v ₁	v ₁	f ₁	f ₁ g f ₂	⊠	⊠	⊠	f ₁	⊠	⊠	⊠	f ₁	f ₁ i f ₂	⊠	f ₁ ∘ f ₂	⊠	⊠	⊠	⊠	⊠	f ₁ ∘ f ₂	v ₂	⊠	⊠	⊠	v ₂	f ₂	⊠	f ₂	⊠	⊠	⊠	⊠	⊠	f ₂	v ₃	⊠	⊠	⊠	⊠	⊠	v ₃	<table><tr><td>↗</td><td>v₁</td><td>f₁</td><td>f₁ ∘ f₂</td><td>v₂</td><td>f₂</td><td>v₃</td></tr><tr><td>v₁</td><td>v₁</td><td>f₁</td><td>f₁ ∘ f₂</td><td>⊠</td><td>⊠</td><td>⊠</td></tr><tr><td>f₁</td><td>⊠</td><td>⊠</td><td>⊠</td><td>f₁</td><td>f₁ ∘ f₂</td><td>⊠</td></tr><tr><td>f₁ ∘ f₂</td><td>⊠</td><td>⊠</td><td>⊠</td><td>⊠</td><td>⊠</td><td>f₁ ∘ f₂</td></tr><tr><td>v₂</td><td>⊠</td><td>⊠</td><td>⊠</td><td>v₂</td><td>f₂</td><td>⊠</td></tr><tr><td>f₂</td><td>⊠</td><td>⊠</td><td>⊠</td><td>⊠</td><td>⊠</td><td>f₂</td></tr><tr><td>v₃</td><td>⊠</td><td>⊠</td><td>⊠</td><td>⊠</td><td>⊠</td><td>v₃</td></tr></table>	↗	v ₁	f ₁	f ₁ ∘ f ₂	v ₂	f ₂	v ₃	v ₁	v ₁	f ₁	f ₁ ∘ f ₂	⊠	⊠	⊠	f ₁	⊠	⊠	⊠	f ₁	f ₁ ∘ f ₂	⊠	f ₁ ∘ f ₂	⊠	⊠	⊠	⊠	⊠	f ₁ ∘ f ₂	v ₂	⊠	⊠	⊠	v ₂	f ₂	⊠	f ₂	⊠	⊠	⊠	⊠	⊠	f ₂	v ₃	⊠	⊠	⊠	⊠	⊠	v ₃
↗	v ₁	f ₁	f ₁ ;f ₂	v ₂	f ₂	v ₃																																																																																																	
v ₁	v ₁	f ₁	f ₁ g f ₂	⊠	⊠	⊠																																																																																																	
f ₁	⊠	⊠	⊠	f ₁	f ₁ i f ₂	⊠																																																																																																	
f ₁ ∘ f ₂	⊠	⊠	⊠	⊠	⊠	f ₁ ∘ f ₂																																																																																																	
v ₂	⊠	⊠	⊠	v ₂	f ₂	⊠																																																																																																	
f ₂	⊠	⊠	⊠	⊠	⊠	f ₂																																																																																																	
v ₃	⊠	⊠	⊠	⊠	⊠	v ₃																																																																																																	
↗	v ₁	f ₁	f ₁ ∘ f ₂	v ₂	f ₂	v ₃																																																																																																	
v ₁	v ₁	f ₁	f ₁ ∘ f ₂	⊠	⊠	⊠																																																																																																	
f ₁	⊠	⊠	⊠	f ₁	f ₁ ∘ f ₂	⊠																																																																																																	
f ₁ ∘ f ₂	⊠	⊠	⊠	⊠	⊠	f ₁ ∘ f ₂																																																																																																	
v ₂	⊠	⊠	⊠	v ₂	f ₂	⊠																																																																																																	
f ₂	⊠	⊠	⊠	⊠	⊠	f ₂																																																																																																	
v ₃	⊠	⊠	⊠	⊠	⊠	v ₃																																																																																																	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																
fong-spivak-seven-sketches-ink-052	294	■1.000	$\begin{tabular}{t}{ l l l }\hline \text{Arrow} \& \text{source} \& \text{target} \\\hline \text{Mngr} \& \text{Employee} \& \text{Employee} \\\hline \text{WorksIn} \& \text{Employee} \& \text{Department} \\\hline \text{Secr} \& \text{Department} \& \text{Employee} \\\hline \text{FName} \& \text{Employee} \& \text{string} \\\hline \text{DName} \& \text{Department} \& \text{string} \\\hline\end{tabular}$	<table><tr><th>Arrow</th><th>source</th><th>target</th></tr><tr><td>Mngr</td><td>Employee</td><td>Employee</td></tr><tr><td>WorksIn</td><td>Employee</td><td>Department</td></tr><tr><td>Secr</td><td>Department</td><td>Employee</td></tr><tr><td>FName</td><td>Employee</td><td>string</td></tr><tr><td>DName</td><td>Department</td><td>string</td></tr></table>	Arrow	source	target	Mngr	Employee	Employee	WorksIn	Employee	Department	Secr	Department	Employee	FName	Employee	string	DName	Department	string	<table><tr><th>Arrow</th><th>source</th><th>target</th></tr><tr><td>Mngr</td><td>Employee</td><td>Employee</td></tr><tr><td>WorksIn</td><td>Employee</td><td>Department</td></tr><tr><td>Secr</td><td>Department</td><td>Employee</td></tr><tr><td>FName</td><td>Employee</td><td>string</td></tr><tr><td>DName</td><td>Department</td><td>string</td></tr></table>	Arrow	source	target	Mngr	Employee	Employee	WorksIn	Employee	Department	Secr	Department	Employee	FName	Employee	string	DName	Department	string												
Arrow	source	target																																																			
Mngr	Employee	Employee																																																			
WorksIn	Employee	Department																																																			
Secr	Department	Employee																																																			
FName	Employee	string																																																			
DName	Department	string																																																			
Arrow	source	target																																																			
Mngr	Employee	Employee																																																			
WorksIn	Employee	Department																																																			
Secr	Department	Employee																																																			
FName	Employee	string																																																			
DName	Department	string																																																			
fong-spivak-seven-sketches-ink-053	294	■1.000	$\begin{tabular}{t}{ l }\hline \text{Vertex} \\\hline \text{Department} \text{Employee} \text{string} \\\hline\end{tabular}$	<table><tr><th>Vertex</th></tr><tr><td>Department Employee string</td></tr></table>	Vertex	Department Employee string	<table><tr><th>Vertex</th></tr><tr><td>Department Employee string</td></tr></table>	Vertex	Department Employee string																																												
Vertex																																																					
Department Employee string																																																					
Vertex																																																					
Department Employee string																																																					
fong-spivak-seven-sketches-ink-054	295	■1.000	$\begin{tabular}{t}{ l l l }\hline \text{Arrow} \& \text{source} \& \text{target} \\\hline 1 \& 4 \& 1 \\\hline 2 \& 4 \& 2 \\\hline 3 \& 5 \& 3 \\\hline 4 \& 5 \& 4 \\\hline 5 \& 5 \\\hline 6 \& 7 \& 6 \\\hline 7 \& 6 \& 7 \\\hline\end{tabular}$	<table><tr><th>Arrow</th><th>source</th><th>target</th></tr><tr><td>1</td><td>4</td><td>1</td></tr><tr><td>2</td><td>4</td><td>2</td></tr><tr><td>3</td><td>5</td><td>3</td></tr><tr><td>4</td><td>5</td><td>4</td></tr><tr><td>5</td><td>5</td><td>5</td></tr><tr><td>6</td><td>7</td><td>6</td></tr><tr><td>7</td><td>6</td><td>7</td></tr></table>	Arrow	source	target	1	4	1	2	4	2	3	5	3	4	5	4	5	5	5	6	7	6	7	6	7	<table><tr><th>Arrow</th><th>source</th><th>target</th></tr><tr><td>1</td><td>4</td><td>1</td></tr><tr><td>2</td><td>4</td><td>2</td></tr><tr><td>3</td><td>5</td><td>3</td></tr><tr><td>4</td><td>5</td><td>4</td></tr><tr><td>5</td><td>5</td><td>5</td></tr><tr><td>6</td><td>7</td><td>6</td></tr><tr><td>7</td><td>6</td><td>7</td></tr></table>	Arrow	source	target	1	4	1	2	4	2	3	5	3	4	5	4	5	5	5	6	7	6	7	6	7
Arrow	source	target																																																			
1	4	1																																																			
2	4	2																																																			
3	5	3																																																			
4	5	4																																																			
5	5	5																																																			
6	7	6																																																			
7	6	7																																																			
Arrow	source	target																																																			
1	4	1																																																			
2	4	2																																																			
3	5	3																																																			
4	5	4																																																			
5	5	5																																																			
6	7	6																																																			
7	6	7																																																			
fong-spivak-seven-sketches-ink-055	295	■1.000	$\begin{tabular}{t}{ l }\hline \text{Vertex} \\\hline 1 \\\hline 2 \\\hline 3 \\\hline 4 \\\hline 5 \\\hline 6 \\\hline 7 \\\hline\end{tabular}$	<table><tr><th>Vertex</th></tr><tr><td>1</td></tr><tr><td>2</td></tr><tr><td>3</td></tr><tr><td>4</td></tr><tr><td>5</td></tr><tr><td>6</td></tr><tr><td>7</td></tr></table>	Vertex	1	2	3	4	5	6	7	<table><tr><th>Vertex</th></tr><tr><td>1</td></tr><tr><td>2</td></tr><tr><td>3</td></tr><tr><td>4</td></tr><tr><td>5</td></tr><tr><td>6</td></tr><tr><td>7</td></tr></table>	Vertex	1	2	3	4	5	6	7																																
Vertex																																																					
1																																																					
2																																																					
3																																																					
4																																																					
5																																																					
6																																																					
7																																																					
Vertex																																																					
1																																																					
2																																																					
3																																																					
4																																																					
5																																																					
6																																																					
7																																																					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.																																																					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																												
fong-spivak-seven-sketches-ink-056	295	■1.000	$\begin{tabular}{t}{ l l l l l }\hline Email & sent_by & received_by & Address \\ \hline Em_1 & Bob & Grace & Bob \\ \hline Em_2 & Grace & Pat & Doug \\ \hline Em_3 & Bob & Emory & Emory \\ \hline Em_4 & Sue & Doug & Grace \\ \hline Em_5 & Doug & Sue & Pat \\ \hline Em_6 & Bob & Bob & Sue \\ \hline \end{tabular}$	(not rendered)	<table><tr><th>Email</th><th>sent_by</th><th>received_by</th><th>Address</th></tr><tr><td>Em_1</td><td>Bob</td><td>Grace</td><td>Bob</td></tr><tr><td>Em_2</td><td>Grace</td><td>Pat</td><td>Doug</td></tr><tr><td>Em_3</td><td>Bob</td><td>Emory</td><td>Emory</td></tr><tr><td>Em_4</td><td>Sue</td><td>Doug</td><td>Grace</td></tr><tr><td>Em_5</td><td>Doug</td><td>Sue</td><td>Pat</td></tr><tr><td>Em_6</td><td>Bob</td><td>Bob</td><td>Sue</td></tr></table>	Email	sent_by	received_by	Address	Em_1	Bob	Grace	Bob	Em_2	Grace	Pat	Doug	Em_3	Bob	Emory	Emory	Em_4	Sue	Doug	Grace	Em_5	Doug	Sue	Pat	Em_6	Bob	Bob	Sue																																
Email	sent_by	received_by	Address																																																														
Em_1	Bob	Grace	Bob																																																														
Em_2	Grace	Pat	Doug																																																														
Em_3	Bob	Emory	Emory																																																														
Em_4	Sue	Doug	Grace																																																														
Em_5	Doug	Sue	Pat																																																														
Em_6	Bob	Bob	Sue																																																														
fong-spivak-seven-sketches-ink-057	298	■0.867	$\begin{tabular}{t}{ l l l l l l }\hline \backslash(\Phi\backslash) & \backslash(a\backslash) & \backslash(b\backslash) & \backslash(c\backslash) & \backslash(d\backslash) & \backslash(e\backslash) \\ \hline \backslash(N\backslash) & true & false & true & false & true \\ \hline \backslash(E\backslash) & true & true & true & true & true \\ \hline W & true & false & true & false & true \\ \hline \backslash(S\backslash) & true & true & true & true & true \\ \hline \end{tabular}$	<table><tr><th>Φ</th><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>N</td><td>true</td><td>false</td><td>true</td><td>false</td><td>true</td></tr><tr><td>E</td><td>true</td><td>true</td><td>true</td><td>true</td><td>true</td></tr><tr><td>W</td><td>true</td><td>false</td><td>true</td><td>false</td><td>true</td></tr><tr><td>S</td><td>true</td><td>true</td><td>true</td><td>true</td><td>true</td></tr></table>	Φ	a	b	c	d	e	N	true	false	true	false	true	E	true	true	true	true	true	W	true	false	true	false	true	S	true	true	true	true	true	<table><tr><th>Φ</th><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>N</td><td>true</td><td>false</td><td>true</td><td>false</td><td>true</td></tr><tr><td>E</td><td>true</td><td>true</td><td>true</td><td>true</td><td>true</td></tr><tr><td>W</td><td>true</td><td>false</td><td>true</td><td>false</td><td>true</td></tr><tr><td>S</td><td>true</td><td>true</td><td>true</td><td>true</td><td>true</td></tr></table>	Φ	a	b	c	d	e	N	true	false	true	false	true	E	true	true	true	true	true	W	true	false	true	false	true	S	true	true	true	true	true
Φ	a	b	c	d	e																																																												
N	true	false	true	false	true																																																												
E	true	true	true	true	true																																																												
W	true	false	true	false	true																																																												
S	true	true	true	true	true																																																												
Φ	a	b	c	d	e																																																												
N	true	false	true	false	true																																																												
E	true	true	true	true	true																																																												
W	true	false	true	false	true																																																												
S	true	true	true	true	true																																																												
fong-spivak-seven-sketches-ink-058	299	■0.546	$\begin{tabular}{t}{ l l l l l }\hline \backslash(\Phi\backslash) & \backslash(x\backslash) & \backslash(y\backslash) & \backslash(z\backslash) \\ \hline \backslash(A\backslash) & 17 & 20 & 20 \\ \hline \backslash(B\backslash) & 11 & 14 & 14 \\ \hline \backslash(C\backslash) & 14 & 17 & 17 \\ \hline D & 12 & 9 & 15 \\ \hline \end{tabular}$	<table><tr><th>Φ</th><th>x</th><th>y</th><th>z</th></tr><tr><td>A</td><td>17</td><td>20</td><td>20</td></tr><tr><td>B</td><td>11</td><td>14</td><td>14</td></tr><tr><td>C</td><td>14</td><td>17</td><td>17</td></tr><tr><td>D</td><td>12</td><td>9</td><td>15</td></tr></table>	Φ	x	y	z	A	17	20	20	B	11	14	14	C	14	17	17	D	12	9	15	<table><tr><th>Φ</th><th>x</th><th>y</th><th>z</th></tr><tr><td>A</td><td>17</td><td>20</td><td>20</td></tr><tr><td>B</td><td>11</td><td>14</td><td>14</td></tr><tr><td>C</td><td>14</td><td>17</td><td>17</td></tr><tr><td>D</td><td>12</td><td>9</td><td>15</td></tr></table>	Φ	x	y	z	A	17	20	20	B	11	14	14	C	14	17	17	D	12	9	15																				
Φ	x	y	z																																																														
A	17	20	20																																																														
B	11	14	14																																																														
C	14	17	17																																																														
D	12	9	15																																																														
Φ	x	y	z																																																														
A	17	20	20																																																														
B	11	14	14																																																														
C	14	17	17																																																														
D	12	9	15																																																														
fong-spivak-seven-sketches-ink-059	299	■0.777	$\begin{tabular}{t}{ l l l l l l }\hline \backslash(\Phi\circ\Psi\backslash) & \backslash(p\backslash) & 9 & \backslash(r\backslash) & \backslash(s\backslash) \\ \hline \backslash(A\backslash) & 22 & 24 & 20 & 21 \\ \hline B & 16 & 18 & 14 & 15 \\ \hline \backslash(C\backslash) & 19 & 21 & 17 & 18 \\ \hline D & 11 & 13 & 9 & 10 \\ \hline \end{tabular}$	<table><tr><th>$\Phi\circ\Psi$</th><th>p</th><th>9</th><th>r</th><th>s</th></tr><tr><td>A</td><td>22</td><td>24</td><td>20</td><td>21</td></tr><tr><td>B</td><td>16</td><td>18</td><td>14</td><td>15</td></tr><tr><td>C</td><td>19</td><td>21</td><td>17</td><td>18</td></tr><tr><td>D</td><td>11</td><td>13</td><td>9</td><td>10</td></tr></table>	$\Phi\circ\Psi$	p	9	r	s	A	22	24	20	21	B	16	18	14	15	C	19	21	17	18	D	11	13	9	10	<table><tr><th>$\Phi\circ\Psi$</th><th>p</th><th>q</th><th>r</th><th>s</th></tr><tr><td>A</td><td>22</td><td>24</td><td>20</td><td>21</td></tr><tr><td>B</td><td>16</td><td>18</td><td>14</td><td>15</td></tr><tr><td>C</td><td>19</td><td>21</td><td>17</td><td>18</td></tr><tr><td>D</td><td>11</td><td>13</td><td>9</td><td>10</td></tr></table>	$\Phi\circ\Psi$	p	q	r	s	A	22	24	20	21	B	16	18	14	15	C	19	21	17	18	D	11	13	9	10										
$\Phi\circ\Psi$	p	9	r	s																																																													
A	22	24	20	21																																																													
B	16	18	14	15																																																													
C	19	21	17	18																																																													
D	11	13	9	10																																																													
$\Phi\circ\Psi$	p	q	r	s																																																													
A	22	24	20	21																																																													
B	16	18	14	15																																																													
C	19	21	17	18																																																													
D	11	13	9	10																																																													
fong-spivak-seven-sketches-ink-060	321	■0.936	$\begin{tabular}{t}{ l l l l l l }\hline & r1 & r2 & r3 & c1 & i1 \\ \hline \backslash(s(-)\backslash) & dl & ul & ur & ul & dl \\ \hline \backslash(t(-)\backslash) & ul & ur & dr & ur & dr \\ \hline \backslash(\ell(-)\backslash) & 1\Omega & 2\Omega & 1\Omega & 3F & 1H \\ \hline \end{tabular}$	<table><tr><th></th><th>r1</th><th>r2</th><th>r3</th><th>c1</th><th>i1</th></tr><tr><td>$s(-)$</td><td>dl</td><td>ul</td><td>ur</td><td>ul</td><td>dl</td></tr><tr><td>$t(-)$</td><td>ul</td><td>ur</td><td>dr</td><td>ur</td><td>dr</td></tr><tr><td>$\ell(-)$</td><td>1Ω</td><td>2Ω</td><td>1Ω</td><td>3F</td><td>1H</td></tr></table>		r1	r2	r3	c1	i1	$s(-)$	dl	ul	ur	ul	dl	$t(-)$	ul	ur	dr	ur	dr	$\ell(-)$	1Ω	2Ω	1Ω	3F	1H	<table><tr><th></th><th>r1</th><th>r2</th><th>r3</th><th>c1</th><th>i1</th></tr><tr><td>$s(-)$</td><td>dl</td><td>ul</td><td>ur</td><td>ul</td><td>dl</td></tr><tr><td>$t(-)$</td><td>ul</td><td>ur</td><td>dr</td><td>ur</td><td>dr</td></tr><tr><td>$\ell(-)$</td><td>1Ω</td><td>2Ω</td><td>1Ω</td><td>3F</td><td>1H</td></tr></table>		r1	r2	r3	c1	i1	$s(-)$	dl	ul	ur	ul	dl	$t(-)$	ul	ur	dr	ur	dr	$\ell(-)$	1Ω	2Ω	1Ω	3F	1H												
	r1	r2	r3	c1	i1																																																												
$s(-)$	dl	ul	ur	ul	dl																																																												
$t(-)$	ul	ur	dr	ur	dr																																																												
$\ell(-)$	1Ω	2Ω	1Ω	3F	1H																																																												
	r1	r2	r3	c1	i1																																																												
$s(-)$	dl	ul	ur	ul	dl																																																												
$t(-)$	ul	ur	dr	ur	dr																																																												
$\ell(-)$	1Ω	2Ω	1Ω	3F	1H																																																												

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

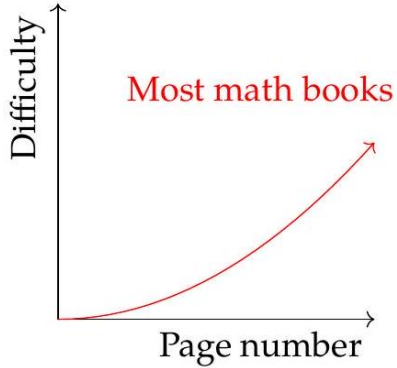
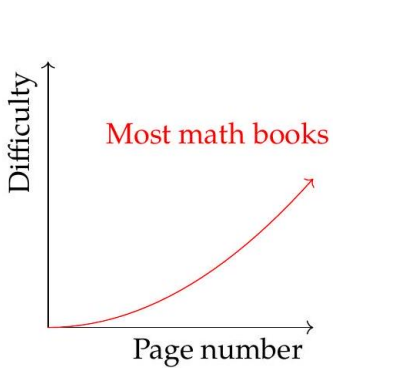
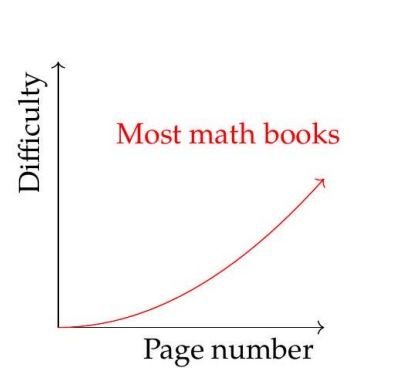
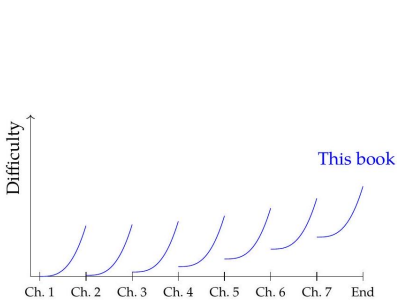
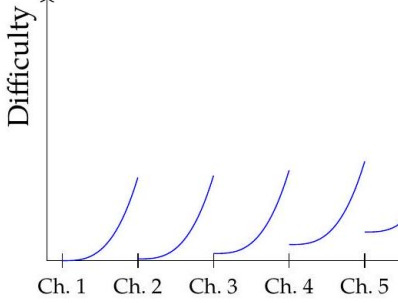
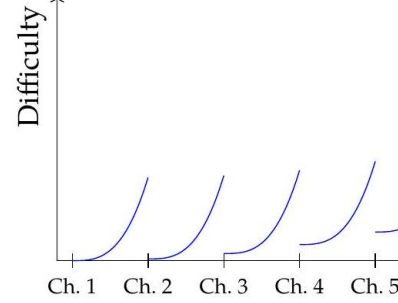
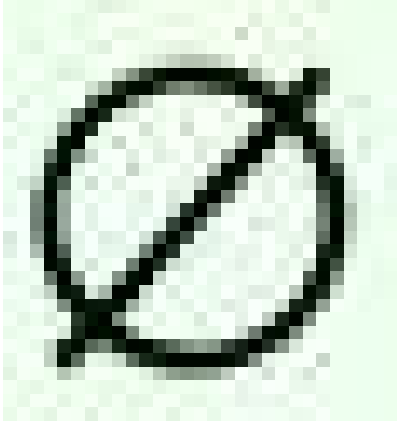
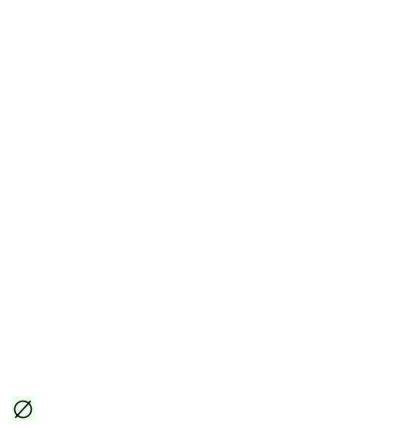
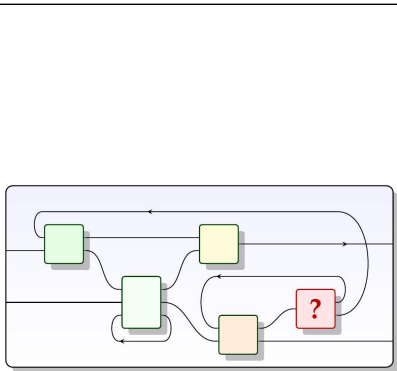
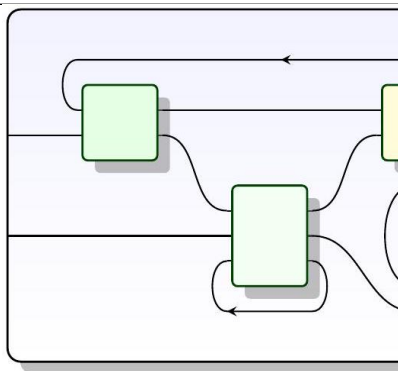
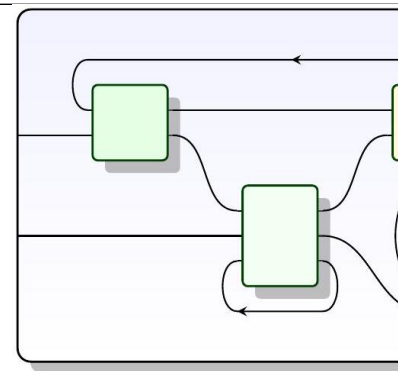
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

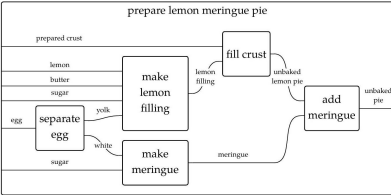
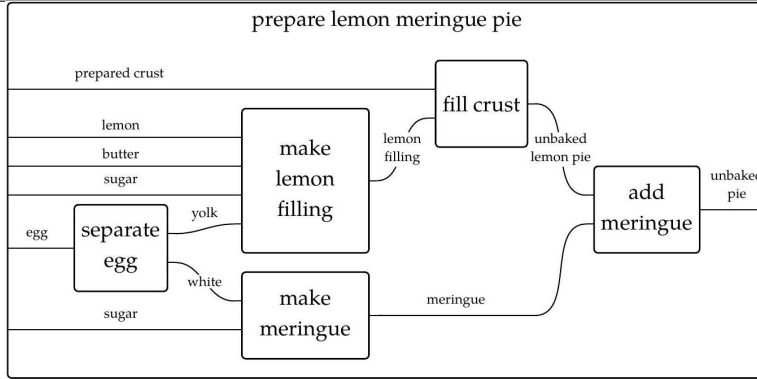
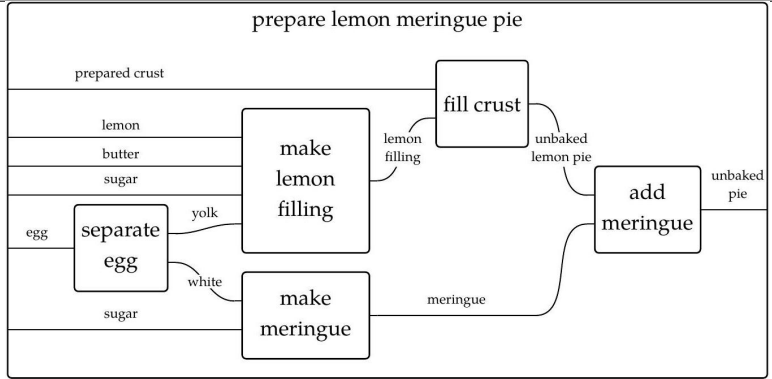
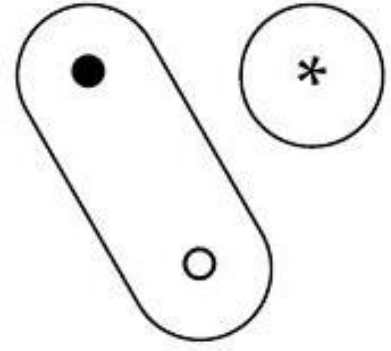
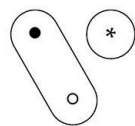
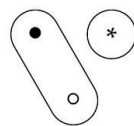
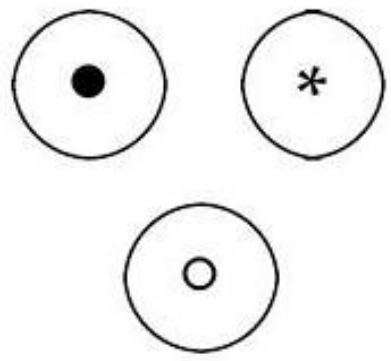
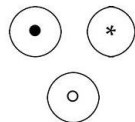
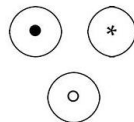
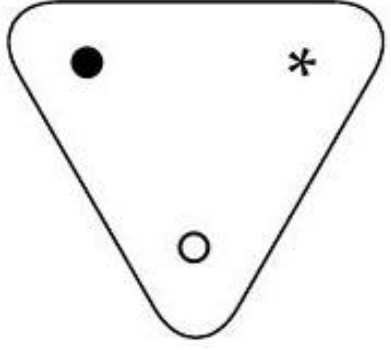
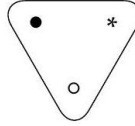
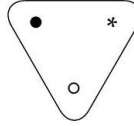
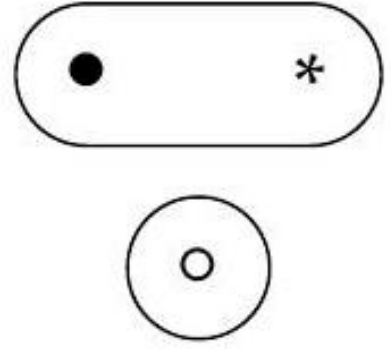
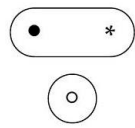
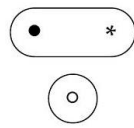
[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																																
fong-spivak-seven-sketches-2022-061	322	■0.975	$\begin{tabular}{t}{ l l l }\hline & 1 & 2 \\ \hline (f^{\prime}(-)) & l & d \\ \hline (g^{\prime}(-)) & r & r \\ \hline \end{tabular}$	<table><tr><td></td><td>1</td><td>2</td></tr><tr><td>$f'(-)$</td><td>l</td><td>d</td></tr><tr><td>$g'(-)$</td><td>r</td><td>r</td></tr></table>		1	2	$f'(-)$	l	d	$g'(-)$	r	r	<table><tr><td></td><td>1</td><td>2</td></tr><tr><td>$f'(-)$</td><td>l</td><td>d</td></tr><tr><td>$g'(-)$</td><td>r</td><td>r</td></tr></table>		1	2	$f'(-)$	l	d	$g'(-)$	r	r																																														
	1	2																																																																			
$f'(-)$	l	d																																																																			
$g'(-)$	r	r																																																																			
	1	2																																																																			
$f'(-)$	l	d																																																																			
$g'(-)$	r	r																																																																			
fong-spivak-seven-sketches-2022-062	322	■0.786	$\begin{tabular}{t}{ l l l }\hline & r1' & r2' \\ \hline (s(-)) & l & r \\ \hline (t(-)) & r & d \\ \hline (\ell(-)) & 5\Omega & 8\Omega \\ \hline \end{tabular}$	<table><tr><td></td><td>r1'</td><td>r2'</td></tr><tr><td>$s(-)$</td><td>l</td><td>r</td></tr><tr><td>$t(-)$</td><td>r</td><td>d</td></tr><tr><td>$\ell(-)$</td><td>5Ω</td><td>8Ω</td></tr></table>		r1'	r2'	$s(-)$	l	r	$t(-)$	r	d	$\ell(-)$	5Ω	8Ω	<table><tr><td></td><td>r1'</td><td>r2'</td></tr><tr><td>$s(-)$</td><td>l</td><td>r</td></tr><tr><td>$t(-)$</td><td>r</td><td>d</td></tr><tr><td>$\ell(-)$</td><td>5Ω</td><td>8Ω</td></tr></table>		r1'	r2'	$s(-)$	l	r	$t(-)$	r	d	$\ell(-)$	5Ω	8Ω																																								
	r1'	r2'																																																																			
$s(-)$	l	r																																																																			
$t(-)$	r	d																																																																			
$\ell(-)$	5Ω	8Ω																																																																			
	r1'	r2'																																																																			
$s(-)$	l	r																																																																			
$t(-)$	r	d																																																																			
$\ell(-)$	5Ω	8Ω																																																																			
fong-spivak-seven-sketches-2022-063	322	■1.000	$\begin{tabular}{t}{ l l l l l l l }\hline & r1 & r2 & r3 & c1 & i1 & r1' & r2' \\ \hline (s^{\prime} \prime (-)) & dl & ul & m & ul & dl & m & r \\ \hline (t^{\prime} \prime (-)) & ul & m & dr & m & dr & r & m \\ \hline (\ell^{\prime} \prime (-)) & 1\Omega & 2\Omega & 1\Omega & 3F & 1H & 5\Omega & 8\Omega \\ \hline \end{tabular}$	<table><tr><td></td><td>r1</td><td>r2</td><td>r3</td><td>c1</td><td>i1</td><td>r1'</td><td>r2'</td></tr><tr><td>$s''(-)$</td><td>dl</td><td>ul</td><td>m</td><td>ul</td><td>dl</td><td>m</td><td>r</td></tr><tr><td>$t''(-)$</td><td>ul</td><td>m</td><td>dr</td><td>m</td><td>dr</td><td>r</td><td>m</td></tr><tr><td>$\ell''(-)$</td><td>1Ω</td><td>2Ω</td><td>1Ω</td><td>3F</td><td>1H</td><td>5Ω</td><td>8Ω</td></tr></table>		r1	r2	r3	c1	i1	r1'	r2'	$s''(-)$	dl	ul	m	ul	dl	m	r	$t''(-)$	ul	m	dr	m	dr	r	m	$\ell''(-)$	1Ω	2Ω	1Ω	3F	1H	5Ω	8Ω	<table><tr><td></td><td>r1</td><td>r2</td><td>r3</td><td>c1</td><td>i1</td><td>r1'</td><td>r2'</td></tr><tr><td>$s''(-)$</td><td>dl</td><td>ul</td><td>m</td><td>ul</td><td>dl</td><td>m</td><td>r</td></tr><tr><td>$t''(-)$</td><td>ul</td><td>m</td><td>dr</td><td>m</td><td>dr</td><td>r</td><td>m</td></tr><tr><td>$\ell''(-)$</td><td>1Ω</td><td>2Ω</td><td>1Ω</td><td>3F</td><td>1H</td><td>5Ω</td><td>8Ω</td></tr></table>		r1	r2	r3	c1	i1	r1'	r2'	$s''(-)$	dl	ul	m	ul	dl	m	r	$t''(-)$	ul	m	dr	m	dr	r	m	$\ell''(-)$	1Ω	2Ω	1Ω	3F	1H	5Ω	8Ω
	r1	r2	r3	c1	i1	r1'	r2'																																																														
$s''(-)$	dl	ul	m	ul	dl	m	r																																																														
$t''(-)$	ul	m	dr	m	dr	r	m																																																														
$\ell''(-)$	1Ω	2Ω	1Ω	3F	1H	5Ω	8Ω																																																														
	r1	r2	r3	c1	i1	r1'	r2'																																																														
$s''(-)$	dl	ul	m	ul	dl	m	r																																																														
$t''(-)$	ul	m	dr	m	dr	r	m																																																														
$\ell''(-)$	1Ω	2Ω	1Ω	3F	1H	5Ω	8Ω																																																														
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value																																																																					
LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ																																																																					
Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured																																																																					
[refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.																																																																					

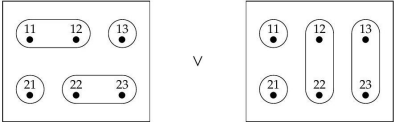
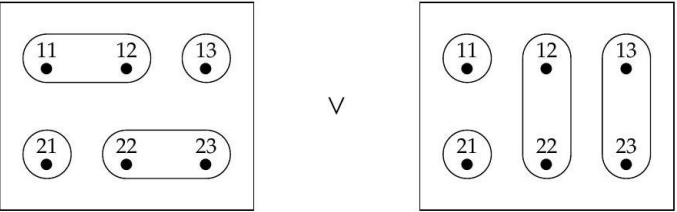
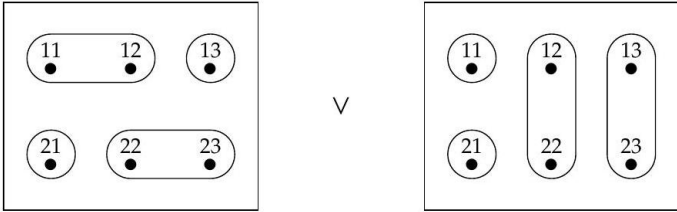
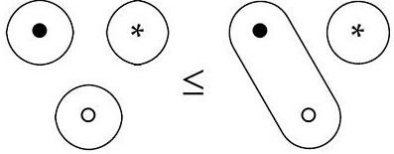
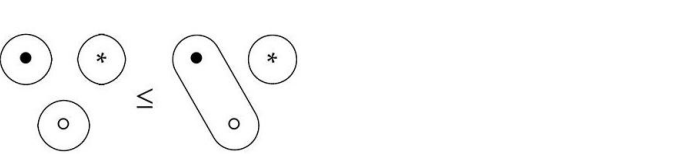
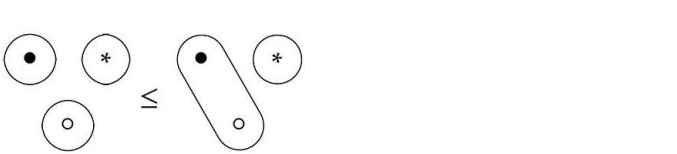
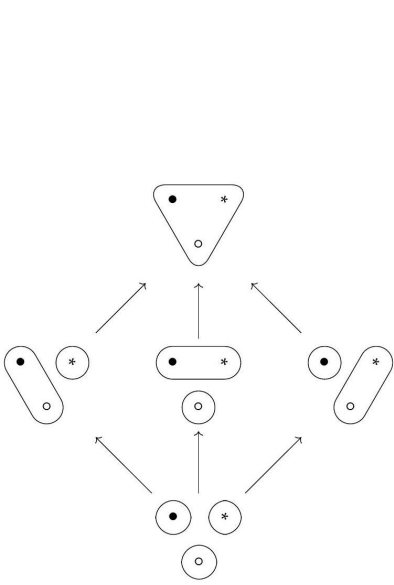
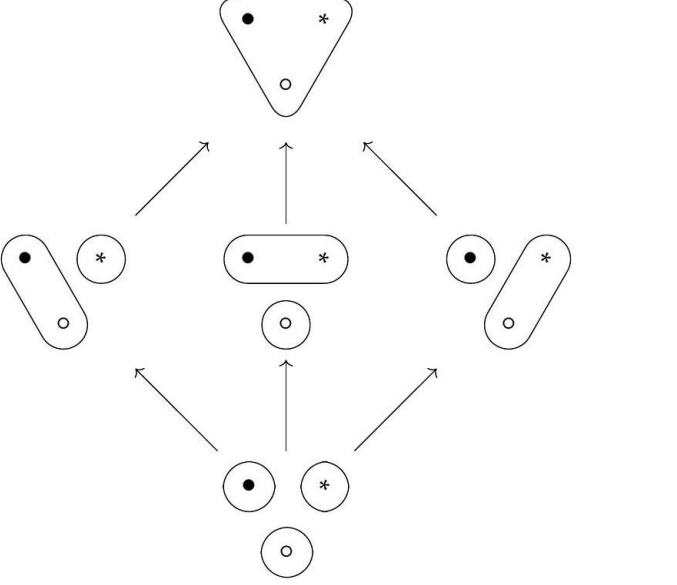
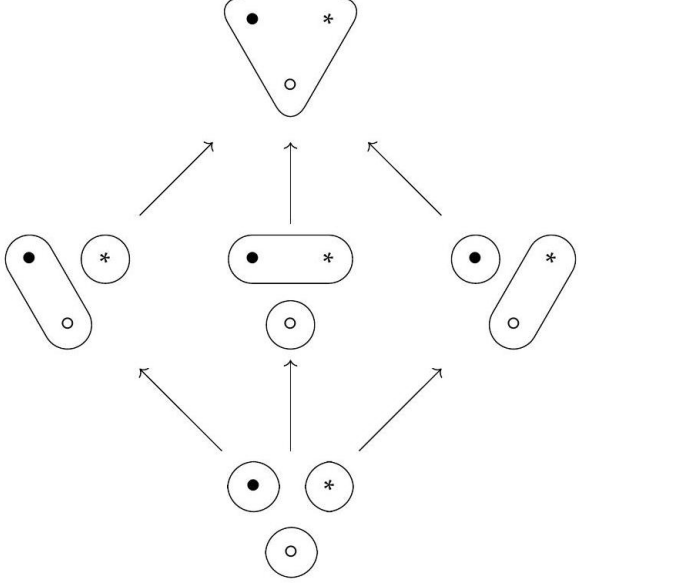
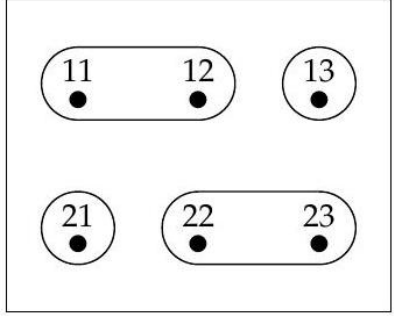
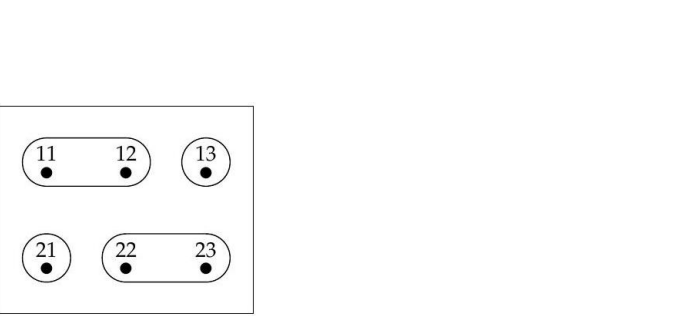
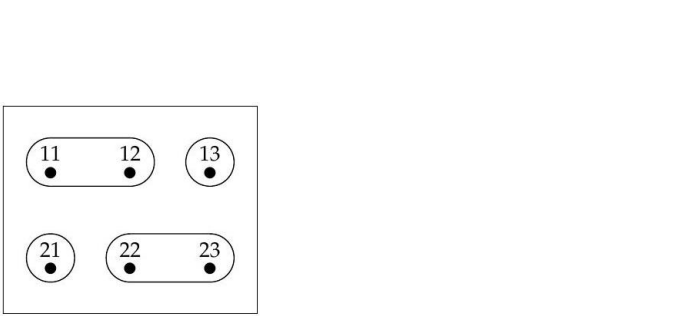
Image regions — rendered against scan

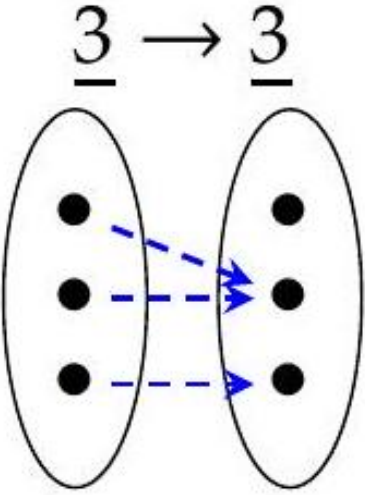
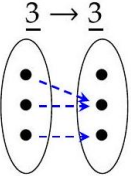
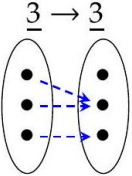
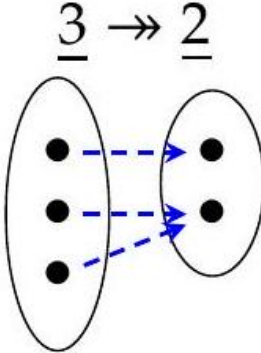
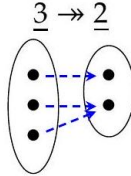
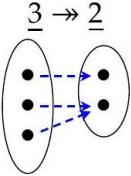
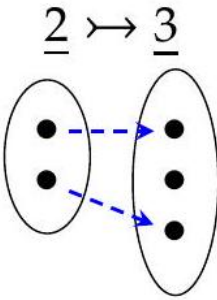
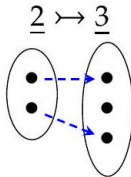
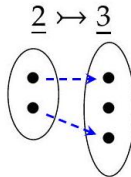
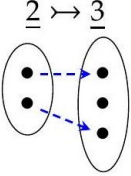
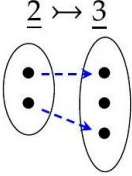
Two comparable cells per row. **Rendered**: the region’s own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan**: the tex.zip image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix’s confidence, as the equation rows do. Setting a region’s LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with pdfdrill vision; verify any LLM result against the real ink with inkdrill.

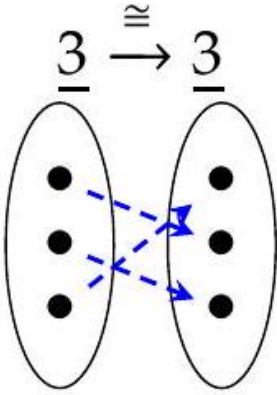
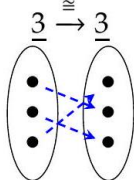
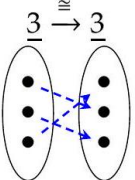
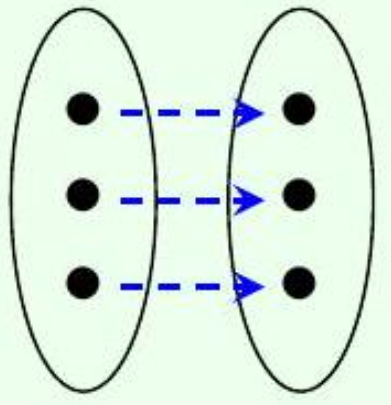
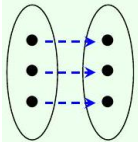
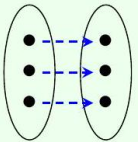
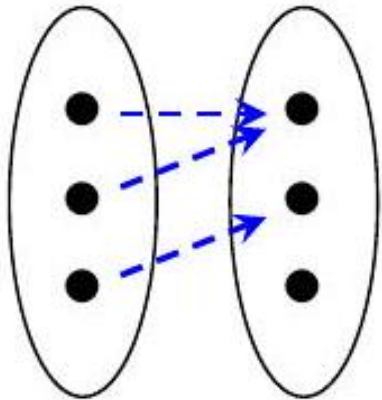
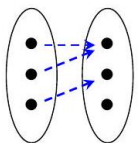
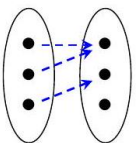
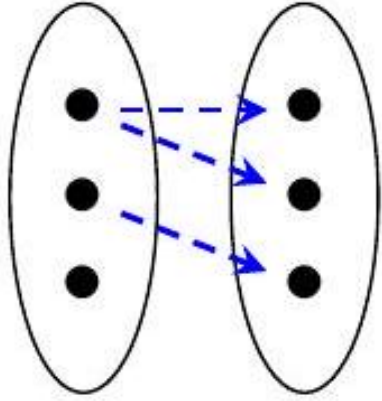
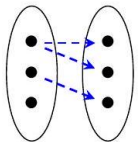
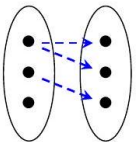
Identifier	Page	Class	Source	Author source	Rendered	Scan
fong-spivak-seven-sketches-005_0001	005	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-005_407_430_1561_351.jpg			
fong-spivak-seven-sketches-005_0002	005	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-005_403_842_1561_923.jpg			
fong-spivak-seven-sketches-029_0003	029	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-029_32_29_1683_1045.jpg			
fong-spivak-seven-sketches-317_0004	317	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-317_41_1294_1900_484.jpg	a) The map $[id_A + \iota_B, \iota_C] = [[\iota_A, \iota_B \circ \iota_{B+C}], \iota_C \circ \iota_{B+C}]: (A+B)+C \rightarrow A+(B+C)$ is an	a) The map $[id_A + \iota_B, \iota_C] = [[\iota_A, \iota_B \circ \iota_{B+C}], \iota_C \circ \iota_{B+C}]: (A+B)+C \rightarrow A+(B+C)$ is an	a) The map $[id_A + \iota_B, \iota_C] = [[\iota_A, \iota_B \circ \iota_{B+C}], \iota_C \circ \iota_{B+C}]: (A+B)+C \rightarrow A+(B+C)$ is an
fong-spivak-seven-sketches-001_0001	001	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-001_483_1004_899_560.jpg			

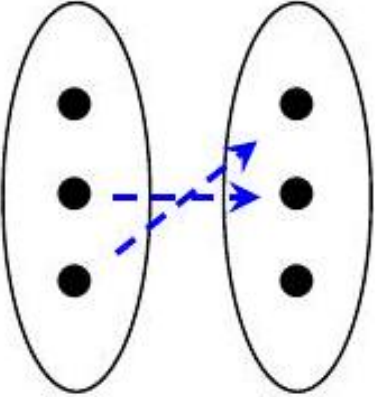
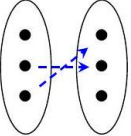
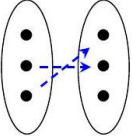
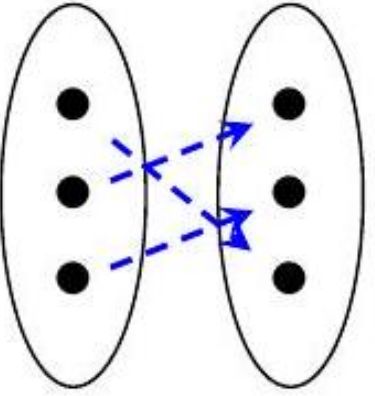
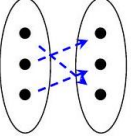
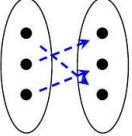
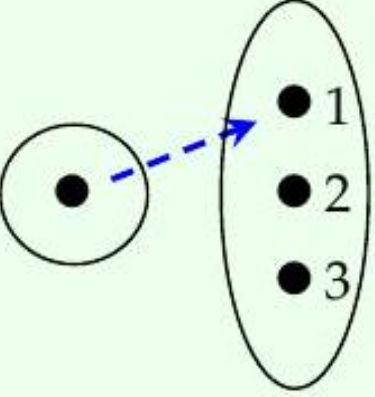
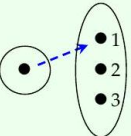
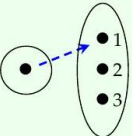
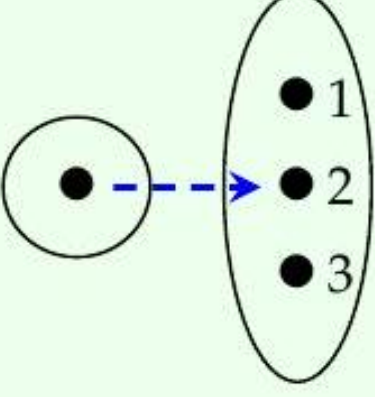
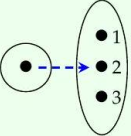
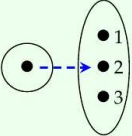
fong-spivak-seven-sketches_110_0002	004 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-004_670_1353_919_380.jpg			
fong-spivak-seven-sketches_110_0003	015 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-015_161_174_344_977.jpg			
fong-spivak-seven-sketches_110_0004	015 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-015_167_174_977_742.jpg			
fong-spivak-seven-sketches_110_0005	015 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-015_165_181_984_1181.jpg			
fong-spivak-seven-sketches_110_0006	015 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-015_161_174_1633_742.jpg			

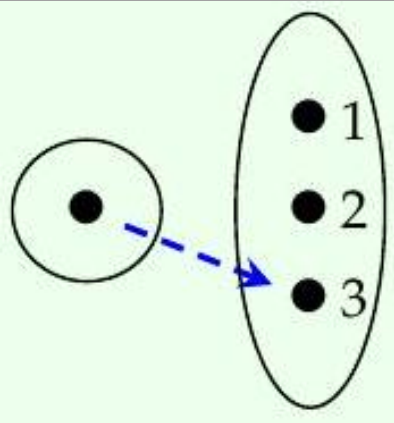
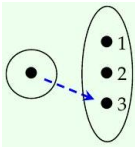
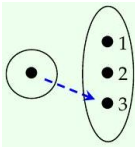
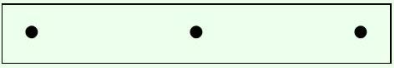
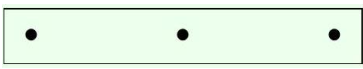
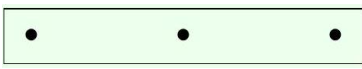
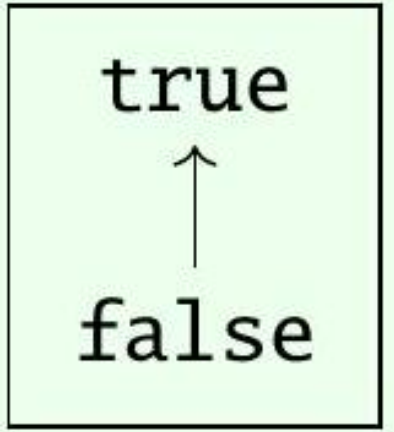
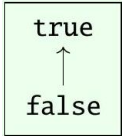
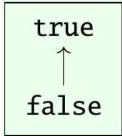
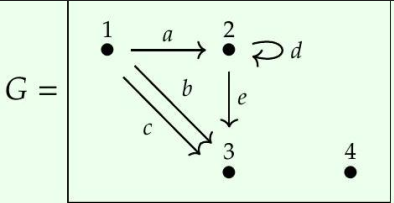
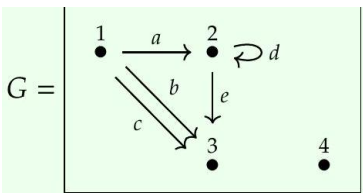
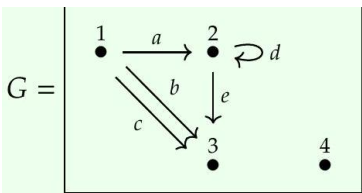
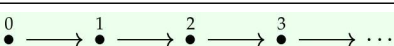
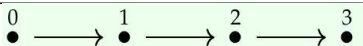
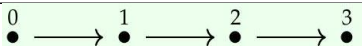
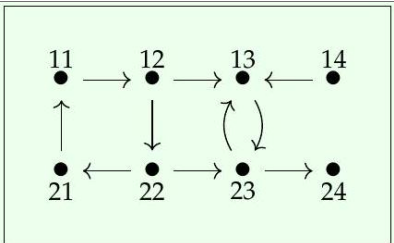
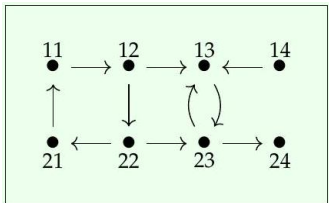
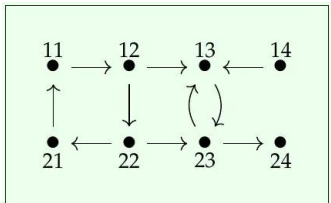
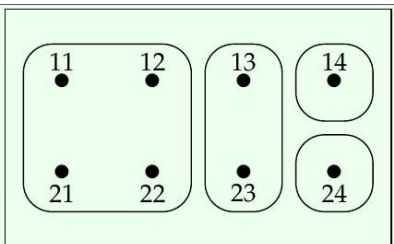
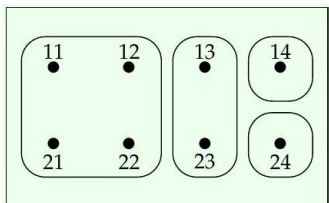
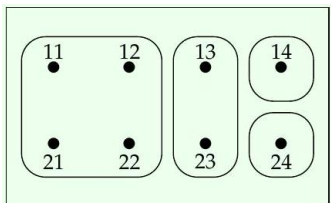
fong-spivak-seven-sketches_110_0007 — (dup)	015		7f1e85c6-4350-401f-a960-eb8a0176e786-015_165_183_1638_1179.jpg			
fong-spivak-seven-sketches_110_0008 — (dup)	015		7f1e85c6-4350-401f-a960-eb8a0176e786-015_165_174_1968_624.jpg			
fong-spivak-seven-sketches_110_0009 — (dup)	015		7f1e85c6-4350-401f-a960-eb8a0176e786-015_158_172_1973_968.jpg			
fong-spivak-seven-sketches_110_0010 — (dup)	015		7f1e85c6-4350-401f-a960-eb8a0176e786-015_158_174_1973_1308.jpg			
fong-spivak-seven-sketches_110_0011 — (dup)	016		7f1e85c6-4350-401f-a960-eb8a0176e786-016_167_742_738_688.jpg			
fong-spivak-seven-sketches_110_0012 — (dup)	016		7f1e85c6-4350-401f-a960-eb8a0176e786-016_170_751_971_683.jpg			

fong-spivak-seven-sketches_117_0013 (dup)	016	7f1e85c6-4350-401f-a960-eb8a0176e786-016_281_887_1231_615.jpg			
fong-spivak-seven-sketches_117_0014 (dup)	017	7f1e85c6-4350-401f-a960-eb8a0176e786-017_165_398_742_860.jpg			
fong-spivak-seven-sketches_117_0015 (dup)	017	7f1e85c6-4350-401f-a960-eb8a0176e786-017_767_753_1050_681.jpg			
fong-spivak-seven-sketches_117_0016 (dup)	018	no image for this region (498 in zip)	(no tex.zip crop for this row)	true ↑ false	true ↑ false
fong-spivak-seven-sketches_117_0017 (dup)	021	7f1e85c6-4350-401f-a960-eb8a0176e786-021_279_339_1306_889.jpg			

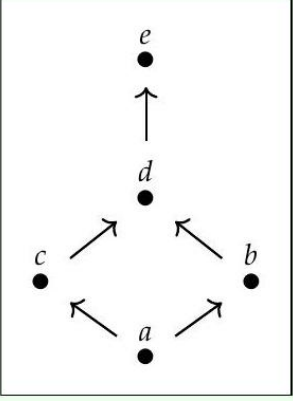
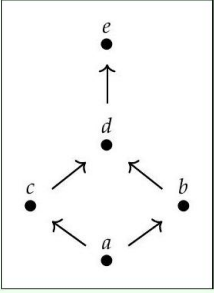
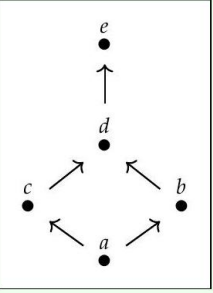
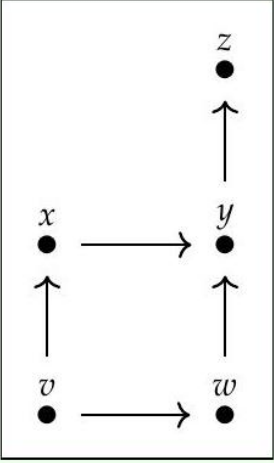
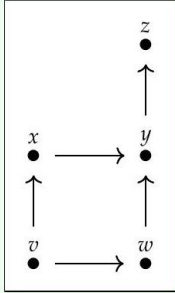
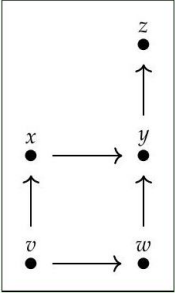
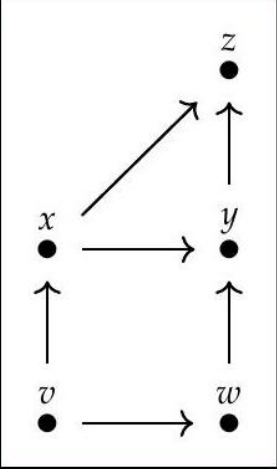
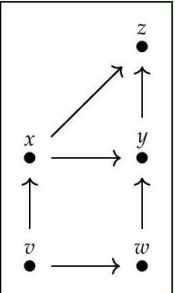
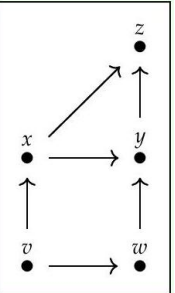
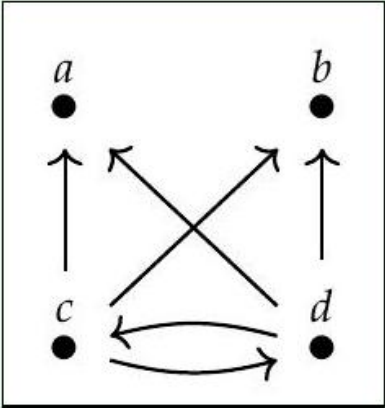
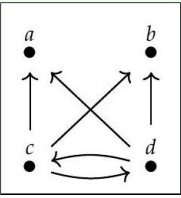
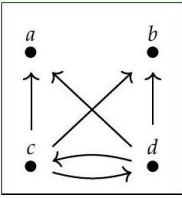
fong-spivak-seven-sketches_023_0018 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-023_238_181_493_437.jpg			
fong-spivak-seven-sketches_023_0019 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-023_294_251_437_733.jpg	surjective func 	surjective func 	surjective func 
fong-spivak-seven-sketches_023_0020 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-023_296_303_434_1109.jpg	injective function 	injective function 	injective function 
fong-spivak-seven-sketches_023_0021 — (dup)	no image for this region (498 in zip)	(no tex.zip crop for this row)		

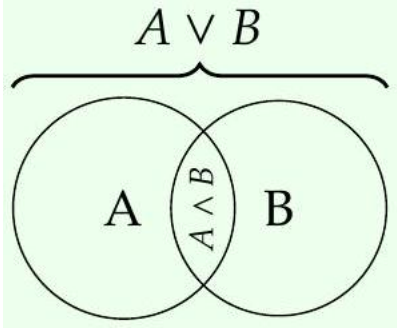
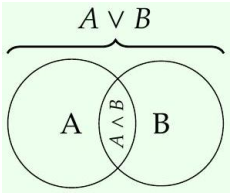
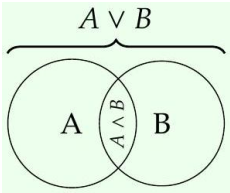
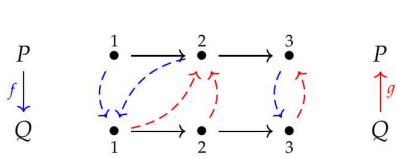
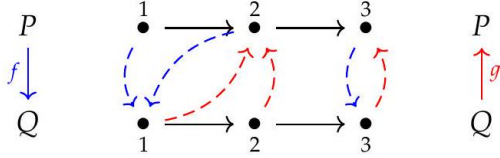
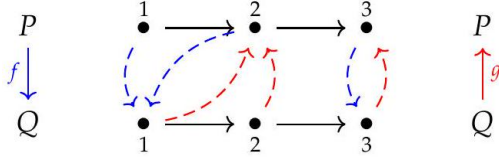
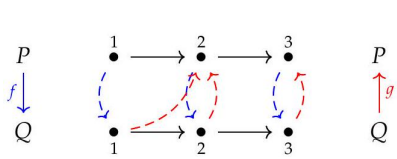
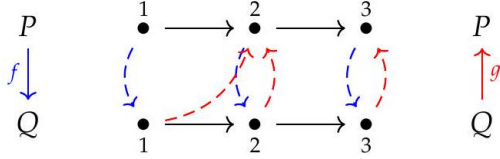
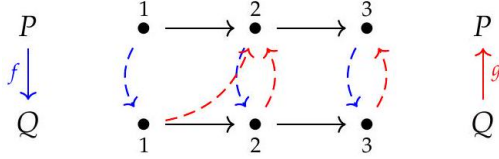
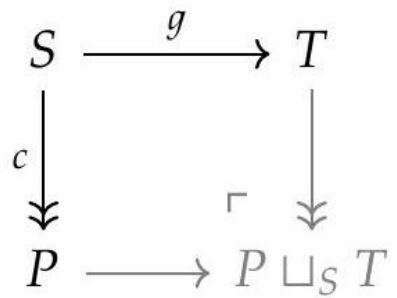
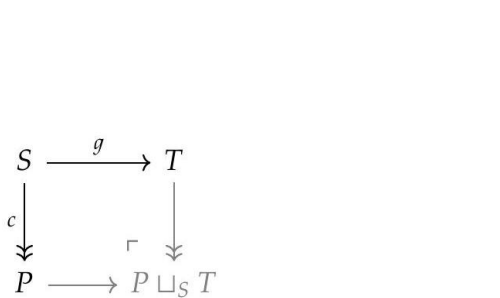
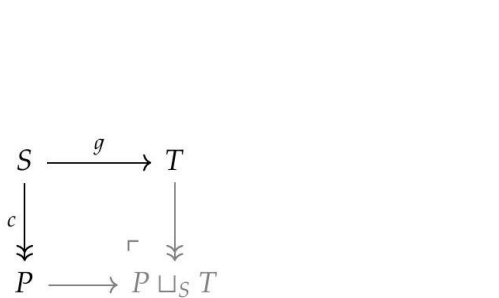
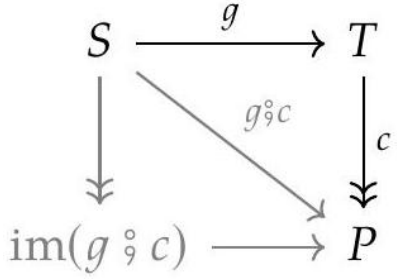
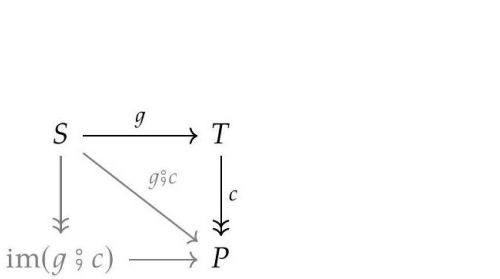
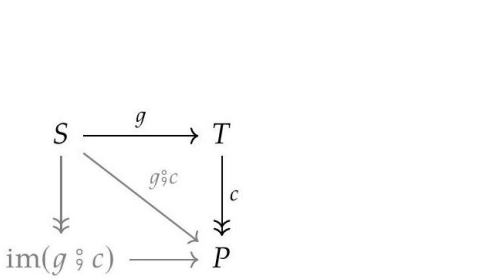
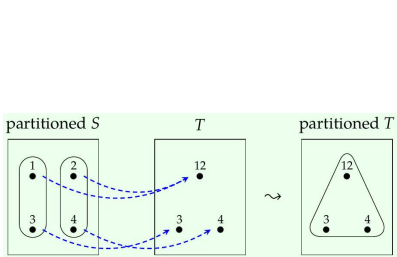
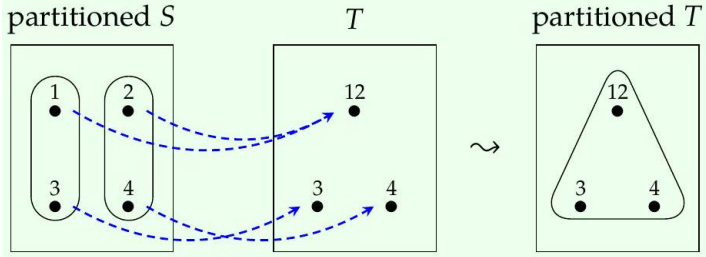
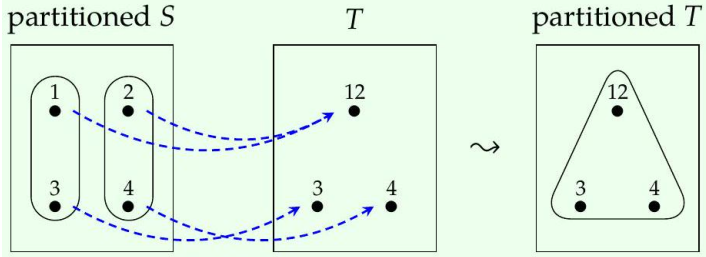
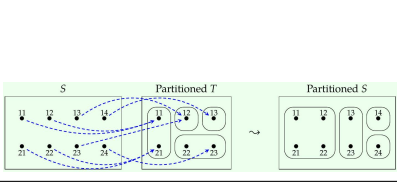
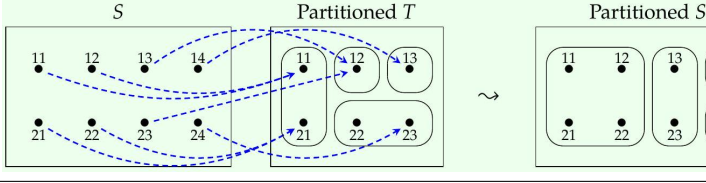
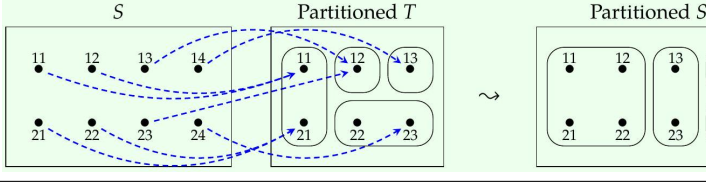
fong-spivak-seven-sketches_210_0022 — (dup)	023	7f1e85c6-4350-401f-a960-eb8a0176e786-023_299_240_432_1475.jpg	bijjective funct 	bijjective funct 	bijjective funct 
fong-spivak-seven-sketches_210_0023 — (dup)	023	7f1e85c6-4350-401f-a960-eb8a0176e786-023_188_179_950_971.jpg			
fong-spivak-seven-sketches_210_0024 — (dup)	023	7f1e85c6-4350-401f-a960-eb8a0176e786-023_190_176_1692_529.jpg			
fong-spivak-seven-sketches_210_0025 — (dup)	023	7f1e85c6-4350-401f-a960-eb8a0176e786-023_188_176_1695_824.jpg			

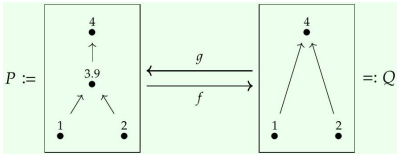
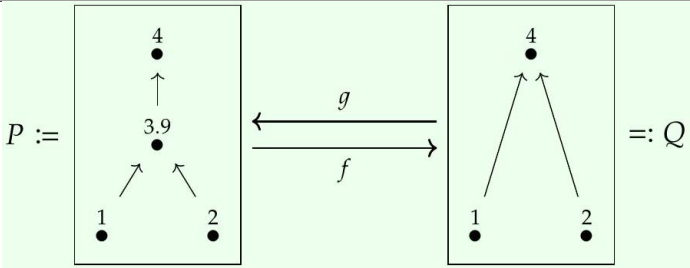
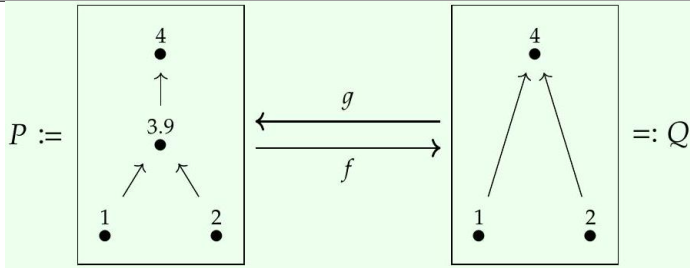
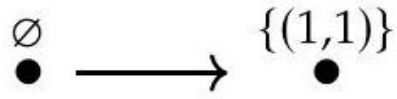
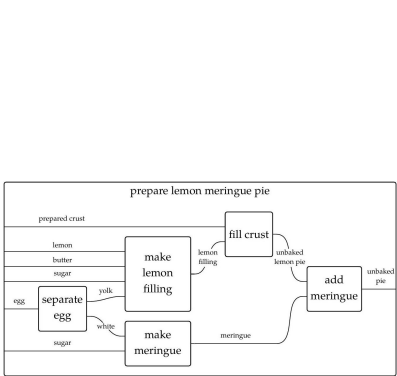
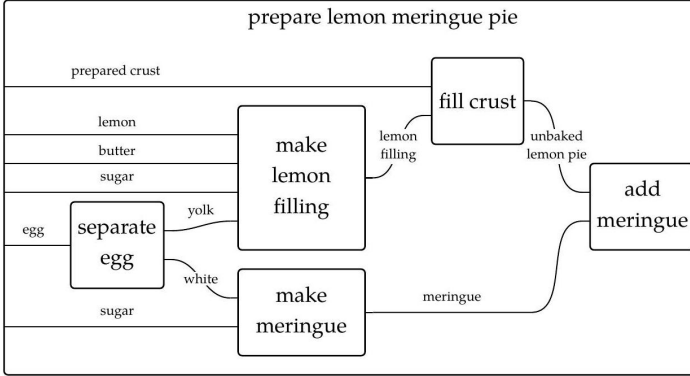
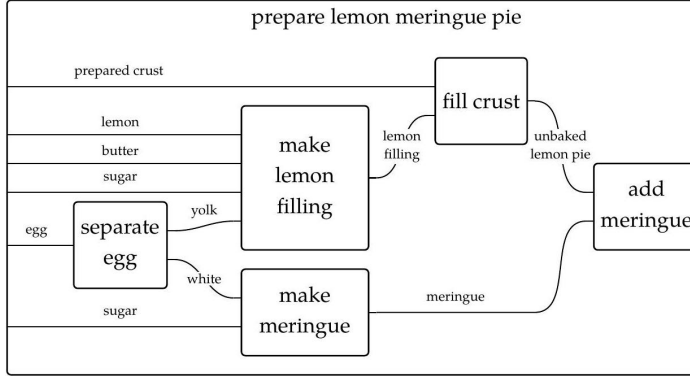
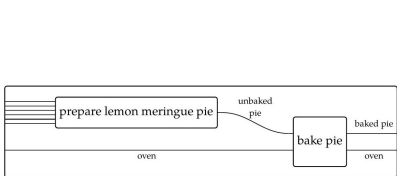
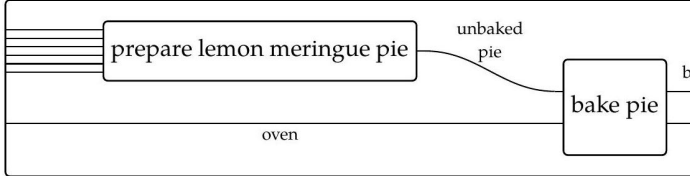
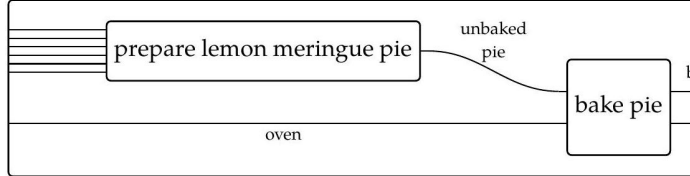
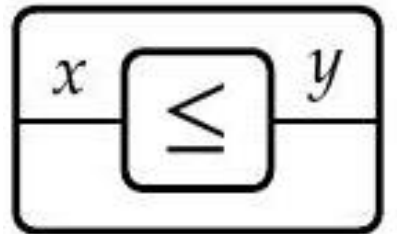
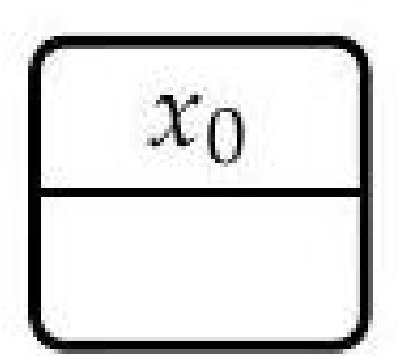
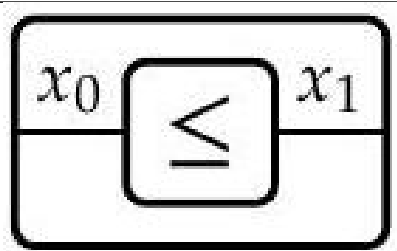
fong-spivak-seven-sketches-023	023 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-023_188_176_1695_1118.jpg			
fong-spivak-seven-sketches-023	023 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-023_188_179_1695_1414.jpg			
fong-spivak-seven-sketches-024	024 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-024_183_176_1561_674.jpg			
fong-spivak-seven-sketches-024	024 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-024_186_176_1561_968.jpg			

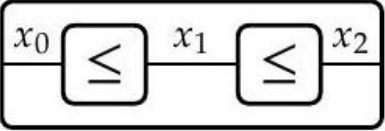
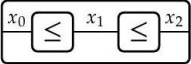
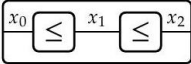
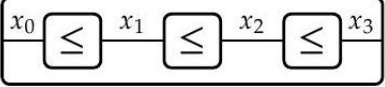
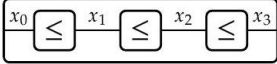
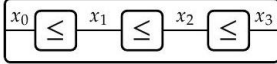
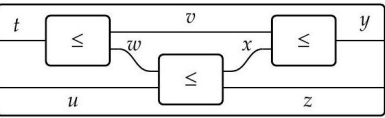
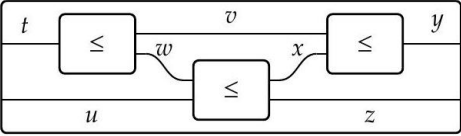
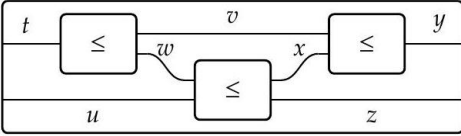
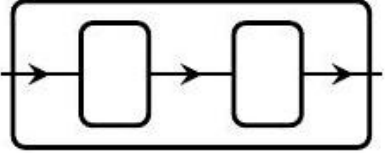
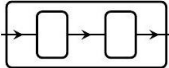
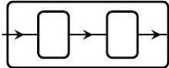
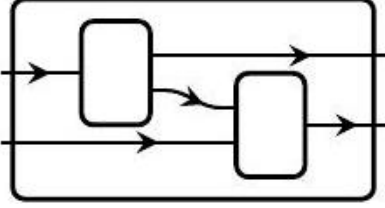
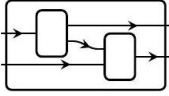
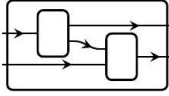
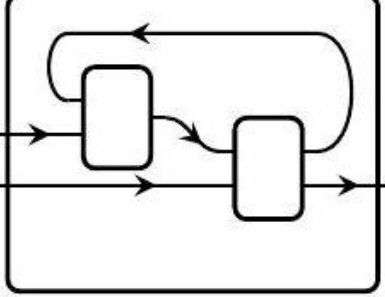
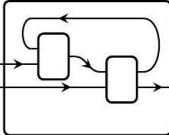
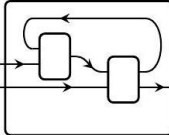
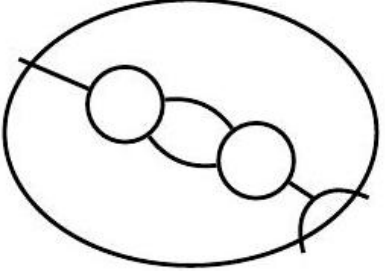
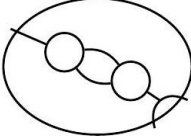
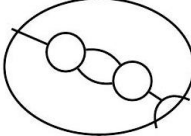
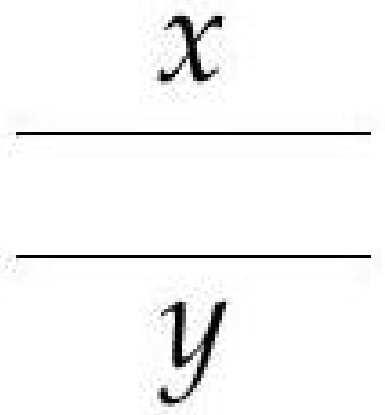
fong-spivak-seven-sketches_214_0030 — (dup)	024	7f1e85c6-4350-401f-a960-eb8a0176e786-024_186_174_1561_1265.jpg			
fong-spivak-seven-sketches_215_0031 — (dup)	025	7f1e85c6-4350-401f-a960-eb8a0176e786-025_81_473_826_824.jpg			
fong-spivak-seven-sketches_216_0032 — (dup)	025	7f1e85c6-4350-401f-a960-eb8a0176e786-025_181_161_1416_980.jpg			
fong-spivak-seven-sketches_217_0033 — (dup)	026	7f1e85c6-4350-401f-a960-eb8a0176e786-026_244_471_919_824.jpg			
fong-spivak-seven-sketches_217_0034 — (dup)	027	7f1e85c6-4350-401f-a960-eb8a0176e786-027_59_629_1717_744.jpg			
fong-spivak-seven-sketches_218_0035 — (dup)	028	7f1e85c6-4350-401f-a960-eb8a0176e786-028_267_428_1054_550.jpg			
fong-spivak-seven-sketches_218_0036 — (dup)	028	7f1e85c6-4350-401f-a960-eb8a0176e786-028_267_428_1054_1140.jpg			

fong-spivak-seven-sketches_00037 — (dup)	028	7f1e85c6-4350-401f-a960-eb8a0176e786-028_346_536_1744_792.jpg			
fong-spivak-seven-sketches_00038 — (dup)	029	no image for this region (498 in zip)	(no tex.zip crop for this row)		
fong-spivak-seven-sketches_00039 — (dup)	030	7f1e85c6-4350-401f-a960-eb8a0176e786-030_253_396_287_862.jpg			
fong-spivak-seven-sketches_00040 — (dup)	030	7f1e85c6-4350-401f-a960-eb8a0176e786-030_396_846_1989_636.jpg			
fong-spivak-seven-sketches_00041 — (dup)	031	7f1e85c6-4350-401f-a960-eb8a0176e786-031_167_1063_480_527.jpg			
fong-spivak-seven-sketches_00042 — (dup)	031	7f1e85c6-4350-401f-a960-eb8a0176e786-031_713_1326_1052_407.jpg			

fong-spivak-seven-sketches_014_0043	034 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-034_387_373_414_387.jpg	$P :=$ 	$P :=$ 	$P :=$ 
fong-spivak-seven-sketches_014_0044	034 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-034_389_324_414_923.jpg	$Q :=$ 	$Q :=$ 	$Q :=$ 
fong-spivak-seven-sketches_014_0045	034 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-034_387_317_414_1412.jpg	$R :=$ 	$R :=$ 	$R :=$ 
fong-spivak-seven-sketches_014_0046	036 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-036_253_242_1597_939.jpg			

fong-spivak-seven-sketches-037_0047 — (dup)	037	7f1e85c6-4350-401f-a960-eb8a0176e786-037_247_296_715_912.jpg			
fong-spivak-seven-sketches-039_0048 — (dup)	039	7f1e85c6-4350-401f-a960-eb8a0176e786-039_204_654_2158_778.jpg			
fong-spivak-seven-sketches-040_0049 — (dup)	040	7f1e85c6-4350-401f-a960-eb8a0176e786-040_213_656_348_778.jpg			
fong-spivak-seven-sketches-040_0050 — (dup)	040	7f1e85c6-4350-401f-a960-eb8a0176e786-040_217_285_1860_624.jpg			
fong-spivak-seven-sketches-040_0051 — (dup)	040	7f1e85c6-4350-401f-a960-eb8a0176e786-040_222_312_1860_1181.jpg			
fong-spivak-seven-sketches-041_0052 — (dup)	041	7f1e85c6-4350-401f-a960-eb8a0176e786-041_339_919_828_600.jpg			
fong-spivak-seven-sketches-041_0053 — (dup)	041	7f1e85c6-4350-401f-a960-eb8a0176e786-041_319_1419_1600_351.jpg			

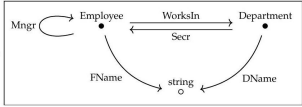
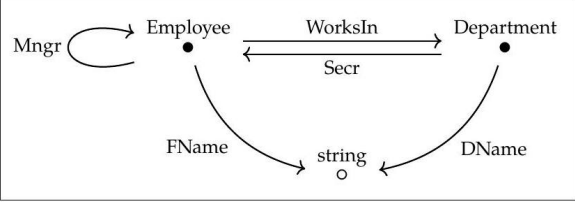
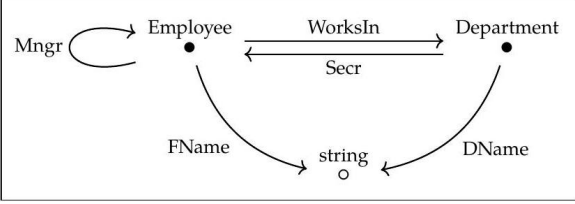
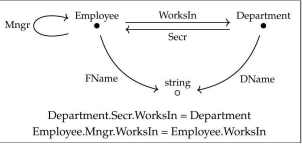
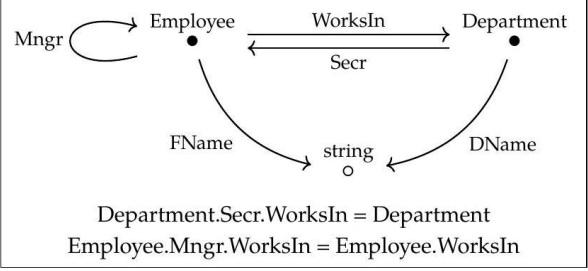
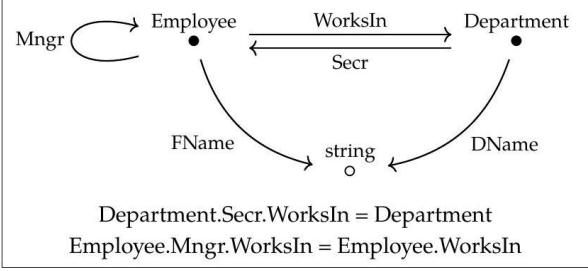
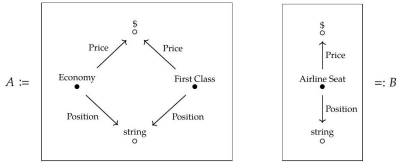
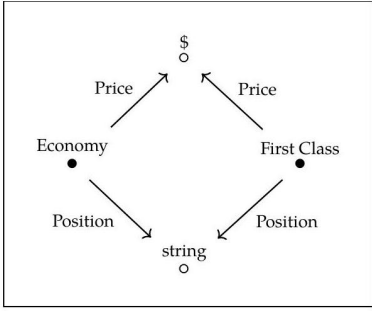
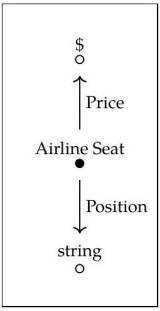
fong-spivak-seven-sketches-043_0054 — (dup)	043	7f1e85c6-4350-401f-a960-eb8a0176e786-043_348_896_1887_611.jpg			
fong-spivak-seven-sketches-047_0055 — (dup)	047	7f1e85c6-4350-401f-a960-eb8a0176e786-047_61_233_1466_941.jpg			
fong-spivak-seven-sketches-052_0056 — (dup)	052	7f1e85c6-4350-401f-a960-eb8a0176e786-052_671_1354_566_381.jpg			
fong-spivak-seven-sketches-052_0057 — (dup)	052	7f1e85c6-4350-401f-a960-eb8a0176e786-052_293_1228_1507_444.jpg			
fong-spivak-seven-sketches-055_0058 — (dup)	055	7f1e85c6-4350-401f-a960-eb8a0176e786-055_90_143_2072_989.jpg			
fong-spivak-seven-sketches-055_0059 — (dup)	055	7f1e85c6-4350-401f-a960-eb8a0176e786-055_86_93_2364_394.jpg			
fong-spivak-seven-sketches-055_0060 — (dup)	055	7f1e85c6-4350-401f-a960-eb8a0176e786-055_90_140_2364_593.jpg			

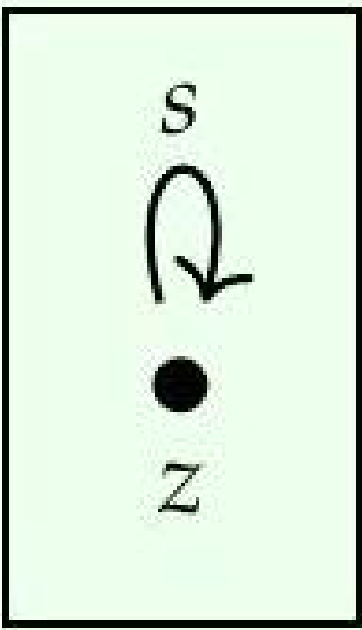
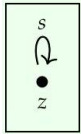
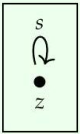
fong-spivak-seven-sketches-055-0061 — (dup)	055		7f1e85c6-4350-401f-a960-eb8a0176e786-055_90_253_2364_842.jpg			
fong-spivak-seven-sketches-055-0062 — (dup)	055		7f1e85c6-4350-401f-a960-eb8a0176e786-055_90_369_2364_1201.jpg			
fong-spivak-seven-sketches-056-0063 — (dup)	056		7f1e85c6-4350-401f-a960-eb8a0176e786-056_179_609_394_756.jpg			
fong-spivak-seven-sketches-056-0064 — (dup)	056		7f1e85c6-4350-401f-a960-eb8a0176e786-056_100_229_1041_394.jpg			
fong-spivak-seven-sketches-056-0065 — (dup)	056		7f1e85c6-4350-401f-a960-eb8a0176e786-056_124_226_1029_729.jpg			
fong-spivak-seven-sketches-056-0066 — (dup)	056		7f1e85c6-4350-401f-a960-eb8a0176e786-056_186_231_1000_1086.jpg			
fong-spivak-seven-sketches-056-0067 — (dup)	056		7f1e85c6-4350-401f-a960-eb8a0176e786-056_188_260_1002_1462.jpg			
fong-spivak-seven-sketches-056-0068 — (dup)	056		7f1e85c6-4350-401f-a960-eb8a0176e786-056_120_109_2222_853.jpg			

fong-spivak-seven-sketches-00069 — (dup)	056	7f1e85c6-4350-401f-a960-eb8a0176e786-056_61_104_2272_1149.jpg		$\overline{x \otimes y}$	$\overline{x \otimes y}$
fong-spivak-seven-sketches-00070 — (dup)	057	7f1e85c6-4350-401f-a960-eb8a0176e786-057_72_464_299_821.jpg	$\overline{I}$ nothing	$\overline{I}$ nothing	$\overline{I}$ nothing
fong-spivak-seven-sketches-00071 — (dup)	057	7f1e85c6-4350-401f-a960-eb8a0176e786-057_163_984_814_593.jpg			
fong-spivak-seven-sketches-00072 — (dup)	057	7f1e85c6-4350-401f-a960-eb8a0176e786-057_127_1330_1471_389.jpg	$\frac{3.14}{3.14} \frac{-1}{-1} \frac{\frac{3.14}{-1}}{-1} \} = \frac{2.14}{0} = \text{nothing}$	$\frac{3.14}{3.14} \frac{-1}{-1} \frac{\frac{3.14}{-1}}{-1} \} = \frac{2.14}{0} = \text{nothing}$	$\frac{3.14}{3.14} \frac{-1}{-1} \frac{\frac{3.14}{-1}}{-1} \} = \frac{2.14}{0} = \text{nothing}$
fong-spivak-seven-sketches-00073 — (dup)	057	7f1e85c6-4350-401f-a960-eb8a0176e786-057_143_561_1762_792.jpg			
fong-spivak-seven-sketches-00074 — (dup)	057	7f1e85c6-4350-401f-a960-eb8a0176e786-057_41_305_2204_907.jpg	$\overline{x}$	$\overline{x}$	$\overline{x}$
fong-spivak-seven-sketches-00075 — (dup)	057	7f1e85c6-4350-401f-a960-eb8a0176e786-057_81_1061_2430_554.jpg			
fong-spivak-seven-sketches-00076 — (dup)	058	7f1e85c6-4350-401f-a960-eb8a0176e786-058_81_688_346_740.jpg			
fong-spivak-seven-sketches-00077 — (dup)	058	7f1e85c6-4350-401f-a960-eb8a0176e786-058_199_615_667_751.jpg			
fong-spivak-seven-sketches-00078 — (dup)	058	7f1e85c6-4350-401f-a960-eb8a0176e786-058_113_403_1050_857.jpg	nothing	nothing	nothing
fong-spivak-seven-sketches-00079 — (dup)	058	7f1e85c6-4350-401f-a960-eb8a0176e786-058_75_192_2258_767.jpg			
fong-spivak-seven-sketches-00080 — (dup)	058	7f1e85c6-4350-401f-a960-eb8a0176e786-058_75_195_2258_1158.jpg			
fong-spivak-seven-sketches-00081 — (dup)	059	7f1e85c6-4350-401f-a960-eb8a0176e786-059_93_439_439_864.jpg			
fong-spivak-seven-sketches-00082 — (dup)	059	7f1e85c6-4350-401f-a960-eb8a0176e786-059_265_896_767_611.jpg			

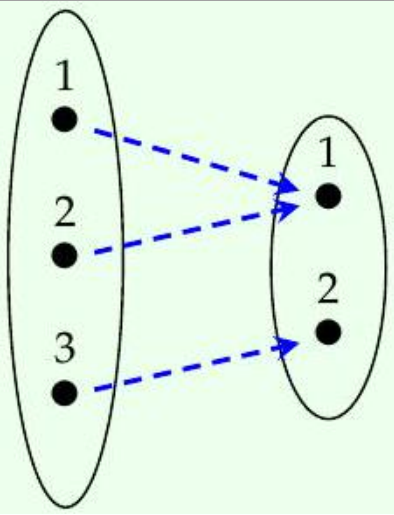
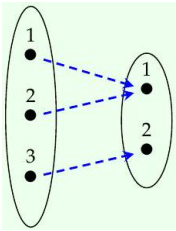
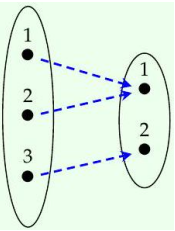
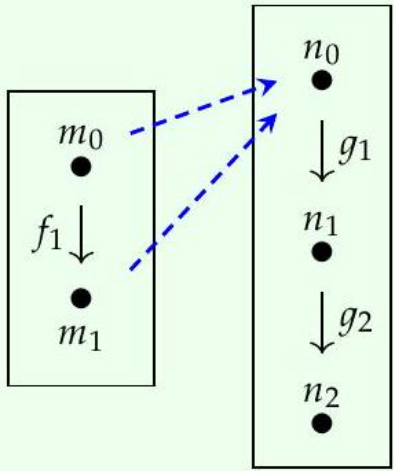
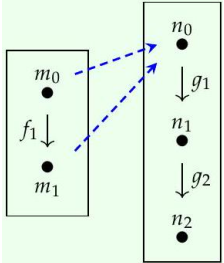
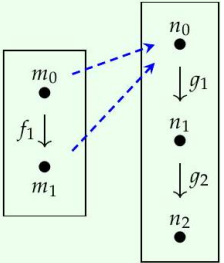
fong-spivak-seven-sketches-061_0083 — (dup)	061	7f1e85c6-4350-401f-a960-eb8a0176e786-061_253_699_2002_710.jpg			
fong-spivak-seven-sketches-062_0084 — (dup)	062	7f1e85c6-4350-401f-a960-eb8a0176e786-062_671_1351_819_381.jpg			
fong-spivak-seven-sketches-062_0085 — (dup)	062	7f1e85c6-4350-401f-a960-eb8a0176e786-062_264_648_1723_734.jpg			
fong-spivak-seven-sketches-063_0086 — (dup)	063	7f1e85c6-4350-401f-a960-eb8a0176e786-063_174_731_1249_695.jpg			
fong-spivak-seven-sketches-063_0087 — (dup)	063	7f1e85c6-4350-401f-a960-eb8a0176e786-063_174_851_1588_633.jpg			
fong-spivak-seven-sketches-063_0088 — (dup)	063	7f1e85c6-4350-401f-a960-eb8a0176e786-063_172_534_2328_792.jpg			
fong-spivak-seven-sketches-064_0089 — (dup)	064	7f1e85c6-4350-401f-a960-eb8a0176e786-064_265_973_342_572.jpg			
fong-spivak-seven-sketches-070_0090 — (dup)	070	7f1e85c6-4350-401f-a960-eb8a0176e786-070_367_450_335_835.jpg			

fong-spivak-seven-sketches-072_0091 — (dup)	072	7f1e85c6-4350-401f-a960-eb8a0176e786-072_256_414_878_851.jpg			
fong-spivak-seven-sketches-074_0092 — (dup)	074	7f1e85c6-4350-401f-a960-eb8a0176e786-074_335_391_799_550.jpg	$X :=$	$X :=$	$X :=$
fong-spivak-seven-sketches-074_0093 — (dup)	074	7f1e85c6-4350-401f-a960-eb8a0176e786-074_337_351_799_1217.jpg			
fong-spivak-seven-sketches-079_0094 — (dup)	079	7f1e85c6-4350-401f-a960-eb8a0176e786-079_299_805_1534_656.jpg	$\mathcal{X} :=$	$\mathcal{X} :=$	$\mathcal{X} :=$
fong-spivak-seven-sketches-080_0095 — (dup)	080	7f1e85c6-4350-401f-a960-eb8a0176e786-080_330_654_333_738.jpg	$\mathcal{X} \times \mathcal{Y} =$	$\mathcal{X} \times \mathcal{Y} =$	$\mathcal{X} \times \mathcal{Y} =$
fong-spivak-seven-sketches-086_0096 — (dup)	086	7f1e85c6-4350-401f-a960-eb8a0176e786-086_339_351_1982_495.jpg	$Y :=$	$Y :=$	$Y :=$

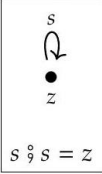
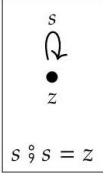
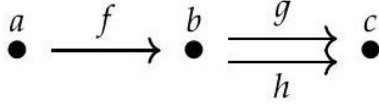
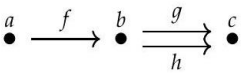
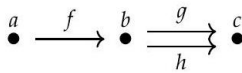
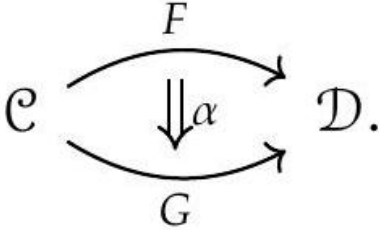
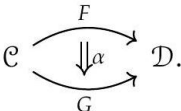
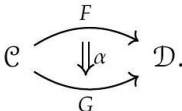
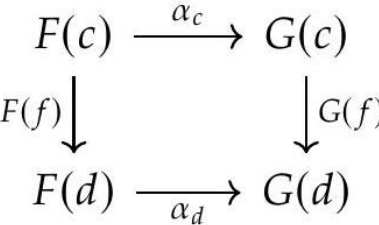
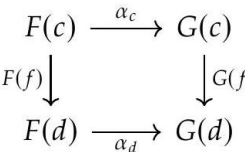
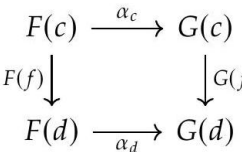
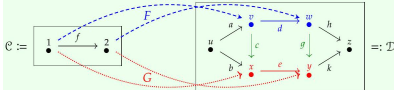
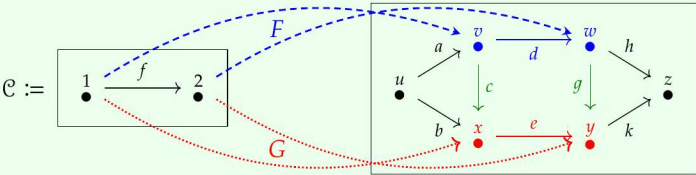
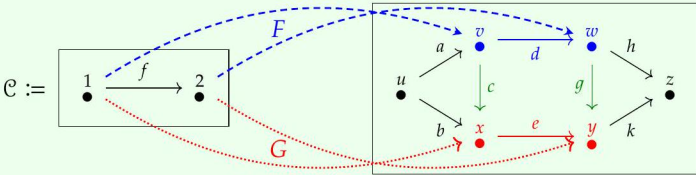
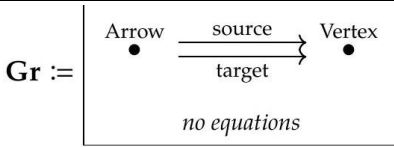
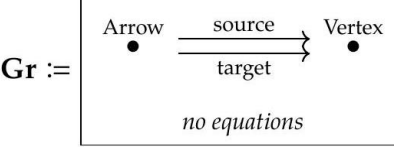
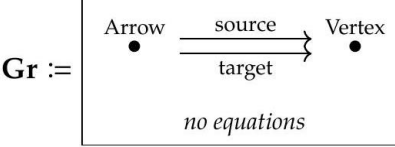
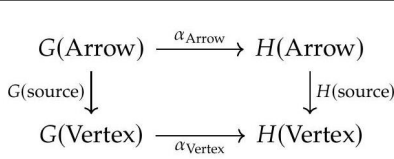
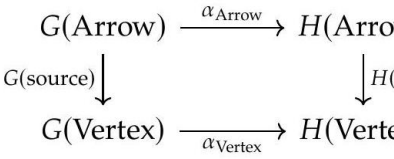
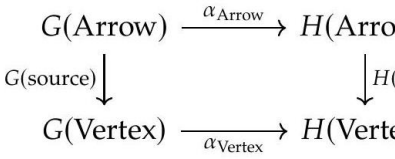
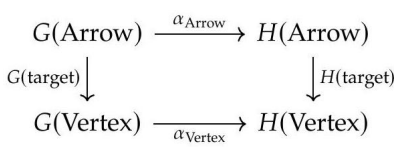
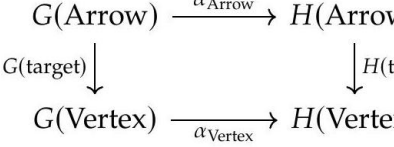
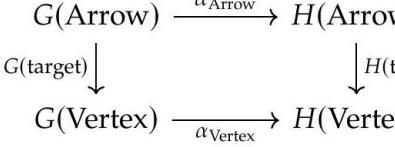
fong-spivak-seven-sketches_img_0090 — (dup)	090	7f1e85c6-4350-401f-a960-eb8a0176e786-090_299_1097_1321_509.jpg		easySchema := 	easySchema := 
fong-spivak-seven-sketches_img_0091 — (dup)	091	7f1e85c6-4350-401f-a960-eb8a0176e786-091_395_1079_559_520.jpg		mySchema := 	mySchema := 
fong-spivak-seven-sketches_img_0091 — (dup)	091	no image for this region (498 in zip)	(no tex.zip crop for this row)	schema mySchema = { nodes Employee, Department; attributes DName : Department -> string, FName : Employee -> string; arrows Mngr : Employee -> Employee, WorksIn : Employee -> Department, Secr : Department -> Employee; equations Department.Scr.WorksIn = Department, Employee.Mngr.WorksIn = Employee.WorksIn; }	schema mySchema = { nodes Employee, Department; attributes DName : Department -> string, FName : Employee -> string; arrows Mngr : Employee -> Employee, WorksIn : Employee -> Department, Secr : Department -> Employee; equations Department.Scr.WorksIn = Department, Employee.Mngr.WorksIn = Employee.WorksIn; }
fong-spivak-seven-sketches_img_0100 — (dup)	092	7f1e85c6-4350-401f-a960-eb8a0176e786-092_500_1217_972_412.jpg			

fong-spivak-seven-sketches_01011011 — (dup)	095	7f1e85c6-4350-401f-a960-eb8a0176e786-095_174_106_1269_1007.jpg			
fong-spivak-seven-sketches_01011012 — (dup)	096	7f1e85c6-4350-401f-a960-eb8a0176e786-096_312_514_683_430.jpg	Free_square := $\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & & \downarrow h \\ C & \xrightarrow{i} & D \end{array}$ no equations	Free_square := $\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & & \downarrow h \\ C & \xrightarrow{i} & D \end{array}$ no equations	Free_square := $\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & & \downarrow h \\ C & \xrightarrow{i} & D \end{array}$ no equations
fong-spivak-seven-sketches_01011013 — (dup)	096	7f1e85c6-4350-401f-a960-eb8a0176e786-096_312_552_683_1131.jpg	Comm_square := $\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & & \downarrow h \\ C & \xrightarrow{i} & D \end{array}$ $f \circ h = g \circ i$	Comm_square := $\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & & \downarrow h \\ C & \xrightarrow{i} & D \end{array}$ $f \circ h = g \circ i$	Comm_square := $\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & & \downarrow h \\ C & \xrightarrow{i} & D \end{array}$ $f \circ h = g \circ i$
fong-spivak-seven-sketches_01011014 — (dup)	096	7f1e85c6-4350-401f-a960-eb8a0176e786-096_308_244_1900_937.jpg	$\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & \searrow j & \downarrow h \\ C & \xrightarrow{i} & D \end{array}$ $f \circ h = j = g \circ i$	$\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & \searrow j & \downarrow h \\ C & \xrightarrow{i} & D \end{array}$ $f \circ h = j = g \circ i$	$\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & \searrow j & \downarrow h \\ C & \xrightarrow{i} & D \end{array}$ $f \circ h = j = g \circ i$

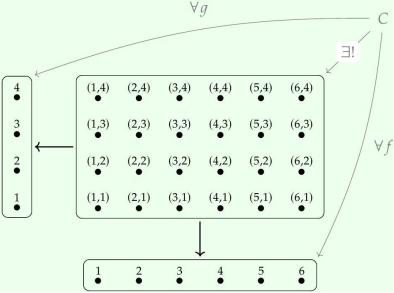
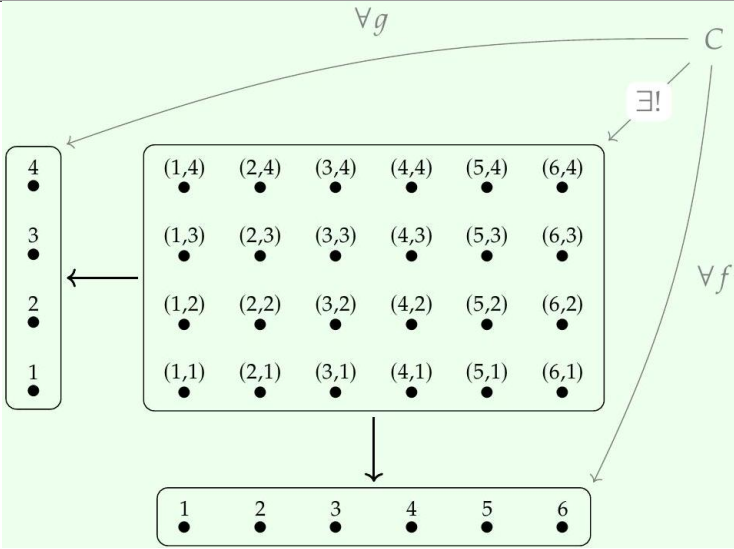
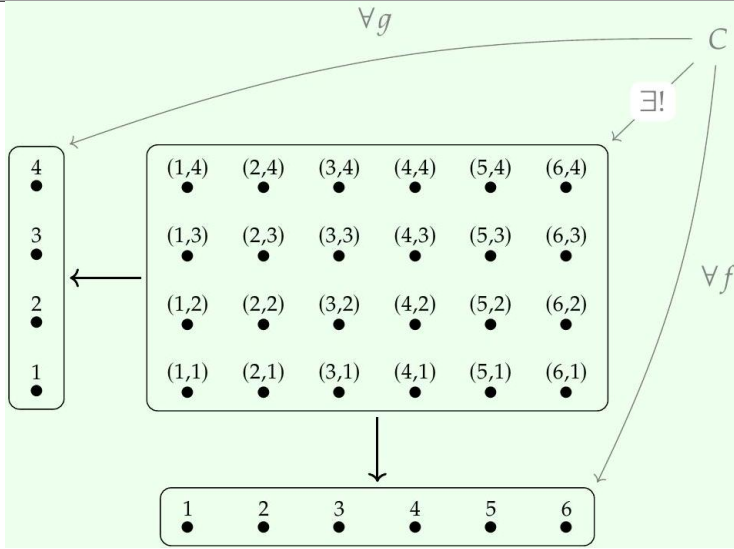
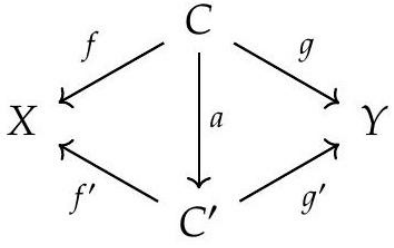
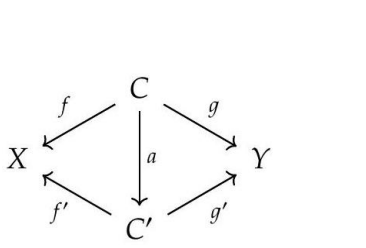
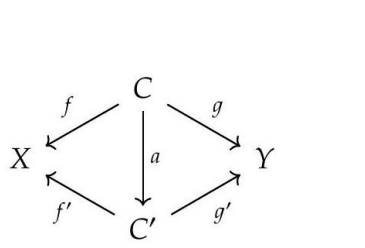
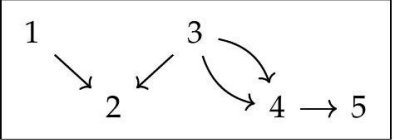
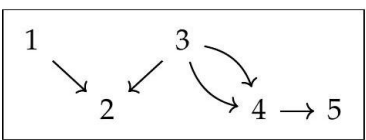
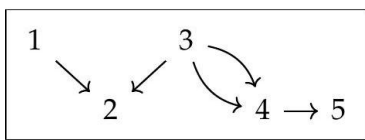
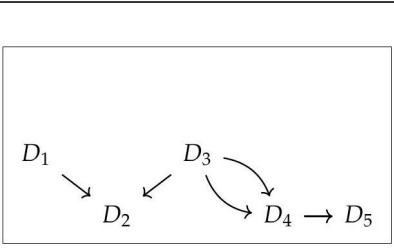
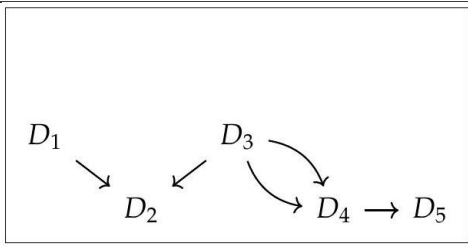
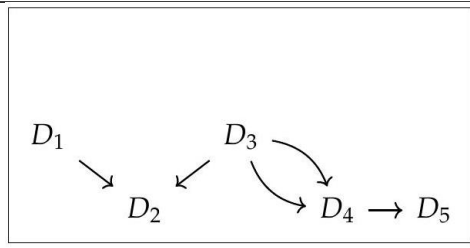
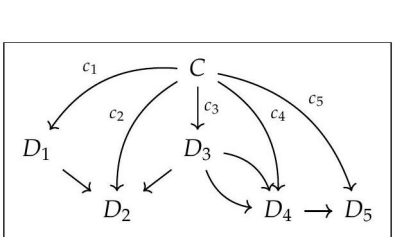
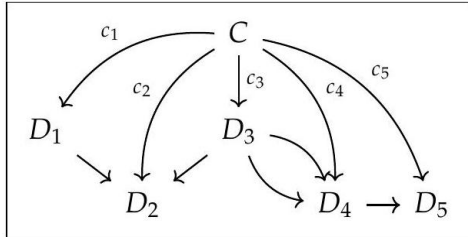
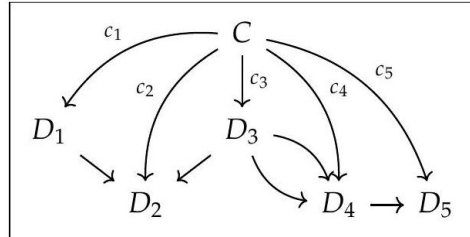
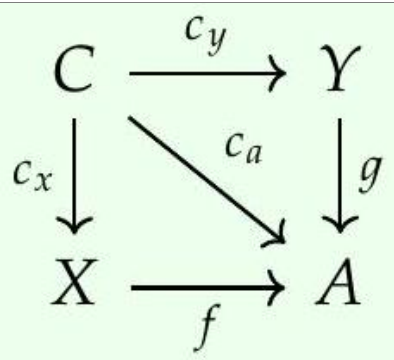
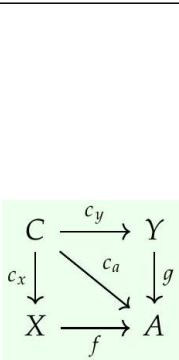
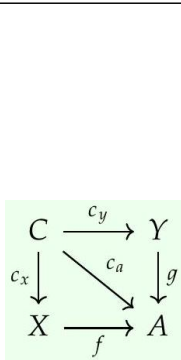
fong-spivak-seven-sketches_0105 — (dup)	097	7f1e85c6-4350-401f-a960-eb8a0176e786-097_278_373_2215_355.jpg			
fong-spivak-seven-sketches_0106 — (dup)	097	7f1e85c6-4350-401f-a960-eb8a0176e786-097_238_342_2199_878.jpg			
fong-spivak-seven-sketches_0107 — (dup)	097	7f1e85c6-4350-401f-a960-eb8a0176e786-097_326_373_2195_1389.jpg			
fong-spivak-seven-sketches_0108 — (dup)	098	7f1e85c6-4350-401f-a960-eb8a0176e786-098_235_1423_733_357.jpg	$G_1 = \begin{array}{ c } \hline \bullet \xrightarrow{f} \bullet \\ \hline \end{array}$ $G_2 = \begin{array}{ c } \hline \bullet \xrightarrow{f} \bullet \\ \hline \end{array}$ $G_3 = \begin{array}{ c } \hline \bullet \xrightarrow{f} \bullet \\ \hline \end{array}$ $G_4 = \begin{array}{ c } \hline \bullet \xrightarrow{f} \bullet \\ \hline \end{array}$ $\diamond$	$G_1 = \begin{array}{ c } \hline \bullet \xrightarrow{f} \bullet \\ \hline \end{array}$ $G_2 = \begin{array}{ c } \hline \bullet \xrightarrow{f} \bullet \\ \hline \end{array}$ $G_3 = \begin{array}{ c } \hline \bullet \xrightarrow{f} \bullet \\ \hline \end{array}$ $G_4 = \begin{array}{ c } \hline \bullet \xrightarrow{f} \bullet \\ \hline \end{array}$ $\diamond$	$G_1 = \begin{array}{ c } \hline \bullet \xrightarrow{f} \bullet \\ \hline \end{array}$ $G_2 = \begin{array}{ c } \hline \bullet \xrightarrow{f} \bullet \\ \hline \end{array}$ $G_3 = \begin{array}{ c } \hline \bullet \xrightarrow{f} \bullet \\ \hline \end{array}$ $G_4 = \begin{array}{ c } \hline \bullet \xrightarrow{f} \bullet \\ \hline \end{array}$ $\diamond$
fong-spivak-seven-sketches_0109 — (dup)	101	7f1e85c6-4350-401f-a960-eb8a0176e786-101_294_224_742_701.jpg			

fong-spivak-seven-sketches-101_0110 — (dup)	101	7f1e85c6-4350-401f-a960-eb8a0176e786-101_294_226_742_1190.jpg			
fong-spivak-seven-sketches-102_0111 — (dup)	102	7f1e85c6-4350-401f-a960-eb8a0176e786-102_114_236_722_463.jpg	Planet of Sol	Planet of Sol	Planet of Sol
fong-spivak-seven-sketches-102_0112 — (dup)	102	7f1e85c6-4350-401f-a960-eb8a0176e786-102_111_259_725_944.jpg	Prime number	Prime number	Prime number
fong-spivak-seven-sketches-102_0113 — (dup)	102	7f1e85c6-4350-401f-a960-eb8a0176e786-102_117_210_719_1444.jpg	Flying pig	Flying pig	Flying pig
fong-spivak-seven-sketches-102_0114 — (dup)	102	7f1e85c6-4350-401f-a960-eb8a0176e786-102_136_543_2081_788.jpg	Beatle $\xrightarrow{\text{Played}}$ Rock-and-roll instrument	Beatle $\xrightarrow{\text{Played}}$ Rock-and-roll instrument	Beatle $\xrightarrow{\text{Played}}$ Rock-and-roll instrument
fong-spivak-seven-sketches-103_0115 — (dup)	103	7f1e85c6-4350-401f-a960-eb8a0176e786-103_346_287_1188_387.jpg			

fong-spivak-seven-sketches_0116 — (dup)	103	7f1e85c6-4350-401f-a960-eb8a0176e786-103_344_287_1190_912.jpg			
fong-spivak-seven-sketches_0117 — (dup)	103	7f1e85c6-4350-401f-a960-eb8a0176e786-103_344_287_1190_1437.jpg			
fong-spivak-seven-sketches_0118 — (dup)	104	7f1e85c6-4350-401f-a960-eb8a0176e786-104_314_538_330_466.jpg	Free_square := $\begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ g' \downarrow & & \downarrow h' \\ C' & \xrightarrow{i'} & D' \end{array}$ no equations	Free_square := $\begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ g' \downarrow & & \downarrow h' \\ C' & \xrightarrow{i'} & D' \end{array}$ no equations	Free_square := $\begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ g' \downarrow & & \downarrow h' \\ C' & \xrightarrow{i'} & D' \end{array}$ no equations
fong-spivak-seven-sketches_0119 — (dup)	104	7f1e85c6-4350-401f-a960-eb8a0176e786-104_310_554_333_1093.jpg	Comm_square := $\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & & \downarrow h \\ C & \xrightarrow{i} & D \end{array}$ $f \circ h = g \circ i$	Comm_square := $\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & & \downarrow h \\ C & \xrightarrow{i} & D \end{array}$ $f \circ h = g \circ i$	Comm_square := $\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & & \downarrow h \\ C & \xrightarrow{i} & D \end{array}$ $f \circ h = g \circ i$

fong-spivak-seven-sketches_01120 — (dup)	106	7f1e85c6-4350-401f-a960-eb8a0176e786-106_226_181_1966_733.jpg	1. 	1. 	1. 
fong-spivak-seven-sketches_01121 — (dup)	106	7f1e85c6-4350-401f-a960-eb8a0176e786-106_93_324_1998_1217.jpg			
fong-spivak-seven-sketches_01122 — (dup)	106	7f1e85c6-4350-401f-a960-eb8a0176e786-106_43_172_2124_1294.jpg	$f \circ g = f \circ h$	$f \circ g = f \circ h$	$f \circ g = f \circ h$
fong-spivak-seven-sketches_01123 — (dup)	107	7f1e85c6-4350-401f-a960-eb8a0176e786-107_152_244_527_937.jpg			
fong-spivak-seven-sketches_01124 — (dup)	107	7f1e85c6-4350-401f-a960-eb8a0176e786-107_199_339_1638_889.jpg			
fong-spivak-seven-sketches_01125 — (dup)	108	7f1e85c6-4350-401f-a960-eb8a0176e786-108_276_1195_380_462.jpg			
fong-spivak-seven-sketches_01126 — (dup)	109	7f1e85c6-4350-401f-a960-eb8a0176e786-109_190_529_2330_792.jpg	$\mathbf{Gr} :=$ 	$\mathbf{Gr} :=$ 	$\mathbf{Gr} :=$ 
fong-spivak-seven-sketches_01127 — (dup)	110	7f1e85c6-4350-401f-a960-eb8a0176e786-110_206_611_2172_353.jpg			
fong-spivak-seven-sketches_01128 — (dup)	110	7f1e85c6-4350-401f-a960-eb8a0176e786-110_206_591_2172_1172.jpg			

fong-spivak-seven-sketches_112_0129 — (dup)	112	7f1e85c6-4350-401f-a960-eb8a0176e786-112_330_543_1955_787.jpg	
fong-spivak-seven-sketches_113_0130 — (dup)	113	7f1e85c6-4350-401f-a960-eb8a0176e786-113_373_767_1600_676.jpg	
fong-spivak-seven-sketches_114_0131 — (dup)	114	7f1e85c6-4350-401f-a960-eb8a0176e786-114_163_1007_568_557.jpg	
fong-spivak-seven-sketches_115_0132 — (dup)	115	7f1e85c6-4350-401f-a960-eb8a0176e786-115_206_480_2258_819.jpg	
fong-spivak-seven-sketches_116_0133 — (dup)	116	7f1e85c6-4350-401f-a960-eb8a0176e786-116_172_498_2342_810.jpg	
fong-spivak-seven-sketches_117_0134 — (dup)	117	7f1e85c6-4350-401f-a960-eb8a0176e786-117_389_1296_1389_410.jpg	
fong-spivak-seven-sketches_121_0135 — (dup)	121	7f1e85c6-4350-401f-a960-eb8a0176e786-121_219_416_353_851.jpg	

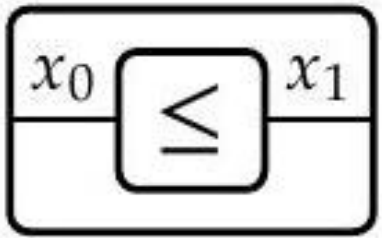
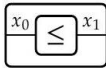
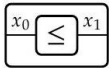
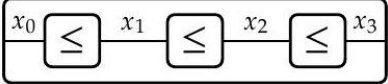
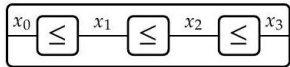
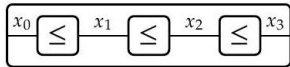
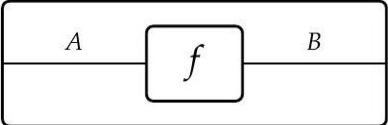
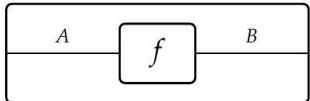
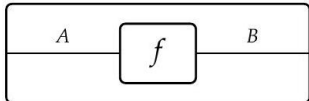
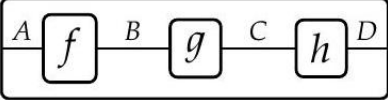
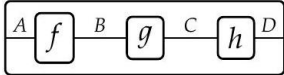
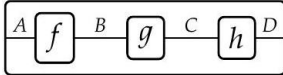
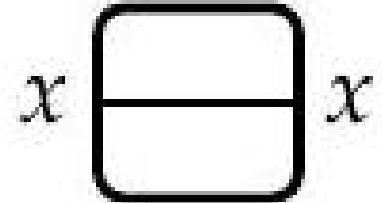
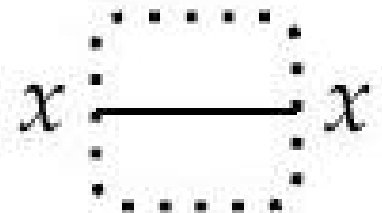
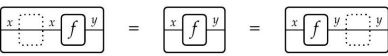
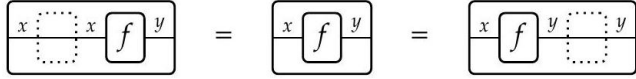
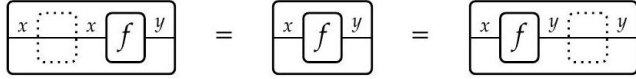
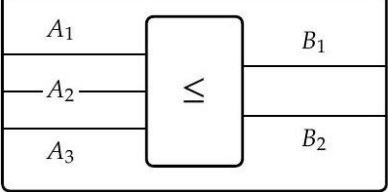
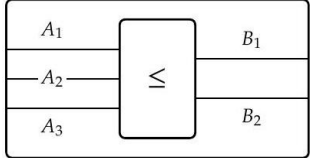
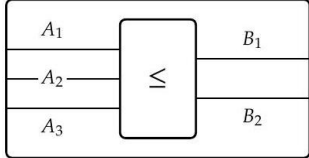
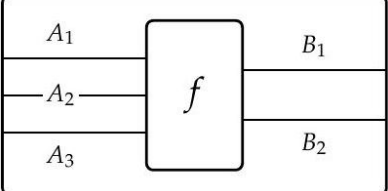
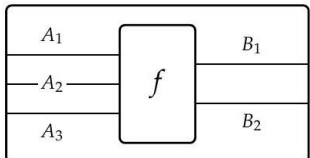
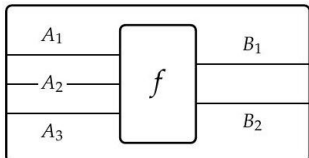
fong-spivak-seven-sketches-2010-0136 — (dup)	121	7f1e85c6-4350-401f-a960-eb8a0176e786-121_710_955_1529_575.jpg			
fong-spivak-seven-sketches-2010-0137 — (dup)	122	7f1e85c6-4350-401f-a960-eb8a0176e786-122_222_355_1837_882.jpg			
fong-spivak-seven-sketches-2010-0138 — (dup)	123	7f1e85c6-4350-401f-a960-eb8a0176e786-123_176_475_1095_824.jpg			
fong-spivak-seven-sketches-2010-0139 — (dup)	123	7f1e85c6-4350-401f-a960-eb8a0176e786-123_317_613_1441_351.jpg			
fong-spivak-seven-sketches-2010-0140 — (dup)	123	7f1e85c6-4350-401f-a960-eb8a0176e786-123_314_615_1443_1154.jpg			
fong-spivak-seven-sketches-2010-0141 — (dup)	125	7f1e85c6-4350-401f-a960-eb8a0176e786-125_208_229_330_663.jpg			

fong-spivak-seven-sketches_0142 — (dup)	125	7f1e85c6-4350-401f-a960-eb8a0176e786-125_213_296_328_1156.jpg			
fong-spivak-seven-sketches_0143 — (dup)	125	7f1e85c6-4350-401f-a960-eb8a0176e786-125_509_787_876_665.jpg			
fong-spivak-seven-sketches_0144 — (dup)	130	7f1e85c6-4350-401f-a960-eb8a0176e786-130_461_1345_1308_387.jpg			
fong-spivak-seven-sketches_0145 — (dup)	132	7f1e85c6-4350-401f-a960-eb8a0176e786-132_233_1136_1649_498.jpg			
fong-spivak-seven-sketches_0146 — (dup)	134	7f1e85c6-4350-401f-a960-eb8a0176e786-134_486_1041_760_538.jpg			
fong-spivak-seven-sketches_0147 — (dup)	135	7f1e85c6-4350-401f-a960-eb8a0176e786-135_367_1014_559_552.jpg			

fong-spivak-seven-sketches_0148 — (dup)	136		7f1e85c6-4350-401f-a960-eb8a0176e786-136_217_407_611_855.jpg	loadvelocityChassis torquespeed\$	loadvelocityChassis torquespeed\$	loadvelocityChassis torquespeed\$
fong-spivak-seven-sketches_0149 — (dup)	136		7f1e85c6-4350-401f-a960-eb8a0176e786-136_165_285_1860_916.jpg	$T$$E$$\Phi$\$	$T$$E$$\Phi$\$	$T$$E$$\Phi$\$
fong-spivak-seven-sketches_0150 — (dup)	136		7f1e85c6-4350-401f-a960-eb8a0176e786-136_296_319_2172_575.jpg	$T :=$ good-natured • ↑ mean-spirited •	$T :=$ good-natured • ↑ mean-spirited •	$T :=$ good-natured • ↑ mean-spirited •
fong-spivak-seven-sketches_0151 — (dup)	136		7f1e85c6-4350-401f-a960-eb8a0176e786-136_292_226_2176_973.jpg	$E :=$ funny • ↑ boring •	$E :=$ funny • ↑ boring •	$E :=$ funny • ↑ boring •

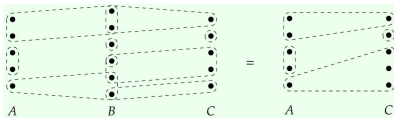
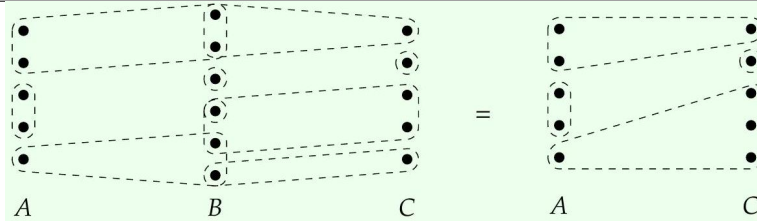
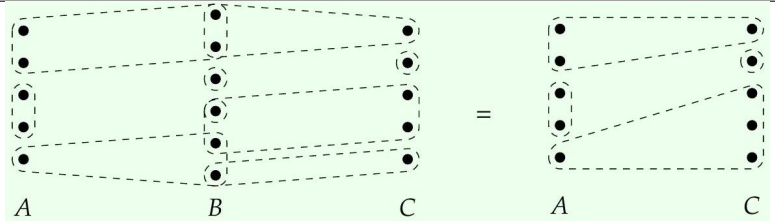
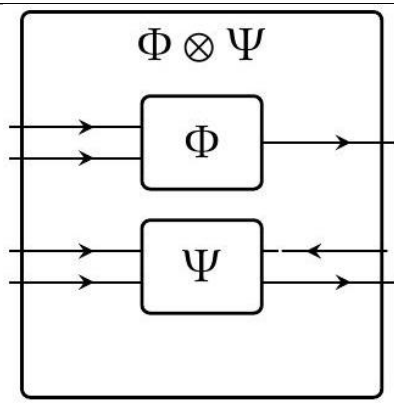
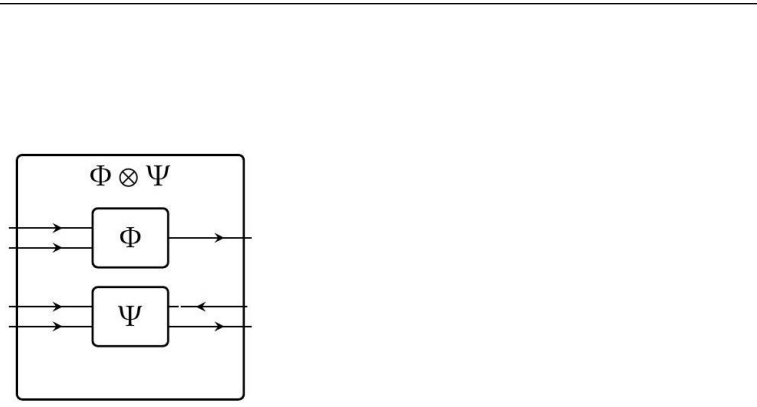
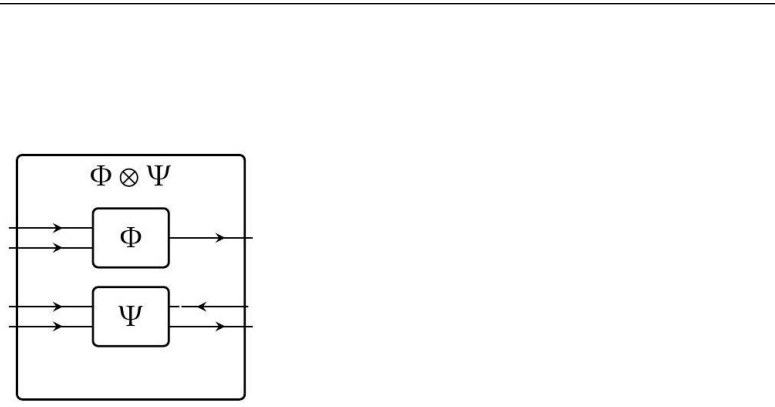
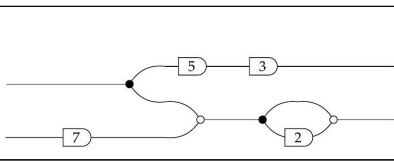
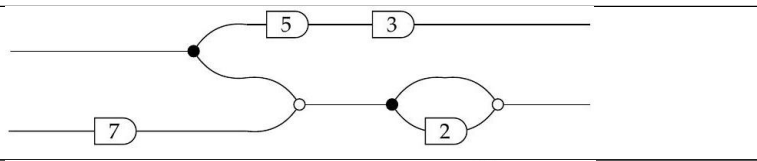
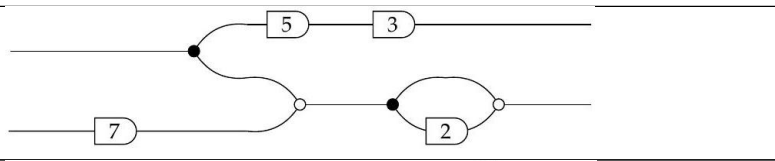
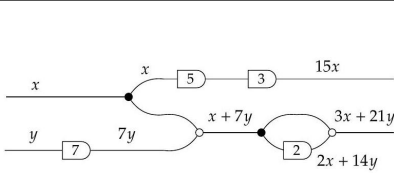
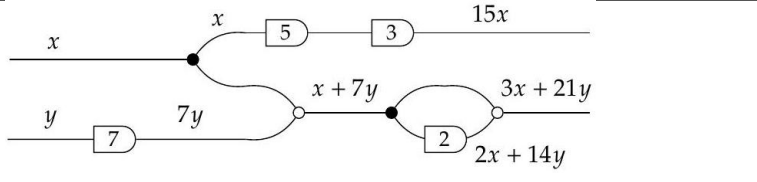
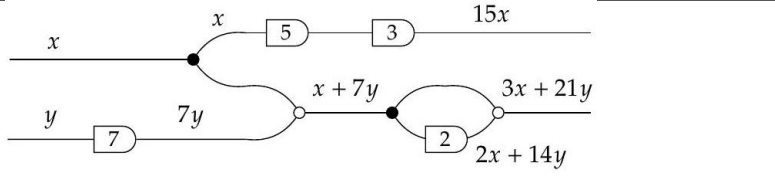
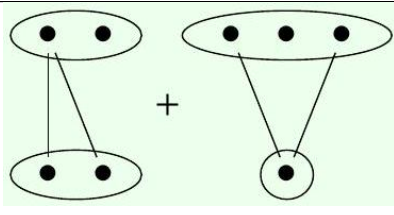
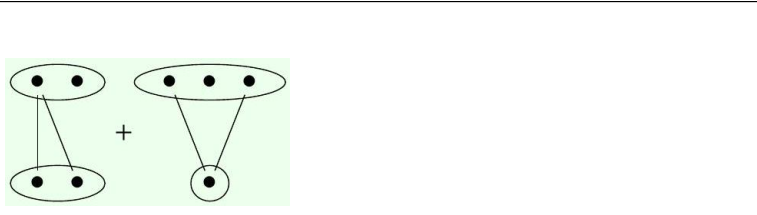
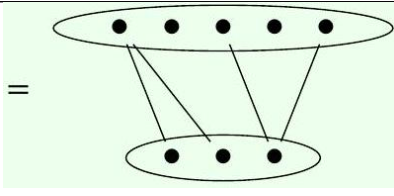
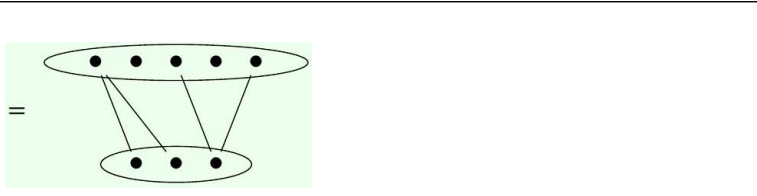
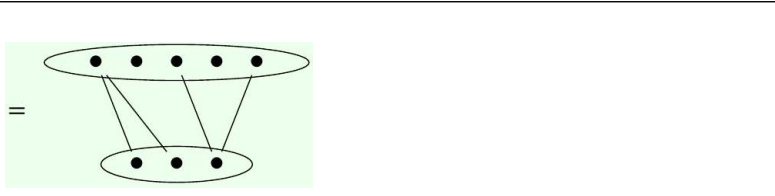
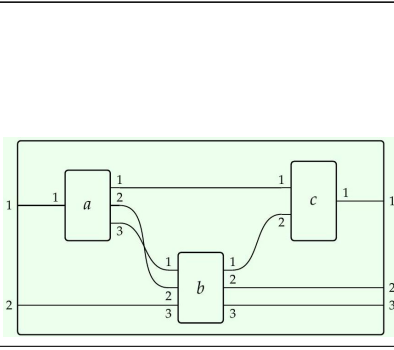
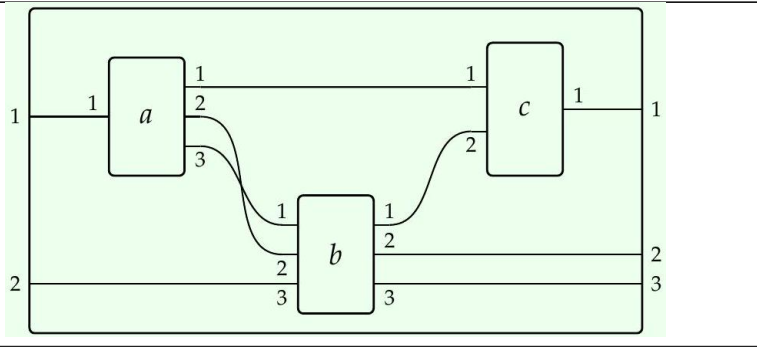
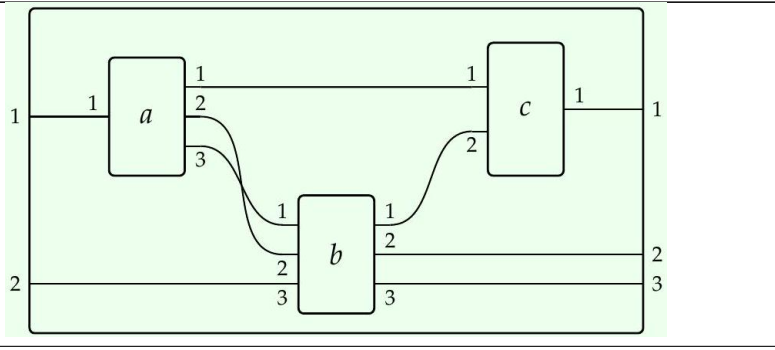
fong-spivak-seven-sketches-136_0152 — (dup)	136	7f1e85c6-4350-401f-a960-eb8a0176e786-136_357_215_2145_1339.jpg			
fong-spivak-seven-sketches-137_0153 — (dup)	137	7f1e85c6-4350-401f-a960-eb8a0176e786-137_423_993_369_563.jpg			
fong-spivak-seven-sketches-138_0154 — (dup)	138	7f1e85c6-4350-401f-a960-eb8a0176e786-138_452_1213_346_407.jpg			
fong-spivak-seven-sketches-139_0155 — (dup)	139	7f1e85c6-4350-401f-a960-eb8a0176e786-139_351_1118_692_500.jpg			
fong-spivak-seven-sketches-140_0156 — (dup)	140	7f1e85c6-4350-401f-a960-eb8a0176e786-140_376_1335_665_391.jpg			
fong-spivak-seven-sketches-141_0157 — (dup)	141	7f1e85c6-4350-401f-a960-eb8a0176e786-141_136_919_1009_600.jpg			

fong-spivak-seven-sketches-141_0158 — (dup)	141	7f1e85c6-4350-401f-a960-eb8a0176e786-141_459_1346_1373_387.jpg			
fong-spivak-seven-sketches-142_0159 — (dup)	142	7f1e85c6-4350-401f-a960-eb8a0176e786-142_398_1156_1602_480.jpg			
fong-spivak-seven-sketches-144_0160 — (dup)	144	7f1e85c6-4350-401f-a960-eb8a0176e786-144_314_760_523_679.jpg			
fong-spivak-seven-sketches-144_0161 — (dup)	144	7f1e85c6-4350-401f-a960-eb8a0176e786-144_294_1131_1416_500.jpg			
fong-spivak-seven-sketches-144_0162 — (dup)	144	7f1e85c6-4350-401f-a960-eb8a0176e786-144_367_1014_1860_552.jpg			
fong-spivak-seven-sketches-145_0163 — (dup)	145	7f1e85c6-4350-401f-a960-eb8a0176e786-145_645_950_1335_586.jpg			

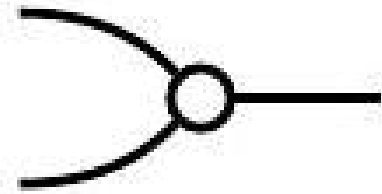
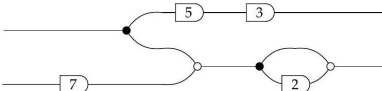
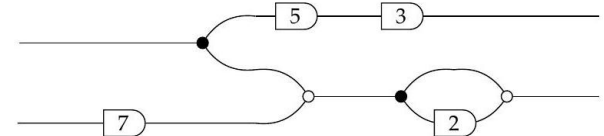
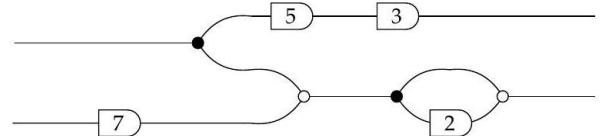
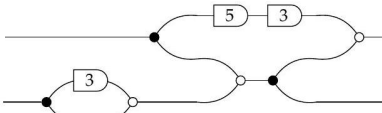
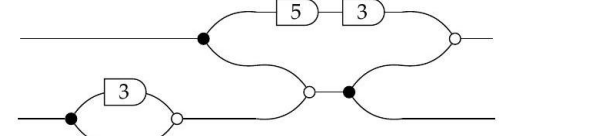
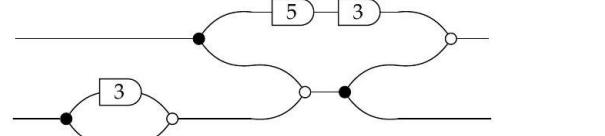
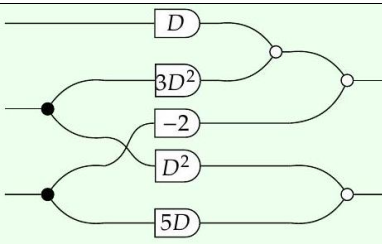
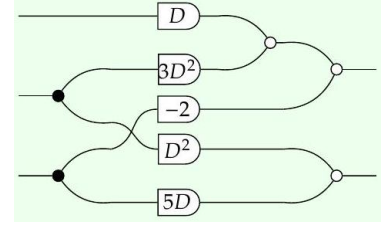
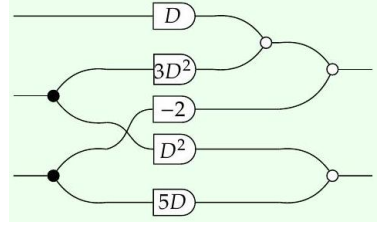
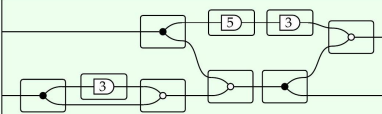
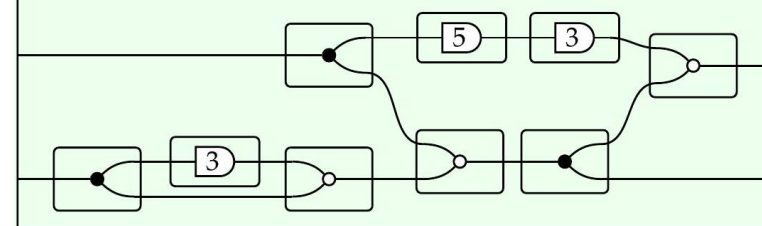
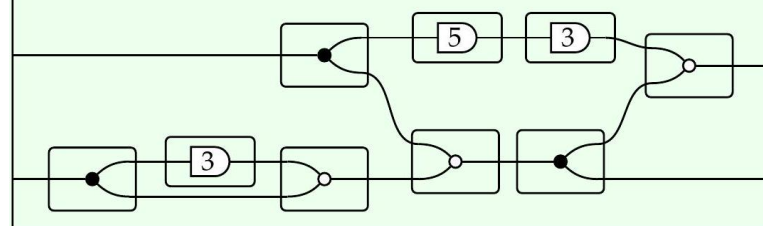
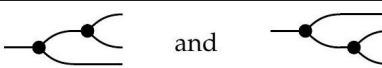
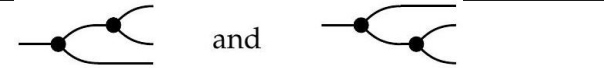
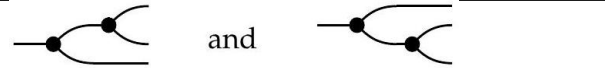
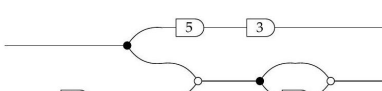
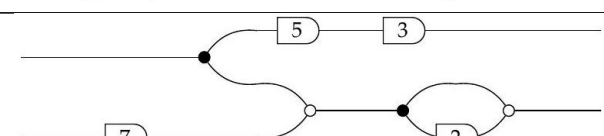
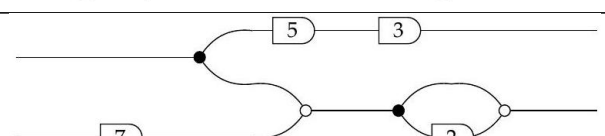
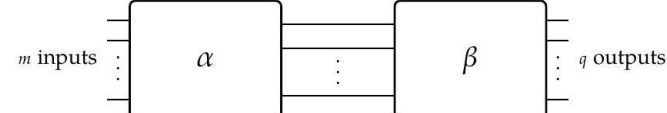
fong-spivak-seven-sketches-414_0164 — (dup)	146		7f1e85c6-4350-401f-a960-eb8a0176e786-146_90_143_1000_989.jpg			
fong-spivak-seven-sketches-414_0165 — (dup)	146		7f1e85c6-4350-401f-a960-eb8a0176e786-146_88_380_1242_869.jpg			
fong-spivak-seven-sketches-414_0166 — (dup)	146		7f1e85c6-4350-401f-a960-eb8a0176e786-146_138_405_1719_857.jpg			
fong-spivak-seven-sketches-414_0167 — (dup)	146		7f1e85c6-4350-401f-a960-eb8a0176e786-146_102_371_2054_873.jpg			
fong-spivak-seven-sketches-414_0168 — (dup)	147		7f1e85c6-4350-401f-a960-eb8a0176e786-147_68_104_387_636.jpg			
fong-spivak-seven-sketches-414_0169 — (dup)	147		7f1e85c6-4350-401f-a960-eb8a0176e786-147_68_104_387_1007.jpg			
fong-spivak-seven-sketches-414_0170 — (dup)	147		7f1e85c6-4350-401f-a960-eb8a0176e786-147_32_102_405_1378.jpg			
fong-spivak-seven-sketches-414_0171 — (dup)	147		7f1e85c6-4350-401f-a960-eb8a0176e786-147_106_833_701_643.jpg			
fong-spivak-seven-sketches-414_0172 — (dup)	147		7f1e85c6-4350-401f-a960-eb8a0176e786-147_213_405_1274_857.jpg			
fong-spivak-seven-sketches-414_0173 — (dup)	147		7f1e85c6-4350-401f-a960-eb8a0176e786-147_213_405_1753_857.jpg			

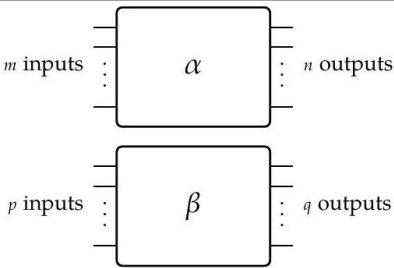
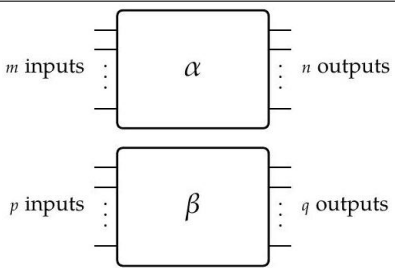
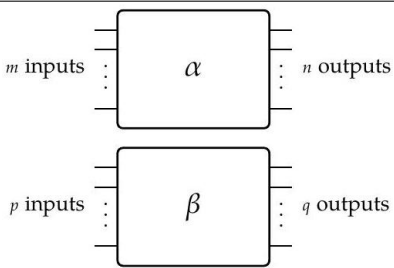
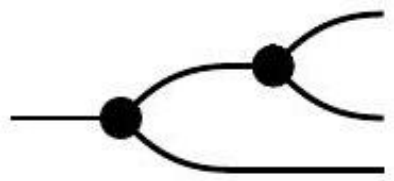
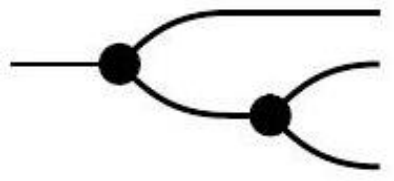
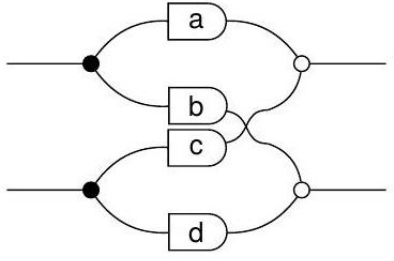
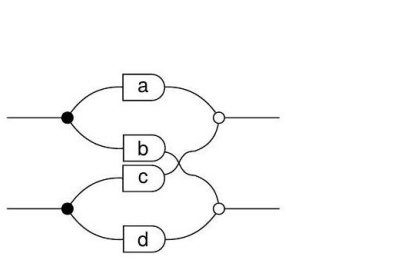
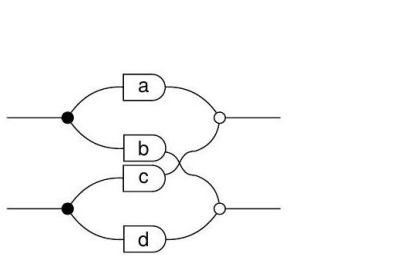
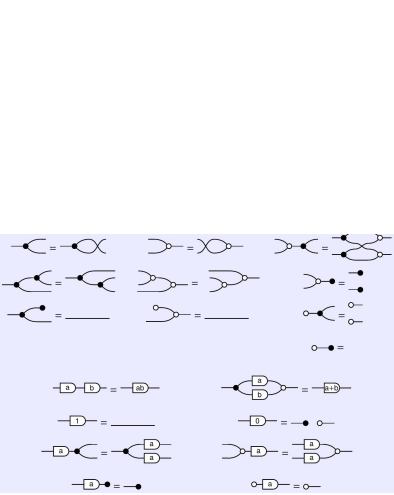
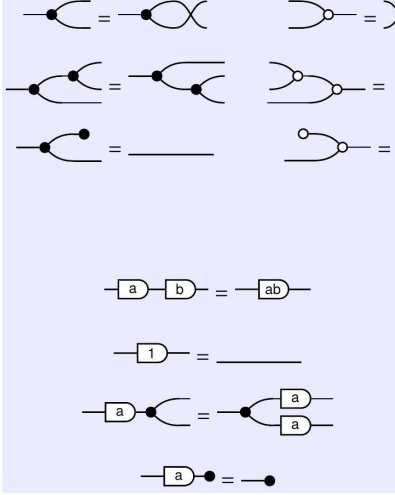
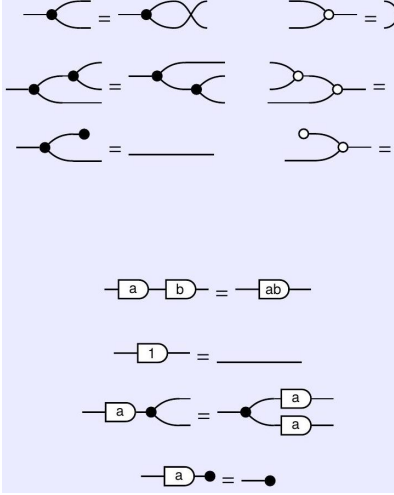
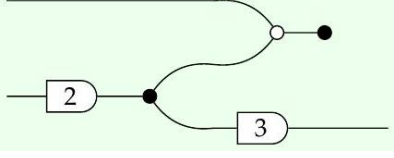
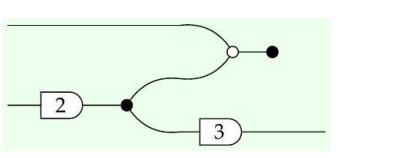
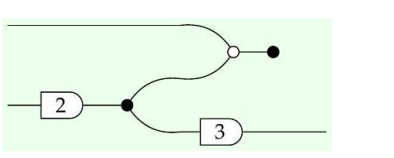
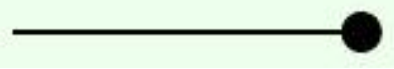
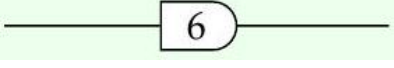
fong-spivak-seven-sketches-147_0174 — (dup)	147	7f1e85c6-4350-401f-a960-eb8a0176e786-147_355_805_2147_656.jpg			
fong-spivak-seven-sketches-150_0175 — (dup)	150	7f1e85c6-4350-401f-a960-eb8a0176e786-150_355_808_529_656.jpg			
fong-spivak-seven-sketches-152_0176 — (dup)	152	7f1e85c6-4350-401f-a960-eb8a0176e786-152_370_995_336_563.jpg			
fong-spivak-seven-sketches-152_0177 — (dup)	152	7f1e85c6-4350-401f-a960-eb8a0176e786-152_176_836_1365_640.jpg			
fong-spivak-seven-sketches-152_0178 — (dup)	152	7f1e85c6-4350-401f-a960-eb8a0176e786-152_182_404_1712_862.jpg			
fong-spivak-seven-sketches-152_0179 — (dup)	152	7f1e85c6-4350-401f-a960-eb8a0176e786-152_227_836_2019_640.jpg			
fong-spivak-seven-sketches-153_0180 — (dup)	153	7f1e85c6-4350-401f-a960-eb8a0176e786-153_301_502_903_387.jpg			
fong-spivak-seven-sketches-153_0181 — (dup)	153	7f1e85c6-4350-401f-a960-eb8a0176e786-153_305_525_900_1113.jpg			

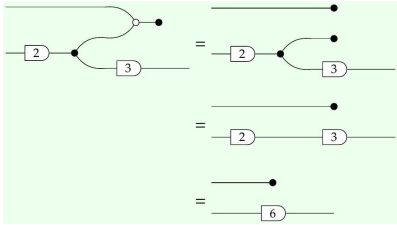
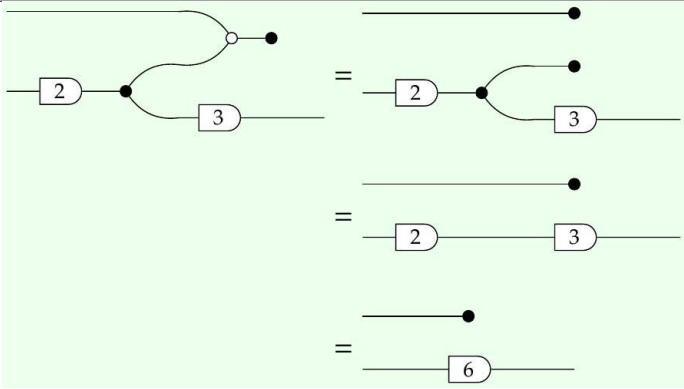
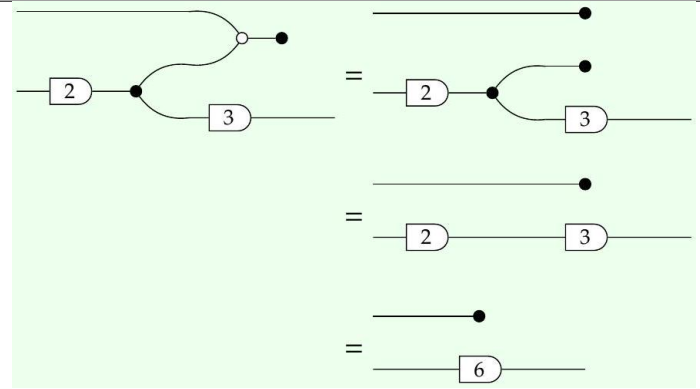
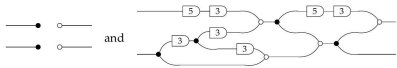
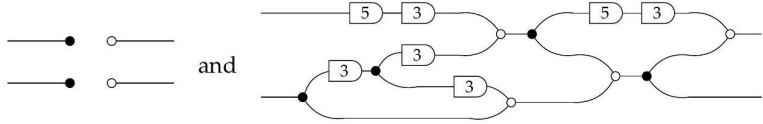
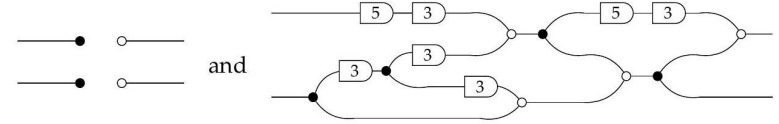
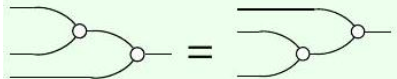
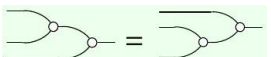
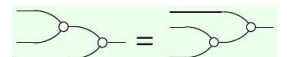
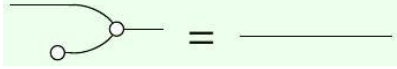
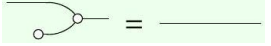
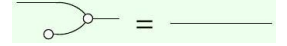
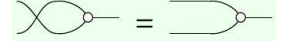
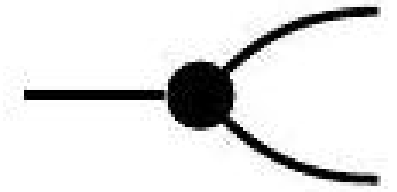
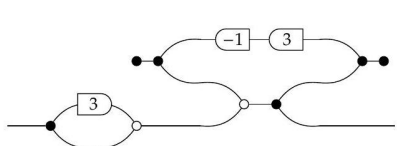
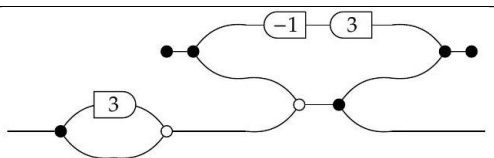
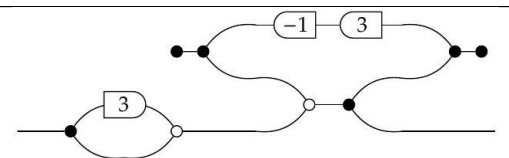
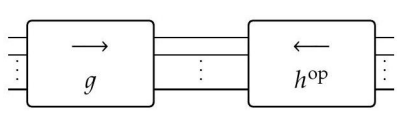
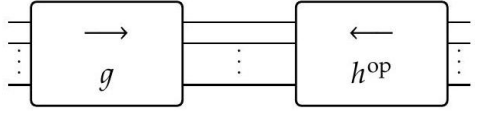
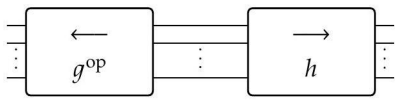
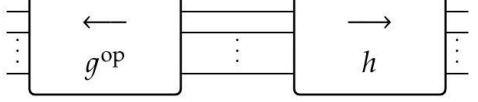
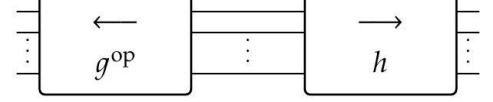
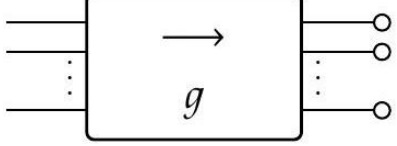
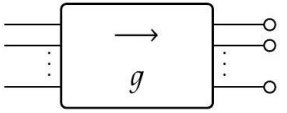
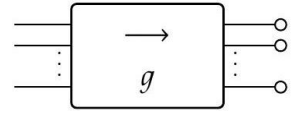
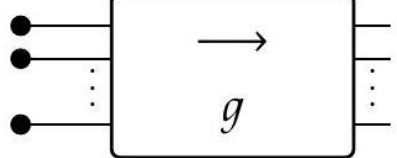
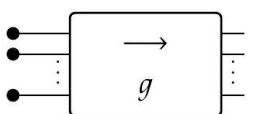
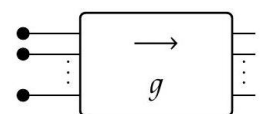
fong-spivak-seven-sketches-153-0182 — (dup)	153	7f1e85c6-4350-401f-a960-eb8a0176e786-153_176_208_1814_710.jpg			
fong-spivak-seven-sketches-153-0183 — (dup)	153	7f1e85c6-4350-401f-a960-eb8a0176e786-153_174_201_1814_1204.jpg			
fong-spivak-seven-sketches-153-0184 — (dup)	153	7f1e85c6-4350-401f-a960-eb8a0176e786-153_333_502_2158_437.jpg			
fong-spivak-seven-sketches-153-0185 — (dup)	153	7f1e85c6-4350-401f-a960-eb8a0176e786-153_335_502_2156_1181.jpg			
fong-spivak-seven-sketches-154-0186 — (dup)	154	7f1e85c6-4350-401f-a960-eb8a0176e786-154_290_480_1570_817.jpg			
fong-spivak-seven-sketches-154-0187 — (dup)	154	7f1e85c6-4350-401f-a960-eb8a0176e786-154_285_477_2145_821.jpg			

fong-spivak-seven-sketches-155-0198 — (dup)	155	7f1e85c6-4350-401f-a960-eb8a0176e786-155_337_1176_317_471.jpg			
fong-spivak-seven-sketches-156-0199 — (dup)	156	7f1e85c6-4350-401f-a960-eb8a0176e786-156_328_324_810_880.jpg			
fong-spivak-seven-sketches-160-0190 — (dup)	160	7f1e85c6-4350-401f-a960-eb8a0176e786-160_186_767_337_676.jpg			
fong-spivak-seven-sketches-160-0191 — (dup)	160	7f1e85c6-4350-401f-a960-eb8a0176e786-160_224_769_1100_670.jpg			
fong-spivak-seven-sketches-162-0192 — (dup)	162	7f1e85c6-4350-401f-a960-eb8a0176e786-162_192_371_2294_656.jpg			
fong-spivak-seven-sketches-162-0193 — (dup)	162	7f1e85c6-4350-401f-a960-eb8a0176e786-162_188_400_2299_1061.jpg			
fong-spivak-seven-sketches-164-0194 — (dup)	164	7f1e85c6-4350-401f-a960-eb8a0176e786-164_434_860_330_629.jpg			

fong-spivak-seven-sketches_014_0195 — (dup)	164	7f1e85c6-4350-401f-a960-eb8a0176e786-164_376_1186_1109_466.jpg			
fong-spivak-seven-sketches_014_0196 — (dup)	164	7f1e85c6-4350-401f-a960-eb8a0176e786-164_285_480_1640_819.jpg			
fong-spivak-seven-sketches_014_0197 — (dup)	165	7f1e85c6-4350-401f-a960-eb8a0176e786-165_167_208_905_955.jpg			
fong-spivak-seven-sketches_014_0198 — (dup)	168	7f1e85c6-4350-401f-a960-eb8a0176e786-168_317_810_751_654.jpg			
fong-spivak-seven-sketches_014_0199 — (dup)	170	7f1e85c6-4350-401f-a960-eb8a0176e786-170_317_810_491_654.jpg			
fong-spivak-seven-sketches_014_0200 — (dup)	172	7f1e85c6-4350-401f-a960-eb8a0176e786-172_43_208_2197_955.jpg			
fong-spivak-seven-sketches_014_0201 — (dup)	173	7f1e85c6-4350-401f-a960-eb8a0176e786-173_63_769_385_674.jpg			
fong-spivak-seven-sketches_014_0202 — (dup)	173	7f1e85c6-4350-401f-a960-eb8a0176e786-173_66_115_593_1002.jpg			
fong-spivak-seven-sketches_014_0203 — (dup)	173	7f1e85c6-4350-401f-a960-eb8a0176e786-173_34_72_943_995.jpg			

fong-spivak-seven-sketches_710_0204 — (dup)	173	7f1e85c6-4350-401f-a960-eb8a0176e786-173_61_113_1217_1002.jpg			
fong-spivak-seven-sketches_710_0205 — (dup)	173	7f1e85c6-4350-401f-a960-eb8a0176e786-173_27_72_1439_1050.jpg			
fong-spivak-seven-sketches_710_0206 — (dup)	173	7f1e85c6-4350-401f-a960-eb8a0176e786-173_188_767_1855_676.jpg			
fong-spivak-seven-sketches_710_0207 — (dup)	174	7f1e85c6-4350-401f-a960-eb8a0176e786-174_204_631_729_744.jpg			
fong-spivak-seven-sketches_710_0208 — (dup)	174	7f1e85c6-4350-401f-a960-eb8a0176e786-174_290_477_1955_821.jpg			
fong-spivak-seven-sketches_710_0209 — (dup)	175	7f1e85c6-4350-401f-a960-eb8a0176e786-175_355_1075_1072_523.jpg			
fong-spivak-seven-sketches_710_0210 — (dup)	175	7f1e85c6-4350-401f-a960-eb8a0176e786-175_90_584_2005_769.jpg			
fong-spivak-seven-sketches_710_0211 — (dup)	177	7f1e85c6-4350-401f-a960-eb8a0176e786-177_186_769_452_674.jpg			
fong-spivak-seven-sketches_710_0212 — (dup)	177	7f1e85c6-4350-401f-a960-eb8a0176e786-177_172_611_1009_751.jpg	$\begin{array}{c cc} & a & b \\ \hline x & 15 & 3 \\ y & 0 & 21 \end{array} \quad \begin{pmatrix} 15 & 3 \\ 0 & 21 \end{pmatrix}$	$\begin{array}{c cc} & a & b \\ \hline x & 15 & 3 \\ y & 0 & 21 \end{array} \quad \begin{pmatrix} 15 & 3 \\ 0 & 21 \end{pmatrix}$	$\begin{array}{c cc} & a & b \\ \hline x & 15 & 3 \\ y & 0 & 21 \end{array} \quad \begin{pmatrix} 15 & 3 \\ 0 & 21 \end{pmatrix}$
fong-spivak-seven-sketches_710_0213 — (dup)	179	7f1e85c6-4350-401f-a960-eb8a0176e786-179_170_853_337_633.jpg			

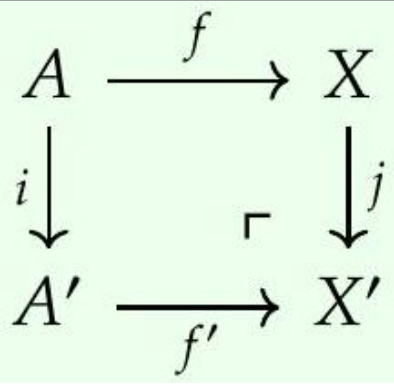
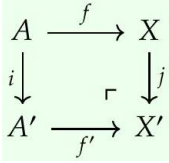
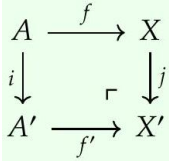
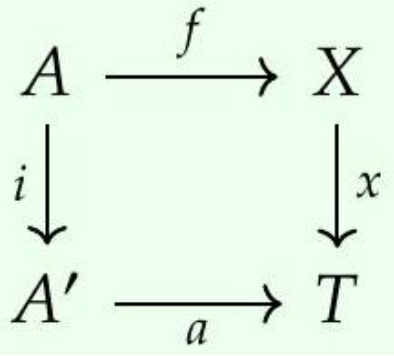
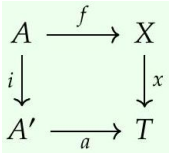
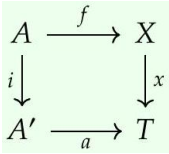
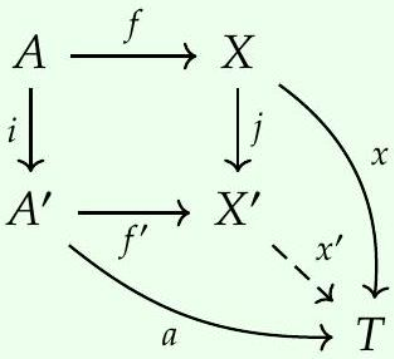
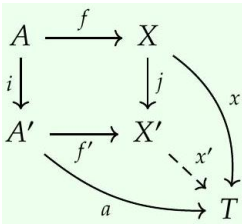
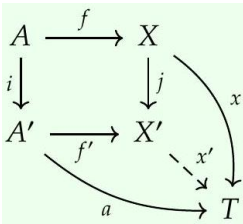
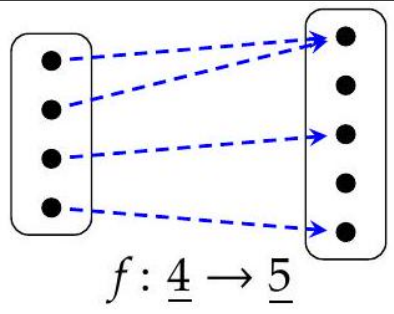
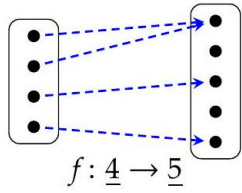
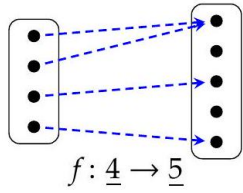
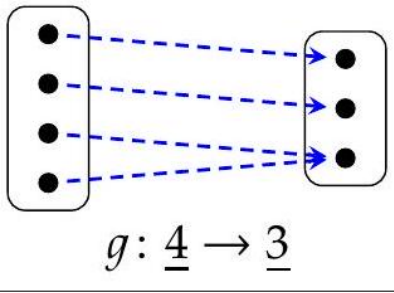
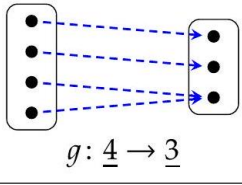
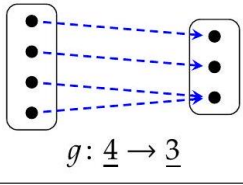
fong-spivak-seven-sketches-010_0214 — (dup)	179	7f1e85c6-4350-401f-a960-eb8a0176e786-179_344_507_1434_805.jpg			
fong-spivak-seven-sketches-010_0215 — (dup)	179	7f1e85c6-4350-401f-a960-eb8a0176e786-179_88_186_2328_1014.jpg			
fong-spivak-seven-sketches-010_0216 — (dup)	180	7f1e85c6-4350-401f-a960-eb8a0176e786-180_88_188_337_1014.jpg			
fong-spivak-seven-sketches-010_0217 — (dup)	181	7f1e85c6-4350-401f-a960-eb8a0176e786-181_244_369_1138_876.jpg			
fong-spivak-seven-sketches-010_0218 — (dup)	182	7f1e85c6-4350-401f-a960-eb8a0176e786-182_916_1396_663_364.jpg			
fong-spivak-seven-sketches-010_0219 — (dup)	183	7f1e85c6-4350-401f-a960-eb8a0176e786-183_172_428_520_636.jpg			
fong-spivak-seven-sketches-010_0220 — (dup)	183	7f1e85c6-4350-401f-a960-eb8a0176e786-183_32_156_548_1195.jpg			
fong-spivak-seven-sketches-010_0221 — (dup)	183	7f1e85c6-4350-401f-a960-eb8a0176e786-183_48_283_611_1197.jpg			

fong-spivak-seven-sketches-0222 — (dup)	183	7f1e85c6-4350-401f-a960-eb8a0176e786-183_502_889_864_615.jpg			
fong-spivak-seven-sketches-0223 — (dup)	183	7f1e85c6-4350-401f-a960-eb8a0176e786-183_208_1262_1873_382.jpg			
fong-spivak-seven-sketches-0224 — (dup)	184	7f1e85c6-4350-401f-a960-eb8a0176e786-184_75_344_2115_441.jpg			
fong-spivak-seven-sketches-0225 — (dup)	184	7f1e85c6-4350-401f-a960-eb8a0176e786-184_61_342_2217_441.jpg			
fong-spivak-seven-sketches-0226 — (dup)	184	7f1e85c6-4350-401f-a960-eb8a0176e786-184_57_342_2303_441.jpg			
fong-spivak-seven-sketches-0227 — (dup)	186	7f1e85c6-4350-401f-a960-eb8a0176e786-186_66_115_1061_1002.jpg			
fong-spivak-seven-sketches-0228 — (dup)	187	7f1e85c6-4350-401f-a960-eb8a0176e786-187_206_633_351_742.jpg			
fong-spivak-seven-sketches-0229 — (dup)	188	7f1e85c6-4350-401f-a960-eb8a0176e786-188_149_615_389_751.jpg			
fong-spivak-seven-sketches-0230 — (dup)	188	7f1e85c6-4350-401f-a960-eb8a0176e786-188_147_613_919_753.jpg			
fong-spivak-seven-sketches-0231 — (dup)	188	7f1e85c6-4350-401f-a960-eb8a0176e786-188_145_362_1805_928.jpg			
fong-spivak-seven-sketches-0232 — (dup)	188	7f1e85c6-4350-401f-a960-eb8a0176e786-188_145_321_2145_946.jpg			

fong-spivak-seven-sketches-189-0233 — (dup)	189	7f1e85c6-4350-401f-a960-eb8a0176e786-189_82_119_1700_756.jpg			
fong-spivak-seven-sketches-189-0234 — (dup)	189	7f1e85c6-4350-401f-a960-eb8a0176e786-189_82_122_1700_1240.jpg			
fong-spivak-seven-sketches-189-0235 — (dup)	189	7f1e85c6-4350-401f-a960-eb8a0176e786-189_154_318_2209_478.jpg			
fong-spivak-seven-sketches-189-0236 — (dup)	189	7f1e85c6-4350-401f-a960-eb8a0176e786-189_154_313_2212_938.jpg			
fong-spivak-seven-sketches-189-0237 — (dup)	189	7f1e85c6-4350-401f-a960-eb8a0176e786-189_154_253_2212_1390.jpg			
fong-spivak-seven-sketches-190-0238 — (dup)	190	7f1e85c6-4350-401f-a960-eb8a0176e786-190_88_274_520_606.jpg			
fong-spivak-seven-sketches-190-0239 — (dup)	190	7f1e85c6-4350-401f-a960-eb8a0176e786-190_86_271_520_1240.jpg			
fong-spivak-seven-sketches-190-0240 — (dup)	190	7f1e85c6-4350-401f-a960-eb8a0176e786-190_281_860_1231_629.jpg			

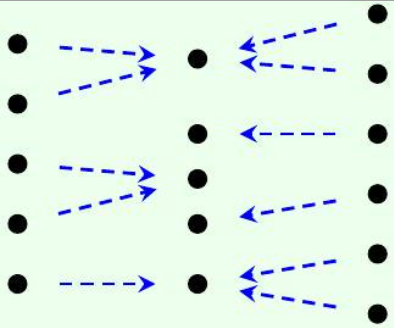
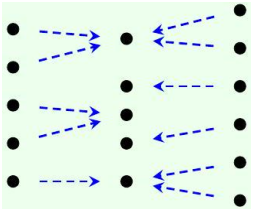
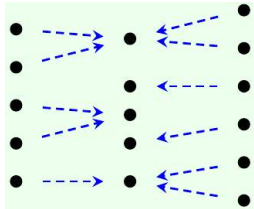
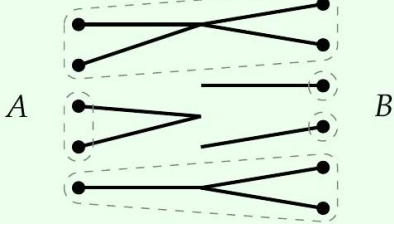
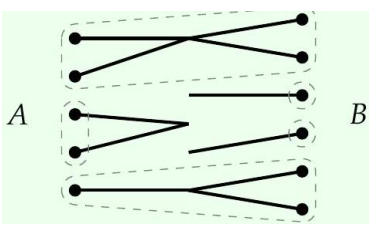
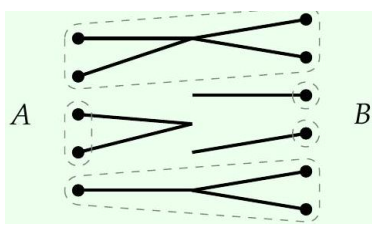
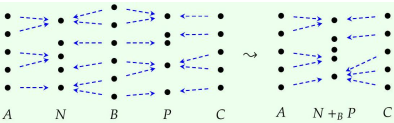
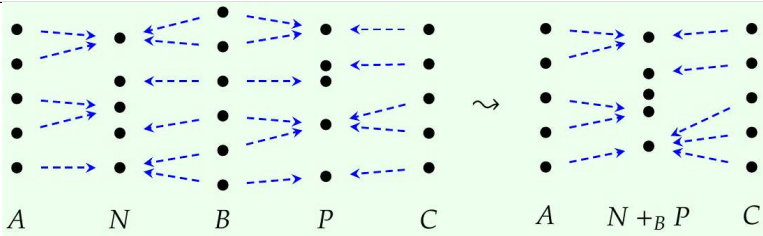
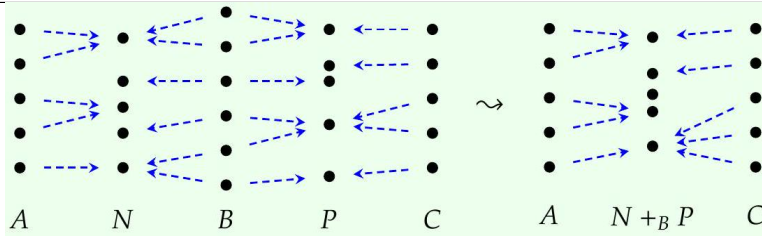
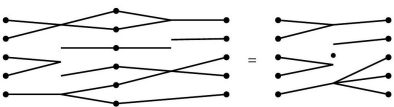
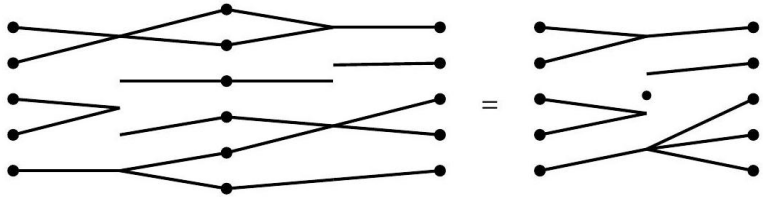
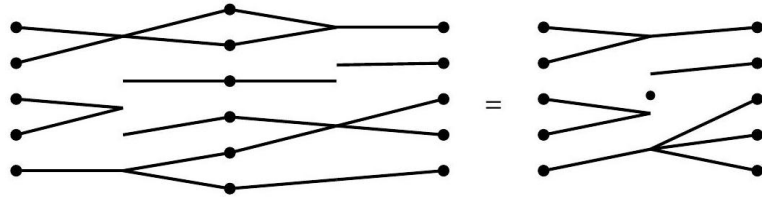
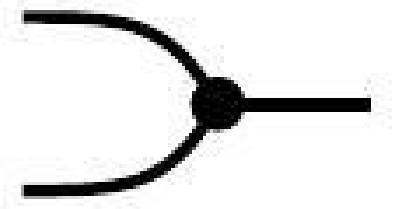
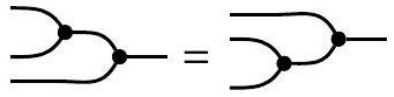
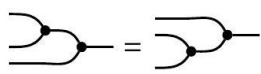
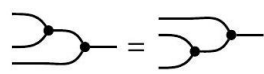
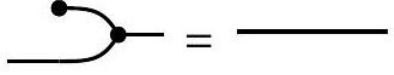
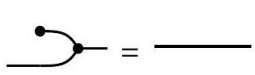
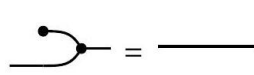
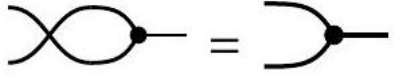
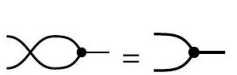
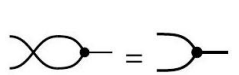
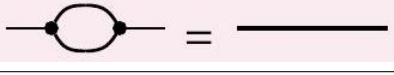
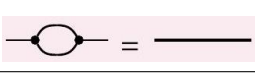
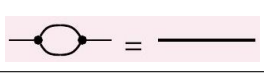
fong-spivak-seven-sketches_014_0241 — (dup)	193	7f1e85c6-4350-401f-a960-eb8a0176e786-193_583_995_1442_569.jpg			
fong-spivak-seven-sketches_014_0242 — (dup)	194	7f1e85c6-4350-401f-a960-eb8a0176e786-194_163_771_993_674.jpg			
fong-spivak-seven-sketches_014_0243 — (dup)	194	7f1e85c6-4350-401f-a960-eb8a0176e786-194_158_814_1769_654.jpg			
fong-spivak-seven-sketches_014_0244 — (dup)	194	7f1e85c6-4350-401f-a960-eb8a0176e786-194_154_817_2213_652.jpg			
fong-spivak-seven-sketches_014_0245 — (dup)	195	7f1e85c6-4350-401f-a960-eb8a0176e786-195_158_602_385_758.jpg			
fong-spivak-seven-sketches_014_0246 — (dup)	195	7f1e85c6-4350-401f-a960-eb8a0176e786-195_81_115_2176_1002.jpg			
fong-spivak-seven-sketches_014_0247 — (dup)	196	7f1e85c6-4350-401f-a960-eb8a0176e786-196_122_455_387_833.jpg			
fong-spivak-seven-sketches_014_0248 — (dup)	198	7f1e85c6-4350-401f-a960-eb8a0176e786-198_208_410_1509_853.jpg			

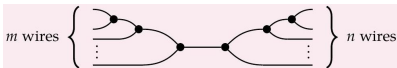
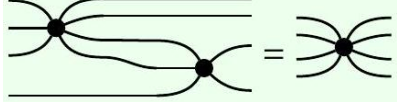
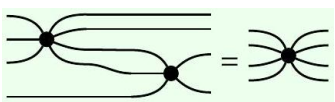
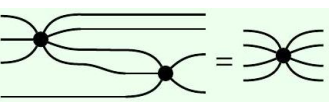
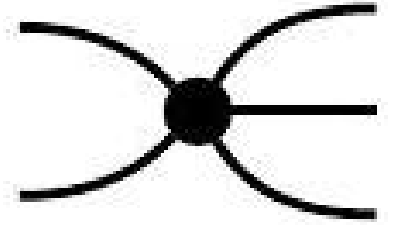
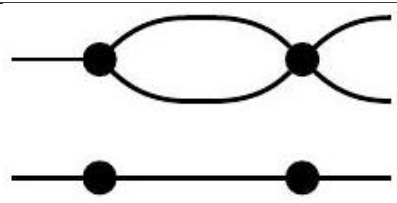
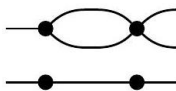
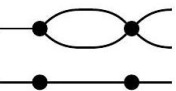
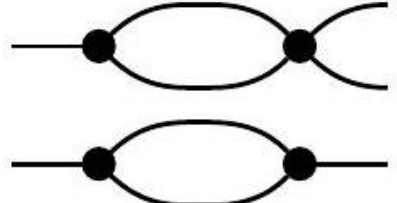
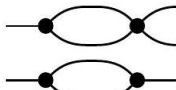
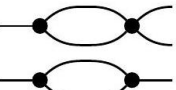
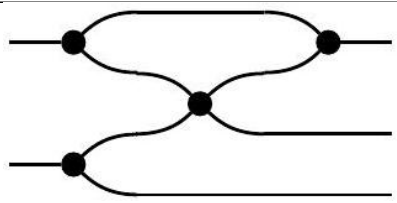
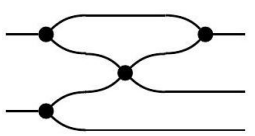
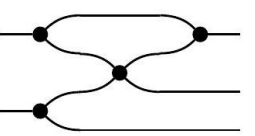
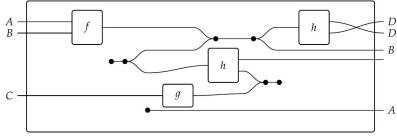
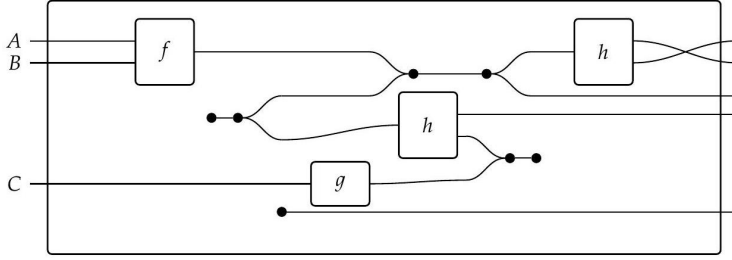
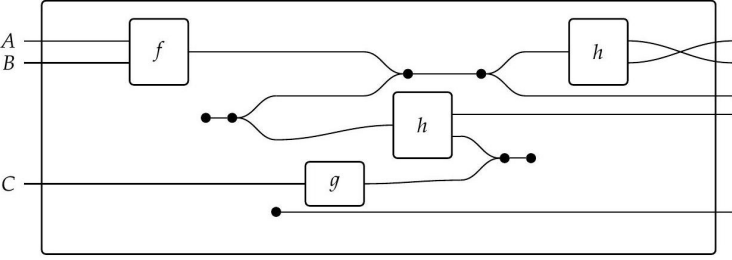
fong-spivak-seven-sketches-01_0249 — (dup)	199	7f1e85c6-4350-401f-a960-eb8a0176e786-199_505_968_337_575.jpg			
fong-spivak-seven-sketches-01_0250 — (dup)	200	7f1e85c6-4350-401f-a960-eb8a0176e786-200_186_181_1014_964.jpg			
fong-spivak-seven-sketches-01_0251 — (dup)	200	7f1e85c6-4350-401f-a960-eb8a0176e786-200_199_294_1439_955.jpg			
fong-spivak-seven-sketches-01_0252 — (dup)	200	7f1e85c6-4350-401f-a960-eb8a0176e786-200_204_215_1805_1000.jpg			
fong-spivak-seven-sketches-01_0253 — (dup)	200	7f1e85c6-4350-401f-a960-eb8a0176e786-200_292_396_2136_905.jpg			

fong-spivak-seven-sketches-011_0254 — (dup)	201	7f1e85c6-4350-401f-a960-eb8a0176e786-201_210_217_878_950.jpg			
fong-spivak-seven-sketches-011_0255 — (dup)	201	7f1e85c6-4350-401f-a960-eb8a0176e786-201_197_217_1222_955.jpg			
fong-spivak-seven-sketches-011_0256 — (dup)	201	7f1e85c6-4350-401f-a960-eb8a0176e786-201_287_314_1588_903.jpg			
fong-spivak-seven-sketches-011_0257 — (dup)	202	7f1e85c6-4350-401f-a960-eb8a0176e786-202_251_317_588_570.jpg			
fong-spivak-seven-sketches-011_0258 — (dup)	202	7f1e85c6-4350-401f-a960-eb8a0176e786-202_224_314_609_1233.jpg			

fong-spivak-seven-sketches-014_0259 — (dup)	202	7f1e85c6-4350-401f-a960-eb8a0176e786-202_201_278_1348_679.jpg	$ \begin{array}{ccc} \emptyset & \xrightarrow{f} & X \\ g \downarrow & & \downarrow \iota_X \\ Y & \xrightarrow{\iota_Y} & X + Y \end{array} $	$ \begin{array}{ccc} \emptyset & \xrightarrow{f} & X \\ g \downarrow & & \downarrow \iota_X \\ Y & \xrightarrow{\iota_Y} & X + Y \end{array} $	$ \begin{array}{ccc} \emptyset & \xrightarrow{f} & X \\ g \downarrow & & \downarrow \iota_X \\ Y & \xrightarrow{\iota_Y} & X + Y \end{array} $
fong-spivak-seven-sketches-014_0260 — (dup)	202	7f1e85c6-4350-401f-a960-eb8a0176e786-202_204_217_1348_1222.jpg	$ \begin{array}{ccc} \emptyset & \xrightarrow{f} & X \\ g \downarrow & & \downarrow x \\ Y & \xrightarrow{y} & T \end{array} $	$ \begin{array}{ccc} \emptyset & \xrightarrow{f} & X \\ g \downarrow & & \downarrow x \\ Y & \xrightarrow{y} & T \end{array} $	$ \begin{array}{ccc} \emptyset & \xrightarrow{f} & X \\ g \downarrow & & \downarrow x \\ Y & \xrightarrow{y} & T \end{array} $
fong-spivak-seven-sketches-014_0261 — (dup)	203	7f1e85c6-4350-401f-a960-eb8a0176e786-203_278_1129_342_491.jpg			
fong-spivak-seven-sketches-014_0262 — (dup)	203	7f1e85c6-4350-401f-a960-eb8a0176e786-203_308_609_2000_355.jpg			
fong-spivak-seven-sketches-014_0263 — (dup)	203	7f1e85c6-4350-401f-a960-eb8a0176e786-203_310_613_1998_1154.jpg			
fong-spivak-seven-sketches-014_0264 — (dup)	204	7f1e85c6-4350-401f-a960-eb8a0176e786-204_287_347_1666_884.jpg	$ \begin{array}{ccc} & B & \longrightarrow Z \\ & \downarrow & \\ A & \longrightarrow & C \\ \downarrow & & \\ D & & \end{array} $	$ \begin{array}{ccc} & B & \longrightarrow Z \\ & \downarrow & \\ A & \longrightarrow & C \\ \downarrow & & \\ D & & \end{array} $	$ \begin{array}{ccc} & B & \longrightarrow Z \\ & \downarrow & \\ A & \longrightarrow & C \\ \downarrow & & \\ D & & \end{array} $

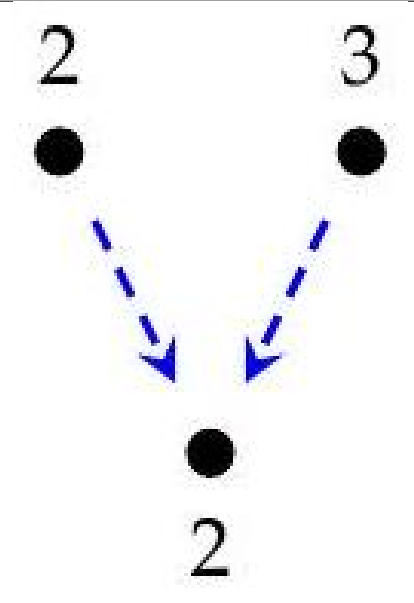
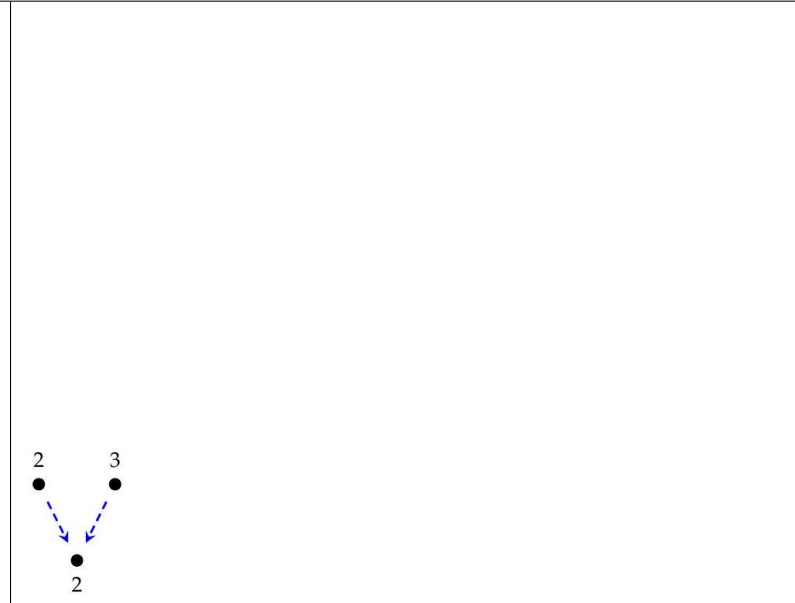
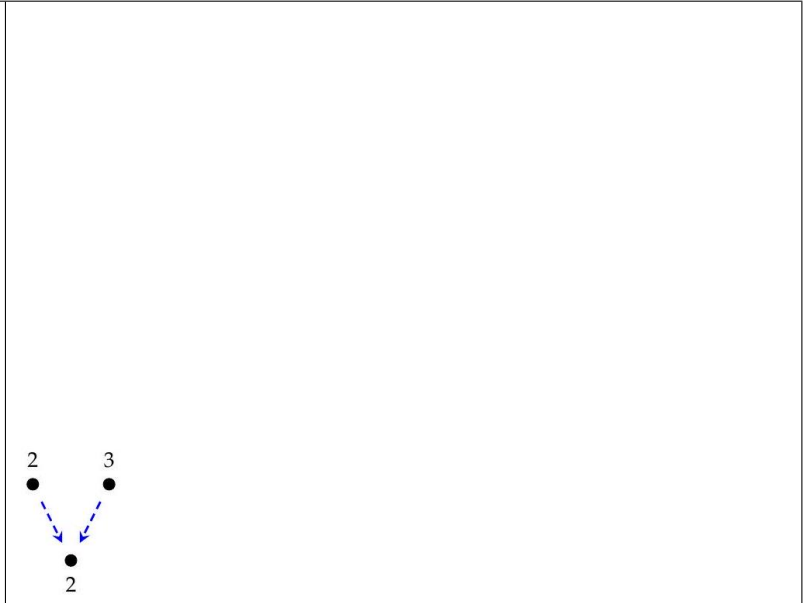
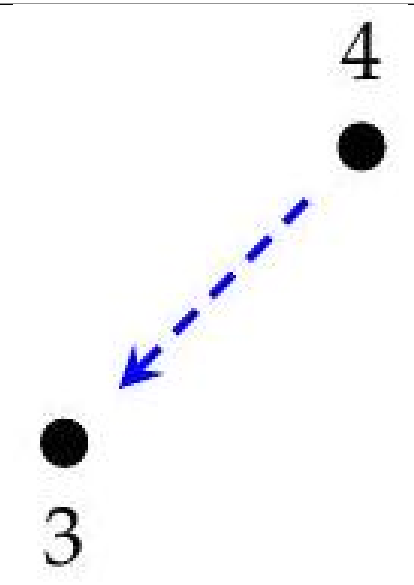
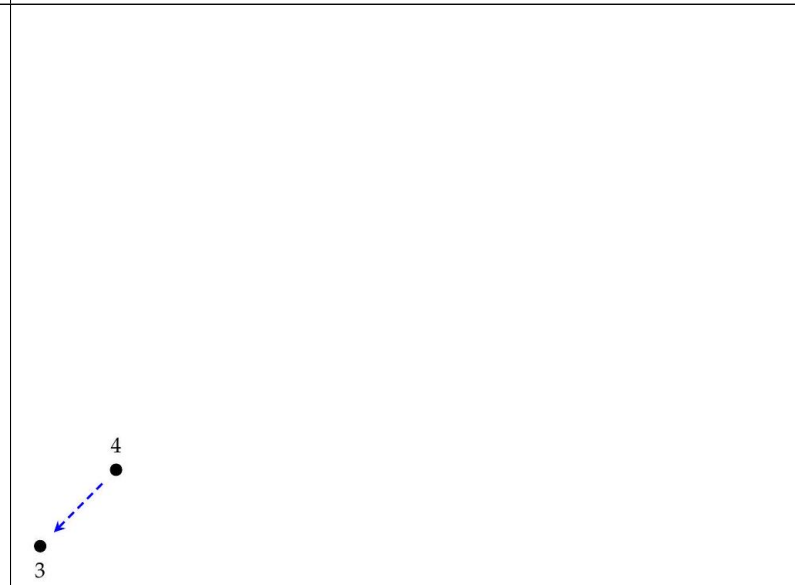
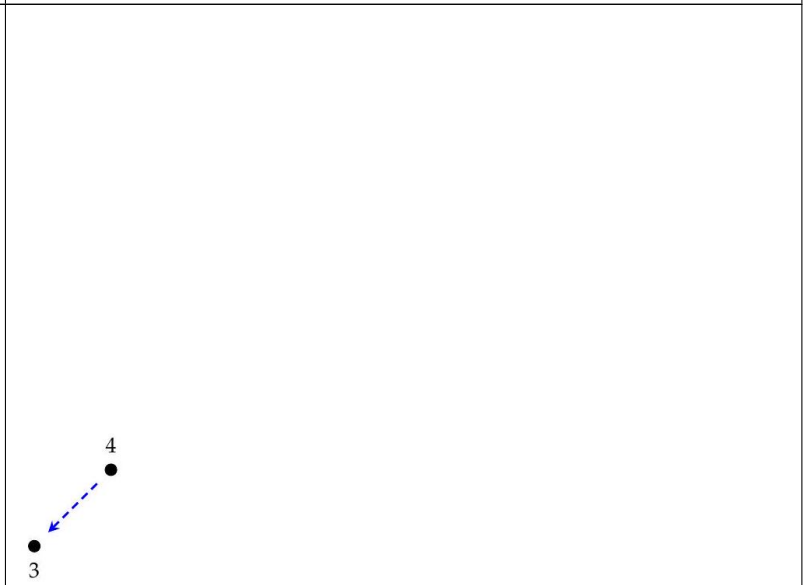
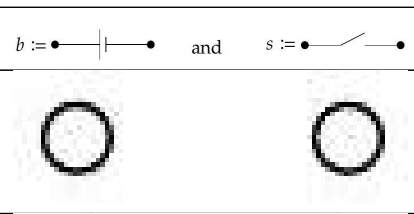
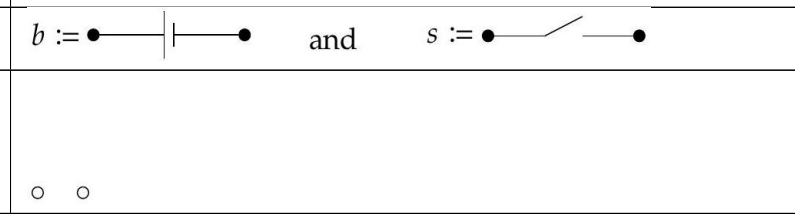
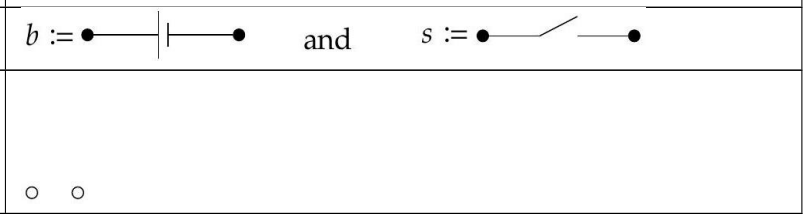
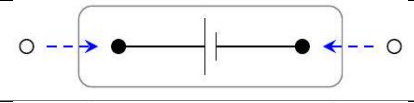
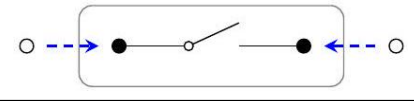
fong-spivak-seven-sketches-0265 — (dup)	204	7f1e85c6-4350-401f-a960-eb8a0176e786-204_293_350_2161_572.jpg			
fong-spivak-seven-sketches-0266 — (dup)	204	7f1e85c6-4350-401f-a960-eb8a0176e786-204_293_353_2161_1194.jpg			
fong-spivak-seven-sketches-0267 — (dup)	206	7f1e85c6-4350-401f-a960-eb8a0176e786-206_158_817_1321_652.jpg			
fong-spivak-seven-sketches-0268 — (dup)	207	7f1e85c6-4350-401f-a960-eb8a0176e786-207_131_351_394_593.jpg			
fong-spivak-seven-sketches-0269 — (dup)	207	7f1e85c6-4350-401f-a960-eb8a0176e786-207_131_337_394_1183.jpg			
fong-spivak-seven-sketches-0270 — (dup)	207	7f1e85c6-4350-401f-a960-eb8a0176e786-207_133_652_699_733.jpg			
fong-spivak-seven-sketches-0271 — (dup)	207	7f1e85c6-4350-401f-a960-eb8a0176e786-207_231_1276_1027_439.jpg			
fong-spivak-seven-sketches-0272 — (dup)	207	7f1e85c6-4350-401f-a960-eb8a0176e786-207_156_260_1993_661.jpg			
fong-spivak-seven-sketches-0273 — (dup)	207	7f1e85c6-4350-401f-a960-eb8a0176e786-207_152_258_1995_1195.jpg			

fong-spivak-seven-sketches_110_0274 — (dup)	208	7f1e85c6-4350-401f-a960-eb8a0176e786-208_269_324_1328_520.jpg			
fong-spivak-seven-sketches_110_0275 — (dup)	208	7f1e85c6-4350-401f-a960-eb8a0176e786-208_276_477_1355_1122.jpg			
fong-spivak-seven-sketches_110_0276 — (dup)	209	7f1e85c6-4350-401f-a960-eb8a0176e786-209_333_1095_376_466.jpg			
fong-spivak-seven-sketches_110_0277 — (dup)	209	7f1e85c6-4350-401f-a960-eb8a0176e786-209_276_1057_1133_532.jpg			
fong-spivak-seven-sketches_110_0278 — (dup)	210	7f1e85c6-4350-401f-a960-eb8a0176e786-210_63_113_536_824.jpg			
fong-spivak-seven-sketches_110_0279 — (dup)	210	7f1e85c6-4350-401f-a960-eb8a0176e786-210_36_143_554_1204.jpg			
fong-spivak-seven-sketches_110_0280 — (dup)	210	7f1e85c6-4350-401f-a960-eb8a0176e786-210_84_342_817_425.jpg			
fong-spivak-seven-sketches_110_0281 — (dup)	210	7f1e85c6-4350-401f-a960-eb8a0176e786-210_66_330_828_869.jpg			
fong-spivak-seven-sketches_110_0282 — (dup)	210	7f1e85c6-4350-401f-a960-eb8a0176e786-210_61_294_828_1308.jpg			
fong-spivak-seven-sketches_110_0283 — (dup)	210	7f1e85c6-4350-401f-a960-eb8a0176e786-210_97_523_2163_579.jpg			
fong-spivak-seven-sketches_110_0284 — (dup)	210	7f1e85c6-4350-401f-a960-eb8a0176e786-210_59_330_2181_1208.jpg			

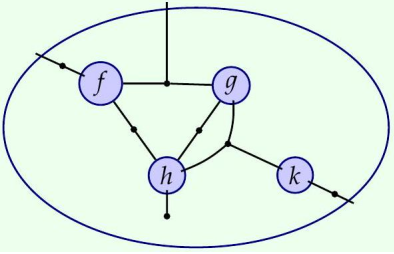
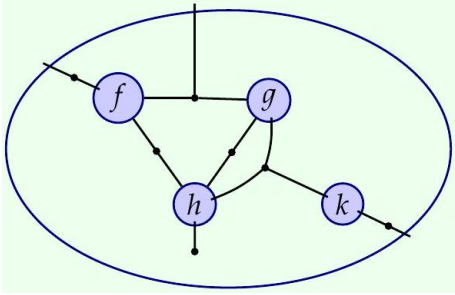
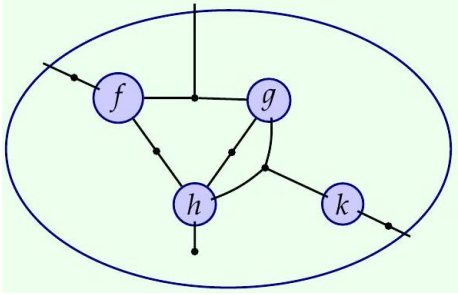
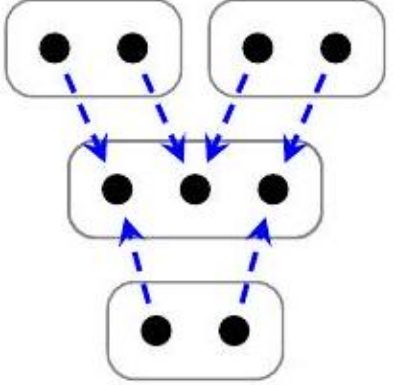
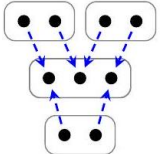
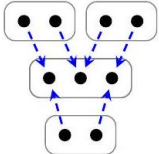
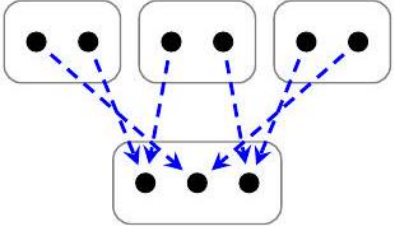
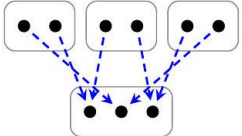
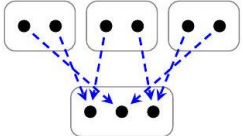
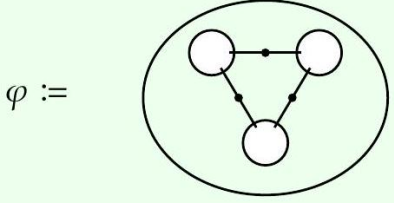
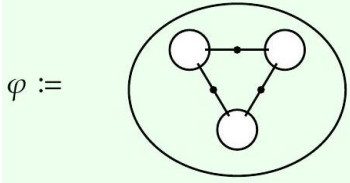
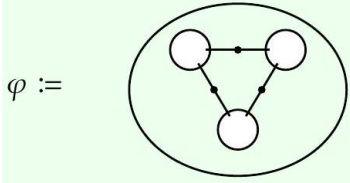
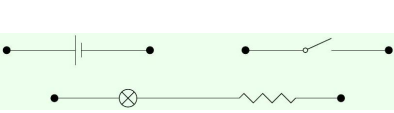
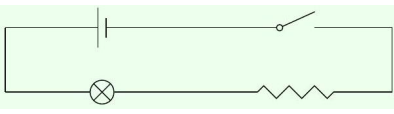
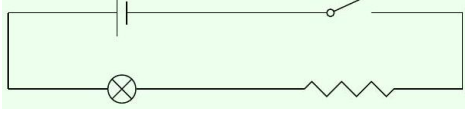
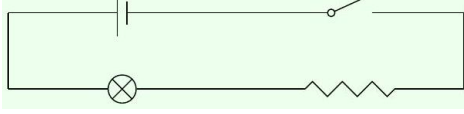
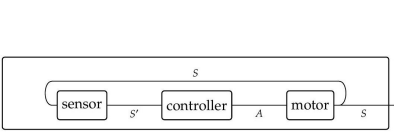
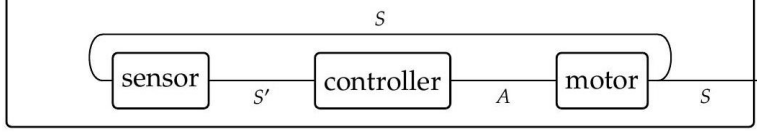
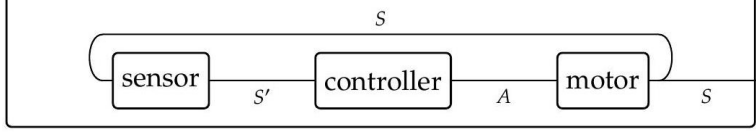
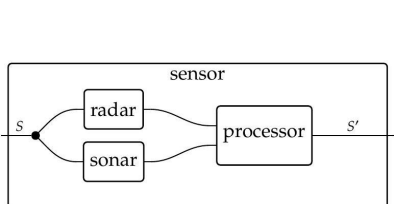
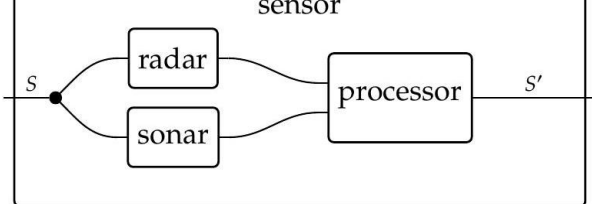
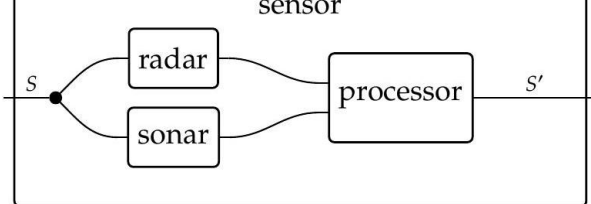
fong-spivak-seven-sketches_114_0285 — (dup)	211		7f1e85c6-4350-401f-a960-eb8a0176e786-211_165_980_636_568.jpg		m wires $\left\{ \begin{array}{c} \text{AND gate} \\ \vdots \end{array} \right\} \rightarrow \left\{ \begin{array}{c} \text{OR gate} \\ \vdots \end{array} \right\} n$ wires	m wires $\left\{ \begin{array}{c} \text{AND gate} \\ \vdots \end{array} \right\} \rightarrow \left\{ \begin{array}{c} \text{OR gate} \\ \vdots \end{array} \right\} n$ wires
fong-spivak-seven-sketches_114_0286 — (dup)	211		7f1e85c6-4350-401f-a960-eb8a0176e786-211_122_432_2095_844.jpg			
fong-spivak-seven-sketches_114_0287 — (dup)	211		7f1e85c6-4350-401f-a960-eb8a0176e786-211_72_115_2432_1050.jpg			
fong-spivak-seven-sketches_114_0288 — (dup)	212		7f1e85c6-4350-401f-a960-eb8a0176e786-212_75_195_287_1005.jpg			
fong-spivak-seven-sketches_114_0289 — (dup)	212		7f1e85c6-4350-401f-a960-eb8a0176e786-212_120_231_428_991.jpg			
fong-spivak-seven-sketches_114_0290 — (dup)	212		7f1e85c6-4350-401f-a960-eb8a0176e786-212_72_231_615_991.jpg			
fong-spivak-seven-sketches_114_0291 — (dup)	212		7f1e85c6-4350-401f-a960-eb8a0176e786-212_133_233_758_991.jpg			
fong-spivak-seven-sketches_114_0292 — (dup)	212		7f1e85c6-4350-401f-a960-eb8a0176e786-212_161_321_959_946.jpg			
fong-spivak-seven-sketches_114_0293 — (dup)	213		7f1e85c6-4350-401f-a960-eb8a0176e786-213_438_1285_660_415.jpg			

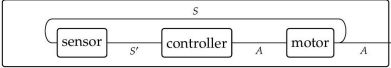
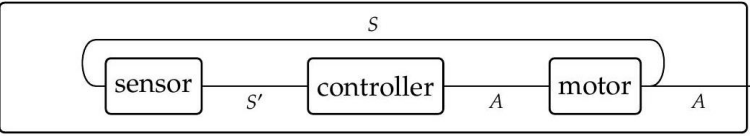
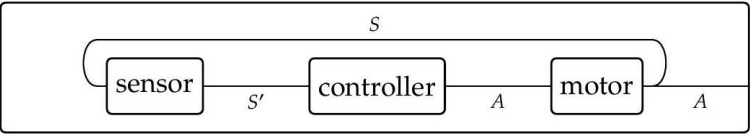
fong-spivak-seven-sketches_114_0294 — (dup)	214	7f1e85c6-4350-401f-a960-eb8a0176e786-214_484_1382_324_348.jpg			
fong-spivak-seven-sketches_114_0295 — (dup)	215	7f1e85c6-4350-401f-a960-eb8a0176e786-215_88_907_925_606.jpg			
fong-spivak-seven-sketches_114_0296 — (dup)	215	7f1e85c6-4350-401f-a960-eb8a0176e786-215_525_1294_1181_489.jpg			
fong-spivak-seven-sketches_114_0297 — (dup)	216	7f1e85c6-4350-401f-a960-eb8a0176e786-216_208_581_1851_765.jpg			
fong-spivak-seven-sketches_114_0298 — (dup)	217	7f1e85c6-4350-401f-a960-eb8a0176e786-217_351_475_378_824.jpg			
fong-spivak-seven-sketches_114_0299 — (dup)	217	7f1e85c6-4350-401f-a960-eb8a0176e786-217_425_925_1158_595.jpg			
fong-spivak-seven-sketches_114_0300 — (dup)	217	7f1e85c6-4350-401f-a960-eb8a0176e786-217_242_785_2104_667.jpg			
fong-spivak-seven-sketches_114_0301 — (dup)	218	7f1e85c6-4350-401f-a960-eb8a0176e786-218_283_1102_342_462.jpg			

fong-spivak-seven-sketches-218_0302 — (dup)	218	7f1e85c6-4350-401f-a960-eb8a0176e786-218_335_1140_819_443.jpg			
fong-spivak-seven-sketches-219_0303 — (dup)	219	7f1e85c6-4350-401f-a960-eb8a0176e786-219_111_224_2382_948.jpg			
fong-spivak-seven-sketches-220_0304 — (dup)	220	7f1e85c6-4350-401f-a960-eb8a0176e786-220_109_226_1335_799.jpg			
fong-spivak-seven-sketches-220_0305 — (dup)	220	7f1e85c6-4350-401f-a960-eb8a0176e786-220_109_224_1335_1095.jpg			
fong-spivak-seven-sketches-220_0306 — (dup)	220	7f1e85c6-4350-401f-a960-eb8a0176e786-220_195_127_1561_799.jpg			

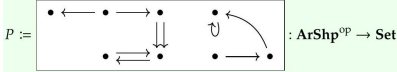
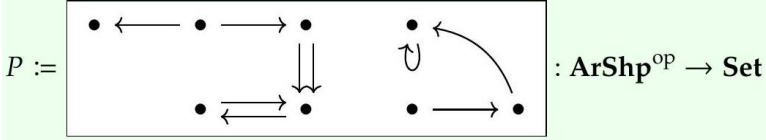
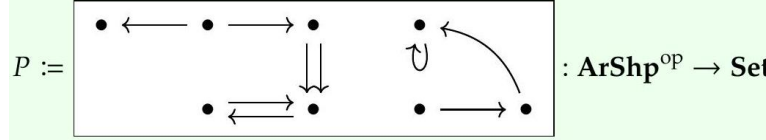
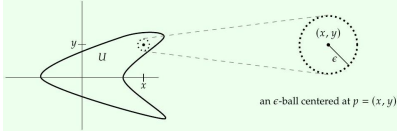
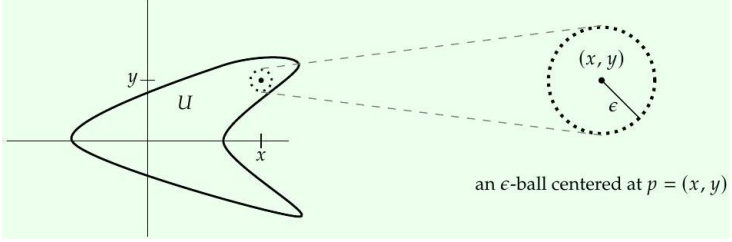
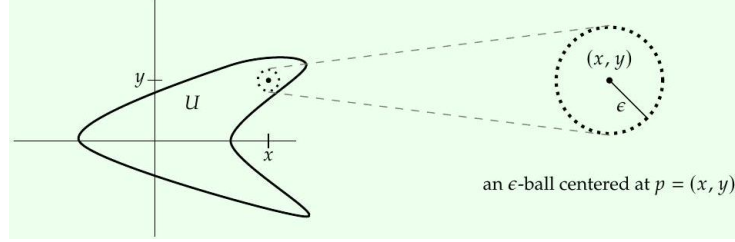
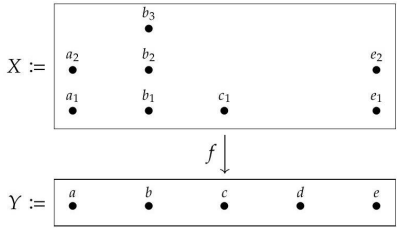
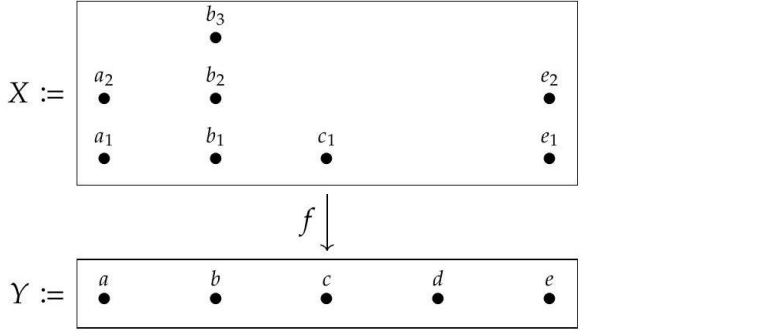
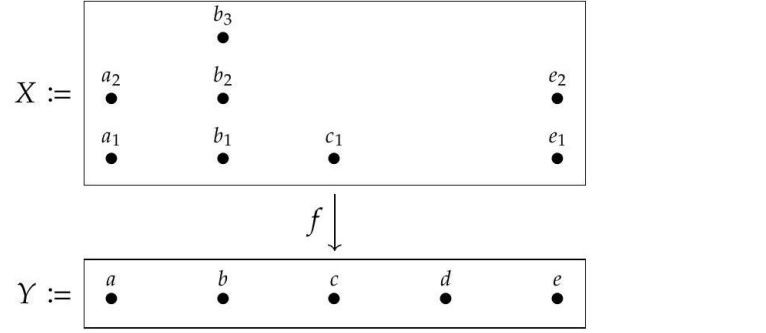
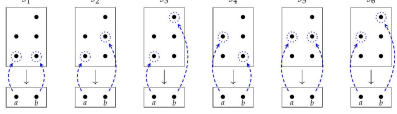
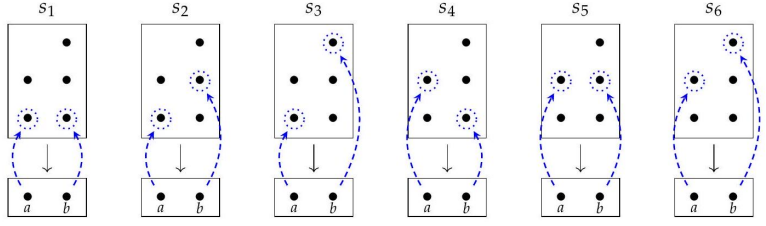
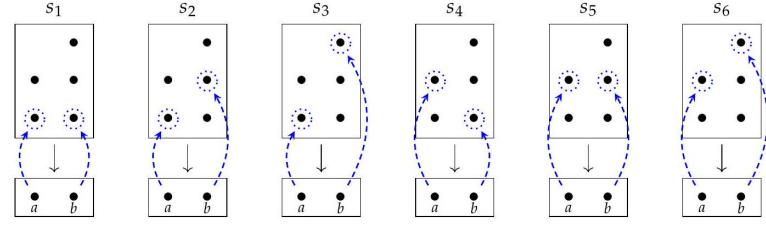
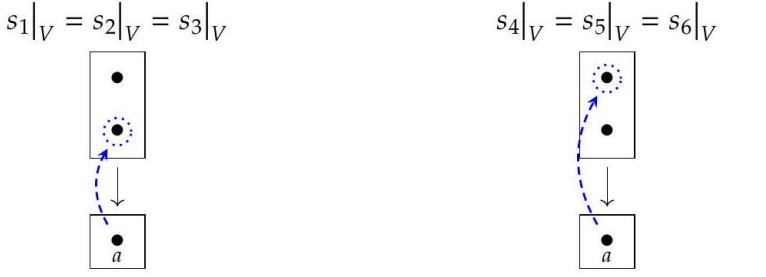
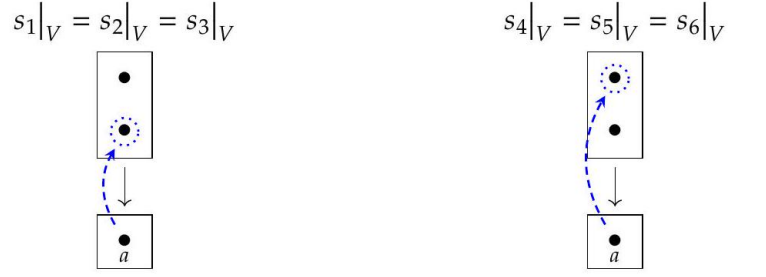
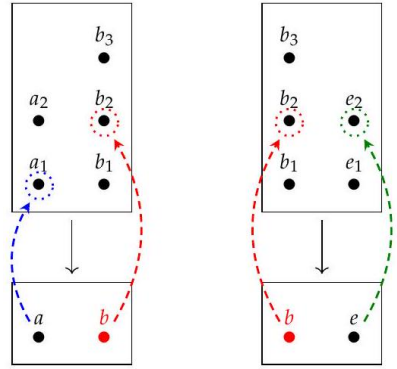
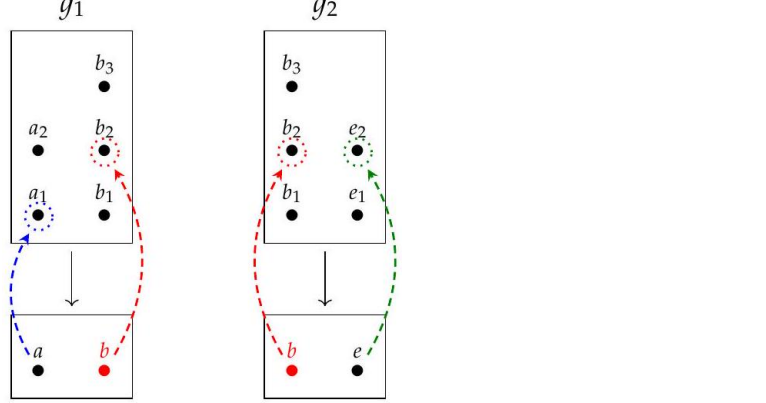
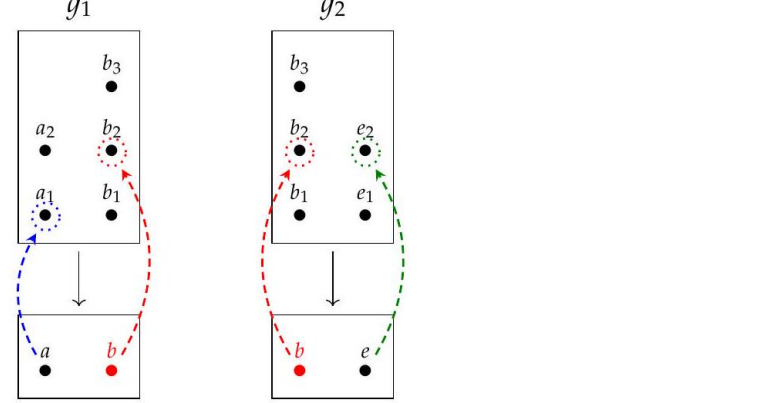
fong-spivak-seven-sketches_220_0307 — (dup)	220	7f1e85c6-4350-401f-a960-eb8a0176e786-220_195_129_1561_995.jpg			
fong-spivak-seven-sketches_220_0308 — (dup)	220	7f1e85c6-4350-401f-a960-eb8a0176e786-220_192_131_1563_1190.jpg			
fong-spivak-seven-sketches_220_0309 — (dup)	220	7f1e85c6-4350-401f-a960-eb8a0176e786-220_68_812_2312_661.jpg	$b := \bullet \text{---} \text{---} \bullet$ and $s := \bullet \text{---} \diagup \text{---} \bullet$	$b := \bullet \text{---} \text{---} \bullet$ and $s := \bullet \text{---} \diagup \text{---} \bullet$	$b := \bullet \text{---} \text{---} \bullet$ and $s := \bullet \text{---} \diagup \text{---} \bullet$
fong-spivak-seven-sketches_221_0310 — (dup)	221	7f1e85c6-4350-401f-a960-eb8a0176e786-221_29_86_715_1016.jpg			
fong-spivak-seven-sketches_221_0311 — (dup)	221	7f1e85c6-4350-401f-a960-eb8a0176e786-221_97_423_1186_848.jpg			
fong-spivak-seven-sketches_221_0312 — (dup)	221	7f1e85c6-4350-401f-a960-eb8a0176e786-221_97_421_1878_848.jpg			
fong-spivak-seven-sketches_221_0313 — (dup)	221	7f1e85c6-4350-401f-a960-eb8a0176e786-221_100_423_2407_414.jpg			
fong-spivak-seven-sketches_221_0314 — (dup)	221	7f1e85c6-4350-401f-a960-eb8a0176e786-221_97_419_2410_1283.jpg			

fong-spivak-seven-sketches-222_0315 — (dup)	222	7f1e85c6-4350-401f-a960-eb8a0176e786-222_262_817_378_649.jpg			
fong-spivak-seven-sketches-222_0316 — (dup)	222	7f1e85c6-4350-401f-a960-eb8a0176e786-222_95_421_751_848.jpg			
fong-spivak-seven-sketches-222_0317 — (dup)	222	7f1e85c6-4350-401f-a960-eb8a0176e786-222_72_615_1070_751.jpg			
fong-spivak-seven-sketches-222_0318 — (dup)	222	7f1e85c6-4350-401f-a960-eb8a0176e786-222_68_419_1421_851.jpg			
fong-spivak-seven-sketches-222_0319 — (dup)	222	7f1e85c6-4350-401f-a960-eb8a0176e786-222_95_615_1606_751.jpg			
fong-spivak-seven-sketches-222_0320 — (dup)	222	7f1e85c6-4350-401f-a960-eb8a0176e786-222_95_615_2407_751.jpg			
fong-spivak-seven-sketches-223_0321 — (dup)	223	7f1e85c6-4350-401f-a960-eb8a0176e786-223_100_618_285_749.jpg			
fong-spivak-seven-sketches-223_0322 — (dup)	223	7f1e85c6-4350-401f-a960-eb8a0176e786-223_197_620_486_749.jpg			
fong-spivak-seven-sketches-223_0323 — (dup)	223	7f1e85c6-4350-401f-a960-eb8a0176e786-223_129_362_962_579.jpg	$\eta := \emptyset$	$\eta := \emptyset$	$\eta := \emptyset$
fong-spivak-seven-sketches-223_0324 — (dup)	223	7f1e85c6-4350-401f-a960-eb8a0176e786-223_124_362_964_1176.jpg			
fong-spivak-seven-sketches-224_0325 — (dup)	224	7f1e85c6-4350-401f-a960-eb8a0176e786-224_438_1299_1237_409.jpg			
fong-spivak-seven-sketches-224_0326 — (dup)	224	7f1e85c6-4350-401f-a960-eb8a0176e786-224_375_594_2127_762.jpg			
fong-spivak-seven-sketches-226_0327 — (dup)	226	7f1e85c6-4350-401f-a960-eb8a0176e786-226_224_1133_495_493.jpg			

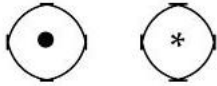
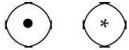
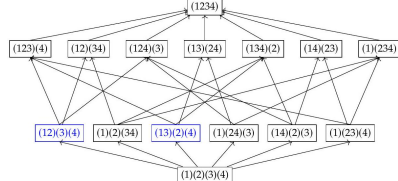
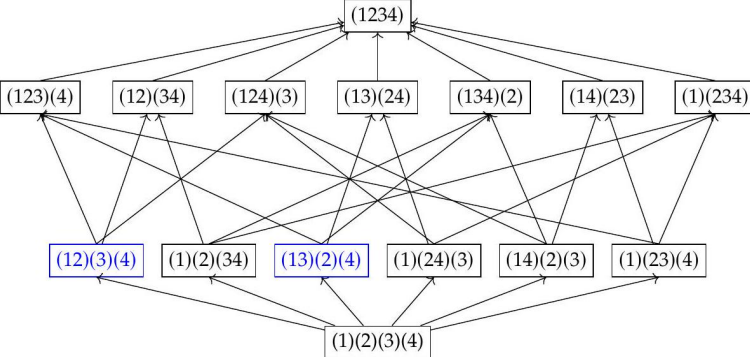
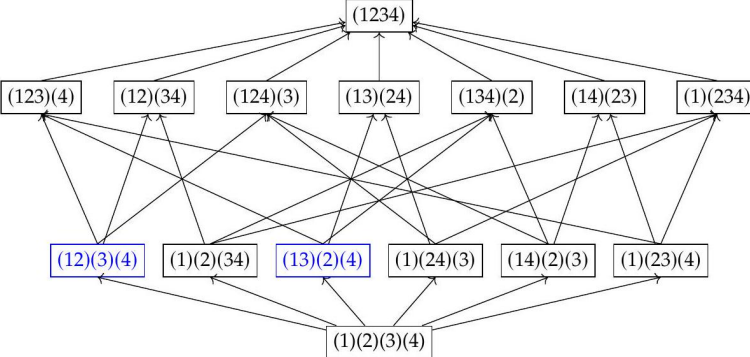
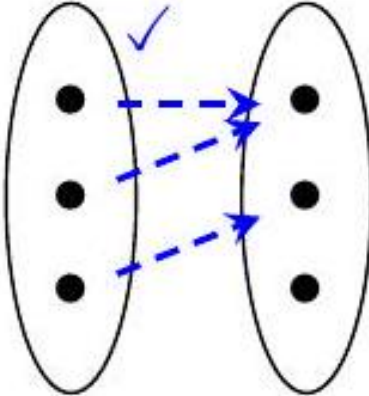
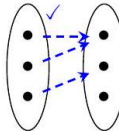
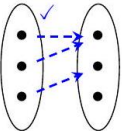
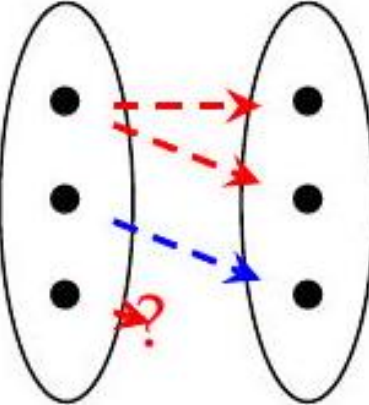
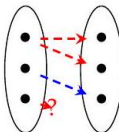
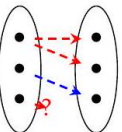
fong-spivak-seven-sketches_204_0328 — (dup)	227	7f1e85c6-4350-401f-a960-eb8a0176e786-227_382_593_1466_765.jpg			
fong-spivak-seven-sketches_204_0329 — (dup)	228	7f1e85c6-4350-401f-a960-eb8a0176e786-228_210_206_527_1005.jpg			
fong-spivak-seven-sketches_204_0330 — (dup)	228	7f1e85c6-4350-401f-a960-eb8a0176e786-228_186_310_980_952.jpg			
fong-spivak-seven-sketches_204_0331 — (dup)	229	7f1e85c6-4350-401f-a960-eb8a0176e786-229_242_455_1699_839.jpg	$\varphi :=$ 	$\varphi :=$ 	$\varphi :=$ 
fong-spivak-seven-sketches_204_0332 — (dup)	229	7f1e85c6-4350-401f-a960-eb8a0176e786-229_158_814_2312_652.jpg			
fong-spivak-seven-sketches_204_0333 — (dup)	230	7f1e85c6-4350-401f-a960-eb8a0176e786-230_159_603_333_756.jpg			
fong-spivak-seven-sketches_204_0334 — (dup)	234	7f1e85c6-4350-401f-a960-eb8a0176e786-234_204_1061_446_527.jpg			
fong-spivak-seven-sketches_204_0335 — (dup)	234	7f1e85c6-4350-401f-a960-eb8a0176e786-234_292_774_1305_672.jpg			

fong-spivak-seven-sketches_000_0336 — (dup)	236	7f1e85c6-4350-401f-a960-eb8a0176e786-236_190_1061_301_527.jpg			
fong-spivak-seven-sketches_000_0337 — (dup)	237	7f1e85c6-4350-401f-a960-eb8a0176e786-237_165_344_1532_602.jpg	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & & \downarrow & \lrcorner & \downarrow \\ D & \longrightarrow & E & \longrightarrow & F \end{array}$	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & & \downarrow & \lrcorner & \downarrow \\ D & \longrightarrow & E & \longrightarrow & F \end{array}$	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & & \downarrow & \lrcorner & \downarrow \\ D & \longrightarrow & E & \longrightarrow & F \end{array}$
fong-spivak-seven-sketches_000_0338 — (dup)	237	7f1e85c6-4350-401f-a960-eb8a0176e786-237_163_344_1534_1172.jpg	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & \lrcorner & \downarrow & & \downarrow \\ D & \longrightarrow & E & \longrightarrow & F \end{array}$	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & \lrcorner & \downarrow & & \downarrow \\ D & \longrightarrow & E & \longrightarrow & F \end{array}$	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & \lrcorner & \downarrow & & \downarrow \\ D & \longrightarrow & E & \longrightarrow & F \end{array}$
fong-spivak-seven-sketches_000_0339 — (dup)	237	7f1e85c6-4350-401f-a960-eb8a0176e786-237_167_371_2131_873.jpg	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\ A' & \longrightarrow & B' & \longrightarrow & C' \end{array}$	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\ A' & \longrightarrow & B' & \longrightarrow & C' \end{array}$	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\ A' & \longrightarrow & B' & \longrightarrow & C' \end{array}$
fong-spivak-seven-sketches_000_0340 — (dup)	238	7f1e85c6-4350-401f-a960-eb8a0176e786-238_210_606_701_753.jpg	$\begin{array}{ccc} A & \xrightarrow{\text{id}_A} & A \\ \text{id}_A \downarrow & \lrcorner & \downarrow f \\ A & \xrightarrow{f} & B \end{array} \quad \begin{array}{ccc} A & \xrightarrow{f} & B \\ f \downarrow & \lrcorner & \downarrow \text{id}_B \\ B & \xrightarrow{\text{id}_B} & B \end{array}$	$\begin{array}{ccc} A & \xrightarrow{\text{id}_A} & A \\ \text{id}_A \downarrow & \lrcorner & \downarrow f \\ A & \xrightarrow{f} & B \end{array} \quad \begin{array}{ccc} A & \xrightarrow{f} & B \\ f \downarrow & \lrcorner & \downarrow \text{id}_B \\ B & \xrightarrow{\text{id}_B} & B \end{array}$	$\begin{array}{ccc} A & \xrightarrow{\text{id}_A} & A \\ \text{id}_A \downarrow & \lrcorner & \downarrow f \\ A & \xrightarrow{f} & B \end{array} \quad \begin{array}{ccc} A & \xrightarrow{f} & B \\ f \downarrow & \lrcorner & \downarrow \text{id}_B \\ B & \xrightarrow{\text{id}_B} & B \end{array}$
fong-spivak-seven-sketches_000_0341 — (dup)	238	7f1e85c6-4350-401f-a960-eb8a0176e786-238_210_217_1419_995.jpg	$\begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ i' \downarrow \cong & \lrcorner & \cong \downarrow i \\ A & \xrightarrow{f} & B \end{array}$	$\begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ i' \downarrow \cong & \lrcorner & \cong \downarrow i \\ A & \xrightarrow{f} & B \end{array}$	$\begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ i' \downarrow \cong & \lrcorner & \cong \downarrow i \\ A & \xrightarrow{f} & B \end{array}$
fong-spivak-seven-sketches_000_0342 — (dup)	238	7f1e85c6-4350-401f-a960-eb8a0176e786-238_208_190_1747_1009.jpg	$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \parallel & \lrcorner & \parallel \\ A & \xrightarrow{f} & B \end{array}$	$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \parallel & \lrcorner & \parallel \\ A & \xrightarrow{f} & B \end{array}$	$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \parallel & \lrcorner & \parallel \\ A & \xrightarrow{f} & B \end{array}$

fong-spivak-seven-sketches-2010-0343 — (dup)	238	7f1e85c6-4350-401f-a960-eb8a0176e786-238_195_231_2077_943.jpg			
fong-spivak-seven-sketches-2010-0344 — (dup)	239	7f1e85c6-4350-401f-a960-eb8a0176e786-239_186_274_796_923.jpg			
fong-spivak-seven-sketches-2010-0345 — (dup)	240	7f1e85c6-4350-401f-a960-eb8a0176e786-240_226_312_1448_579.jpg			
fong-spivak-seven-sketches-2010-0346 — (dup)	240	7f1e85c6-4350-401f-a960-eb8a0176e786-240_235_380_1446_1158.jpg			
fong-spivak-seven-sketches-2010-0347 — (dup)	245	7f1e85c6-4350-401f-a960-eb8a0176e786-245_158_253_344_932.jpg			
fong-spivak-seven-sketches-2010-0348 — (dup)	245	7f1e85c6-4350-401f-a960-eb8a0176e786-245_133_629_683_744.jpg			
fong-spivak-seven-sketches-2010-0349 — (dup)	245	7f1e85c6-4350-401f-a960-eb8a0176e786-245_299_450_928_835.jpg			

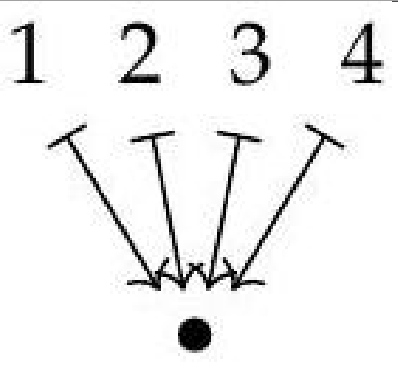
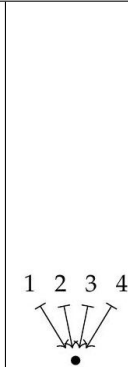
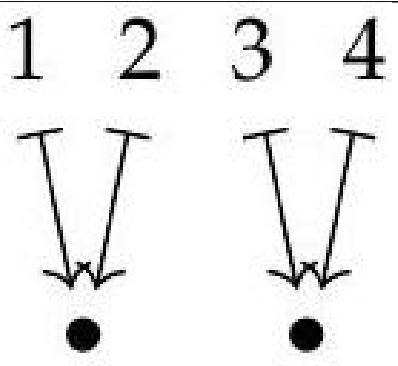
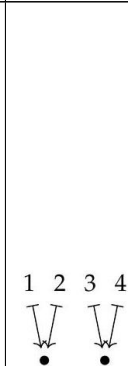
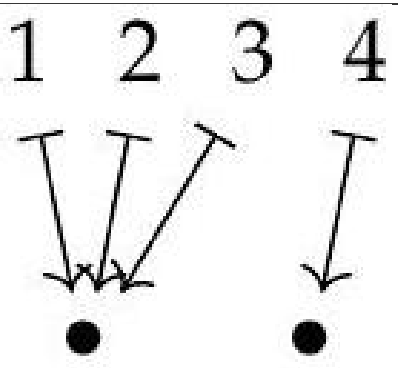
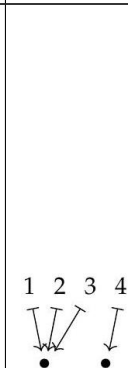
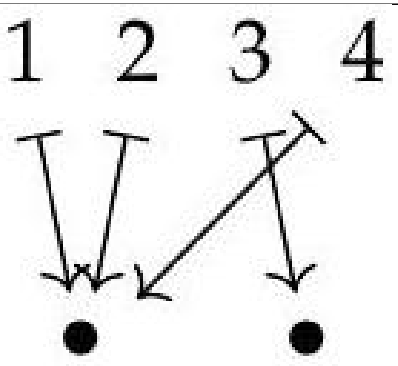
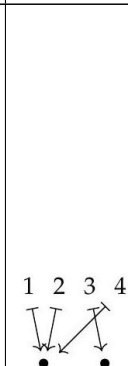
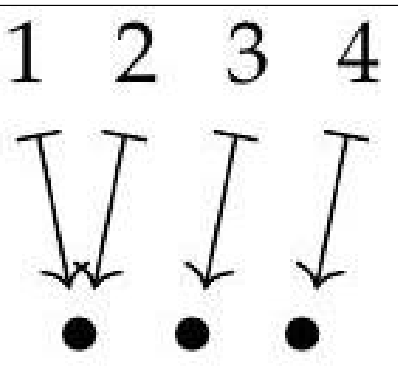
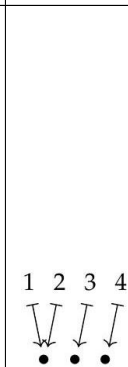
fong-spivak-seven-sketches-245_0350 — (dup)	245		7f1e85c6-4350-401f-a960-eb8a0176e786-245_195_1043_1484_538.jpg			
fong-spivak-seven-sketches-247_0351 — (dup)	247		7f1e85c6-4350-401f-a960-eb8a0176e786-247_316_947_466_591.jpg			
fong-spivak-seven-sketches-250_0352 — (dup)	250		7f1e85c6-4350-401f-a960-eb8a0176e786-250_434_753_466_688.jpg			
fong-spivak-seven-sketches-251_0353 — (dup)	251		7f1e85c6-4350-401f-a960-eb8a0176e786-251_407_1394_367_357.jpg			
fong-spivak-seven-sketches-251_0354 — (dup)	251		7f1e85c6-4350-401f-a960-eb8a0176e786-251_362_934_1602_604.jpg			
fong-spivak-seven-sketches-252_0355 — (dup)	252		7f1e85c6-4350-401f-a960-eb8a0176e786-252_534_520_339_796.jpg			

				glued section		
fong-spivak-seven-sketches-011_0356 — (dup)	252	7f1e85c6-4350-401f-a960-eb8a0176e786-252_541_271_1029_923.jpg		glued section		glued section
fong-spivak-seven-sketches-011_0357 — (dup)	253	7f1e85c6-4350-401f-a960-eb8a0176e786-253_441_620_1191_748.jpg				
fong-spivak-seven-sketches-011_0358 — (dup)	261	7f1e85c6-4350-401f-a960-eb8a0176e786-261_206_301_1722_912.jpg	$\begin{array}{ccc} \forall_t p & \longrightarrow & 1 \\ \downarrow \lrcorner & & \downarrow \text{true}^T \\ S & \xrightarrow{p'} & \Omega^T \end{array}$	$\begin{array}{ccc} \forall_t p & \longrightarrow & 1 \\ \downarrow \lrcorner & & \downarrow \text{true}^T \\ S & \xrightarrow{p'} & \Omega^T \end{array}$	$\begin{array}{ccc} \forall_t p & \longrightarrow & 1 \\ \downarrow \lrcorner & & \downarrow \text{true}^T \\ S & \xrightarrow{p'} & \Omega^T \end{array}$	
fong-spivak-seven-sketches-011_0359 — (dup)	262	7f1e85c6-4350-401f-a960-eb8a0176e786-262_186_403_860_860.jpg	$\begin{array}{ccc} \{S \times T \mid p\} & \twoheadrightarrow & S \times T \\ \downarrow & & \downarrow \pi_S \\ \exists_t p & \twoheadrightarrow & S \end{array}$	$\begin{array}{ccc} \{S \times T \mid p\} & \twoheadrightarrow & S \times T \\ \downarrow & & \downarrow \pi_S \\ \exists_t p & \twoheadrightarrow & S \end{array}$	$\begin{array}{ccc} \{S \times T \mid p\} & \twoheadrightarrow & S \times T \\ \downarrow & & \downarrow \pi_S \\ \exists_t p & \twoheadrightarrow & S \end{array}$	
fong-spivak-seven-sketches-011_0360 — (dup)	271	7f1e85c6-4350-401f-a960-eb8a0176e786-271_262_1253_1774_450.jpg				

fong-spivak-seven-sketches_711_0361 — (dup)	271	7f1e85c6-4350-401f-a960-eb8a0176e786-271_224_296_2272_971.jpg	(12) =  (1)(2) = 	(12) =  (1)(2) = 	(12) =  (1)(2) = 
fong-spivak-seven-sketches_712_0362 — (dup)	272	7f1e85c6-4350-401f-a960-eb8a0176e786-272_574_1203_322_516.jpg			
fong-spivak-seven-sketches_713_0363 — (dup)	273	7f1e85c6-4350-401f-a960-eb8a0176e786-273_174_165_1946_593.jpg	 		
fong-spivak-seven-sketches_714_0364 — (dup)	273	7f1e85c6-4350-401f-a960-eb8a0176e786-273_176_161_1946_891.jpg	 		

fong-spivak-seven-sketches_710_0365 — (dup)	273	7f1e85c6-4350-401f-a960-eb8a0176e786-273_176_163_1946_1186.jpg			
fong-spivak-seven-sketches_710_0366 — (dup)	273	7f1e85c6-4350-401f-a960-eb8a0176e786-273_176_163_1946_1480.jpg			
fong-spivak-seven-sketches_710_0367 — (dup)	274	7f1e85c6-4350-401f-a960-eb8a0176e786-274_321_1360_398_389.jpg			
fong-spivak-seven-sketches_710_0368 — (dup)	275	7f1e85c6-4350-401f-a960-eb8a0176e786-275_355_557_337_799.jpg			
fong-spivak-seven-sketches_710_0369 — (dup)	275	7f1e85c6-4350-401f-a960-eb8a0176e786-275_276_645_1050_753.jpg			
fong-spivak-seven-sketches_710_0370 — (dup)	275	7f1e85c6-4350-401f-a960-eb8a0176e786-275_570_1489_1937_371.jpg			

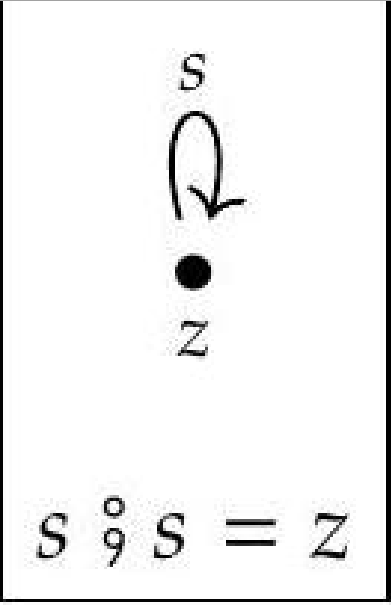
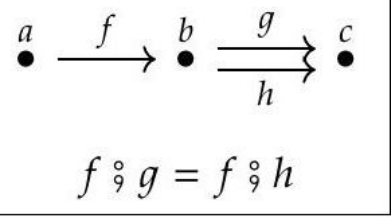
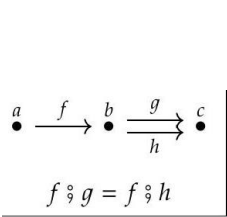
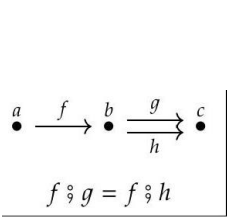
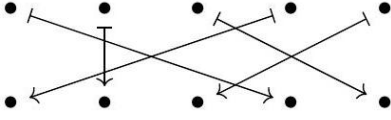
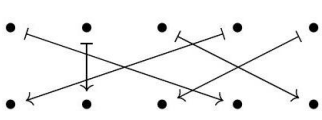
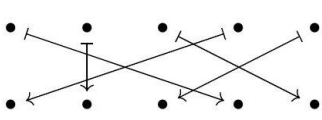
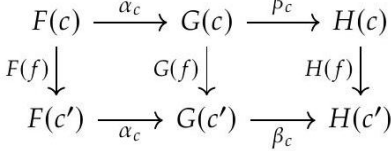
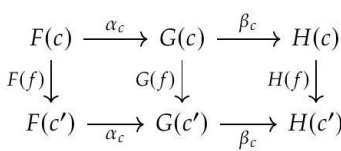
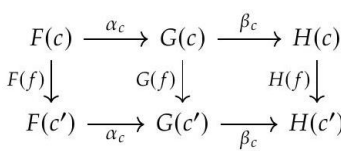
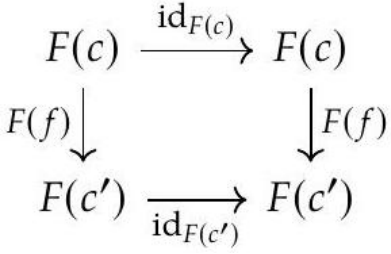
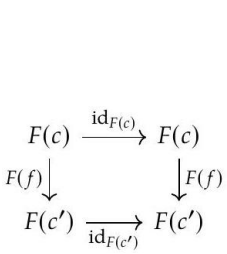
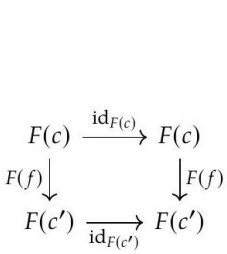
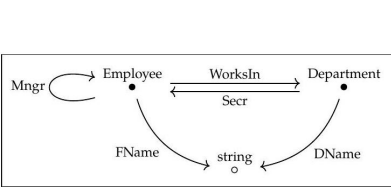
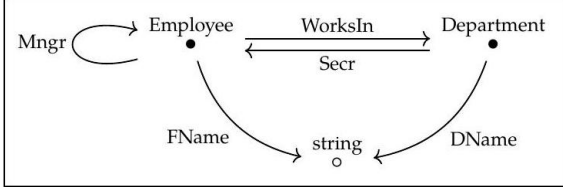
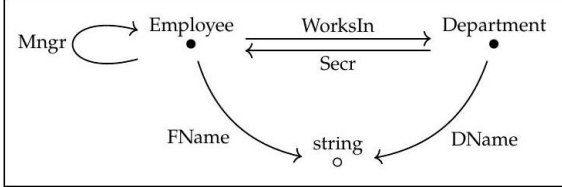
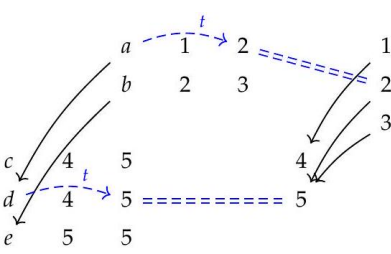
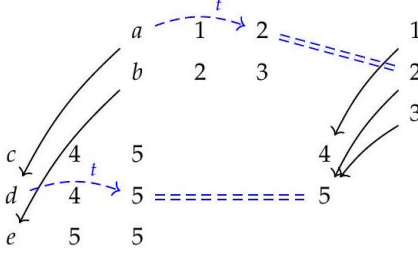
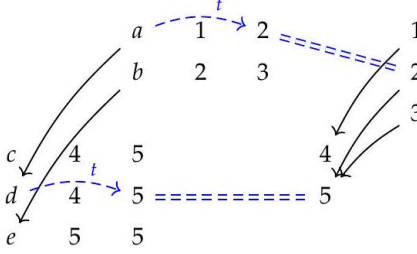
fong-spivak-seven-sketches-710_0371 — (dup)	276		7f1e85c6-4350-401f-a960-eb8a0176e786-276_369_826_391_663.jpg			
fong-spivak-seven-sketches-710_0372 — (dup)	276		7f1e85c6-4350-401f-a960-eb8a0176e786-276_274_878_887_638.jpg			
fong-spivak-seven-sketches-710_0373 — (dup)	276		7f1e85c6-4350-401f-a960-eb8a0176e786-276_281_869_1609_683.jpg			
fong-spivak-seven-sketches-710_0374 — (dup)	278		7f1e85c6-4350-401f-a960-eb8a0176e786-278_172_541_1041_846.jpg			
fong-spivak-seven-sketches-710_0375 — (dup)	278		7f1e85c6-4350-401f-a960-eb8a0176e786-278_174_543_1520_846.jpg			
fong-spivak-seven-sketches-710_0376 — (dup)	278		7f1e85c6-4350-401f-a960-eb8a0176e786-278_131_222_2353_448.jpg			
fong-spivak-seven-sketches-710_0377 — (dup)	278		7f1e85c6-4350-401f-a960-eb8a0176e786-278_129_247_2355_769.jpg			
fong-spivak-seven-sketches-710_0378 — (dup)	278		7f1e85c6-4350-401f-a960-eb8a0176e786-278_129_244_2355_1115.jpg			
fong-spivak-seven-sketches-710_0379 — (dup)	278		7f1e85c6-4350-401f-a960-eb8a0176e786-278_129_247_2355_1459.jpg			

fong-spivak-seven-sketches_0110_0380 — (dup)	279	7f1e85c6-4350-401f-a960-eb8a0176e786-279_131_140_674_534.jpg			
fong-spivak-seven-sketches_0110_0381 — (dup)	279	7f1e85c6-4350-401f-a960-eb8a0176e786-279_129_138_676_771.jpg			
fong-spivak-seven-sketches_0110_0382 — (dup)	279	7f1e85c6-4350-401f-a960-eb8a0176e786-279_129_138_676_1007.jpg			
fong-spivak-seven-sketches_0110_0383 — (dup)	279	7f1e85c6-4350-401f-a960-eb8a0176e786-279_129_138_676_1244.jpg			
fong-spivak-seven-sketches_0110_0384 — (dup)	279	7f1e85c6-4350-401f-a960-eb8a0176e786-279_129_140_676_1480.jpg			

fong-spivak-seven-sketches_010_0385 — (dup)	279	7f1e85c6-4350-401f-a960-eb8a0176e786-279_129_229_1025_765.jpg	$c =$	$c =$	$c =$
fong-spivak-seven-sketches_010_0386 — (dup)	279	7f1e85c6-4350-401f-a960-eb8a0176e786-279_129_258_1025_1217.jpg	$g!(c) =$	$g!(c) =$	$g!(c) =$
fong-spivak-seven-sketches_010_0387 — (dup)	279	7f1e85c6-4350-401f-a960-eb8a0176e786-279_127_188_1281_1025.jpg	$d =$	$d =$	$d =$
fong-spivak-seven-sketches_010_0388 — (dup)	279	7f1e85c6-4350-401f-a960-eb8a0176e786-279_127_210_1538_1014.jpg	$e =$	$e =$	$e =$
fong-spivak-seven-sketches_010_0389 — (dup)	279	7f1e85c6-4350-401f-a960-eb8a0176e786-279_131_262_1792_790.jpg	$g^*(d) =$	$g^*(d) =$	$g^*(d) =$
fong-spivak-seven-sketches_010_0390 — (dup)	279	7f1e85c6-4350-401f-a960-eb8a0176e786-279_131_283_1792_1163.jpg	$g^*(e) =$	$g^*(e) =$	$g^*(e) =$
fong-spivak-seven-sketches_010_0391 — (dup)	280	7f1e85c6-4350-401f-a960-eb8a0176e786-280_333_828_1005_661.jpg	$P :=$ $\xleftrightarrow{g} \xleftrightarrow{f} =: Q$	$P :=$ $\xleftrightarrow{g} \xleftrightarrow{f} =: Q$	$P :=$ $\xleftrightarrow{g} \xleftrightarrow{f} =: Q$

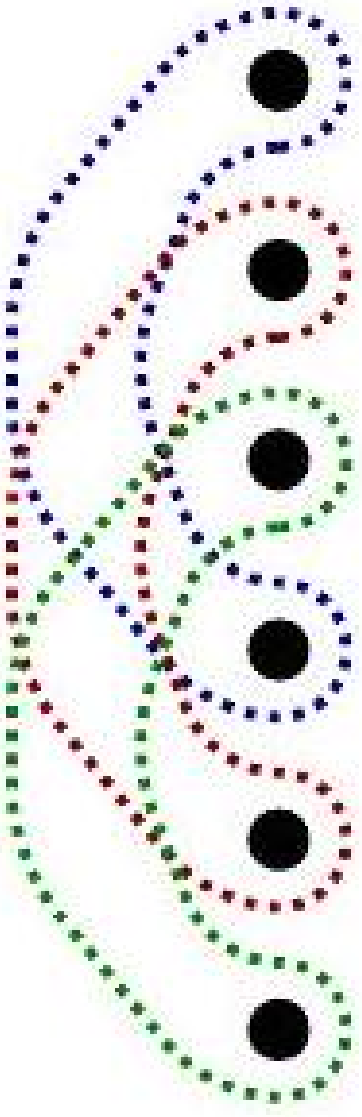
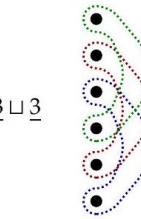
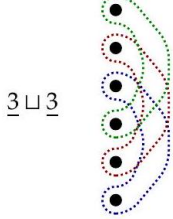
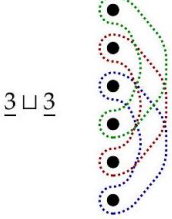
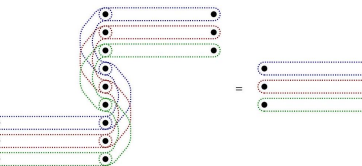
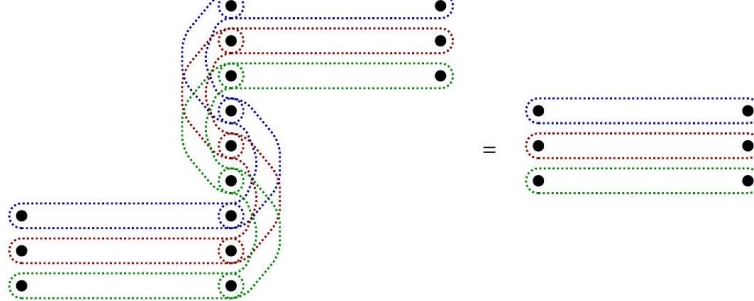
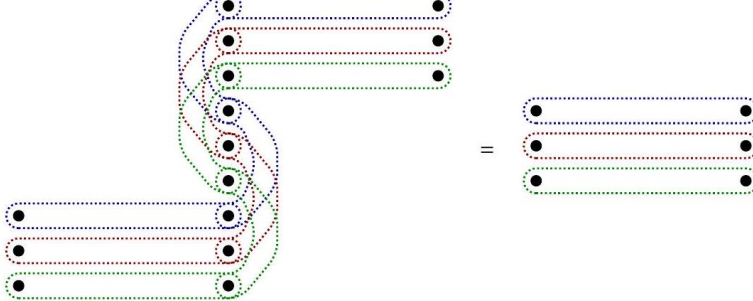
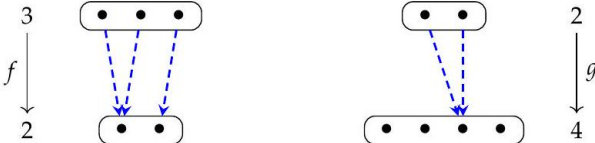
fong-spivak-seven-sketches-280_0392 — (dup)	280	7f1e85c6-4350-401f-a960-eb8a0176e786-280_328_419_2192_873.jpg	$X := \begin{array}{ c } \hline a_1 \bullet \qquad \qquad \qquad c_2 \bullet \\ \hline \qquad \qquad \qquad c_1 \bullet \\ \hline \end{array}$ $f \downarrow$ $Y := \begin{array}{ c } \hline a \bullet \qquad b \bullet \qquad c \bullet \\ \hline \end{array}$	$X := \begin{array}{ c } \hline a_1 \bullet \qquad \qquad \qquad c_2 \bullet \\ \hline \qquad \qquad \qquad c_1 \bullet \\ \hline \end{array}$ $f \downarrow$ $Y := \begin{array}{ c } \hline a \bullet \qquad b \bullet \qquad c \bullet \\ \hline \end{array}$	$X := \begin{array}{ c } \hline a_1 \bullet \qquad \qquad \qquad c_2 \bullet \\ \hline \qquad \qquad \qquad c_1 \bullet \\ \hline \end{array}$ $f \downarrow$ $Y := \begin{array}{ c } \hline a \bullet \qquad b \bullet \qquad c \bullet \\ \hline \end{array}$
fong-spivak-seven-sketches-281_0393 — (dup)	281	7f1e85c6-4350-401f-a960-eb8a0176e786-281_473_1278_982_437.jpg			
fong-spivak-seven-sketches-285_0394 — (dup)	285	7f1e85c6-4350-401f-a960-eb8a0176e786-285_326_550_290_844.jpg			
fong-spivak-seven-sketches-285_0395 — (dup)	285	7f1e85c6-4350-401f-a960-eb8a0176e786-285_229_314_1367_962.jpg			
fong-spivak-seven-sketches-285_0396 — (dup)	285	7f1e85c6-4350-401f-a960-eb8a0176e786-285_104_432_2396_860.jpg			
fong-spivak-seven-sketches-290_0397 — (dup)	290	7f1e85c6-4350-401f-a960-eb8a0176e786-290_274_428_523_860.jpg	$\text{Free_square} := \begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & & \downarrow h \\ C & \xrightarrow{i} & D \end{array}$ no equations	$\text{Free_square} := \begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & & \downarrow h \\ C & \xrightarrow{i} & D \end{array}$ no equations	$\text{Free_square} := \begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & & \downarrow h \\ C & \xrightarrow{i} & D \end{array}$ no equations
fong-spivak-seven-sketches-290_0398 — (dup)	290	7f1e85c6-4350-401f-a960-eb8a0176e786-290_208_328_1557_912.jpg	$\mathcal{D} := \begin{array}{ c } \hline s \curvearrowright \\ \hline \bullet \\ \hline z \\ \hline \end{array}$ $s \circ s \circ s \circ s \circ s = s \circ s$	$\mathcal{D} := \begin{array}{ c } \hline s \curvearrowright \\ \hline \bullet \\ \hline z \\ \hline \end{array}$ $s \circ s \circ s \circ s \circ s = s \circ s$	$\mathcal{D} := \begin{array}{ c } \hline s \curvearrowright \\ \hline \bullet \\ \hline z \\ \hline \end{array}$ $s \circ s \circ s \circ s \circ s = s \circ s$

fong-spivak-seven-sketches_0400.jpg	290 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-290_247_1285_1968_434.jpg					
fong-spivak-seven-sketches_0400.jpg	291 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-291_335_285_1774_405.jpg					
fong-spivak-seven-sketches_0401.jpg	291 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-291_337_283_1774_932.jpg					
fong-spivak-seven-sketches_0402.jpg	291 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-291_337_287_1774_1455.jpg					

fong-spivak-seven-sketches_0443 — (dup)	292	7f1e85c6-4350-401f-a960-eb8a0176e786-292_204_133_1991_410.jpg			
fong-spivak-seven-sketches_0444 — (dup)	292	7f1e85c6-4350-401f-a960-eb8a0176e786-292_165_294_2016_837.jpg			
fong-spivak-seven-sketches_0445 — (dup)	292	7f1e85c6-4350-401f-a960-eb8a0176e786-292_124_416_2376_910.jpg			
fong-spivak-seven-sketches_0446 — (dup)	293	7f1e85c6-4350-401f-a960-eb8a0176e786-293_186_448_1271_889.jpg			
fong-spivak-seven-sketches_0447 — (dup)	293	7f1e85c6-4350-401f-a960-eb8a0176e786-293_190_292_1638_973.jpg			
fong-spivak-seven-sketches_0448 — (dup)	294	7f1e85c6-4350-401f-a960-eb8a0176e786-294_251_733_373_710.jpg			
fong-spivak-seven-sketches_0449 — (dup)	294	7f1e85c6-4350-401f-a960-eb8a0176e786-294_330_550_1529_844.jpg			

fong-spivak-seven-sketches-294_0410 — (dup)	294	7f1e85c6-4350-401f-a960-eb8a0176e786-294_353_830_2140_656.jpg			
fong-spivak-seven-sketches-295_0411 — (dup)	295	7f1e85c6-4350-401f-a960-eb8a0176e786-295_333_627_333_1138.jpg			
fong-spivak-seven-sketches-295_0412 — (dup)	295	7f1e85c6-4350-401f-a960-eb8a0176e786-295_176_692_2156_733.jpg			
fong-spivak-seven-sketches-298_0413 — (dup)	298	7f1e85c6-4350-401f-a960-eb8a0176e786-298_372_672_438_782.jpg			
fong-spivak-seven-sketches-300_0414 — (dup)	300	7f1e85c6-4350-401f-a960-eb8a0176e786-300_326_851_1357_652.jpg			
fong-spivak-seven-sketches-301_0415 — (dup)	301	7f1e85c6-4350-401f-a960-eb8a0176e786-301_276_663_1767_747.jpg			

fong-spivak-seven-sketches_0416	302	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-302_310_59_2195_914.jpg		105	
--	----------------	------------------	---	---	----------------	--

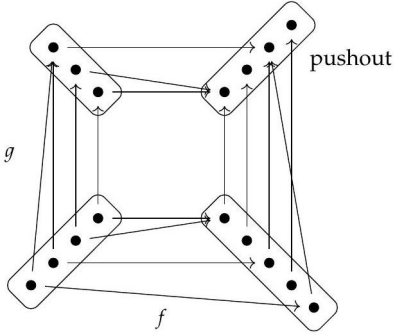
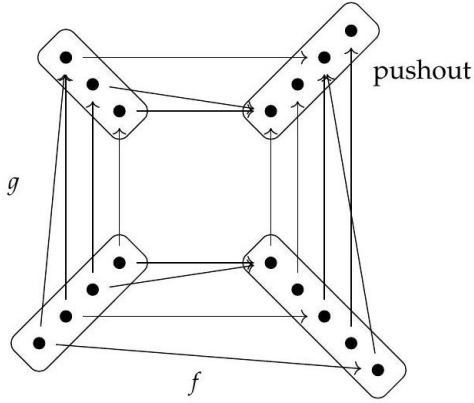
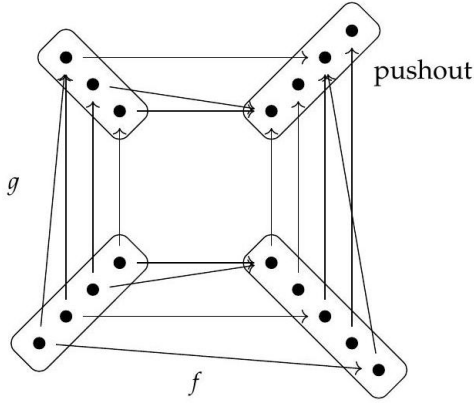
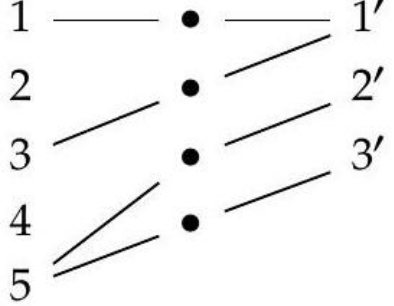
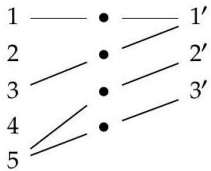
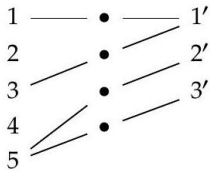
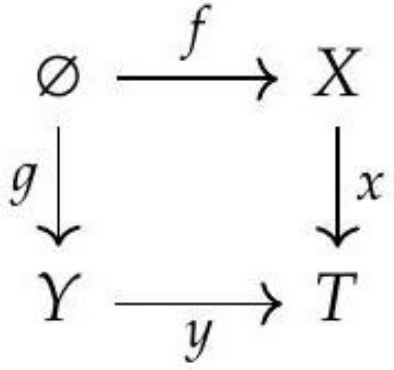
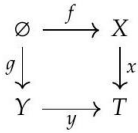
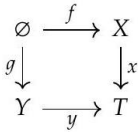
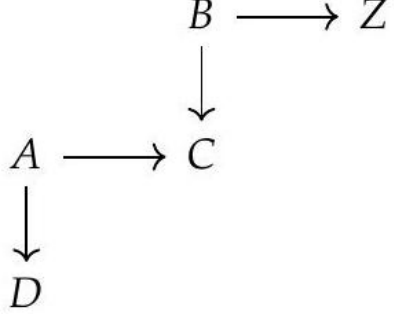
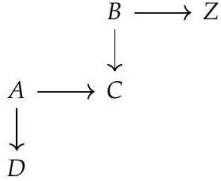
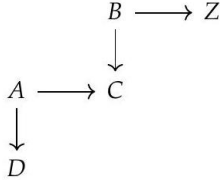
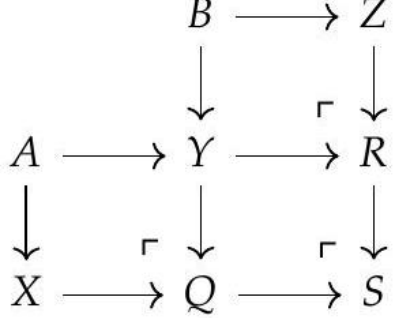
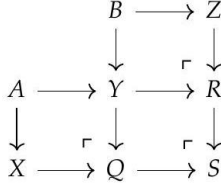
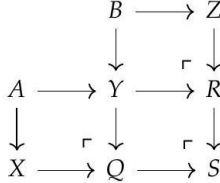
fong-spivak-seven-sketches-0417	302 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-302_294_102_2204_1163.jpg			
fong-spivak-seven-sketches-0418	303 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-303_308_534_324_851.jpg			
fong-spivak-seven-sketches-0419	303 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-303_443_1077_729_579.jpg			
fong-spivak-seven-sketches-0420	305 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-305_195_783_452_726.jpg			

fong-spivak-seven-sketches-0421 — (dup)	305	7f1e85c6-4350-401f-a960-eb8a0176e786-305_192_462_729_891.jpg			
fong-spivak-seven-sketches-0422 — (dup)	305	7f1e85c6-4350-401f-a960-eb8a0176e786-305_195_1097_1102_568.jpg			
fong-spivak-seven-sketches-0423 — (dup)	305	7f1e85c6-4350-401f-a960-eb8a0176e786-305_192_545_1407_839.jpg			
fong-spivak-seven-sketches-0424 — (dup)	306	7f1e85c6-4350-401f-a960-eb8a0176e786-306_167_561_2353_523.jpg			
fong-spivak-seven-sketches-0425 — (dup)	306	7f1e85c6-4350-401f-a960-eb8a0176e786-306_138_491_2378_1138.jpg			
fong-spivak-seven-sketches-0426 — (dup)	307	7f1e85c6-4350-401f-a960-eb8a0176e786-307_170_932_317_611.jpg			
fong-spivak-seven-sketches-0427 — (dup)	307	7f1e85c6-4350-401f-a960-eb8a0176e786-307_136_262_808_686.jpg			
fong-spivak-seven-sketches-0428 — (dup)	307	7f1e85c6-4350-401f-a960-eb8a0176e786-307_235_276_708_1190.jpg			
fong-spivak-seven-sketches-0429 — (dup)	309	7f1e85c6-4350-401f-a960-eb8a0176e786-309_188_652_389_751.jpg			
fong-spivak-seven-sketches-0430 — (dup)	309	7f1e85c6-4350-401f-a960-eb8a0176e786-309_201_627_1880_765.jpg			

fong-spivak-seven-sketches-310_0431 — (dup)	310		7f1e85c6-4350-401f-a960-eb8a0176e786-310_86_187_370_739.jpg			
fong-spivak-seven-sketches-310_0432 — (dup)	310		7f1e85c6-4350-401f-a960-eb8a0176e786-310_86_187_554_739.jpg			
fong-spivak-seven-sketches-310_0433 — (dup)	310		7f1e85c6-4350-401f-a960-eb8a0176e786-310_142_408_834_915.jpg	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$
fong-spivak-seven-sketches-310_0434 — (dup)	310		7f1e85c6-4350-401f-a960-eb8a0176e786-310_88_432_1016_904.jpg	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$
fong-spivak-seven-sketches-310_0435 — (dup)	310		7f1e85c6-4350-401f-a960-eb8a0176e786-310_86_146_1242_700.jpg	$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$	$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$	$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$
fong-spivak-seven-sketches-310_0436 — (dup)	310		7f1e85c6-4350-401f-a960-eb8a0176e786-310_271_537_1143_1005.jpg			
fong-spivak-seven-sketches-311_0437 — (dup)	311		7f1e85c6-4350-401f-a960-eb8a0176e786-311_143_498_611_905.jpg	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$
fong-spivak-seven-sketches-311_0438 — (dup)	311		7f1e85c6-4350-401f-a960-eb8a0176e786-311_88_511_799_900.jpg	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$
fong-spivak-seven-sketches-311_0439 — (dup)	311		7f1e85c6-4350-401f-a960-eb8a0176e786-311_229_609_928_1005.jpg			
fong-spivak-seven-sketches-311_0440 — (dup)	311		7f1e85c6-4350-401f-a960-eb8a0176e786-311_152_1005_1294_652.jpg			
fong-spivak-seven-sketches-311_0441 — (dup)	311		7f1e85c6-4350-401f-a960-eb8a0176e786-311_63_878_1493_715.jpg			

fong-spivak-seven-sketches-311_0442 — (dup)	311	7f1e85c6-4350-401f-a960-eb8a0176e786-311_538_950_1624_710.jpg			
fong-spivak-seven-sketches-311_0443 — (dup)	311	7f1e85c6-4350-401f-a960-eb8a0176e786-311_97_287_2357_502.jpg			
fong-spivak-seven-sketches-311_0444 — (dup)	311	7f1e85c6-4350-401f-a960-eb8a0176e786-311_199_837_2305_896.jpg			
fong-spivak-seven-sketches-312_0445 — (dup)	312	7f1e85c6-4350-401f-a960-eb8a0176e786-312_43_344_410_946.jpg			
fong-spivak-seven-sketches-312_0446 — (dup)	312	7f1e85c6-4350-401f-a960-eb8a0176e786-312_100_149_679_1045.jpg			
fong-spivak-seven-sketches-312_0447 — (dup)	312	7f1e85c6-4350-401f-a960-eb8a0176e786-312_181_389_1271_554.jpg			
fong-spivak-seven-sketches-312_0448 — (dup)	312	7f1e85c6-4350-401f-a960-eb8a0176e786-312_181_319_1271_1020.jpg			
fong-spivak-seven-sketches-312_0449 — (dup)	312	7f1e85c6-4350-401f-a960-eb8a0176e786-312_154_249_1290_1425.jpg			
fong-spivak-seven-sketches-312_0450 — (dup)	312	7f1e85c6-4350-401f-a960-eb8a0176e786-312_181_378_1588_559.jpg			

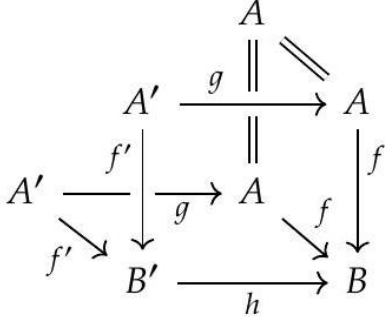
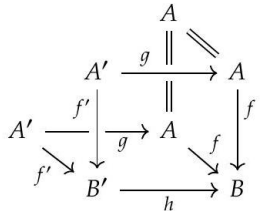
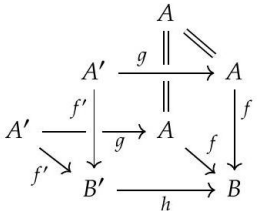
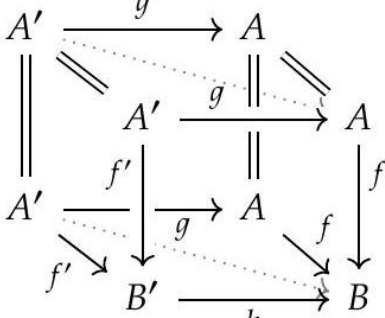
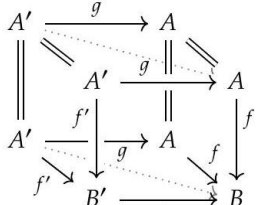
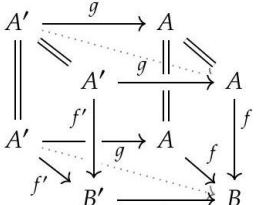
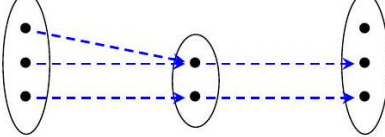
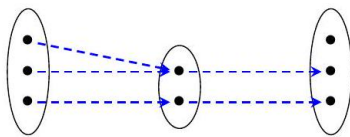
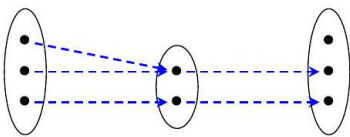
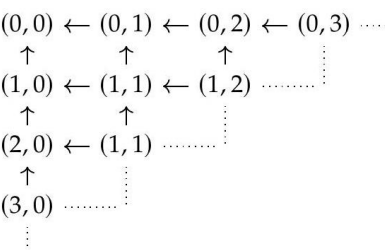
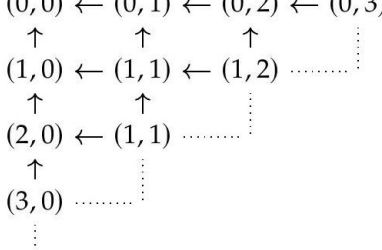
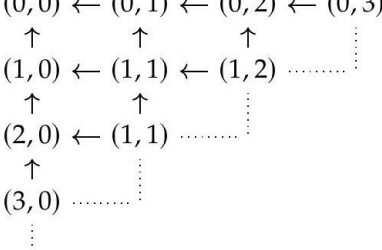
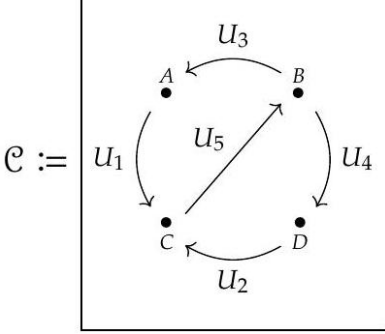
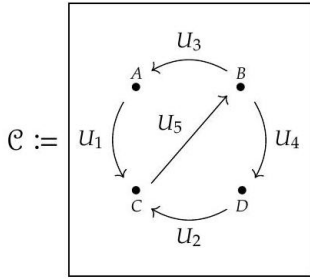
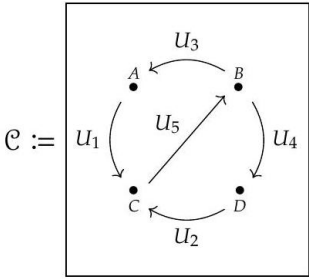
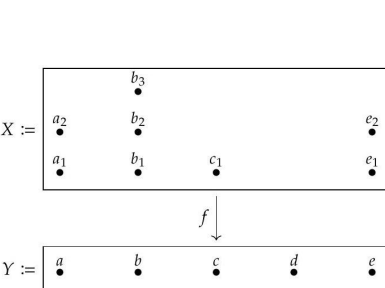
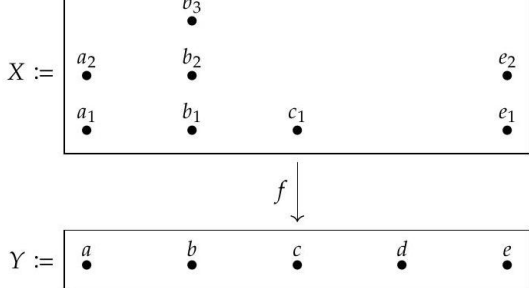
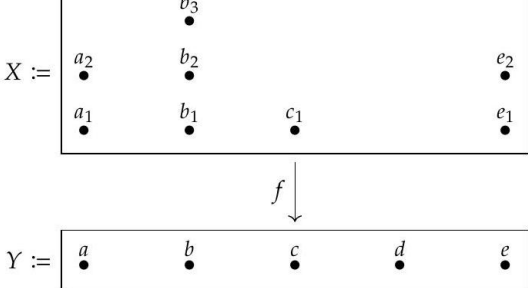
fong-spivak-seven-sketches-114_0451 — (dup)	312	7f1e85c6-4350-401f-a960-eb8a0176e786-312_179_324_1593_1016.jpg			
fong-spivak-seven-sketches-114_0452 — (dup)	312	7f1e85c6-4350-401f-a960-eb8a0176e786-312_154_249_1611_1421.jpg			
fong-spivak-seven-sketches-114_0453 — (dup)	316	no image for this region (498 in zip)	(no tex.zip crop for this row)	$\begin{aligned} [f, g]: A \sqcup B &\longrightarrow T \\ \text{apple1} &\mapsto a \\ \text{banana1} &\mapsto b \\ \text{pear1} &\mapsto p \\ \text{cherry1} &\mapsto c \\ \text{orange1} &\mapsto o \\ \text{apple2} &\mapsto e \\ \text{tomato2} &\mapsto o \\ \text{mango2} &\mapsto o. \end{aligned}$	$\begin{aligned} [f, g]: A \sqcup B &\longrightarrow T \\ \text{apple1} &\mapsto a \\ \text{banana1} &\mapsto b \\ \text{pear1} &\mapsto p \\ \text{cherry1} &\mapsto c \\ \text{orange1} &\mapsto o \\ \text{apple2} &\mapsto e \\ \text{tomato2} &\mapsto o \\ \text{mango2} &\mapsto o. \end{aligned}$
fong-spivak-seven-sketches-114_0454 — (dup)	316	7f1e85c6-4350-401f-a960-eb8a0176e786-316_308_391_1785_921.jpg			
fong-spivak-seven-sketches-114_0455 — (dup)	316	7f1e85c6-4350-401f-a960-eb8a0176e786-316_192_317_2258_959.jpg			
fong-spivak-seven-sketches-114_0456 — (dup)	317	7f1e85c6-4350-401f-a960-eb8a0176e786-317_283_597_1088_819.jpg			

fong-spivak-seven-sketches-110_0457 — (dup)	318	7f1e85c6-4350-401f-a960-eb8a0176e786-318_527_615_387_769.jpg			
fong-spivak-seven-sketches-110_0458 — (dup)	318	7f1e85c6-4350-401f-a960-eb8a0176e786-318_222_274_1100_939.jpg			
fong-spivak-seven-sketches-110_0459 — (dup)	318	7f1e85c6-4350-401f-a960-eb8a0176e786-318_172_179_1824_1029.jpg			
fong-spivak-seven-sketches-110_0460 — (dup)	318	7f1e85c6-4350-401f-a960-eb8a0176e786-318_238_287_2269_654.jpg			
fong-spivak-seven-sketches-110_0461 — (dup)	318	7f1e85c6-4350-401f-a960-eb8a0176e786-318_242_290_2267_1208.jpg			

fong-spivak-seven-sketches-319_0462 — (dup)	319	7f1e85c6-4350-401f-a960-eb8a0176e786-319_240_287_376_932.jpg			
fong-spivak-seven-sketches-319_0463 — (dup)	319	7f1e85c6-4350-401f-a960-eb8a0176e786-319_597_457_1290_846.jpg			
fong-spivak-seven-sketches-320_0464 — (dup)	320	7f1e85c6-4350-401f-a960-eb8a0176e786-320_310_1260_609_446.jpg			
fong-spivak-seven-sketches-320_0465 — (dup)	320	7f1e85c6-4350-401f-a960-eb8a0176e786-320_59_704_1131_726.jpg			
fong-spivak-seven-sketches-320_0466 — (dup)	320	7f1e85c6-4350-401f-a960-eb8a0176e786-320_172_278_1308_939.jpg			
fong-spivak-seven-sketches-320_0467 — (dup)	320	7f1e85c6-4350-401f-a960-eb8a0176e786-320_172_269_1602_943.jpg			

fong-spivak-seven-sketches-240_0468 — (dup)	320	7f1e85c6-4350-401f-a960-eb8a0176e786-320_240_308_1964_923.jpg			
fong-spivak-seven-sketches-240_0469 — (dup)	320	7f1e85c6-4350-401f-a960-eb8a0176e786-320_93_353_2416_900.jpg			
fong-spivak-seven-sketches-321_0470 — (dup)	321	7f1e85c6-4350-401f-a960-eb8a0176e786-321_95_421_1910_867.jpg			
fong-spivak-seven-sketches-321_0471 — (dup)	321	7f1e85c6-4350-401f-a960-eb8a0176e786-321_70_618_2213_769.jpg			
fong-spivak-seven-sketches-322_0472 — (dup)	322	7f1e85c6-4350-401f-a960-eb8a0176e786-322_410_930_1256_613.jpg			
fong-spivak-seven-sketches-322_0473 — (dup)	322	7f1e85c6-4350-401f-a960-eb8a0176e786-322_410_968_1738_588.jpg			
fong-spivak-seven-sketches-322_0474 — (dup)	322	7f1e85c6-4350-401f-a960-eb8a0176e786-322_215_1032_2294_611.jpg			
fong-spivak-seven-sketches-323_0475 — (dup)	323	7f1e85c6-4350-401f-a960-eb8a0176e786-323_136_312_480_896.jpg			
fong-spivak-seven-sketches-323_0476 — (dup)	323	7f1e85c6-4350-401f-a960-eb8a0176e786-323_240_294_452_1348.jpg			

fong-spivak-seven-sketches-270_0484 — (dup)	324	7f1e85c6-4350-401f-a960-eb8a0176e786-324_249_382_2131_885.jpg			
fong-spivak-seven-sketches-270_0485 — (dup)	325	7f1e85c6-4350-401f-a960-eb8a0176e786-325_251_385_367_882.jpg			
fong-spivak-seven-sketches-270_0486 — (dup)	325	7f1e85c6-4350-401f-a960-eb8a0176e786-325_233_265_998_948.jpg			
fong-spivak-seven-sketches-270_0487 — (dup)	325	7f1e85c6-4350-401f-a960-eb8a0176e786-325_179_183_1697_1025.jpg			
fong-spivak-seven-sketches-270_0488 — (dup)	325	7f1e85c6-4350-401f-a960-eb8a0176e786-325_179_156_2183_1038.jpg			

fong-spivak-seven-sketches-326_0489 — (dup)	326_0489 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-326_283_335_376_615.jpg			
fong-spivak-seven-sketches-326_0490 — (dup)	326_0490 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-326_296_333_364_1208.jpg			
fong-spivak-seven-sketches-326_0491 — (dup)	326_0491 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-326_176_459_1122_846.jpg			
fong-spivak-seven-sketches-326_0492 — (dup)	326_0492 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-326_344_559_1914_833.jpg			
fong-spivak-seven-sketches-328_0493 — (dup)	328_0493 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-328_362_407_2034_878.jpg			
fong-spivak-seven-sketches-329_0494 — (dup)	329_0494 — (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-329_400_692_697_735.jpg			

fong-spivak-seven-sketches_329_0495 (dup)	329	7f1e85c6-4350-401f-a960-eb8a0176e786-329_398_690_1301_738.jpg			
fong-spivak-seven-sketches_329_0496 (dup)	329	7f1e85c6-4350-401f-a960-eb8a0176e786-329_367_1199_1982_505.jpg			
fong-spivak-seven-sketches_330_0497 (dup)	330	7f1e85c6-4350-401f-a960-eb8a0176e786-330_156_1129_412_511.jpg			
fong-spivak-seven-sketches_330_0498 (dup)	330	7f1e85c6-4350-401f-a960-eb8a0176e786-330_500_1072_787_572.jpg			
fong-spivak-seven-sketches_331_0499 (dup)	331	7f1e85c6-4350-401f-a960-eb8a0176e786-331_229_1118_579_516.jpg			